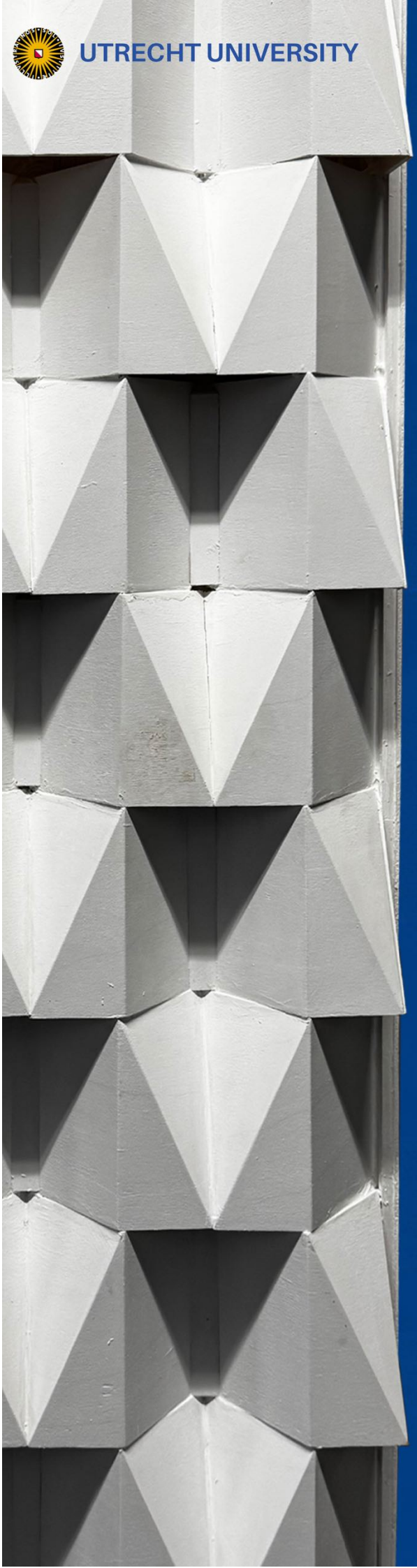


UTRECHT UNIVERSITY

23/10/2026

Graduate School of Life Sciences
MSc Bio Inspired Innovation

MAJOR INNOVATION PROJECT

FunctionalSkin

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Layman summary

Urban heat stress is on the rise thanks to increasing global temperatures and inadequate cooling mechanisms in urban environments. To combat this, new methods are needed that are low energy, easy to implement and sustainable. We investigated how a modular panel installed on a building's façade can help to passively cool the building's outside and the immediate surrounding environment. It was found that a combination of a geometry that promotes air circulation, and a coating that combines solar reflection, with radiating its heat into space, can make for such a functioning panel.

Tests under simulated conditions of a hot Dutch summer day, showed that the temperature of the exterior wall of a building can be up to 30 °C lower with a passively cooling façade panel in front, than when fully exposed to the sunlight. The installation of such a panel could reduce active cooling energy demands and reduce temperatures in surrounding urban areas, which are the area's most susceptible to extreme heat stress.

Furthermore, we investigated how this could be produced from biobased materials, like wood and fungal biomaterials. Some modifications, like coatings are needed to protect these natural materials from the outside conditions, like water and wind exposure. Nonetheless, it was shown through prototyping and experiments that a functioning passively cooling façade panel can be produced from these materials.

Meet the *FunctionalSkin* team

We are the FunctionalSkin team from Utrecht University (left to right, top to bottom): Ruben Langendoen, Ocker Hazelbag, Jochem ten Pas, Felipe Krebs and Alexandra Cooke-Kasyanto. Our five backgrounds and perspectives come together in one shared goal: how can a building's façade function like an urban living skin? Our façade panels help to cool buildings using similar techniques seen in nature, and do so completely passively – functioning without any need for energy!



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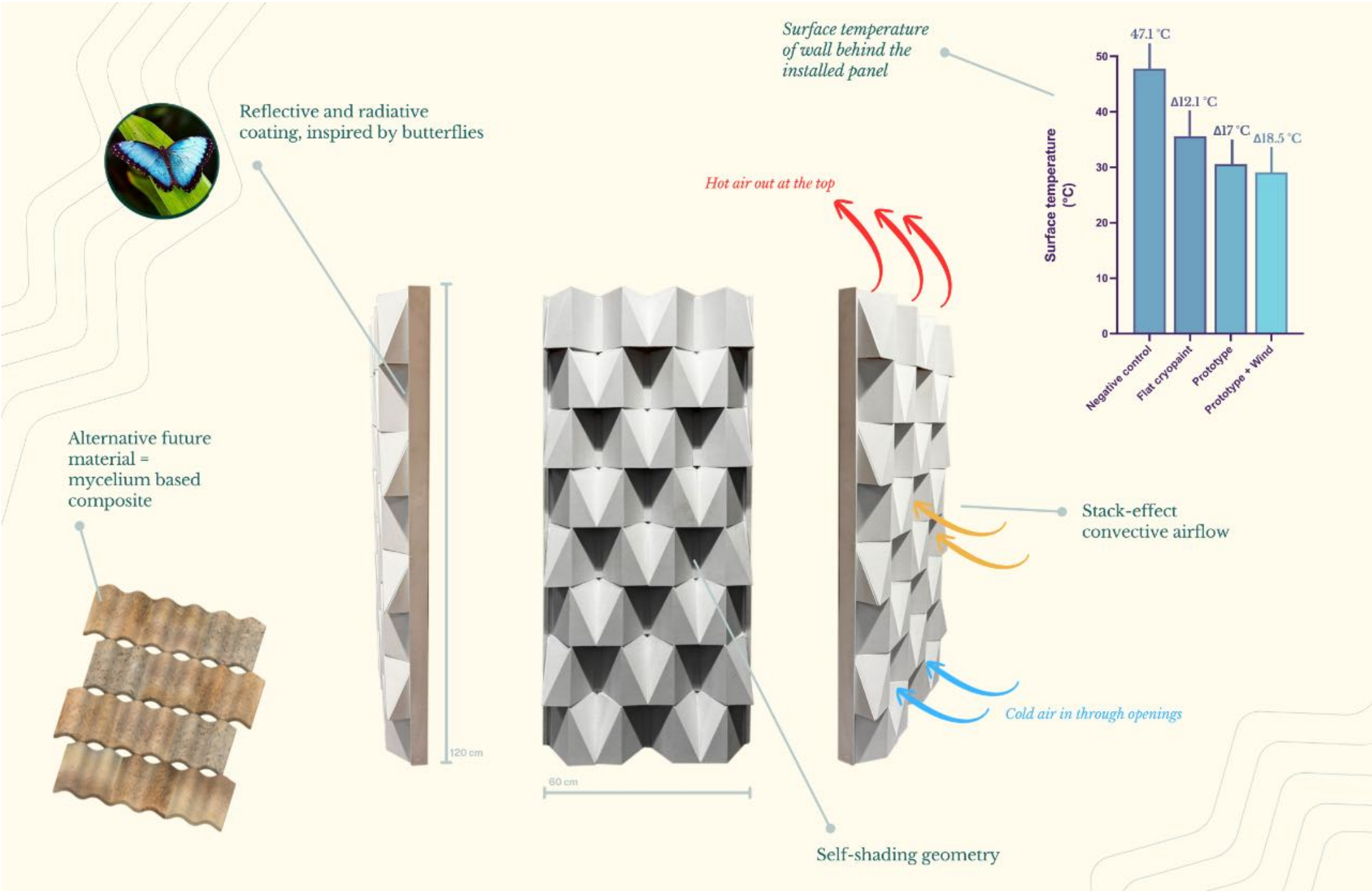
Abstract

Rising global temperatures intensify heat stress in urban environments and increase the need to cool down buildings in the Netherlands. This study investigates whether a façade panel can passively cool the outside of a building and its immediate surroundings. The panel combines a three-dimensional geometry, with cavities that remove hot air by convection, with a radiative cooling coating. Under simulated direct sunlight (800 W m^{-2}, 30 °C ambient temperature), we measured the temperature of the inner backwall behind the panel.

Compared with an unventilated bare backwall as a negative control, a flat panel with a radiative cooling coating already lowered the back-wall temperature from 64.0 °C to 46.2 °C. Adding the three-dimensional geometry and cavities reduced it further to 39.2 °C, and adding external airflow to simulate wind lowered it to 34.2 °C, only 4 °C above the ambient room temperature

Literature research and a Life Cycle Assessment (LCA) identified Douglas fir as the most suitable sustainable, biobased material, but plywood was chosen for prototyping purposes. We also tested heat-pressed mycelium-based composites (MBCs) as an alternative future material and showed that it can be formed into a three-dimensional shape. However, it needs more protection against outdoor weather conditions and less energy-intensive production before it could be implemented as a façade material.

Graphical Abstract



Theoretical background

Urban Heat Island effect

Rising global temperatures are impacting humanity and earth systems in many different ways. One of the challenges that arises in cities is the increasing severity of the Urban Heat Island (UHI) effect (Tong et al., 2021). This phenomenon is characterized by the difference in air- and surface temperatures between urban and rural environments. Several factors drive this mechanism, such as lack of greenery and elevated anthropogenic heat sources in cities (Jabbar et al., 2023; Masson et al., 2020). Another major contributor is the increased concentration of low-reflectivity (low-albedo) surfaces in urban areas (Andrés-Anaya et al., 2021).

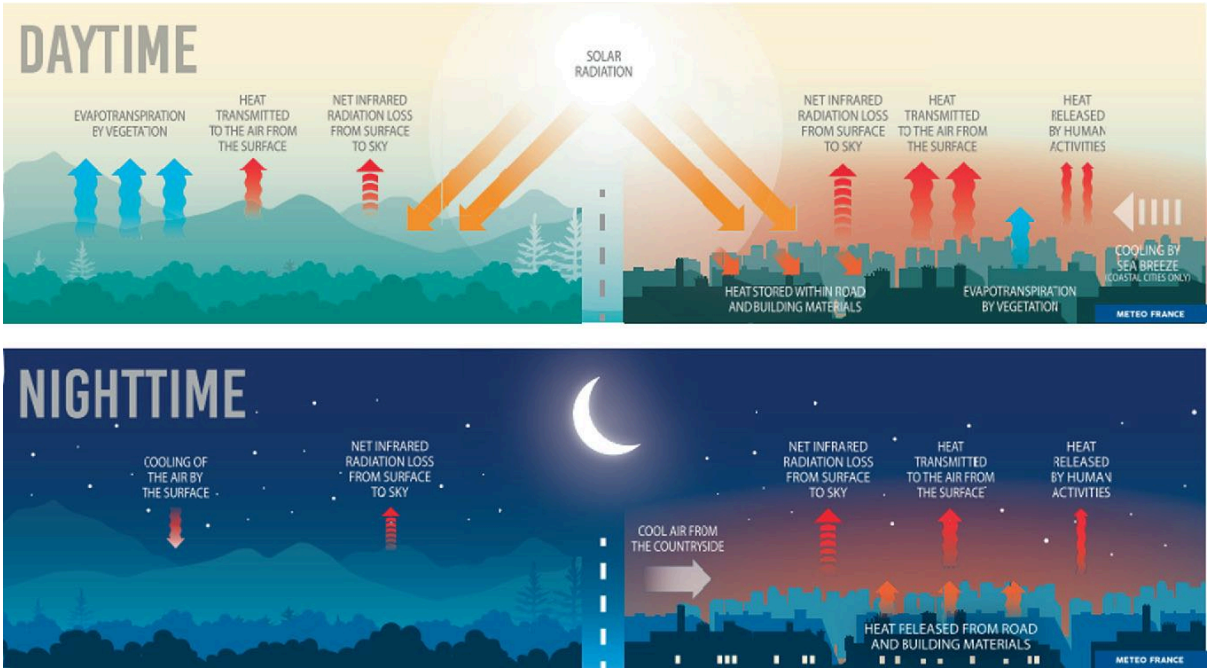


Figure 1: *Processes driving urban heat islands during daytime and nighttime in rural (left) and urban (right) areas.* Urban areas contain more surfaces that absorb solar energy during the day and release it at night, leading to higher temperatures at nighttime compared to the rural areas. Adapted from (Masson et al., 2020)

Surfaces, such as roads, pavements, roofs and façades, have a low albedo, resulting in high absorption of incoming solar radiation (Ziaeemehr et al., 2023). These surfaces are often made from materials that possess a high heat capacity, such as asphalt and concrete, which can absorb more energy per degree of temperature rise than materials with a low heat capacity. This absorbed heat is then released at night, resulting in heated urban environments (Masson et al., 2020; Figure 1).

Researchers are conceptualising ways to mitigate the effects of UHIs, for example by reintroducing vegetation in urban areas, reducing the energy consumption of buildings and introducing high-albedo materials on roofs or façades (US EPA, 2014). The connection between building density and UHIs shows the opportunity that lies in redesigning buildings from drivers to barriers to elevated urban temperatures.

Retrofitting existing façades is one method to reduce the impact of buildings on urban heat, for example by introducing green walls, (retro)reflective coatings, phase change materials or thermochromic paints (Ahluwalia & Khurana, 2025; Han et al., 2015; Santamouris & Yun, 2020). A different way to approach it, is to design a façade cladding panel that also functions as a cooling panel. Currently, façade panels are mainly used for aesthetics, thermal regulation and weather protection (Thevega et al., 2022). However, by introducing an extra function within these panels, it is a possibility to create a multifunctional panel that is aesthetically pleasing, protects the building against weather conditions, and contributes to cooling the urban environment.

Passive cooling strategies

To design a panel that utilizes passive cooling strategies, it is necessary to know what passive cooling strategies exist. Nature serves as a useful guide to examine passive cooling methods. Organisms have to passively regulate their temperature, sometimes under energy constraints (Appendix B). Façade panels also have to deal with this constraint, which highlights the inspiration that can be gained from nature. Five passive cooling strategies of nature have been evaluated using scientific literature, namely evaporation, radiation, reflection, convection and conduction (Appendix B).

The most important findings from this literature research are the possible integrations of reflective, radiative and convective cooling strategies into the façade panel. An example of convection as a passive cooling strategy in nature can be found in termite mounds (King et al., 2015). Inside the mound, the air is warmer than the ambient air, resulting in the inside air rising by buoyancy while the cooler, more dense air sinks. This effect is called the stack (or chimney) effect and is already widely used in the construction industry in the form of ventilated façade systems (Girma & Tariku, 2021).

The potential integration of reflective and radiative cooling strategies into a façade panel also followed from the literature research (Appendix B). Reflection and radiation are closely related: for a surface to cool under direct sunlight, it must reflect most solar radiation, while simultaneously dissipating heat by emitting in the 8-13 μm atmospheric window. In this window, the atmosphere is almost transparent, absorbing close to zero thermal radiation (Pirvaram et al., 2022). Therefore, heat can be released to outer space without being retained in the atmosphere. Together, these two properties enable a net cooling effect during daytime.

Two other passive cooling strategies, conduction and evaporation, are present in nature, but were left out as strategy for cooling in this study. Evaporation has proven to be a successful strategy to cool buildings and the environment, as green walls and roofs have been widely implemented in architecture (Harbiankova & Manso, 2025). However, it often requires some type of active mechanism, such as watering, and is difficult to implement into a façade panel, combined with the other cooling strategies. Similarly, conduction was assessed to be outside of the scope of this project, and we therefore decided to focus on convection, reflection and radiation (Appendix B).

Previous students at Utrecht University and the HU University of Applied Sciences Utrecht have worked on creating a panel that passively cools buildings. This produced two outcomes: a conceptual design of panels inspired by tree bark (Pernice et al., 2022; Figure 2), and the design of a multi-layered panel with a specific three-dimensional geometry that employs convective cooling (Boer et al., 2025; Figure 2). However, what these projects did not assess is the possibility to combine convective cooling with radiative and reflective cooling.

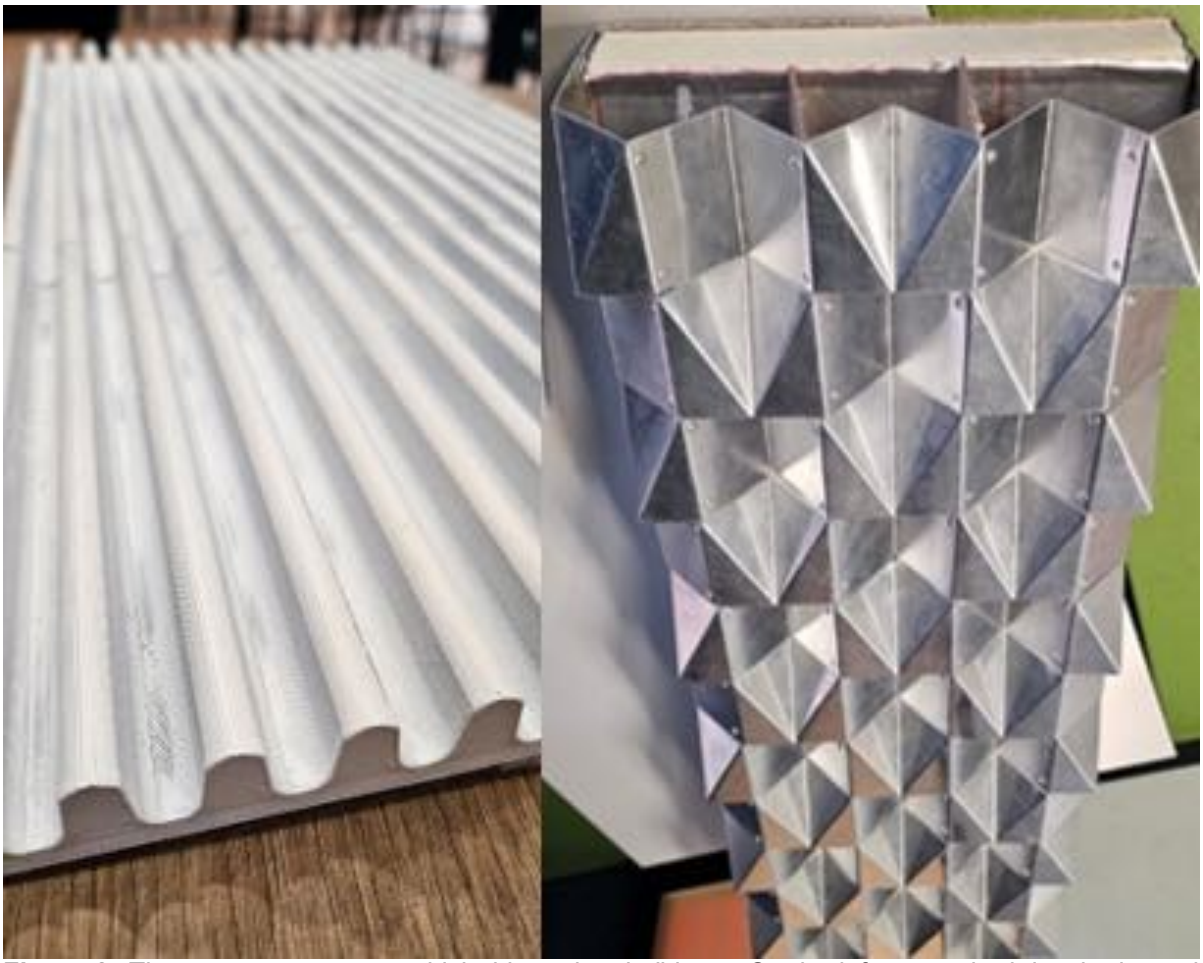


Figure 2: *The two prototypes on which this project builds on.* On the left, a tree bark inspired panel (Pernice et al., 2022), on the right a multilayered panel that utilizes convection as passive cooling strategy (Boer et al., 2025).

Sustainable façade cladding materials

A cooling façade cladding panel can only contribute to a more sustainable urban environment if the material it is made from does not undo its benefits. Conventional façade systems carry a considerable environmental impact compared to their biobased alternatives (Alabdulrazzaq et al., 2026). Sustainable materials can reduce this impact, which is why their application in façade cladding is increasingly being researched (Sandak et al., 2019).

One of these materials is mycelium-based composite (MBC). MBCs consist of plant fibres, such as hemp or grass, bound together by a network of fungal hyphae, also called mycelium, which act as natural glue. MBCs have also been examined in combination with passive

cooling. Soh et al. (2025) produced an MBC façade tile with a texture inspired by elephant skin, which retains water and thereby cools the surface.

However, this concept relies on evaporative cooling, which requires water, while moisture is precisely what limits the use of MBCs outdoors. Uncoated MBCs, such as those tested by Soh et al. (2025), have been shown to degrade after continuous exposure to weather and are therefore unsuitable for outdoor use (Ghazvinian & Gursoy, 2022; Motamedi et al., 2025). A possible intervention could be to grow a fungal skin on the MBC, which has been suggested to improve the hydrophobicity of the material (Appels et al., 2019). Moreover, the mechanical properties of MBCs can also be improved. Heat-pressing, for example, can improve the density, Young's modulus and tensile strength of MBCs (Appels et al., 2019). Most heat-pressed MBC has resulted in flat panels and to our knowledge, heat-pressing MBCs into a three-dimensional shape has not been reported before (Andrade et al., 2026).

Research question and scope

Within our project, we set out to answer the following question: How can a façade panel, made from a sustainable material, passively cool buildings and their environment? One part of this project was therefore dedicated to producing a functioning panel with the highest cooling performance. In this part, we examined whether convection, radiation and reflection can be combined into one panel, starting with the geometry of Boer et al. (2025). We focused on innovating the outer skin by selecting a sustainable material, whose environmental impact was assessed in a Life Cycle Assessment (LCA), and by adding a reflective and radiative coating. The inner wall and insulation are considered outside the scope of this project.

The second part of our research focused on a sustainable material, MBC, and examined whether MBCs could potentially be used as a material for façade cladding panels in the future. By heat-pressing MBC into a three-dimensional shape, we investigated the possibility of producing individual MBC tiles which could be used for a façade panel. Furthermore, we tested what impact the three-dimensional shape had on the flexural strength and water resistance of the MBC. Lastly, we compared MBC to different types of wood to assess how long its lifespan should be to match their environmental performance.

The report is divided into three parts. Part A describes the design and prototyping of the panel and the choices made during this process. The first section of part B presents the outcome of the experiments which were conducted to assess the cooling performance of different panel configurations. The second section of B shows the research conducted on MBCs as a possible future façade cladding material. Part C addresses the implementation and business plan and analyses the intellectual property of our panel. Lastly, the general discussion evaluates the results of the experiments and discusses possible future integrations and research opportunities.

Part A: Design and prototyping

In this part, the process that led to the design and development of our plywood prototype is described. This includes the requirements that were drafted for the design and material selection, the actual choice of material, the reasoning behind it and the chosen three-dimensional geometry of the panel. Lastly, the manufacturing method of the final prototype is described together with its dimensions and specifications. This part does not include the production of anything related to the MBC part of the project, which will be discussed in part B.3.

A.1. Design requirements and criteria

Façade cladding panels need to be multifunctional. As it is the outermost layer of a building, it needs to be both protective and aesthetically pleasing. In this project, we added an important functionality to the requirements: passive cooling. As convection was identified as the main passive cooling strategy, this impacted the design of the panel. The convective heat transfer is dependent on the geometry of the panel, as an increased and channelled surface area improves the buoyancy-driven airflow within the panel.

This led to three design requirements that were formulated in the *Design Report* (Appendix D) (Figure 3). First, the cavity depth of the panel should be large enough to facilitate buoyancy-driven convection. The selected cavity depth ranges between 4 and 8 cm, similar to Boer et al. (2025). This is smaller than the depths of similar designs, such as solar chimneys (20 to 50 cm) or double-skin façades (70-120 cm). This reduced cavity depth offers the opportunity to examine whether convective cooling is also effective using smaller cavities.

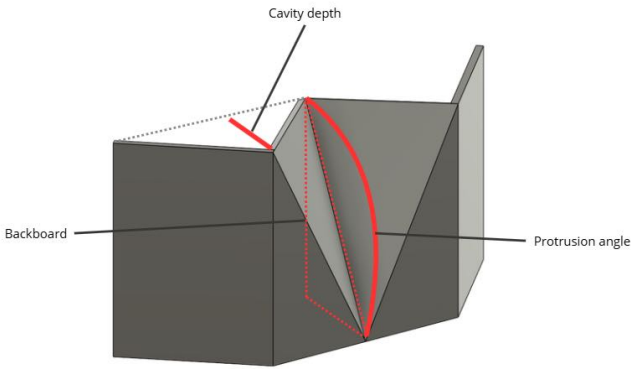


Figure 3*: Visualization of geometric design requirements drafted in the Design Report.* The protrusion angle is the angle between the protruding facet and the backboard. The cavity depth is the maximum distance between the backboard and the tile. The grey and red dashed lines indicate reference planes.

Second, the angle of geometric protrusions needs to improve the natural ventilation. No validated range for the angle has been established (Appendix D). The one study found, which looked different angles (0°, 30°, 45°, 60° and 80°) of venetian style blinds within a double-skin glass façade, reported that an angle of 80° promoted natural ventilation the most (by 35%) (Ji et al., 2007). The limitation of this study was that the blinds were behind glass and in windless conditions, so this study was mostly used as an indication. The precedent design by Boer et al. (2025) used an angle of 72°, which is close to the 80° found by Ji et al. (2007), and therefore also used in our design.

Lastly, the height of the panel must be sufficient to generate effective pressure differences to drive the chimney effect. For an optimised effect, the height of the façade should be more than two storeys (minimum of 6 m) (Appendix D). Therefore, it is important that the design of the panel allows continuous air movement through multiple panels.

These design requirements in turn impose a manufacturing requirement, namely that the individual tiles need to be formed into a specific three-dimensional geometry. Therefore, this became one of the defining criteria for the material selection, alongside several other requirements (Table 1; Appendix C). As a result of the drafted requirements, the material will need to be a thermal barrier, able to be painted or coated, lightweight and recyclable, as well as available to us and formable into the correct shape.

Table 1: *Requirements on material properties*. From left to right, the criterion, type of requirement (threshold or qualitative) and whether it is a hard constraint or ranking criterion. Reasoning behind certain thresholds or constraints can be found in the *Material Selection Report* (Appendix C).

Criterion	**Threshold or qualitative**	**Hard constraint or ranking criterion**
Thermal barrier		
Thermal conductivity	Below 0.5 W m^{-1} K^{-1}	Hard constraint
Surface treatability		
Paint/coating compatibility	Qualitative	Hard constraint
Weather resistance		
UV resistance	Qualitative	Ranking criterion
Water absorption rate	Qualitative	Ranking criterion
Freeze-thaw resistance	Qualitative	Ranking criterion
Weight		
Surface density	Below 10 kg m^{-2}	Minimum requirement – Ranking criterion
Fire resistance		
Fire resistance	Fire class B, s2, d0	Ranking criterion
Sustainability		
Recyclability	Qualitative	Hard constraint
Biodegradability	Qualitative	Ranking criterion

Biobased content	Qualitative	Ranking criterion
Formability & availability		
Material availability	Qualitative	Hard constraint
Material formability	Qualitative	Hard constraint
Instrumentation availability	Qualitative	Hard constraint
Prototyping feasibility	Qualitative	Hard constraint

A.2. Material selection

The set of requirements in Table 1 was used to assess several materials that are commonly used in façade cladding systems (Appendix C). This included metals, glass, plastics, fibre-reinforced composites, Nabasco® (a commercially available bio-composite cladding material used in the Netherlands), mycelium-based composites (MBC) and wood. In the end, wood was selected as material to use for the panel, as it was available, recyclable and lightweight. Additionally, wood can be painted or coated and possessed the required thermal barrier properties (Table 2).

Table 2: *Materials assessed based on the requirements from the* Material Selection Report *and the decisive criterion by which it was rejected.*

Material / material class	**Outcome**	**Decisive criterion**
Metals	Rejected	Thermal conductivity
Glass	Rejected	Weight, formability and thermal conductivity
Plastics	Rejected	Recyclability
Fibre-reinforced composites	Rejected	Weight and recyclability
Nabasco®	Rejected	Weight, availability
Mycelium-based composites	Rejected	Weather durability
Wood	Selected	Passed all hard constraints

The next step was to assess which type of wood would then be most suitable for the façade panel. For this, the wood types most commonly used for façade cladding were assessed on natural durability classification, density, price and sustainability. Additionally, wood modification methods were discussed. Douglas fir was identified as most suitable regular wood type for the panel production, as it is durable, cheap, lightweight and available (Appendix C). A potential other type of wood was thermally modified spruce (TMS), which is more expensive but also more durable and dimensionally stable than Douglas fir (Appendix C).

At the prototyping stage, however, it became apparent that thicker cuts of wood would impose difficulties in manufacturing. The 20 mm thick planks made manufacturing with the computer numerical control (CNC) mill, which we selected for producing the grooves in the individual tiles, more time intensive and increased the weight of the façade significantly. Therefore, plywood was used to make the first prototype of the façade panel, as it is thin, cheap and available in large sheets. Although plywood (excluding exterior/marine plywood) is not suitable for outdoor conditions, the prototype was not tested on weather durability, making plywood a suitable and justified material for this prototype.

To decide on the final material for the design, the four considered types of wood (Douglas fir, indoor and outdoor plywood, TMS) were assessed on environmental impact in a cradle-to-gate LCA, where they were compared with MBCs (Appendix E). The functional unit (FU) of this LCA was 1 m^2 of façade coverage by the panels. This analysis was then related to the lifespan of each material to examine whether increased production impact is justified by longer durability of the material (Appendix E).

Douglas fir was calculated to have the lowest global warming potential (GWP), both in total (~ 4.02 kg CO_2e/FU) and per year (0.27-0.40 kg CO_2e/FU; Table 3). TMS had the second least yearly GWP, whilst having the highest CO_2e output in total (Appendix E). This is mainly because of the increased lifespan resulting from the thermal modification.

MBC had the second lowest GWP (~11.34 kg CO_2e/FU), less than both plywoods and TMS. This suggests that MBC has the potential to be an environmentally competitive construction material, in terms of production impact, if weather durability can be increased. As of yet, the lifespan in outdoor conditions of MBC has not been assessed (Lewandowska et al., 2025), which is why no estimation of the GWP per year is given in Table 3. MBCs need to last for 28-42 years to have the same GWP/year as Douglas fir, or 10-13 years to have a comparable yearly GWP as TMS.

Table 3: *Comparison of global warming potential of each material relative to its service life.* Derived from the *LCA* (Appendix E).

Material type	**GWP (kg CO_2e/FU)**	**Lifespan (in outdoor conditions)**	**GWP/year**
Douglas fir	4.02	10-15 years	0.27-0.40
Indoor plywood	13.53	1 year	13.5
Outdoor plywood	15.52	5-10 years	1.55-3.10
TMS	22.10	20-25 years	0.88-1.11
Mycelium-based composite	11.34	Unknown	Unknown

The research on heat-pressed MBC and plywood as materials for the prototype were conducted in parallel. The development of the plywood panel is described in the next section, whereas the research on heat-pressed MBC is presented in part B.3.

A.3. Design development and selection

The concept and design of our panel started with the panel that was produced by Boer et al. (2025). This panel involved a faceted design incorporating small air gaps, or “chimneys”, to create a convective airflow that passively cools. This multifaceted design simultaneously resulted in a self-shading effect. The measured effect of this panel was small: an air speed of 0.3 m s^{-1} in the cavity and a 3.4 °C reduction in temperature of the backwall was observed.

Studies using comparable designs showed that such faceted designs can have a cooling effect on buildings. A study in Dubai recorded a 10 °C temperature gradient between the upper and the lower part of a façade, indicating a temperature-driven airflow (Pujadas-Gispert et al., 2020). Similarly, Summa et al. (2025) found that a faceted design can reduce the temperature by 10 °C compared to a flat panel, as well as decreasing the temperature of the corresponding backwall by 1.3 °C, highlighting the self-shading effect of the design.

Based on these results, it was decided to develop a panel with the same shape. Inspiration to further innovate the design was acquired from the *Cooling Mechanisms Report* (Appendix B). For example, inspired by cacti, a ridged design was considered, as it could increase the self-shading potential of the panel and act as additional air channels to increase convection. When this iteration was modelled, it proved to be difficult to align the ridges at the points where they cross the facet edges (Figure 4). This could not be resolved within the timeframe into a form which could be cleanly cut by the CNC. Therefore, the ridged design was dropped from the project.

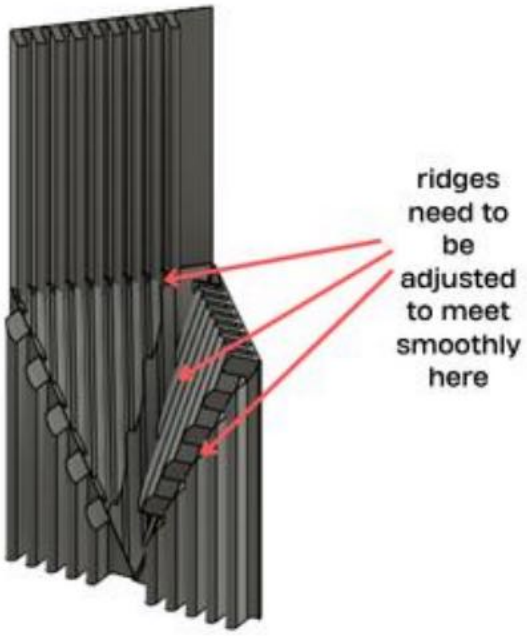


Figure 4*: The design of a ridged tile.* The arrows indicate the points where the facet edges cross, which proved to be difficult to align.

Additionally, reflective and radiative coatings which mimic the cooling capabilities of the Saharan silver ant were examined. The microstructures that this insect utilizes were too small to replicate at this scale, which is why different commercially available reflective and radiative coatings were tested. The outcomes of these experiments will be discussed in Part B.2.

The decision to make modular, 15 x 15 cm tiles, which together would form the larger panel, was inspired by Pujadas-Gispert et al. (2020). In this study, they created larger, 30 x 30 cm tiles of Nabasco®, which they installed onto a plywood backboard. This led us to the decision to make modular tiles of wood which we would then install on a wooden backboard.

To develop the modular, 15 x 15 cm tile, a laser-cut first prototype was assembled from scrap plywood and resulted in gaps between part of the tiles (Figure 5). Therefore, grooves were milled into individual cut plywood, while keeping the outer veneer of the plywood intact. This method, used in other woodworking practices, can result in a seamless outer layer (Steve Szell, timber specialist, personal communication, 24 April 2026). This resulted in tiles that were folded into the correct shape upon heat and steam pressure, resulting in a seamless outer layer.

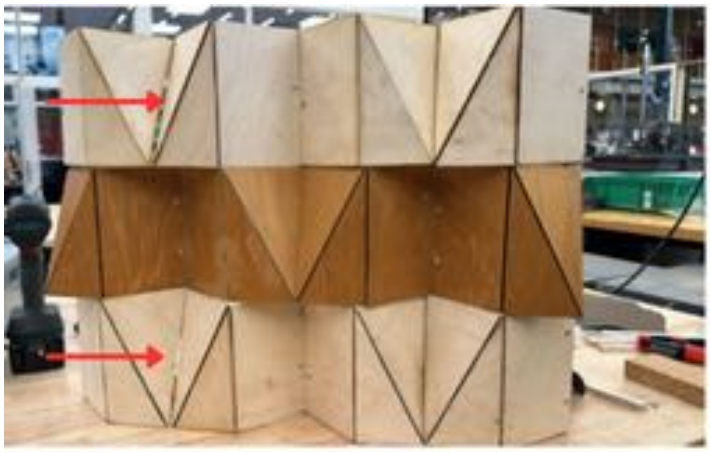


Figure 5*: Laser-cut iteration made at the beginning of the project*. The red arrows point to the gaps between the individually cut pieces.

After 3D-modelling and printing the tile with polylactic acid (PLA) several times, the best way to design the top corner joints, where the three facets meet, was determined. At first, the top corner joints did not align, which imposed difficulties with folding the tile into the correct shape. This led to a reorientation of the angles of the top corner joints, ensuring a clean fold and a 135° angle along the groove (Figure 6). These refinements resulted in the final design of the tile, which was created in Fusion 360 (Autodesk, San Rafael, CA, version 2704.1.15). From this, a toolpath to an open-source CNC router was generated, which could then be used to cut the tiles from plywood sheets using the CNC mill.

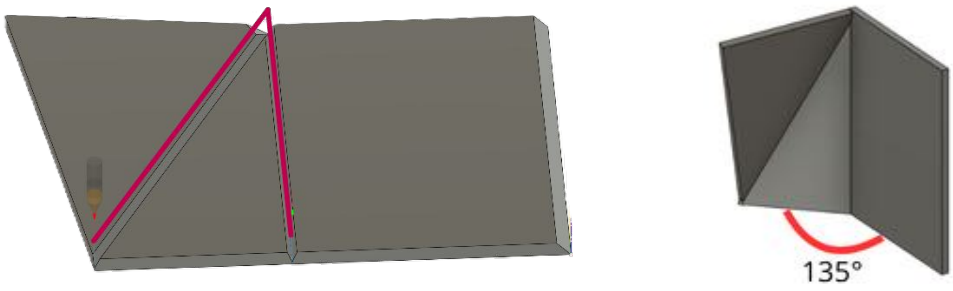


Figure 6: *Toolpath of 45-degree V-groove and the resulting 135° angles after folding the panel along the groove.*

The final design that was used for the prototype was a 15 x 15 cm tile with the shape that was used by Boer et al. (2025). Two inverse versions of the tile were developed, sixteen of each, which would then be mounted onto a 120 x 60 cm backboard. The individual tiles were milled using the CNC and folded into the correct shape. The full manufacturing of the panel will be discussed in Part A.4.

A.4. Manufacturing and assembly

The tiles were produced using a CNC mill (CNC-Step, High-Z S720T). This ensured consistent cuts across all tiles and allowed the exact angles and depth of each groove to be set. It also could demonstrate the potential ability to scale up the production, if proven to be

a successful method of manufacturing. The folding step, however, was completely manual, which could have impacted the accuracy of the folded geometry.

The created toolpath consisted of three milling operations, each with a different bit: to create the angle that needed to be mounted to the backboard, a 90° V-groove milling bit was used. A flat-top milling bit was used to trace the contour of the tiles; and a 45° V-groove milling bit was used to produce the 135° folds needed for the design (Figure 6). The 45° path was cut to a depth leaving approximately 1 mm of material, so that the plywood could be folded at that position while the outer veneer layer stayed intact.

For the two V-groove milling bits, a spindle speed of 26000 rpm was used, while for the flat-top milling bit, a spindle speed of 32000 rpm was used. By posting the toolpath in Fusion 360 (Autodesk, San Rafael, CA, version 2704.1.15) the file was converted into a G-Code file, which could be read by the CNC router. Using this G-code, six tiles at a time were milled out of 65 x 34 cm plywood sheets with a thickness of 5.5 mm **(**Figure 7). To get a seamless connection of the tiles on the backboard, a 5 mm strip was removed from each tile at the 45° angle by sawing.

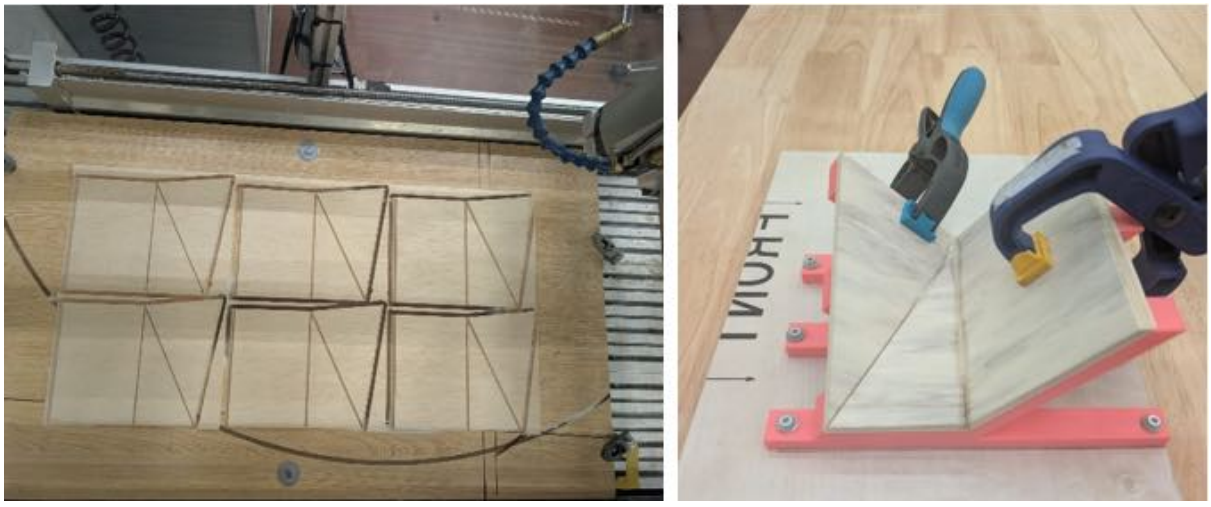


Figure 7*: On the left, a plywood sheet of 65 x 34 cm with six individual milled tiles. On the right, a tile folded and clamped into the right angle (135°), using a 3D-printed jig.*

The finished tiles were moistened with a wet cloth and heated at the groove dedicated for bending with a handheld clothing steamer (Pure POP DT2022, Tefal) for ~ 2 min to induce flexibility of the wood fibres. Under continued heat application via the steaming iron they were folded to 135°, held in this position using a 3D-printed jig (Figure 7) and bonded using polyurethane construction adhesive (Bison Konstruktielijm, EN 204 class D4). Each tile was clamped for four hours, following the manufacturer's manual. The exact dimensions of the panels as they were manufactured in the CNC as well as the dimensions of the 3D-printed jig can be found in the supplementary data (Appendix I).

The backboard was constructed using 48 x 22 mm spruce battens (KONSTA Vuren panlat ruw 22 x 48 x 2100 mm, Hornbach), joined with waterproof wood glue (Pattex, EN 204 class D3) and 35 mm nails using a nail gun (Bostitch Tacker SB-2IN1-E). Five battens run lengthwise and nine across, resulting in a 120 x 60 cm backboard.

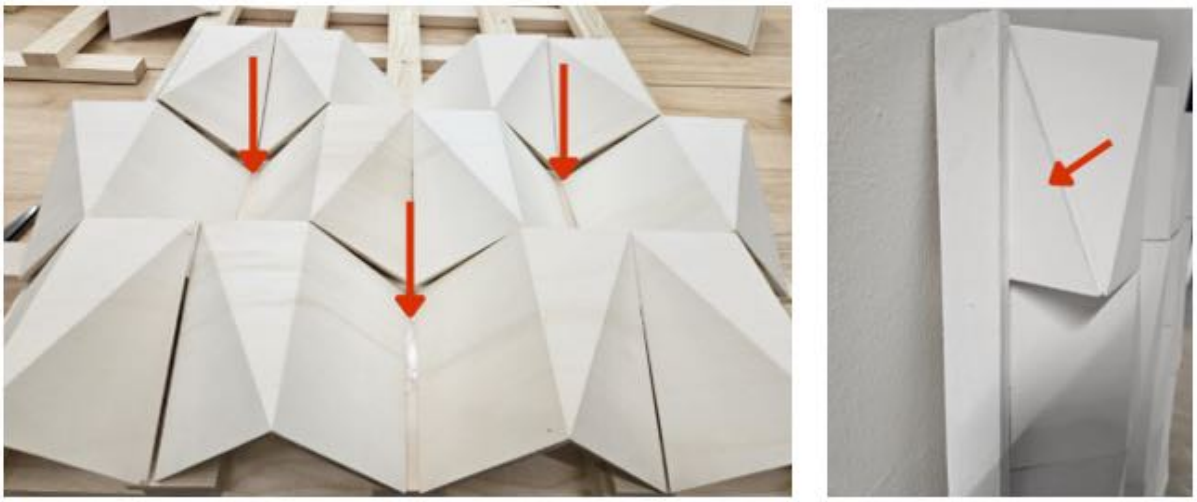


Figure 8: *On the left, the red arrows correspond to the positions where the slats were placed to introduce extra stiffness of the individual tiles. On the right, a triangular piece which was placed on the side of the panel to close the openings that were inherent to the tiles geometry at the panel edge.*

The tiles were attached to the backboard with wood glue (Pattex, EN 204 class D3) and 15 mm nails, using a nail gun (Bostitch Tacker SB-2IN1-E). Between each pair of inverse tiles, slats sawn at a 45° angle, were added to introduce extra stability (Figure 8). Wood glue and nails (15 mm) were also used for this application. At the sides, where the tiles left large openings, sawn triangular pieces of plywood were also added and adhered using construction glue (Bison Konstruktielijm, EN 204 class D4) (Figure 8). Seams that remained after tiles were mounted on the backboard were filled with a white acrylic sealant (Kelfort acrylkit anti-crack) (Figure 9).



Figure 9 *: The seams which were caulked up using a white acrylic sealant after they were mounted to the backboard.*

Finally, the panel was painted with a radiative cooling paint (CryoPaint®). This paint was selected based on the heat transfer and convection experiment reported in Section B.2. Three coating layers were applied with a brush to achieve full coverage. Every layer of applied paint was dried for at least 24 h before the next layer was applied. Only the outside facets of the panel and the sides were painted, leaving the inner cavity side uncoated.

A.5. Final *FunctionalSkin* prototype

The process described in Part A.4 resulted in the final plywood panel (Figure 10). The maximum cavity depth was 8 cm. This closely corresponds to the design requirements drafted in Part A.1, where the desired cavity depth ranged between 4 and 8 cm. The protrusion angle in the final design was 75°. In the design requirements, it was stated that the most effective pressure differences to drive the chimney effect would be reached with a height of more than two storeys. Within the scope of this project, only one panel of 120 x 60 cm was manufactured, which is not high enough to reach the most effective pressure differences. However, if more panels are stacked onto each other, a continuous airflow could be facilitated, thereby reaching the desired effect.

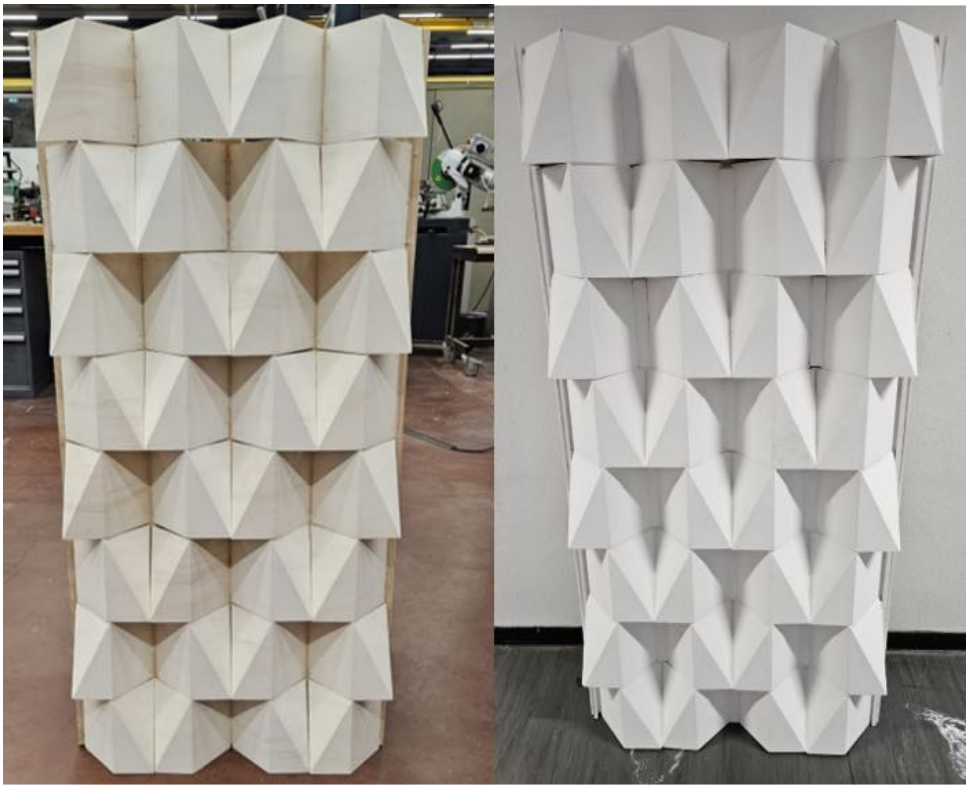


Figure 10: *On the left, the 120 x 60 cm FunctionalSkin panel after mounting the individual tiles onto the backboard. On the right, the final panel with slats between the tiles, caulked up seams and painted with radiative cooling paint.*

Part B: Experiments and material testing

B.1. Introduction to experimental foci

To investigate the design, materials and functionality of the *FunctionalSkin* façade panel, the experimental procedure was undertaken two-fold. Firstly, a timber façade was tested for its ability to transfer heat via convective flows, with iterative tests to seek the best formulation of material and processing. Secondly, the novel 3D heat-pressing of MBC was explored as a potential future material to broaden the future scope of the project. Together, this multidimensional investigation explored the potential of biobased products while maintaining functional capabilities and pushing for boundary exploration.

B.2. Heat transfer and convection

B.2.1. Introduction

The objective of this study was to evaluate the thermal performance of several ventilated façade panel configurations under controlled laboratory conditions. The experimental work focused on quantifying the thermal behaviour of the *FunctionalSkin* façade panel by measuring surface temperature, cavity air temperature, airflow velocity and simulated solar irradiance. Attention was given to the influence of façade geometry, material thickness and surface treatment on the development of buoyancy-driven airflow within the ventilated cavity.

Each ventilated façade panel consisted of an exterior cladding layer, a naturally ventilated air cavity and an interior plywood wall. When exposed to simulated solar radiation, the exterior cladding absorbed thermal energy, increasing its surface temperature. Heat was subsequently transferred to the air inside the cavity, creating vertical temperature gradients. These temperature differences generated buoyancy forces that induced upward airflow through the cavity, drawing cooler ambient air into the lower opening while exhausting warmer air through the upper opening. This natural ventilation mechanism reduces heat transfer towards the inner wall and improves passive cooling performance (Moran et al., 2010; Picallo-Perez & Sala-Lizarraga, 2023).

To characterise these processes, air temperature was measured at multiple locations within the cavity, together with airflow velocity, at the cavity inlet and outlet. Surface temperature of the façade components was continuously monitored, while infrared thermography was used to visualise the spatial temperature distribution across the exterior façade surface. The resulting data were used to quantify temperature gradients, airflow characteristics, and transient thermal behaviour of the ventilated façade panels (Figure 11).

The insights gained from the rectangular shafts informed the design and testing of a full-scale bio-inspired prototype façade panel inside a walk-in environmental climate chamber. The climate chamber experiments evaluated the performance of the complex three-dimensional folded prototype panel, coated with CryoPaint®, against a flat CryoPaint® reference and a

bare unventilated control. Additionally, this setup tested the façade under a fixed ambient summer temperature baseline (30 °C) both with and without the presence of an external airflow stream, resolving whether ambient wind reinforces internal cavity convection and enhances building skin cooling.

B.2.2. Materials and methods

B.2.2.1 Materials and methods: Rectangular shafts

A combination of contact sensors, air probes and non-contact thermal imaging equipment is used to measure airflow within the cavity, cavity air temperature and surface temperature of the outer façade panel and the inner layer.

Ventilated façade panels

Eight ventilated façade panel configurations were investigated (Table 4). Unless noted otherwise, all configurations had nominal dimensions of 600 mm (height) × 300 mm (width) with an 80 mm cavity depth and were mounted on an identical 12 mm plywood backboard.

Table 4 *Structural, Material, and Surface Specifications of the Evaluated Ventilated Façade Panel Configurations.* Unless otherwise specified, all panels measured 600 mm × 300 mm and contained an 80-mm ventilated cavity mounted on a 12-mm plywood backboard. CryoPaint® refers to the radiative cooling surface treatment, whereas the albedo coating refers to the high-albedo reflective coating.

Ventilated facade panel configuration	Description of configuration
Standard panel	Constructed entirely from 5.5 mm plywood with an 80 mm ventilated cavity.
Deep cavity panel	Identical to the standard configuration except for an increased cavity depth of 160 mm.
9 mm panel	Identical geometry to the standard panel but manufactured from 9 mm plywood.
12 mm panel	Identical geometry to the standard panel but manufactured from 12 mm plywood
Negative control	Identical to the standard configuration except that the lower cavity opening was sealed, preventing natural air intake while maintaining the upper cavity opening.
albedo-coated panel	Standard panel coated with four layers of high-albedo reflective paint (Arcareflect 1198, Arcane Industries, Aubagne, France).
CryoPaint®-coated panel	Standard panel coated with three layers of radiative cooling paint (CryoPaint®, Cryo X Co., United States).

Airflow velocity measurements

Airflow velocity within the ventilated cavity was measured using a Testo 410i vane anemometer (Testo, Titisee-Neustadt, Germany). Velocity measurements were performed near the lower and upper outlet openings of the ventilated cavity to quantify the magnitude of buoyancy-induced airflow generated during heating.

Air temperature measurements

Air temperature within the panel cavity was measured using two Testo 605i thermo-hygrometers (Testo, Titisee-Neustadt, Germany). Air temperature data were recorded using the Testo Smart App on an iPad Pro (Apple Inc., model 2020) and exported as comma-separated value (CSV) files. The probes were positioned at the lower and upper sections of the cavity to capture the vertical temperature profile associated with buoyancy-driven airflow. The lower sensor measured the incoming air temperature, and the upper sensor measured the outlet air temperature. These measurements enabled the determination of the cavity temperature gradient and the temperature increase experienced by the air while flowing through the cavity.

Surface temperature measurements

Temperature of the outer and inner surface of the exterior cladding (called STFO and STFI respectively), the inner backwall surface temperature (STBI) and the backwall surface of the plywood backwall (STBB) were measured using a Pico Technology TC-08 thermocouple data logger (Pico Technology Ltd., Cambridgeshire, United Kingdom) connected to Type K thermocouples. The thermocouples were attached to the measurement surfaces using thermally conductive adhesive to ensure good thermal contact and minimise contact resistance. Temperature data were recorded using PicoLog software (Pico Technology Ltd., Cambridgeshire, United Kingdom) on a MacBook Pro (Apple Inc., model 2019) and exported as CSV files for further analysis.

Spatial temperature distributions of the exterior façade surface were recorded using a Testo 860i infrared thermal camera (Testo Titisee-Neustadt, Germany). Infrared images were obtained during heating and after the heat source had been switched off to visualise temperature distributions and cooling behaviour of the different façade configurations (for schematic of sensor placement see Figure 11).

Simulated radiation measurements

Irradiation was simulated using two 1000 W halogen construction lamps (Fenon R7s, Keraf, Rotterdam, Netherlands) positioned at a distance from the façade panel to expose the surface of the cladding with 800 W m^{-2} of light. The simulated solar irradiance on the façade surface was measured using a pyranometer (SL 200 Solarimeter, Kimo Instruments) prior to each experiment to verify light exposure.

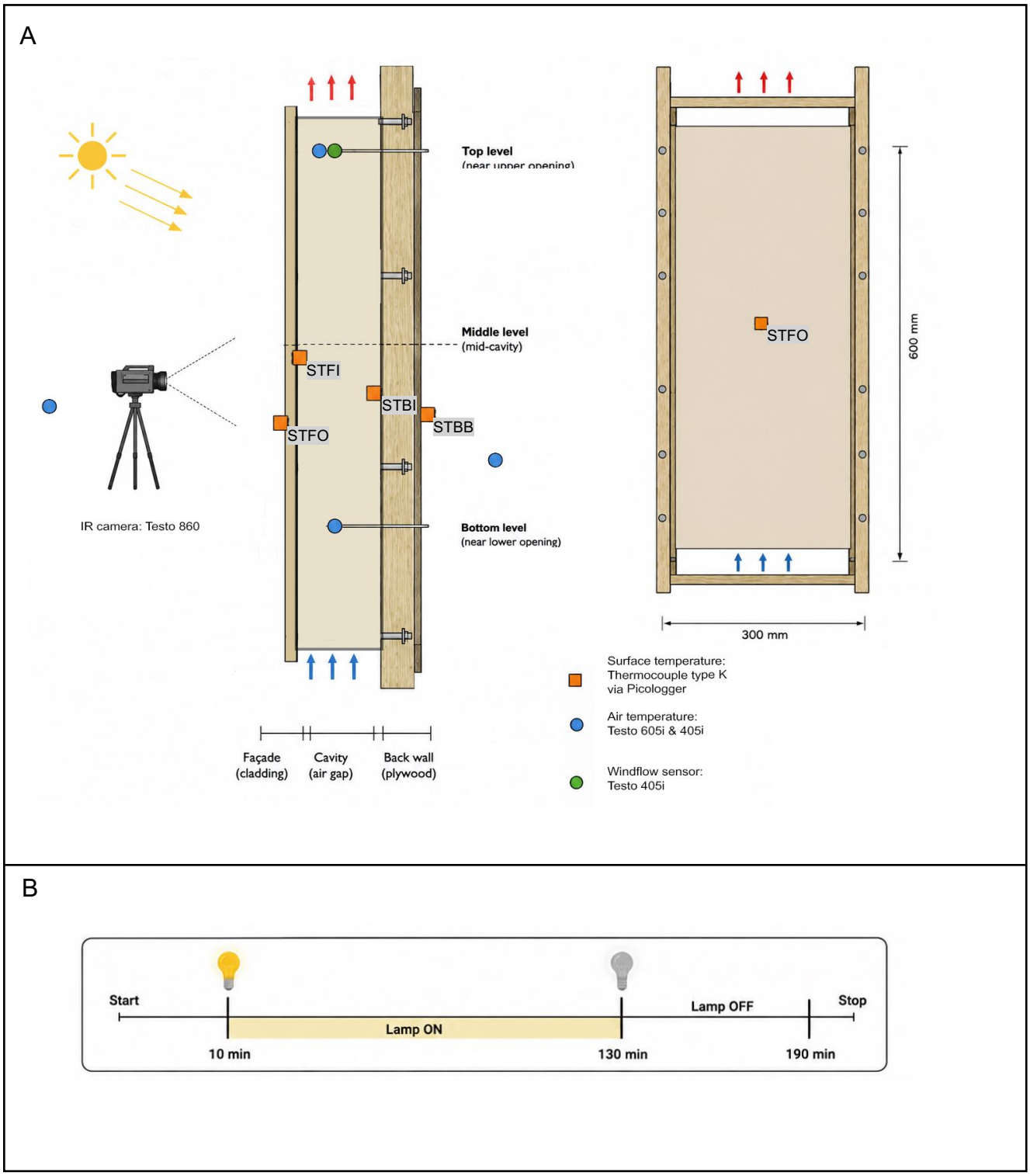


Figure 11: *Schematic layout of the experimental set up and timeline.* (A) Cross-sectional and front elevations of the ventilated façade test panel showing principal dimensions (600×300 mm) and sensor placement. Instrumentation includes Type K thermocouples for surface temperature (STFO, STFI, STBI, STBB), Testo 605i thermo-hygrometers for vertical cavity air temperature (T_{top}, T_{bottom}), hot-wire anemometers (Testo 405i) for cavity airflow velocity, and an infrared thermographic camera. Red arrows denote buoyancy-driven convection. (B) Chronological protocol illustrating the initial ambient baseline (0–10 min), the active heating phase under simulated solar irradiance (10–130 min), and the post-exposure cooling phase (130–190 min).

Data preprocessing and time synchronisation

All data processing and statistical analyses were performed in R. Raw data exported from the Testo and PicoLog software were imported into R. Non-numeric metadata and summary rows contained in the exported files were automatically removed during preprocessing. Since the Testo logger recorded absolute timestamps whereas the PicoLog software stored elapsed measurement times, all datasets were synchronised using the Testo logger as the common temporal reference. Sensor measurements were subsequently converted to elapsed experimental time relative to the start of each experiment. To facilitate comparison between sensors with different sampling frequencies, all datasets were resampled onto a common 10-second time grid. For measurements recorded at higher sampling frequencies, observations within each 10-second interval were averaged before merging the datasets into a single time series.

Phase identification

The thermal response of each ventilated façade panel was divided into three phases: the heating phase, during which temperatures and airflow increased following activation of the heat source; the quasi-steady-state plateau phase, during which the measured variables approached stable values while the heat source remained active; and the cooling phase, which began after the heat source was switched off.

The onset of the plateau phase was identified independently for each sensor using the Kneedle knee-detection algorithm (Satopaa et al., 2011) the algorithm was applied to the transient heating segment, from the start of heating to the peak measured value to identify the knee point based on the deviation of the response curve from a linear reference. This approach accounts for the differences in the thermal response of individual measurement points arising from their location, thermal mass, and measurement principle. As a result, the identified onset of the plateau phase was sensor-specific and did not necessarily occur at the same time for surface temperatures, cavity air temperatures and airflow velocity.

For each measured variable, the mean value was calculated using the datapoints recorded during its identified plateau phase. These mean values were used to compare the thermal and aerodynamic performance of the different façade configurations. In addition, the time required for STBI to reach is identified plateau phase was displayed as $t_{plateau}$ in Table 5.

The sensible heat transport rate ($\dot{Q}_{air}$) was calculated using the mean values of the sensor measurement over their respective sensor-specific plateau phase intervals. Because the plateau phase was identified indecently for each sensor, this interval did not exactly coincide. The calculated sensible heat transport rates should therefore be interpreted as approximate comparative estimates.

Thermal analysis

The thermal measurements were combined with a dynamic energy and exergy analysis methodology adapted from Picallo-Perez et al. (2023). To this end, sensor data were used to

quantify the following parameters that have an influence of the cooling capacity of the ventilated façade.

Vertical air temperature gradient was calculated using formula (1).

$$\Delta T_{cavity} = T_{top} - T_{bottom} \quad (1)$$

Where T_{bottom} and T_{top} are the air temperatures (°C) measured at the lower and upper cavity openings, respectively. This change in air temperature was determined to evaluate whether a higher difference in air temperature along the vertical gradient of the tube would result in an increased wind speed.

The heat removed by natural ventilation within the cavity was estimated from the sensible heat transported by the airflow:

$$\dot{Q}_{air} = \dot{m}_{air} \cdot c_p \cdot \Delta T_{cavity} \quad (2)$$

Where $\dot{Q}_{air}$ is the sensible heat transport rate (W), $\dot{m}_{air}$ is the air mass flow rate (kg s^{-1}), c_p is the specific heat capacity of the air (J kg^{-1} K^{-1}), and ΔT_{cavity} is the air temperature difference (°C) between the upper and lower cavity measurement points as described above.

The air mass flow rate ($\dot{m}_{air}$) was calculated as:

$$\dot{m}_{air} = v_{air} \cdot A_{cav} \cdot \rho_{air} \quad (3)$$

where v_{air} is the measured air velocity (m s^{-1}), A_{cav} is the cavity cross-sectional area (m^2), and ρ_{air} is the air density (kg m^{-3}).

B.2.2.2 Prototype panel

To evaluate convective cooling performance under representative summer ambient conditions, experiments were conducted inside a climate chamber (Lectoraat Nieuwe Energie in de Stad, HU University of Applied Sciences Utrecht) maintained at a constant dry-bulb temperature of 30.0 ± 0.5 °C and 50-60 % relative humidity. The testing frame accommodated three primary test configurations mounted vertically to a 12 mm plywood backwall: (i) an unventilated negative control representing an exposed bare wall without cladding, (ii) a flat panel coated with radiative cooling CryoPaint® and (iii) the prototype panel coated with CryoPaint®. This coated prototype was evaluated under natural convection

(quiescent chamber air, < 0.2 m s^{-1}) and forced convection with a continuous ambient wind field (2 - 4 m s^{-1}) directed across the façade surface.

Solar irradiation was provided by three fixed halogen lamps of the same type as those described in Section B.2.2.1. Irradiance at the exterior panel surface was maintained at approximately 800 W m^{-2} and monitored using a calibrated solar pyranometer. Surface temperatures (STFO, STFI, STBI and STBB) was recorded using the thermocouple system described in Section B.2.2.1 to Type K thermocouples.

As shown in Figure 12, the climate-chamber setup differed from the rectangular-shaft setup (Figure 11) by incorporating the three-dimensional folded prototyped and paired airflow sensors positioned near the lower inlet and upper outlet of the prototype. These two testo 405i thermal hot-wire anemometers positioned recorded both airflow velocity and air temperature at these locations. Two additional testo 605i thermos hygrometers monitored the ambient air temperature in the climate chamber and the indoor-side air temperature behind the backwall.

Data logging, time synchronisation, resampling to a common 10-second time grid, and phase identification followed the procedures described in Section B.2.2.1.

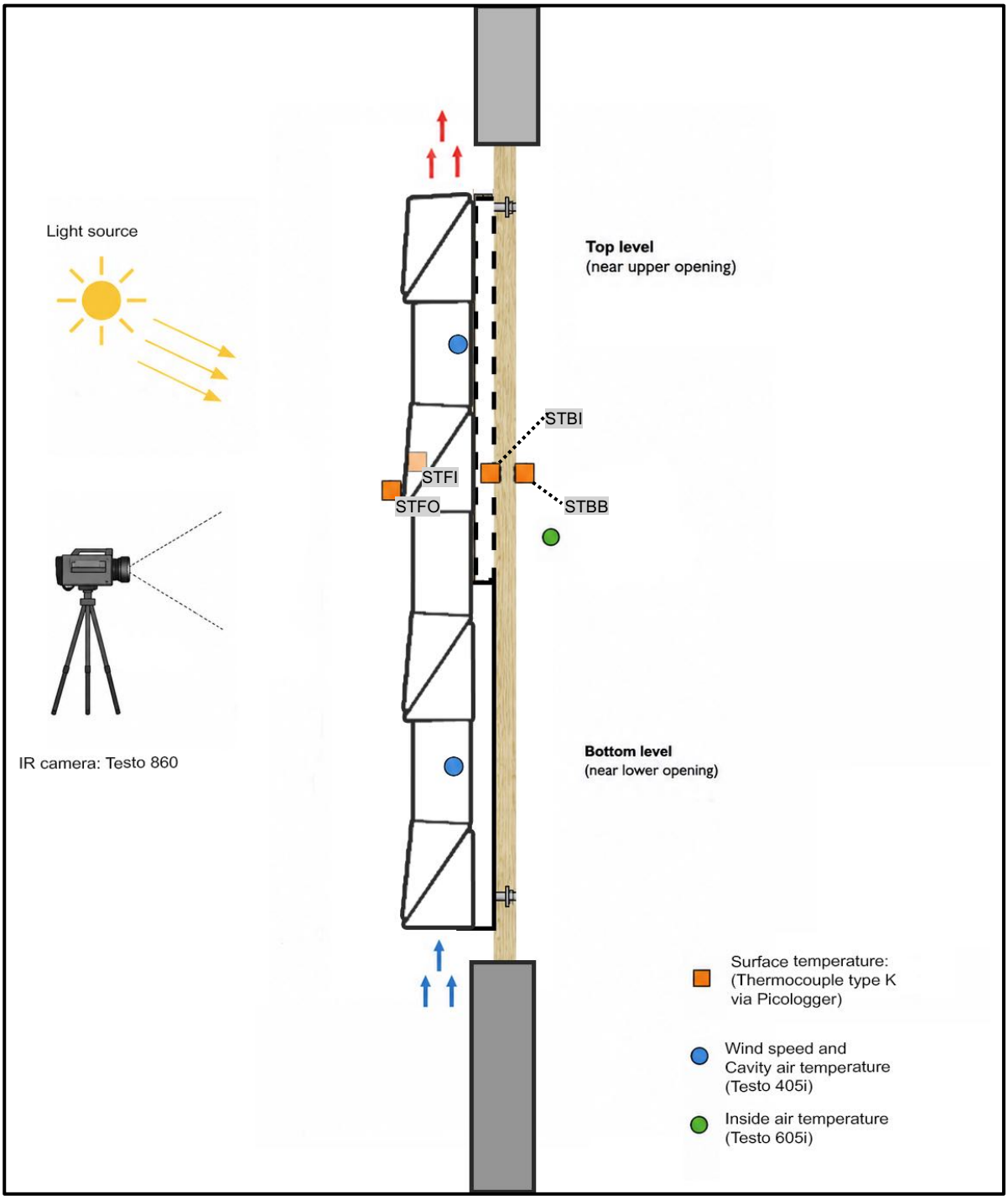


Figure 12: *Schematic layout of the climate-chamber experimental setup for the prototype façade panel*. The schematic shows the three-dimensional folded façade prototype mounted on a plywood backwall within the climate chamber. Type K thermocouples measured surface temperatures, while Testo 405i probes measured cavity airflow velocity and air temperature near the upper and lower openings. A Testo 605i probe measured indoor air temperature behind the backwall. Arrows indicate the expected direction of buoyancy-driven airflow, and the light source represents the simulated solar irradiance applied to the façade.

B.2.3. Results

B.2.3.1 Results: Rectangular shafts

Simulated solar Irradiance measurements

Simulated solar irradiance was broadly comparable across all experimental runs, ranging between 772 W m⁻² and 820 W m⁻² with an average value of 790.7 ± 17.6 W m⁻². The specific irradiance value during the plateau phase per experiment is shown in Table 5. Because solar input was held under near-identical boundary conditions, observed differences in thermal performance, airflow velocities, and temperature distributions can be attributed to the specific façade panel configurations rather than variations in incident solar radiation.

Table 5: *Measured and calculated thermal, aerodynamic parameters during the quasi-steady-state plateau phase*. Values represent the average of measurements (n = 1) obtained during the quasi-steady-state plateau phase. ΔT_{cavity} = vertical cavity air temperature difference; v_{air} = cavity exit air velocity; $\dot{Q}_{air}$ = sensible heat transport rate; *STBI* = inner backwall surface temperature; $t_{plateau}$ = elapsed time to reach thermal equilibrium.

Variable	Negative control	Standard panel	Deep cavity panel	9 mm panel	12 mm panel	albedo-coated panel	CryoPaint®-coated panel
ΔT_{cavity} (°C)	0.6	7.2	7.4	9.3	8.1	7.3	8.1
v_{air} (m s^{-1})	0.452	0.707	0.789	0.714	0.701	0.691	0.721
$\dot{Q}_{air}$ *(W)*	5.9	111	254.8	143.4	123	109.5	126.4
$STBI$ *(°C)*	44.3	39.8	33.9	37.9	35.0	37.5	36.3
$t_{plateau}$ *(h)*	1.2	0.96	1.77	1.43	1.35	1.04	0.97
Irradiance (W m^{-2})	800	778	774	820	804	772	787

Airflow velocity

In the negative control, sealing the lower opening of the tube restricted air intake, yielding a lower mean velocity of 0.452 m s⁻¹ dominated by local turbulent recirculation rather than an upward movement of air. Once the lower opening was opened in the standard panel, a steady buoyancy-driven draft of 0.707 m s⁻¹ was established. Expanding the cavity depth to 160 mm in the deep cavity configuration resulted in the highest wind speed observed across all configurations, reaching 0.789 m s⁻¹ (an 11.6% increase relative to the standard panel). Modifications to cladding thickness and surface treatment had negligible influence on the chimney wind velocity; the 9 mm plywood panel (0.714 m s⁻¹), 12 mm plywood panel (0.701 m s⁻¹), albedo-coated panel (0.691 m s⁻¹), and CryoPaint® panel (0.721 m s⁻¹) all displayed velocity values that were within a narrow range of ±4% of the standard panel. Graphical representation of these results can also be found in Figure 13.

Thermal air gradient and inner backwall surface temperature

The time required for STBI to reach the plateau phase varied considerably between configurations. The standard panel reached the plateau fastest (0.96h). The deep cavity panel showed the longest time to reach the plateau (1.77 h). Thus, the configuration with the highest calculated sensible heat transport rate, reached the plateau phase latest. The plateau phase STBI values showed clear differences between configurations. The negative control reached the highest STBI (44.3 °C), whereas the deep cavity panel showed the lowest value (33.9 °C).

The degree of passive cooling achieved by each façade panel was assessed through the vertical cavity air temperature differential (formula 1) and the plateau phase of STBI. Because $\Delta T_{cavity} = T_{top} - T_{bottom}$, the positive value indicates that the upper cavity air temperature was higher than the lower cavity air temperature in all tested configurations (Table 5). In the sealed negative control, the absence of continuous wind flow suppressed the vertical air temperature gradient to 0.60 °C, leading to thermal accumulation and a backwall temperature of 44.3 °C. Enabling natural ventilation in the standard configuration generated a thermal air temperature gradient of 7.2 °C and reduced STBI to 39.8 °C, resulting in an immediate 4.54 °C benefit. The deep cavity configuration produced a comparable vertical air gradient of 7.4 °C but provided the most effective overall cooling, resulting in a STBI to 33.86 °C (a reduction of 10.48 °C compared to the negative control and 5.94 °C compared to the standard panel). Increasing cladding thickness also reduced backwall (STBI) heating, with the 9 mm and 12 mm panels recording backwall temperatures of 37.91 °C and 35.0 °C while sustaining high vertical gradients of 9.3 °C and 8.1 °C, respectively. Surface treatments similarly reduced backwall heating; the albedo coating and CryoPaint® yielded plateau phase STBI values of 37.5 °C and 36.3 °C alongside cavity gradients of 7.3 °C and 8.1 °C, respectively.

Sensible heat transport

The rate of heat removed by natural convection through the cavity ($\dot{Q}_{air}$) exhibited divergence across the tested panel configurations. In the negative control, convective energy transport was negligible at 5.9 W due to the lack of mass throughput. The standard panel displayed a $\dot{Q}_{air}$ value of 111 W, representing an eighteen-fold increase over the sealed baseline. The deep cavity panel displayed the highest heat removal capacity of all tested assemblies at 254.8 W, representing an increase of 129.5% relative to the standard panel. This increase stemmed from the combined influence of an increased channel cross-sectional area and elevated airflow velocity, which together expanded the cavity mass flow rate ($\dot{m}_{air}$). Among the remaining assemblies, the 9 mm panel evacuated 143.4 W (+29.2% over standard), primarily driven by its pronounced ΔT_{cavity}. The 12 mm panel extracted 123 W, while the CryoPaint® configuration evacuated 126.4 W (+13.9% over standard). The albedo coating evacuated 109.5 W, displaying convective cooling closely comparable to the standard configuration.

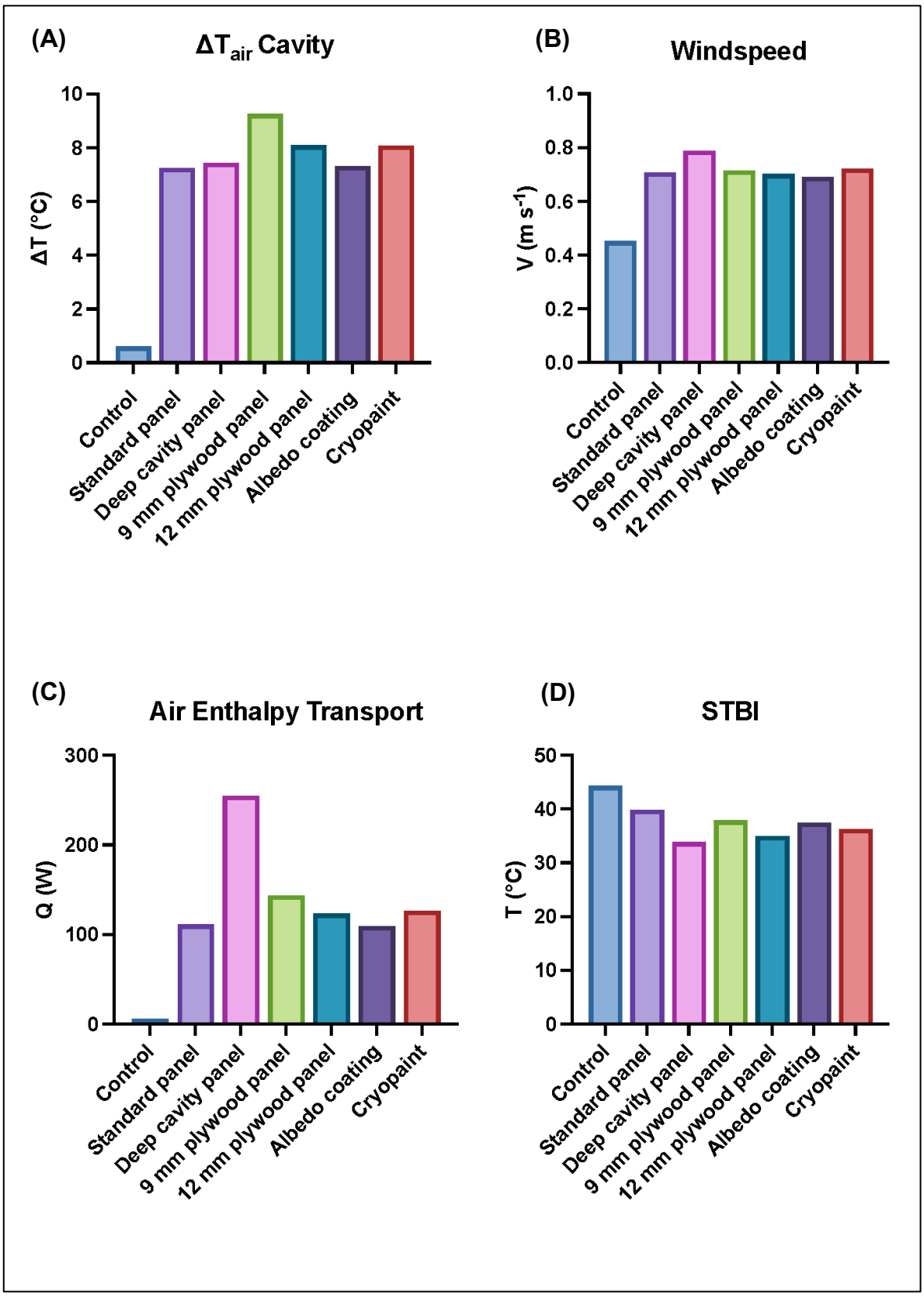


Figure 13: *Comparative thermal and aerodynamic performance of the façade configurations during the quasi-steady-state plateau phase*. The four panels show (a) vertical cavity air temperature difference (ΔT_{cavity}), (b) cavity airflow velocity (v_{air}), (c) sensible heat transport rate through the cavity ($\dot{Q}_{air}$), and (d) inner backwall surface temperature (STBI). Values represent measurements obtained during the quasi-steady-state plateau phase. $\Delta T_{cavity} = T_{top} - T_{bottom}$.

B.2.3.2: Results: prototype in climate chamber

Thermal and aerodynamic parameters measured during the climate chamber experiment are summarized in Figure 14 and the individual time-series segmentation plots (Supplementary figures; Appendix I). Across all trials, steady-state plateau conditions were successfully achieved within 1.5 - 2 h of continuous irradiation in the controlled climate chamber.

In the prototype under natural convection, buoyancy forces produced an entry airspeed of 0.29 m s^{-1} at the bottom and an exit velocity of 0.47 m s^{-1} at the upper aperture. Cavity air entered at 40.0 °C and exited at 45.5 °C, establishing a vertical thermal gradient of ΔT_{cavity} =5.5 °C. When external wind was introduced (Prototype + Wind), cavity aerodynamics were profoundly altered. The air velocity at the upper cavity exhaust accelerated nearly fourfold to 1.77 m s^{-1}, while the lower inlet registered 0.32 m s^{-1}. This increase in air movement in the cavity removed accumulated heat, depressing cavity air temperatures to 34.7 °C at the inlet and 35.5 °C at the outlet collapsing the vertical air gradient to ΔT_{cavity} =0.8 °C.

The STBI served as the benchmark for building thermal protection. The unventilated negative control reached severe overheating at STBI=64.0 °C and a backwall backside surface temperature (STBB) of 47.5 °C. Installing the flat CryoPaint® panel lowered STBI to 46.2 °C (Δ=−17.8 °C vs. control) and STBB to 35.4 °C. The prototype panel under natural convection provided superior shielding, further depressing STBI to 39.2 °C (Δ=−24.8 °C vs. control; −7.0 °C vs. flat CryoPaint®) and STBB to 30.5 °C. Finally, the addition of external wind to the prototype suppressed STBI to 34.2 °C (approaching the 30 °C climate chamber baseline) and STBB to 29.0 °C.

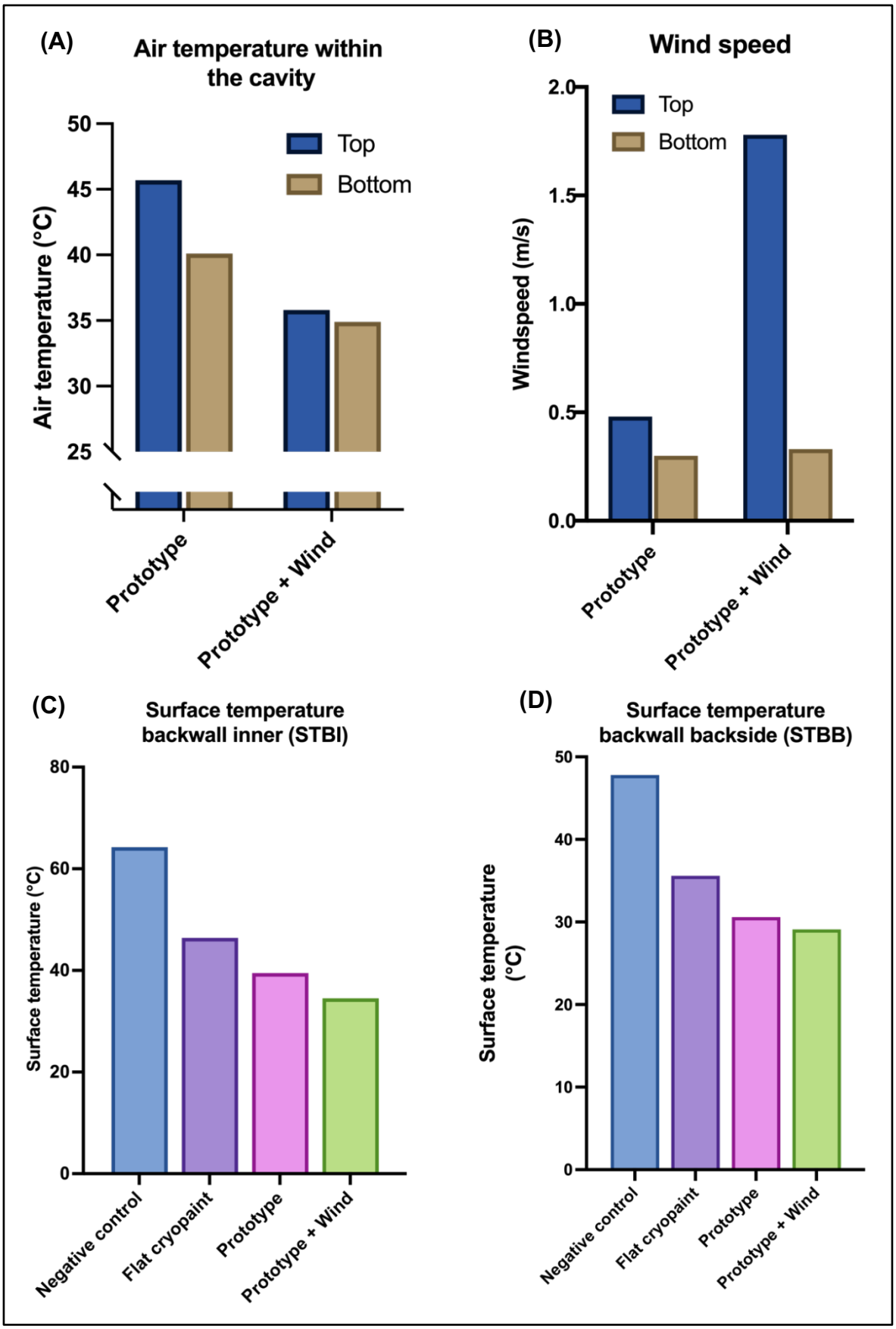


Figure 14: *Thermal and aerodynamic performance of the folded prototype façade under natural and forced convection.* The panels show (a) cavity air temperature at the upper and lower measurement locations, (b) airflow velocity at the upper and lower cavity openings, (c) inner backwall surface temperature (STBI), and (d) backside backwall surface temperature (STBB). Comparisons are shown for the folded prototype under natural convection and external wind conditions, together with the negative control and flat CryoPaint® reference where applicable. Cavity temperature and airflow measurements were obtained during the quasi-steady-state phase of the climate-chamber experiments.

B.2.4. Discussion

The findings of this study demonstrate the potential of integrating passive radiative coatings with a specific ventilated cavity geometry to limit heat transfer through a building envelope. Across both experimental campaigns, ventilation provided a mechanism for transporting sensible heat away from the façade cavity. In the rectangular shaft experiments, the ventilated configurations exhibited airflow velocities of approximately 0.69–0.79 m s^{-1}, with calculated sensible heat transport rates ranging from approximately 109 to 255 W. The deep cavity configuration achieved the highest calculated heat transport rate (254.8 W) and the lowest STBI (33.9 °C). These findings indicate that cavity geometry can strongly influence the thermal performance of ventilated façades. Previous research on naturally ventilated double-skin façades has similarly identified cavity geometry and external environmental conditions as important determinants of ventilation and thermal performance (Barbosa & Ip, 2014). However, although the dept cavity exhibited higher airflow velocity and calculated sensible heat transport than the standard configuration, the internal airflow distribution and pressure differences were not measured directly. Therefore, the results are consistent with reduced flow resistance in the deeper cavity, but this mechanism could not be confirmed.

The effects of cavity depth differed from those of cladding thickness and surface treatment. Increasing the plywood thickness reduced the inner backwall temperature, with STBI values of 37.9 °C for the 9 mm panel and 35.0 °C for the 12 mm panel, compared with 39.8 °C for the standard panel. In contrast, airflow velocities for the panels with different thicknesses and coatings remained within approximately 4% of the standard configuration. This suggests that the improved thermal performance of the thicker panels was primarily associated with increased conductive resistance rather than a substantial change in ventilation. Nevertheless, the relationship between material thickness and calculated sensible heat transport was not linear, because heat transport depends on both the air mass flow rate and the temperature difference between the inlet and outlet air. The results therefore indicate that airflow velocity alone is insufficient to describe the thermal performance of a ventilated façade.

Surface treatment also affected the observed thermal response. The albedo-coated and CryoPaint®-coated panels reached STBI values of 37.5 °C and 36.3 °C, respectively, while maintaining airflow velocities like that of the standard panel. The lower backwall temperature of the CryoPaint®-coated configuration suggests that its surface properties contributed to the observed thermal benefit. Research on radiative cooling identifies solar reflectance and thermal emittance as important properties affecting surface cooling performance (Mandal et al., 2020a). However, because the spectral reflectance and thermal emittance of the coatings used in this study were not independently measured, their individual contributions cannot be isolated. The observed differences may reflect the combined effects of radiative properties, heat transfer to the cavity air, and conductive resistance through the façade assembly.

The climate chamber experiments demonstrated that the specific three-dimensional geometry and external airflow can further influence façade thermal performance. Under quiescent chamber conditions, the folded prototype reduced STBI to 39.2 °C, compared with 46.2 °C for the flat CryoPaint® reference, corresponding to a difference of 7.0 °C. The

addition of external wind further reduced STBI to 34.2 °C and STBB to 29.0 °C. These findings suggest that the folded geometry and wind-enhanced ventilation can provide complementary thermal benefits. The folds change the orientation of the external surfaces relative to incident radiation and increase the effective surface area available for heat transfer within the cavity. External wind also increased the measured outlet airflow velocity, while the cavity air temperature difference decreased from 5.5 °C under natural convection to 0.8 °C under forced convection. The smaller temperature difference should not, by itself, be interpreted as reduced cooling performance, because the amount of sensible heat transported also depends on the mass flow rate. Nevertheless, the individual contributions of surface orientation, increased heat-transfer area, buoyancy and wind-driven flow cannot be separated from the present experiments.

Research on termite mounds has demonstrated that mound geometry, porosity, internal structure and diurnal temperature fluctuations can interact to generate passive ventilation (King et al., 2015). The relevant similarity is the use of geometry and environmental temperature differences to facilitate airflow without relying exclusively on mechanical ventilation. However, termite mounds and building façades differ substantially in scale, material properties, opening configurations and environmental boundary conditions. Their absolute airflow velocities should therefore not be compared directly without a consistent basis for measurement. The biological system provides a useful design principle but does not by itself demonstrate that the prototype reproduces the same ventilation mechanism or achieves equivalent performance.

Several experimental and instrumentation limitations should therefore be considered when interpreting the results. First, the shaft experiments relied on a single-point velocity measurement at the upper outlet. This introduces uncertainty into estimates of the mean cavity velocity, air mass flow rate ($\dot{m}_{air}$), and consequently sensible heat transport ($\dot{Q}_{air}$). In addition, the absence of measurements at the lower inlet prevented direct verification of mass continuity between the intake and exhaust streams. Although the observed increase in velocity and thermal performance with cavity depth is consistent with reduced flow resistance, the internal velocity distribution cannot be established from the available measurements. Future experiments should therefore employ multi-point or spatially resolved anemometry across both the inlet and outlet cross-sections, ideally complemented by measurements at several heights within the cavity. This would allow the development of the buoyant boundary layer and the relationship between cavity geometry, flow resistance, and mass transport to be quantified more directly.

Second, the upper cavity air temperature sensor (T_{top}) in the shaft experiments may have been affected by radiation from the solar simulator. Radiative heating of the sensor body directly by the irradiance of the lamps could have caused an overestimation of the measured upper air temperature, thereby affecting the calculated cavity temperature difference and sensible heat transport. The magnitude of this bias cannot be determined retrospectively. Although the experimental conditions were maintained as consistently as possible between configurations, the extent to which this potential error influenced relative differences remains

uncertain. Future experiments should use black-bulb temperature sensor to measure the air temperature and the radiant heat in the environment.

An additional limitation concerns the calculation of the sensible heat transport rate ($\dot{Q}$air) Because the mean temperature and airflow values were obtained from independently identified, sensor-specific plateau intervals, the values used in the calculation did not necessarily represent simultaneous measurements. This temporal mismatch introduces additional uncertainty into the calculated heat transport rates and limits the interpretation of their absolute magnitudes and differences between configurations. For future experiments, it is recommended to define a common plateau phase time interval based on one representative sensor, such as the plateau phase for STBI, which reflects the thermal response of the façade. The mean values of all relevant sensors, including cavity air temperatures and airflow velocity, should then be calculated over this same interval. This approach would ensure that the input variables used to calculate the sensible heat transport rate correspond to the same period of the experiment, reducing the uncertainty associated with combining averages from different sensor-specific plateau intervals.

Finally, the controlled climate-chamber conditions limit the direct applicability of the results to real building façades. A constant ambient temperature of 30 °C provides a controlled warm-condition baseline, but outdoor façades experience changing wind speeds, wind directions, solar irradiance and atmospheric turbulence. External wind pressure can either reinforce or oppose buoyancy-driven flow, depending on façade orientation and opening configuration. Future research should therefore examine the prototype under dynamic wind conditions and realistic outdoor exposure. Independent measurements of the optical properties, conductive resistance and airflow characteristics would also help separate the contributions of radiation, conduction and convection.

Among the rectangular shaft configurations, the deep cavity panel demonstrated the strongest overall thermal performance, combining the highest calculated sensible heat transport rate (254.8 W) with the lowest STBI (33.9 °C). However, it also required the longest time to reach the quasi-steady-state plateau (1.77 h). In the climate-chamber experiments, the folded prototype under external wind achieved the lowest STBI (34.2 °C) and STBB (29.0 °C) among the tested climate-chamber configurations. As the two experimental campaigns differed in geometry and boundary conditions, their best-performing panels cannot be ranked directly against one another. Further development should focus on optimising cavity depth, fold angle, and inlet and outlet geometry to improve ventilation while minimising material use and structural complexity. The prototype should also be tested under varying wind directions and outdoor conditions to establish whether the observed thermal benefits persist in realistic environments.

Further development should focus on optimising cavity depth, fold angle and inlet and outlet geometry to enhance ventilation while limiting material use and structural complexity. Ventilated rainscreen and double-skin façade systems provide relevant existing technologies against which the prototype could be benchmarked. However, the present experiments do not establish whether the proposed design is competitive with commercial alternatives. A

meaningful comparison would require equivalent testing of thermal performance in the case of whole-wall coverage, ventilation effectiveness, durability, moisture management, structural integration, cost and life-cycle environmental impact.

Overall, the findings support the potential of passive ventilated façades as a means of limiting heat accumulation within the building envelope. Establishing their practical value requires validation of the calculated heat transport, further separation of the underlying heat-transfer mechanisms and testing under realistic outdoor conditions.

B.3. Mycelium-Based Composites: Heat-pressing and mechanical properties

B.3.1. Introduction

In the search for a sustainable material for the cooling panel, MBCs were tested as a possibility. For the plywood prototype, individual tiles were produced, which were assembled and formed a final panel. For MBCs, we examined whether we could form individual tiles by heat-pressing into a three-dimensional shape. In investigating this, there were two key questions to consider: can it be heat-pressed into shapes that mimic the bio-models (and timber prototype); and can it function as a material in an urban context?

The design of these tiles was different from the plywood tiles. The sharper angles that were included in the design of the plywood tile were suggested to cause major cracking during heat-pressing the MBC (dr. Jordi Pelkmans, dr. Pieter Kooijman). Therefore, a smoother, waved-shaped design was chosen for the MBC tiles, as it would allow a more evenly distributed pressure over the MBC and therefore have a greater chance of success - so a custom aluminium mould was made (Figure 15). The dimensions of the heat-press were 10 x 10 cm, which resulted in the production of MBC tiles of the same size. The wider angle of the waves also meant that a standard cubic MBC block could be pressed, rather than growing the blocks in a similar, predetermined 3D-shaped mould to match the aluminium press mould. This makes the tests easier to replicate and to scale up in the future. Additionally, heat-pressing was chosen over cold-pressing, as it has been shown that heat-pressing enhances the homogeneity and strength of MBCs (Appels, et al., 2019). After heat-pressing, signs of cracking, change in colour and thickness were assessed to determine what pressing temperature and duration are most effective in achieving the desired shape in a reproducible way.

After MBC blocks were successfully heat-pressed into shape, a series of experiments was conducted to determine flexural strength and water resistance of the material. These tests were undertaken to assess the potential of the tiles to be used for façade panels, which are exposed to wet, windy weather conditions. The outcome of these experiments was intended to indicate which coatings or maintenance might be necessary.

The change in white value (an indication of reflectivity) was also assessed to determine if the heat-pressing affects the natural cooling properties of the fungal skin. It is known that black coloured materials heat up more than their white counterparts because white reflects more wavelengths (visible and solar), and studies show that white roofs can stay cooler than darker roofs in direct sunlight (Berdahl & Bretz, 1997). Therefore, by measuring the change in white value of the MBC blocks after heat-pressing, the change in reflective ability can be estimated. However, white value only covers the visible light and is not a measured reflectivity.

B.3.2. Materials and methods

Species choice

The choice of mycelium species can be further understood in the *Material Selection Report* (Appendix C). Briefly, the similar *Fomes fomentarius* and *Fomes inzengae* (Cristini et al., 2025; Pohl et al., 2022) are relatively new and promising species used in MBC material research. Of which, *Fomes inzengae* was readily available for this project along with expert guidance.

Growth conditions

Fomes inzengae was first inoculated on Potato Dextrose Agar (PDA) plates and then grown in cellulose (Recell, Groningen). This was used to inoculate hemp shives (Welkoop, NL), in which it was grown for 13 days in acrylic moulds of 10 x 10 x 3 cm in an incubator set to 30°C. Following this, they were removed from the moulds to grow a skin in the same incubator. After 7 days the samples were set to air dry (unobstructed on 5/6 sides) at ambient room humidity (50-60%) and temperature (20-22 °C) over two and a half weeks. After this time the blocks were dry to the touch and slightly shrunken. This drying method was chosen due to its low energy (passive) demand and advice from fungal experts.

Heat-pressing

The MBC blocks were pressed for 5 or 20 min at 115 kN at 150 °C and 190 °C using a heat press (Carver, USA) and aluminium mould (Figure 15). The aluminium mould was used to heat-press a flat square shape, and additional aluminium, CNC milled, three-dimensional inserts (positive and negative parts, measuring 100 x 100 mm) were used to achieve the 3D-pressed MBC tiles. Digital callipers were used to measure thickness after heat-pressing at Ha and Hb (Figure 16). The two thicknesses vary slightly due to the geometry of the mould design.

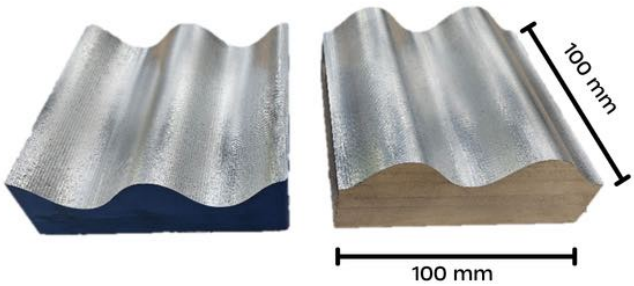


Figure 15: *Aluminium press mould; left is the top, right is the bottom.*

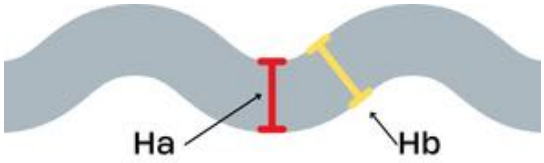


Figure 16: *Side view of pressed block where the thickness was measured at Ha and Hb.*

White value

The flat faces (underside) of the dried blocks were scanned (Epson Perfection V370 Photo) before and after heat-pressing (flat-pressed only) to determine any change in white value. ImageJ (Fiji, NIH Image) was used to measure this change after scanning. Measured region is within the boundaries of the block (avoiding the background interfering with the measurement).

Water contact angle

Six samples were tested for water contact angle. This included four 3D-pressed tiles (two each of 150 °C and 190 °C heat-pressed), and two flat-pressed tiles. Six drops of 5 µl demineralised water were placed on each of the 2 faces of the 3D-pressed tiles (four drops per face used on flat tiles due to their surface being more uniform). These 12 drops were placed one-by-one, and of which, a photo was taken at 1 sec, 1 min and 5 min following drop placement. The left and right angles were measured and averaged using ImageJ (plugin Drop Analysis – DropSnake).

Water absorption

Weight and thickness of six samples were recorded before submerging in demineralised water using a weight to keep them down. The weight and thickness were then recorded after 10, 30, and 60 min and after 2, 4, 24, 49 h. When measuring, the excess water was blotted with filter paper, and sample measured within 10 sec of removing from water.

Flexural strength testing

Twelve samples of heat-pressed tiles were tested for flexural strength in a tensile testing machine (Lloyd, Amtek) at the HU University of Applied Sciences Utrecht. Four tiles were heat-pressed flat, while eight had been 3D-pressed. Additionally, a 10x10 cm plywood tile (5.5 mm thickness) was tested to compare. The machine was set to 5 N mm^{-2} s^{-1}. Each tile was separately loaded into the machine, and value was recorded at failure point. Failure criterion was indicated by a force drop right after achieving peak value (sample fail due to fracture). RStudio (Posit PBC) was utilised to analyse results by generating stress-strain curves and calculating Young's modulus (= $E = \frac{\sigma}{\varepsilon}$ where σ is stress, and ε is strain) after manually selecting for the linear region of each sample's stress-strain curve.

Assembled panel prototype

A final prototype panel was assembled using heat-pressed MBCs (*Fomes inzengae*) which were initially waste from another student's research (blocks provided courtesy of Zachary Leopold). This was due to the difficulty growing enough blocks needed, and priority was given to experimental testing for the ones grown during this project. Using the waste MBCs provided, they were heat-pressed for 10 min at 190 °C and assembled in a 3 x 3 tile configuration. These were then nailed and glued (polyurethane construction adhesive (Bison Konstruktielijm, EN 204 class D4)) to a pine timber frame.

B.3.3. Results

Heat-pressing

MBC blocks of *Fomes inzengae* and hemp shives were grown in 100 x 100 x 30 mm moulds. These blocks were allowed to grow a fungal skin before drying, after which the average thickness of the dried blocks was measured to be 25 mm (reduced from 30 mm due to shrinkage after drying). Important to note, is the choice of growing a fungal skin on all samples. This was decided after initial tests with MBC both with and without skin determined that samples held shape more effectively after heat-pressing when a fungal skin was present. Samples without skin were prone to crumbling edges of substrate and a less uniform surface after heat-pressing. It was however observed that after heat-pressing in a three-dimensional shape, tears along the surface occurred at points of stress at the “peaks” of the design. This was not observed in the flat-pressed MBCs (Figure 17).

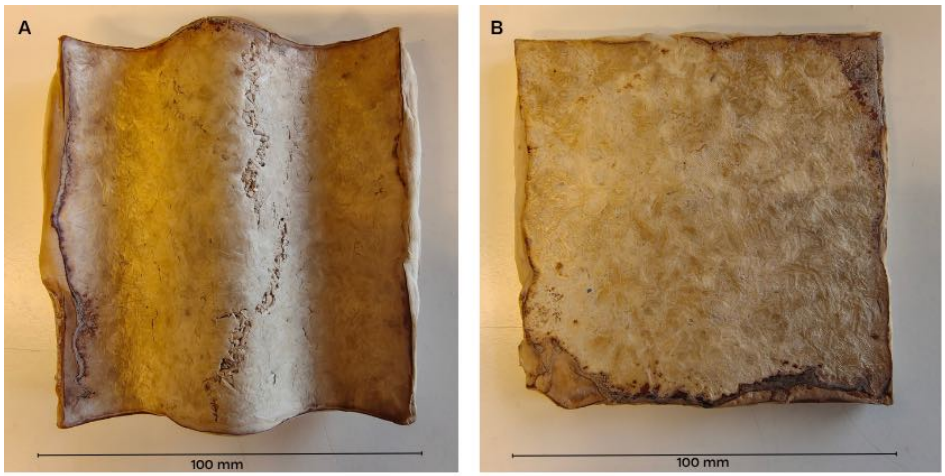


Figure 17: *Fungal skin tear at peak of block #13 after pressing (A). This occurred on all 3D pressed samples and did not occur on the flat pressed samples, such as block #21 (B).*

Using aluminium moulds (flat and 3D-shaped), it was observed that heat-pressing at a higher temperature of 190 °C resulted in thinner tiles, with the thinnest average measuring 4.5 mm (Standard Deviation (SD) = 0.3) after 20 min of heat-pressing and 4.8 mm (SD = 0.6) after only 5 min of heat-pressing (Table 6). This result is reflected in the 3D-pressed samples with the thinnest tile average of 4.3 mm (SD = 0.8) after heat-pressing for 20 min at 190 °C and the thickest measuring 5.3 mm (SD = 1.1) after 5 min at 150 °C (Table 8). For the flat tiles however, the thinnest average thickness of 4.8 mm was measured after heat-pressing for 190 °C for 20 min, and 150 °C for 5 min, and the thickest was after heat-pressing for 5 min at 190 °C (Table 7).

Table 6: *Comparison of temperature and time pressing (regardless of shape) on the MBC thickness after pressing.*

Temperature (°C)	Minutes pressed	n	Thickness (mm)	SD
150	5	4	5.2	0.3
190	5	4	4.8	0.6
150	20	6	5.0	1.3
190	20	5	4.5	0.3

Table 7*: Flat pressed MBC tiles and their thickness after heat pressing.*

Temperature (°C)	Minutes pressed	n	Thickness (mm)	SD
150	5	1	4.8	n/a
190	5	1	5.3	n/a
150	20	3	4.9	1.3
190	20	2	4.8	0.0

Table 8: *Thickness of 3D pressed MBC tiles only.*

Temp (°C)	Mins pressed	n	Ha (mm)	SD	Hb (mm)	SD	Average (Ha & Hb) (mm)	SD
150	5	3	6.2	0.3	4.3	0.2	5.3	1.1
190	5	3	5.4	0.8	3.9	0.6	4.7	1.0
150	20	3	5.9	1.6	4.0	1.4	5.0	1.7
190	20	3	5.0	0.3	3.6	0.3	4.3	0.8

White value

After heat-pressing MBC blocks, the white value for every sample decreased. For samples heat-pressed at 150 °C there was an average decrease in whiteness of 14%, and for samples heat-pressed at 190 °C this was 22%. The mean change between pre and post heat-pressing is quite variable, however (Table 9). The visual change can clearly be identified in **Figure** 18 below.

Table 9: *Measurement of white value before and after heat-pressing MBC blocks (using ImageJ, value of 0 = black, 255 = white).*

			Pre-pressed			Pressed			
Temp pressed	**Press time (min)**	**Block #**	**mean value**	**min**	**max**	**mean value**	**min**	**max**	**% change**
150	20	2	164.15	36	255	154.03	33	255	-6%
150	20	4	193.15	48	255	154.78	22	255	-20%
150	20	5	194.31	35	255	168.22	32	255	-13%
190	20	6	197.62	48	255	131.86	34	255	-33%
190	5	19	198.24	37	255	171.69	50	255	-13%
190	5	21	188.35	49	255	151.58	29	255	-20%
190	20	3	n/a	n/a	n/a	131.83	19	255	n/a

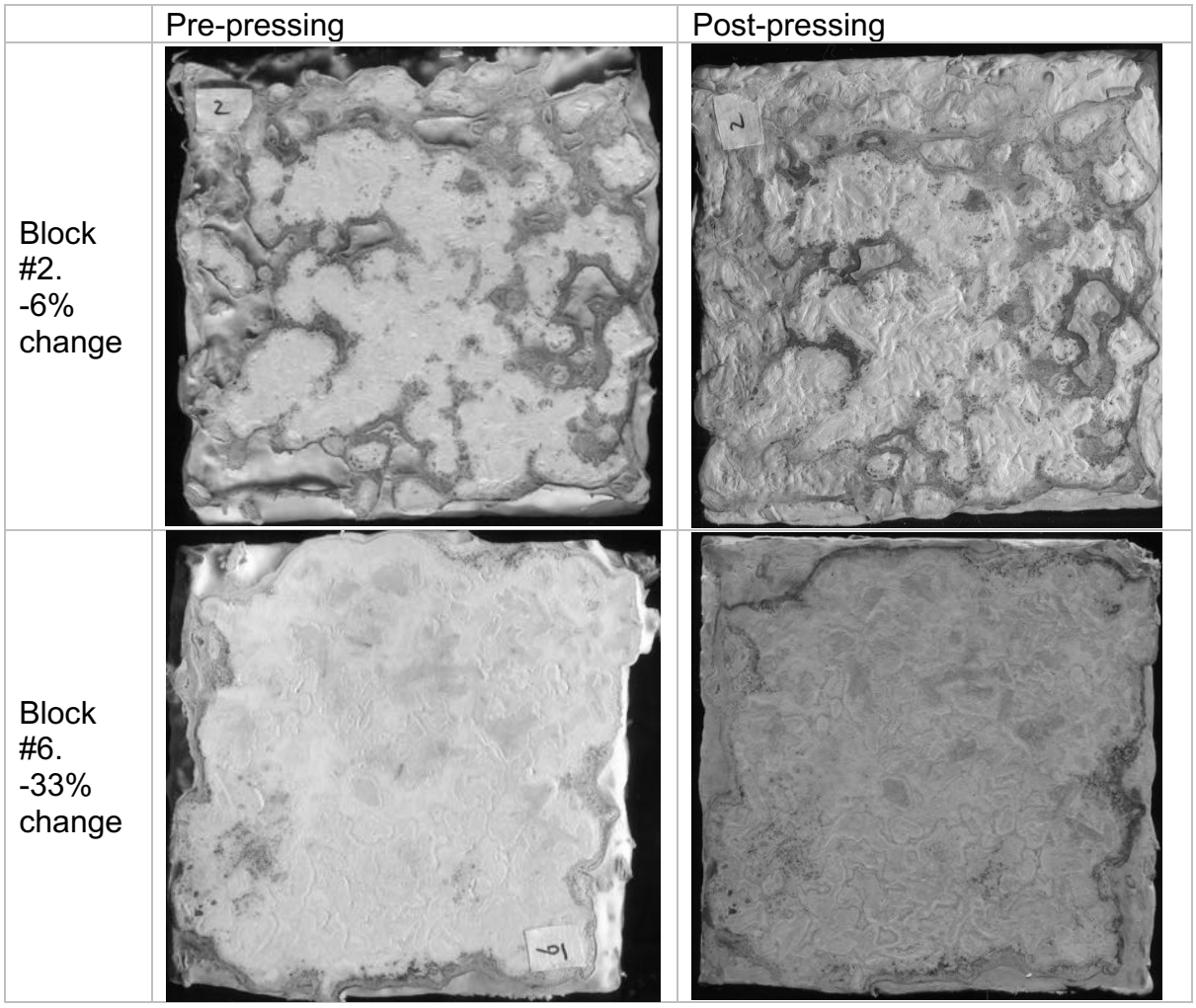


Figure 18: *Scanned images showing biggest and smallest change in white value on MBC samples before and after heat-pressing.* Note that black #2 began with less white area than block #6.

Water contact angle

The water contact angle on every sample recorded at 1 s were greater than 90° which can be defined as hydrophobic (Figure 19). However, every water drop was absorbed by the material (longest remained for ~30 min) eventually. The sample which retained the greatest number of drops after 5 min was block #9 3D-pressed at 150 °C for 5 min (Figure 20), which retained half of the drops placed. The sample which saw the absorption of the most drops at this point was block #15 3D-pressed for 190 °C for 5 min, of which only 1 of 12 drops remained. Block #12 (3D-pressed at 150 °C for 20 min) had the smallest contact angle compared with all other 3D-pressed tiles, and this was also consistent in both its peaks and valleys (see Figure 21). In the 3D-pressed samples, it was observed that there was a 3° difference in angle (at 1 sec) between the peaks (109.99°, SD = 17.75) and valleys (106.63°, SD = 20.29) (Table 11). It's important to note that the material is curved, which may be the cause for this difference in contact angle. For this reason, the flat samples are not compared with the 3D-pressed samples as this discrepancy is again a likely influence on the data. The flat-pressed samples were very consistent with each other despite their different heat-pressing temperatures. Both samples had water contact angles of 114° (150 °C SD = 26, 190 °C SD = 12) and 13 % of drops were maintained at 5 min (Table 10).

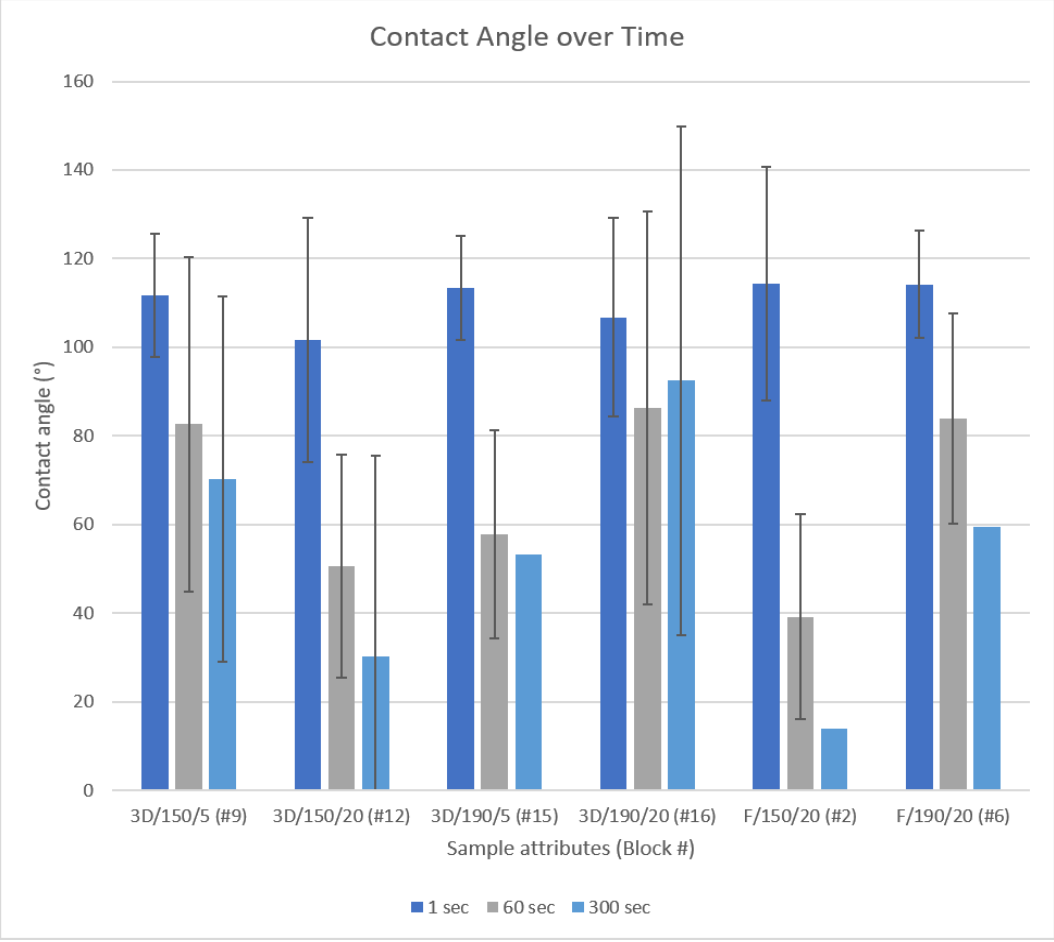


Figure 19*: Water contact angle change over 5 min on heat-pressed MBCs.*

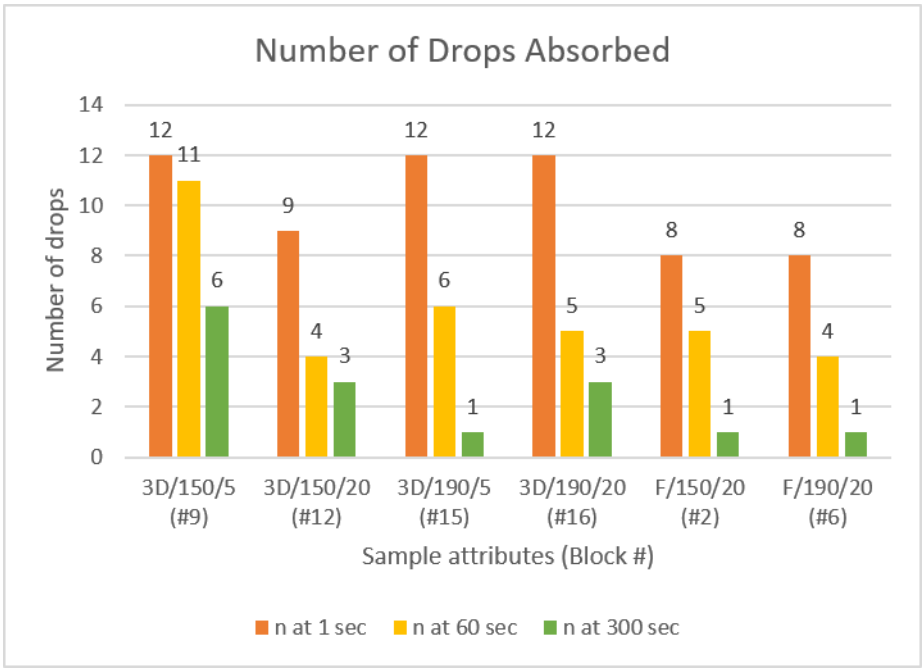


Figure 20: *Number of water drops (n) absorbed over time for each sample.* Tile attributes described as: shape (3D or flat) / temperature (°C) / Minutes heat-pressed.

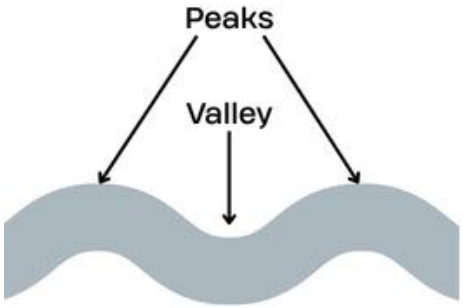


Figure 21: *Grey object is the heat-pressed MBC block from the front view, showing where the measurements were taken as "peaks" and "valleys" of the design.*

Table 10: *Water contact angle of water drops per MBC sample at 1 second, 60 seconds and 300 seconds after placement. Data corresponding to Figure 19.*

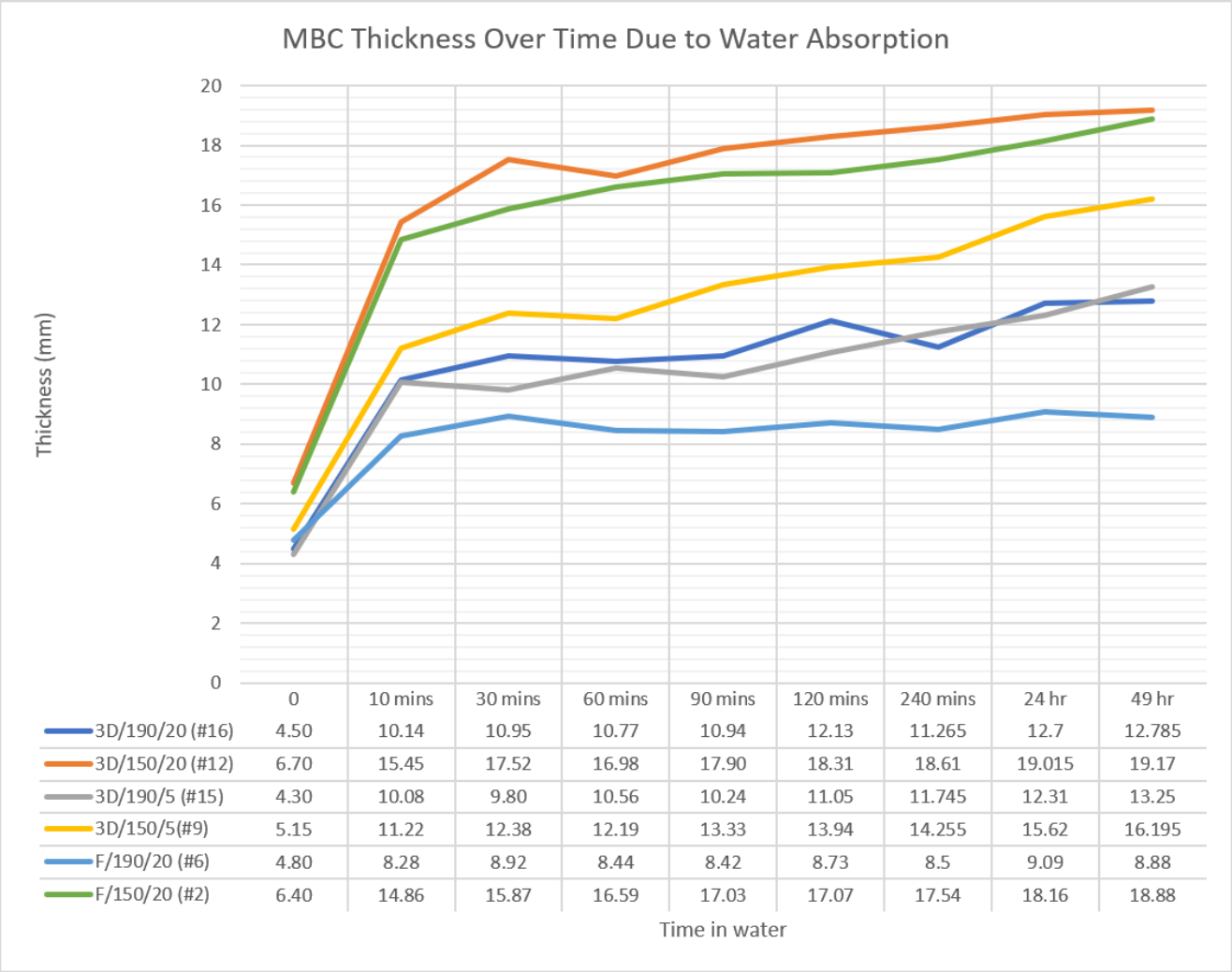
Block #	Attributes	1 sec (°)	n	SD	60 sec (°)	n	SD	300 sec (°)	n	SD
9	3D/150/5	111.71	12	14	82.61	11	38	70.28	6	41
12	3D/150/20	101.54	9	28	50.57	4	25	30.15	3	45
15	3D/190/5	113.39	12	12	57.66	6	23	53.24	1	n/a
16	3D/190/20	106.74	12	22	86.26	5	44	92.50	3	57
2	F/150/20	114.36	8	26	39.18	5	23	13.92	1	n/a
6	F/190/20	114.18	8	12	83.89	4	24	59.34	1	n/a

Table 11: *Water contact angle of 3D-pressed samples only at 1 sec, comparing the peak and valley.*

Block #	Attributes	Peak (°)	SD	Valley (°)	SD
9	3D/150/5	109.12	13.43	114.31	15.14
12	3D/150/20	100.13	26.06	102.67	31.76
15	3D/190/5	119.89	7.70	106.89	11.81
16	3D/190/20	110.81	23.81	102.66	22.45
	Average	109.99	17.75	106.63	20.29

Water absorption

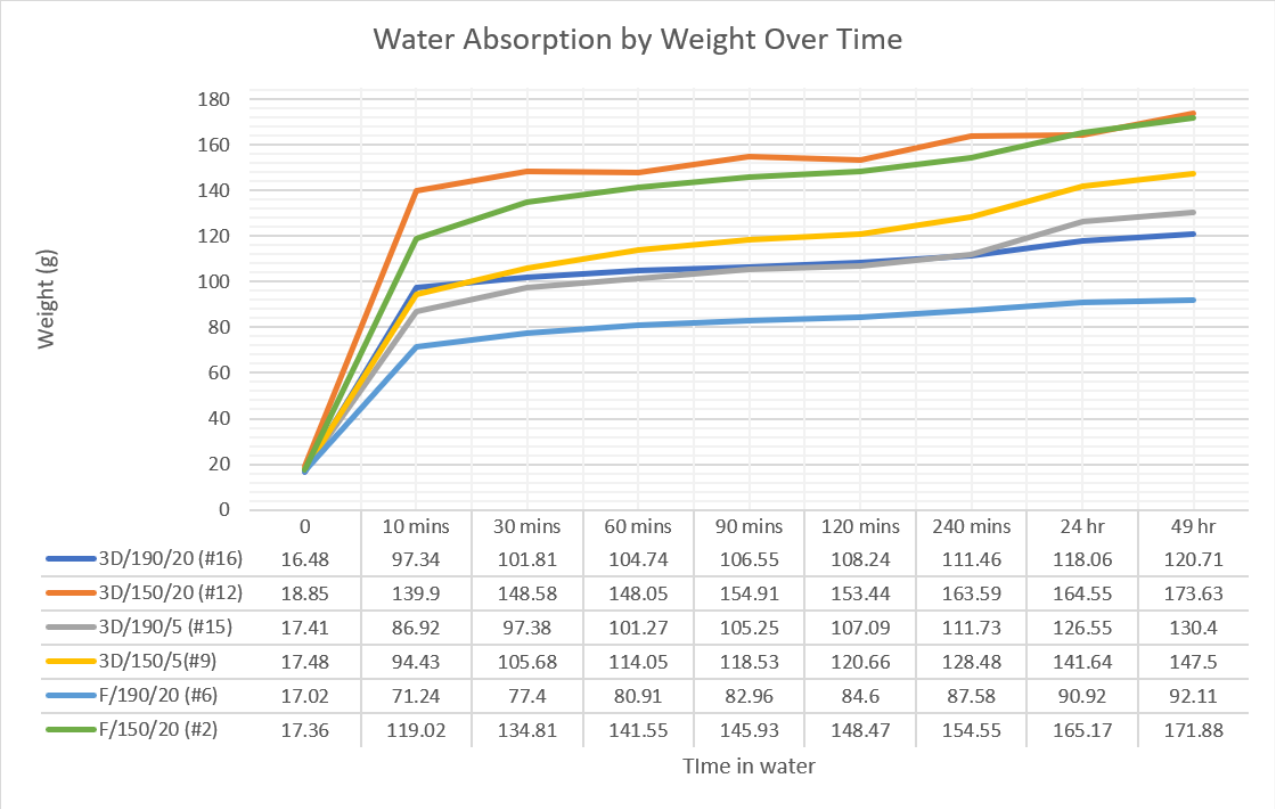
The starting thickness of the heat-pressed MBC blocks was somewhat variable, with the thinnest being ~ 4.3 mm and the thickest ~ 6.7 mm. The 3D-pressed sample at 150 °C for 20 min maintained the largest thickness throughout the experiment, having a thickness of 19.17 mm after 49 h submerged in water. The sample heat-pressed flat at 190 °C for 20 min (block #6) maintained a smaller thickness over the duration of the experiment and saw a change in thickness less than its thicker counterparts (increase of 85% over 49 h in water), the largest increase in thickness was seen in the 3D-pressed sample at 150 °C for 5 min (block #9) of which an increase of 214% (Figure 24) thickness was observed. In assessing the change of thickness over time, the largest changes were seen in the initial 10 min of the experiment (Figure 22) which slowed considerably after 2 h.



	0	10 mins	30 mins	60 mins	90 mins	120 mins	240 mins	24 hr	49 hr
3D/190/20 (#16)	4.50	10.14	10.95	10.77	10.94	12.13	11.265	12.7	12.785
3D/150/20 (#12)	6.70	15.45	17.52	16.98	17.90	18.31	18.61	19.015	19.17
3D/190/5 (#15)	4.30	10.08	9.80	10.56	10.24	11.05	11.745	12.31	13.25
3D/150/5(#9)	5.15	11.22	12.38	12.19	13.33	13.94	14.255	15.62	16.195
F/190/20 (#6)	4.80	8.28	8.92	8.44	8.42	8.73	8.5	9.09	8.88
F/150/20 (#2)	6.40	14.86	15.87	16.59	17.03	17.07	17.54	18.16	18.88

Figure 22: *MBC thickness in mm over time due to water absorption.*

Weight change of samples while submerged in water was also assessed. The highest increase in weight was observed in the first 10 min of submersion (Figure 23). The largest increase in weight was observed in the sample that was flat-pressed at 150 °C for 20 min (block #2) which saw an increase of 890 % since its dry measurement (Figure 24). The smallest change in weight was observed in the sample heat-pressed flat at 190 °C for 20 min (block #6). The three samples heat-pressed at 190 °C (#16, #15, #6) increased in weight collectively less than the three samples heat-pressed at 150 °C. During the experiment, it was observed that block #16 took on water weight at a slower pace than blocks #9 and #15 after initially absorbing more than them in the first 10 min. Comparing #16 and #15, the block which was heat-pressed for a shorter time took on more weight.



	0	10 mins	30 mins	60 mins	90 mins	120 mins	240 mins	24 hr	49 hr
3D/190/20 (#16)	16.48	97.34	101.81	104.74	106.55	108.24	111.46	118.06	120.71
3D/150/20 (#12)	18.85	139.9	148.58	148.05	154.91	153.44	163.59	164.55	173.63
3D/190/5 (#15)	17.41	86.92	97.38	101.27	105.25	107.09	111.73	126.55	130.4
3D/150/5(#9)	17.48	94.43	105.68	114.05	118.53	120.66	128.48	141.64	147.5
F/190/20 (#6)	17.02	71.24	77.4	80.91	82.96	84.6	87.58	90.92	92.11
F/150/20 (#2)	17.36	119.02	134.81	141.55	145.93	148.47	154.55	165.17	171.88

Figure 23: *Water absorption by weight in g over 49 hours.*

Comparing the uptake of water by the heat-pressed MBC samples via weight and thickness showed some variations (Figure 24). Block #9 had a higher (in comparison with the other samples) change in thickness but a lower change in weight. This was observed contrarywise in block #2 which saw the highest weight increase of 890% yet only 195% increase in thickness. A trend can also be identified in that the samples heat-pressed at a lower temperature absorbed more water weight, and samples heat-pressed for only 5 min saw the largest increases in thickness. The sample heat-pressed flat at 190 °C for 20 min showed the smallest change in both weight and thickness, additionally it also had the smallest difference between the two changes (difference of 356%).

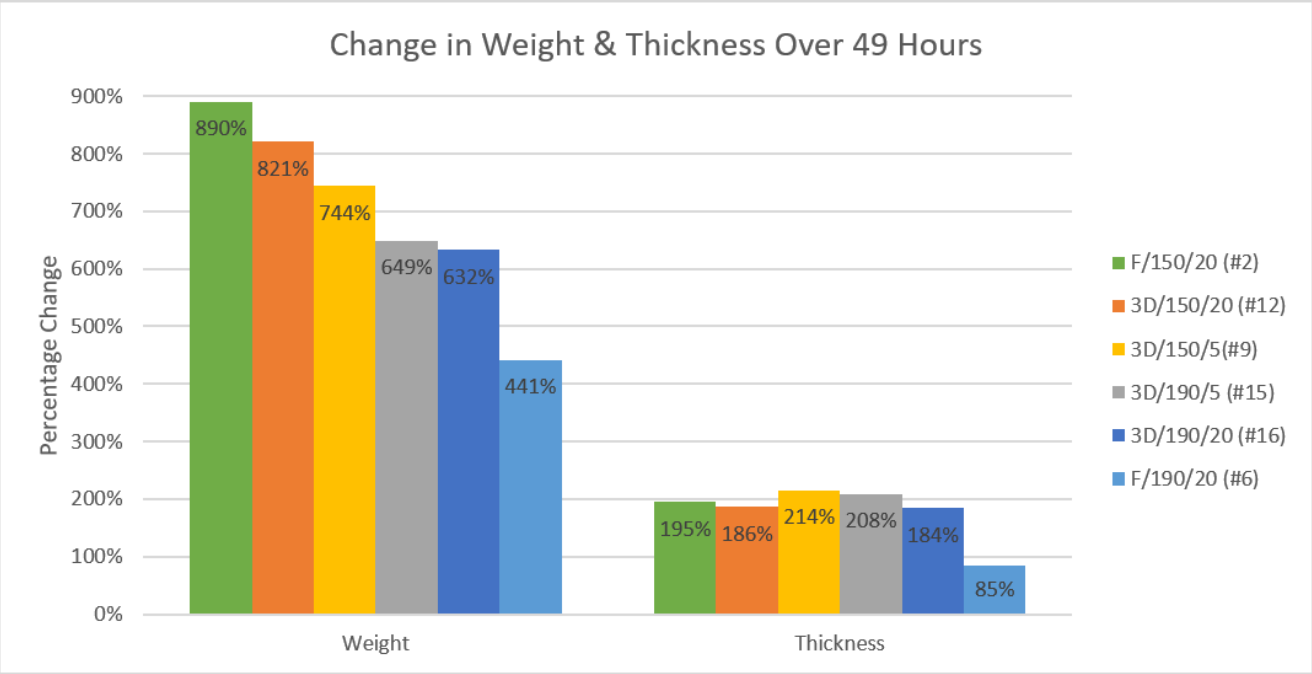


Figure 24: *Change in weight and thickness of heat-pressed MBC blocks from 0 min to 49 h while submerged in water.*

Maintenance of shape from water absorption was observed despite the increase in thickness of all samples. Every 3D-pressed MBC did maintain their shape, including the sample heat-pressed for only 5 min at 150 °C (Figure 25). However, in handling the samples, it was observed that the now thicker MBCs were more delicate and prone to breaking into two pieces, especially along the tear lines that were formed during heat-pressing.



Block #16 (3D/190/20)



Block #9 (3D/150/5)

Figure 25: *The shape of the 3D-pressed MBC blocks after 24 h of water absorption.*

Flexural strength

The way that each 3D-pressed tile deformed was consistent amongst samples. The middle peak which took force from above (Figure 26) deformed whilst the sides of the sample slid outwards and upwards. The skin on the underside tore underneath the failure point but the

whole sample remained in one piece. After the test, the blocks also maintained their changed state indicating plastic deformation.

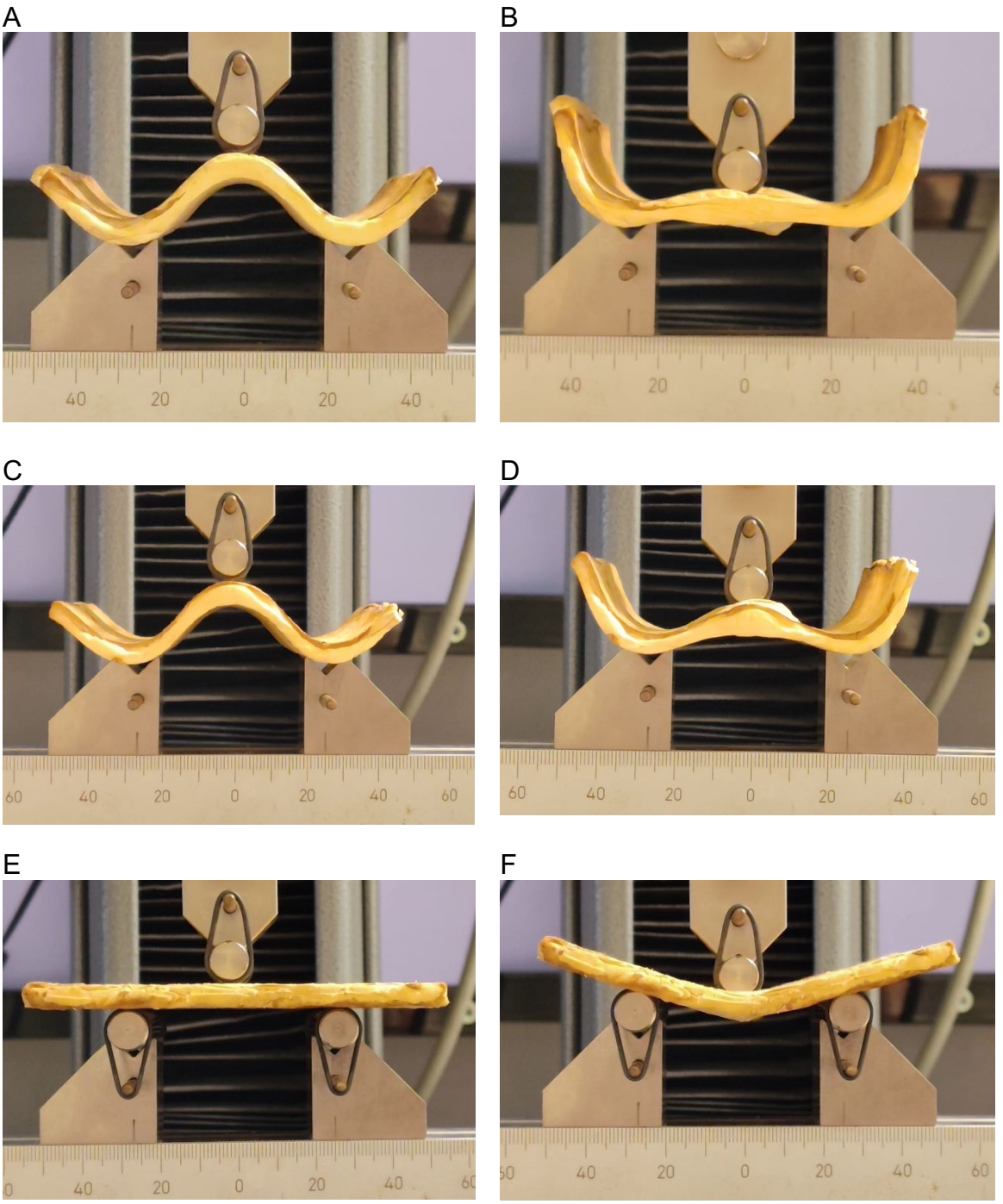


Figure 26: *Images showing block#18 (3D/190/20) at beginning of experiment (A) and deformation during failure (B); block #14 (3D/190/5) before test (C) and at failure point (D); also block #3 (Flat/190/20) at beginning of experiment (E) and deformation at failure (F).*

Of the 3D-pressed samples, the highest flexural strength was 2.88 MPa (pressed at 190 °C for 5 min (Figure 27)). Whilst amongst the flat samples, the highest flexural strength was achieved by the sample heat-pressed at 150 °C for 20 min. Interestingly, the 3D-pressed sample (pressed 190 °C for 5 min) which achieved the best flexural strength has a lower

Young's modulus (54.65 MPa) than its runner up (sample 3D/150/20 with Young's modulus of 84 MPa). The sample 3D-pressed at 190 °C for 20 min had a similar result: flexural strength was in a similar range to its flat-pressed counterpart (3D = 1.89 MPa, flat = 2.04 MPa), whereas the Young's modulus was then lower in comparison (3D = 41.1 MPa, flat = 59 MPa).

The maximum load that could be applied before fracture were lowest for samples heat-pressed both in 3D and flat at 150 °C for 5 min (Table 12), however important to note is the unreliability of the 3D-pressed sample in this instance (3D/150/5), whereby the sample initially slipped during the run and needed repositioning (that is, some force had been applied before this measured result was recorded). Table 12 also shows the number of samples per test which is lower than initially planned for due to hurdles growing the MBC. Therefore, these results are only indicative and further testing is required. Looking at the extension, the 3D-pressed samples are each larger than the flat-pressed, indicating much more movement/deformation before failure of the material.

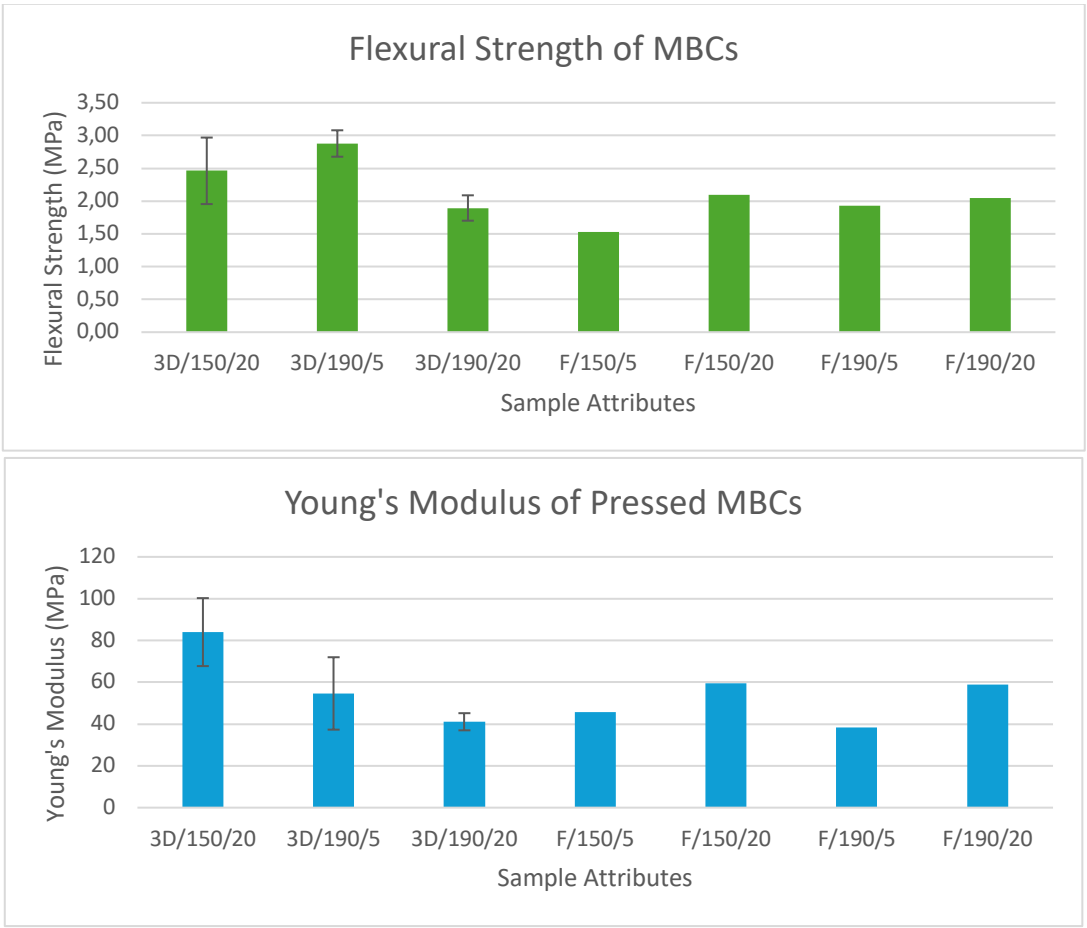


Figure 27: *Flexural strength of heat-pressed MBCs and the Young's modulus of each. Standard deviation for samples n >1.*



Table 12: *Three-point flexural test results for heat-pressed MBCs, showing highest load (N), flexural strength (MPa) and the maximum extension (mm), and Youngs Modulus (MPa).* Highest load value, flexural strength and extension was recorded directly from calibrated testing machine/computer. Young's modulus calculated using RStudio. *Note, data missing from sample 3D/150/5 due to unreliability of sample.

Tile Attributes	n	Highest Load (N) Value & (SD)	Flexural Strength (MPa) & (SD)	Extension At Max Load (mm) & (SD)	Young's Modulus (MPa) & (SD)
3D/150/5	1	43.95	n/a*	10.60	n/a*
3D/150/20	2	78.91 (16.26)	2.46 (0.51)	11.47 (1.85)	84 (16.26)
3D/190/5	2	92.26 (6.46)	2.88 (0.2)	13.68 (1.26)	54.65 (17.32)
3D/190/20	2	60.72 (6.21)	1.89 (0.19)	14.34 (0.67)	41.1 (4.1)
F/150/5	1	48.99	1.53	8.59	45.8
F/150/20	1	67.28	2.10	8.21	59.4
F/190/5	1	61.69	1.92	8.30	38.3
F/190/20	1	65.45	2.04	6.25	59

Flexural strength can be further investigated using the below stress-strain curves (Figure 28). Looking at the graphs below, samples 10 and 11 seem to have an obvious initial yield in the material, before failure under a slightly higher stress. The flat samples have a very clean curve, achieving maximum stress before decline. The 3D-pressed samples experience a prolonged strain with the deformation of the material and its shape. The overlapping stress-strain curves in Figure 29 depict a similar linear incline amongst samples until about 0.5 MPa, at which point the samples begin to behave differently from each other. It's interesting to note that there looks to be some grouping of when the flexural stress begins to increase, however no correlation with sample types can be reasonably determined from this set of data.

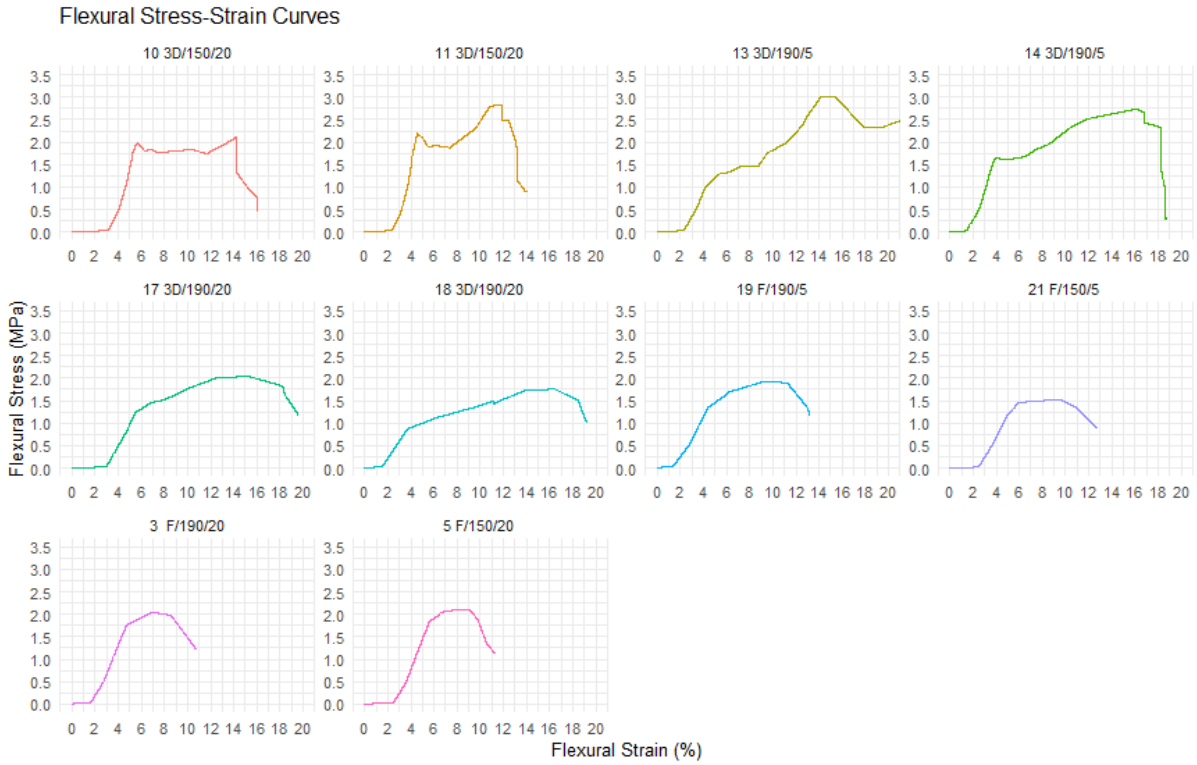


Figure 28: *Stress-strain curves of each MBC sample undergone a 3-point flexural strength test.*

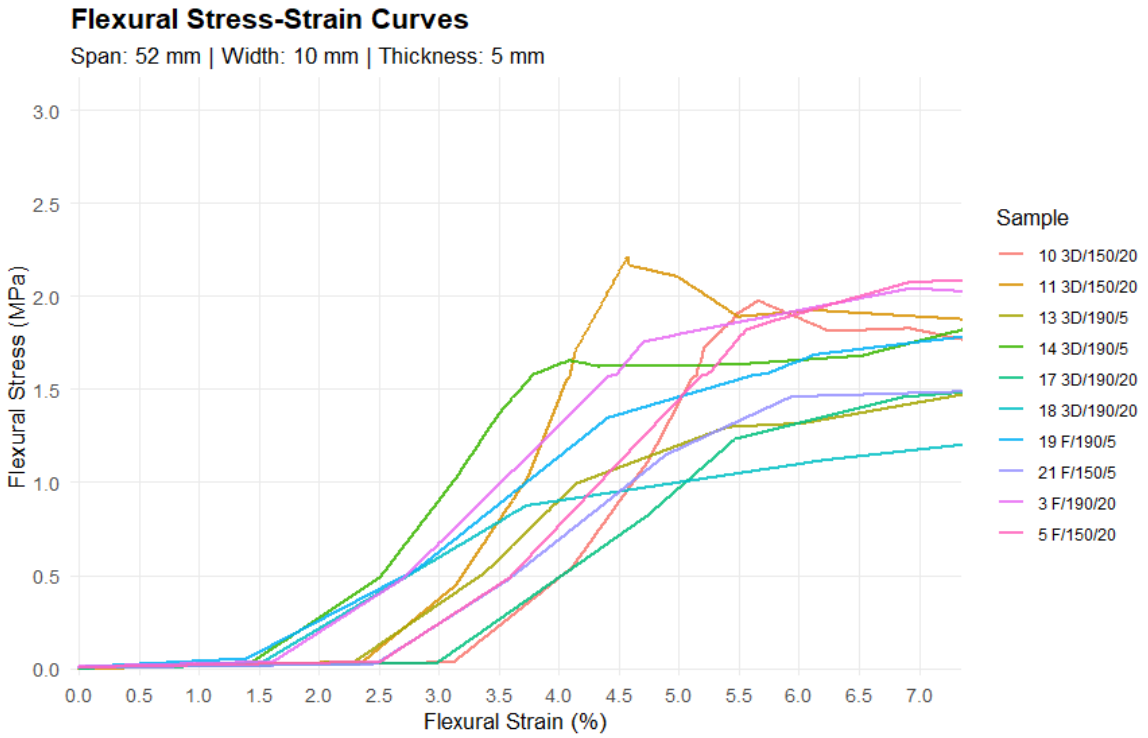


Figure 29: *Overlapping flexural stress strain curves of the heat-pressed MBCs after 3-point strength test.*

Assembled panel prototype

The heat-pressed MBC tiles were assembled in a 3 x 3 tile configuration and attached to a timber frame to determine how they would look and fit together as a panel prototype. This proved successful as the tiles provide air spaces thanks to the curved design allowing airflow to pass through, and the construction conveys a much more biophilic aesthetic (Figure 30). However, this prototype is uncoated and as such would require storage/display indoors in dry environments to avoid moisture absorption.



Figure 30: *Final prototype made from assembled heat-pressed MBC tiles on a timber frame (approx. 30 x 30 cm).* Left: front view, right: angled side view with timber frame visible.

B.3.4. Discussion

Material tests performed on the MBC blocks garnered some indications of the weatherability of the material, as well as proved that they can be heat-pressed into a three-dimensional shape from a cubic starting shape. Heat-pressing the MBC blocks at a higher temperature of 190° and for a longer time of 20 min reduces the average thickness of the material considerably (from 25 mm pre-pressing to 4.5 mm after heat-pressing). Therefore, depending on the desired outcome, one could tailor the method of heat-pressing to achieve it. More importantly, the blocks held the shape they were heat-pressed into, even with observed tearing of the skin at the peak area of the block. This is likely because the block was grown in a cubic shape, which requires some deformation of the surface to meet the three-dimensional form. This could be avoided by either growing the block in a matching three-

dimensional form or heat-pressing into a three-dimensional shape with smaller deviations from the initial cubic form and surface.

Heat-pressing did reduce the white value of all samples which was to be expected (average decrease across samples of 20%). This would indicate a reduction in reflectivity which detracts from the desired cooling function this project is aiming towards but highlights this aspect as a topic for further study as there are other factors such as differing white values before heat-pressing per sample. Additionally, since the surface does not maintain a white colour after heat-pressing, a coating is necessary to optimise reflective cooling. In an outdoor context, the heat-pressed blocks would likely have difficulty maintaining form and function due to wet weather, as indicated by the water tests. Water contact angle tests showed that the untreated material is initially hydrophobic, but this is short lived - after 5 min all samples saw the absorption of drops occur, with the best performing attributes being seen on the sample 3D-pressed at 150 °C for 5 min. This is likely due to the least change of the natural mycelium surface (which is normally quite hydrophobic) in comparison with the other samples.

Interestingly, the opposite effect was seen when investigating water absorption of the heat-pressed MBC blocks. On all accounts, these samples absorbed more than what has been seen in previous studies , however this is likely due to the method of testing and difference in density/thickness. This highlights the need for density tests to be undertaken in conjunction with absorption tests; as well as a more standardised method of comparison testing as also outlined by Andrade, et al. (2026). Due to these samples being three-dimensional (without a flat surface), they were submerged in water – this method was undertaken to achieve an even water exposure over the entire sample. The subsequent tests revealed that the sample heat-pressed flat at 190 °C for 20 min showed the smallest change in both weight and thickness (441% weight increase, 85% thickness swelling). The same can be said for the block heat-pressed into a three-dimensional shape for the same time and temperature in comparison with the other 3D-pressed blocks. These observations indicate the material is better able to resist water absorption if it is heat-pressed under these conditions, likely due to the improved bonding during the heat-pressing process .

In testing the material's flexural strength and deformation behaviour there was much to observe. Among the 3D-pressed samples, the strongest was heat-pressed at 190 °C for 5 min and showed a lower Young's modulus; while among the flat samples, the strongest sample was heat-pressed for 150 °C for 20 min and maintained a higher Young's Modulus than its counterparts. This indicates a potential effect of the geometry of the heat-pressed MBCs for enhancing the flexibility of the material while maintaining strength. More broadly, the flexural strength of the flat samples is similar to the range seen in other studies with various species/substrate combinations (Appels, et al., 2019; Liu et al., 2019; Shakir, et al., 2020). Due to the nature of the 3-point test, it is difficult to distinguish the relationship between the three-dimensional shape bending with the force applied, and the strength of the material itself. Nonetheless, it did give insight into the deformation behaviour, as it could be observed how the shape changed, and in what directions. For the design of the façade panel, it can now be predicted that the tile can deform in one central area, while the adjoining sides

maintain shape (which can in turn affect neighbouring tiles in the larger panel configuration) which was observed in every sample tested. Interestingly, recent researched has indicated semi-wet (30-50%) heat-pressing of MBC has improved flexural strength (Song, et al., 2025), highlighting a possible improved method. However, there is still a lack of research into 3D-pressed forms beyond the variation of flat-pressed studies (Andrade, et al., 2026) due to the large variation in species, substrate and testing protocols. To form a conclusive answer to which heat-pressing methods are best in the context of 3D-shaped pressed samples, more tests of this kind should be undertaken in combination with necessary discussions amongst fellow researchers in this field to improve standardisation for the long-term progress of this biomaterial.

Regarding the tests performed on these MBC blocks, it must be concluded that further testing is required. Results are also influenced by species and substrate choice (Appels et al., 2019; Liu et al., 2019; Shakir et al., 2020). Further research could include tests of reflectivity, UV degradation, long term testing in context, and large-scale prototype testing (convection) in synchronicity with the broader goal of this project. For the purposes of this project, the key goal of determining if heat-pressing into a three-dimensional shape is possible was indeed achieved. And the accompanying tests performed indicate that there are certainly differences in material quality based on how the material is processed. MBC could be integrated into the project panel as heat-pressed MBC in the future, but not in the current state. These experiments revealed that the three-dimensional shape is indeed a viable option and can be assembled into a larger panel, but the importance of surface treatment for outdoor conditions is currently lacking in these samples. These tests proved the concept of heat-pressing rectangular MBC blocks into a three-dimensional shape, which paves the way for a variety of applications beyond this project scope (such as packaging, building materials, and more). Investigating the range of shapes possible would reveal to what extent these applications might be, and improving the strength of the material in these shapes would also enhance this.

Part C: Product Development and Implementation

To successfully translate an innovation into a product that produces real- world change a concrete business and implementation plan is necessary. The following section examines how the façade panel can be turned into a product, which through its technological features reduces energy consumption and mitigates the UHI effect.

To achieve said goal a possible business model, Intellectual property (IP) strategies and implementation strategies are analysed. The section is a summary of three individual reports, all of which can be found in greater detail in their respective reports (Appendices F-H).

C.1. Business Model Analysis

The primary customer base for the *FunctionalSkin* is believed to be professional clients (Appendix F2), rather than individual homeowners. Based on this customer profile it is possible to create a value proposition canvas which highlights how the products and services provided translate into created gains and relieved pains for the customer (Figure 31).

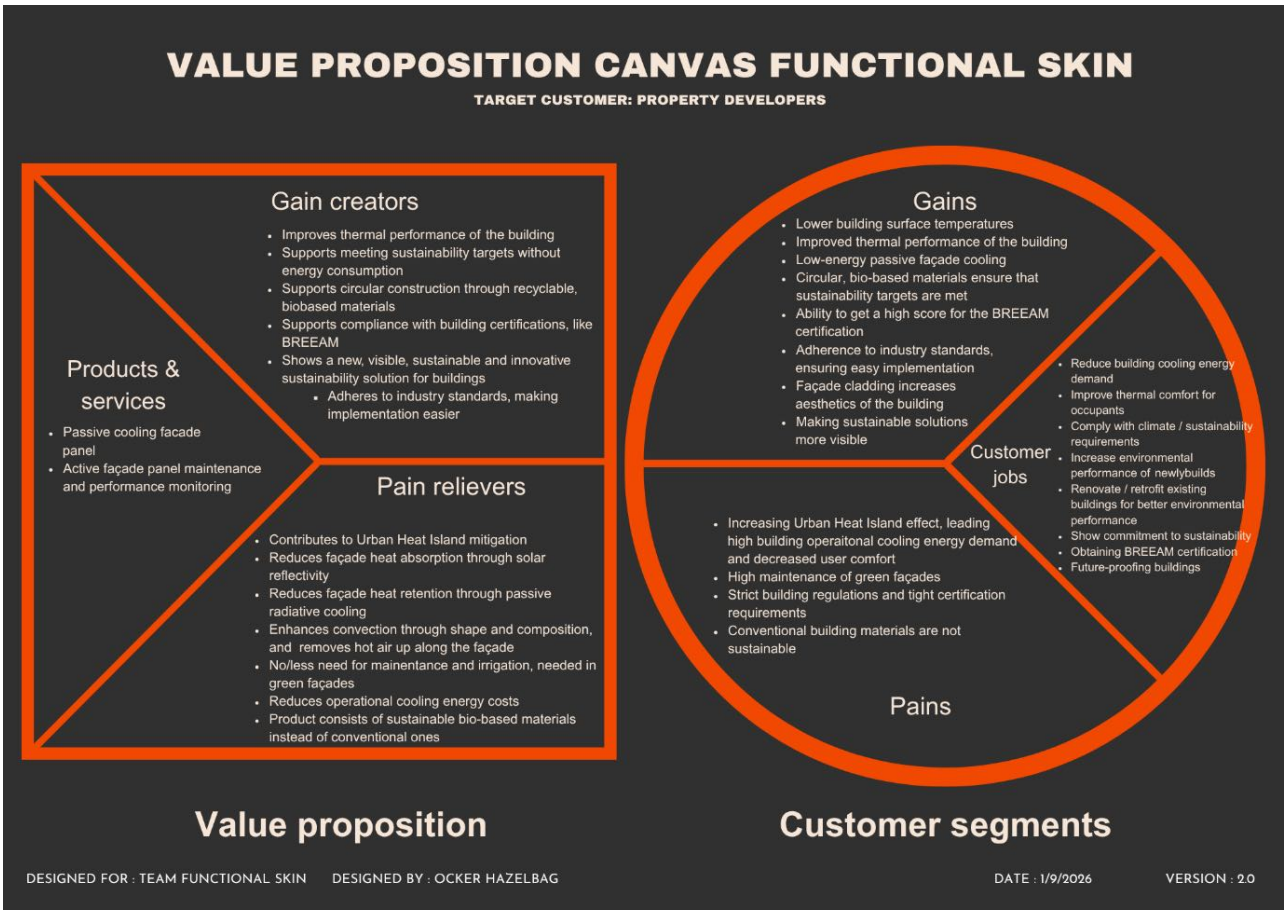


Figure 31: *Value Proposition Canvas with property developers represented as target customer group*

Apart from the primary customers there are other groups whose interest are indirectly represented through the customer. The residents as well as architects or construction workers tasked with installing the panel each give demands to the target customer which he

needs to comply with. To fulfil the target customer's needs, it is therefore important to address the needs of these additional stakeholder groups as well.

In the industry of façade cladding systems with a market share of 270.5 billion USD (Grand View Research, 2026), ventilated façade make up a share of 21.74 billion USD (WiseGuyReports, 2026). The biggest market advantages the *FunctionalSkin* has in this industry is its reliance on sustainable building materials and the passive cooling mechanism. They allow the panel to pass sustainable building regulations and reduce the time and costs needed for maintenance, compared to an active cooling mechanism.

When compared to other passive cooling façades, the *FunctionalSkin* would rank somewhere between a conventional ventilated façade and a living green wall in terms of both instalment and maintenance costs. Its cooling function is more sophisticated than that of a conventional ventilated façade panel. Similar to a Living green wall, its embodied carbon is low if biobased materials are used for its production (Table 13).

Table 13*: Market comparison of the FunctionalSkin panel, a conventional ventilated façade, and a Living Green Wall.*

Criterion	***FunctionalSkin***	Ventilated Façade	Living Green Wall
Installed cost per m^2	€150 - € 200 (assumption)	€50 - €120 (Kurian & Karthi, 2022)	€450 - €1500 (*Cost Comparison of Different Types of Green Facades*, z.d.)
Maintenance costs per m^2 per year	€4 - €6 (assumption)	Low, optional cleaning	€50 - €150
Cooling function	Reflects solar heat, radiates absorbed heat into space, channels warm air upwards	Cavity ventilation	Shading and evapotranspiration by plants
Energy and water input	None	None	Irrigation system required
Embodied carbon	Low (biobased materials in use)	Higher (often non-biobased materials in use)	Low

The main revenue streams for the product is the sale of façade panels to the customers. Maintenance services and performance monitoring services can generate additional revenues. IP Licensing is not a viable revenue stream for our product (Appendices F, G). The business model canvas combines activities, interactions with partners and customers, created values as well as costs and revenues into a concise infographic (Figure 32).

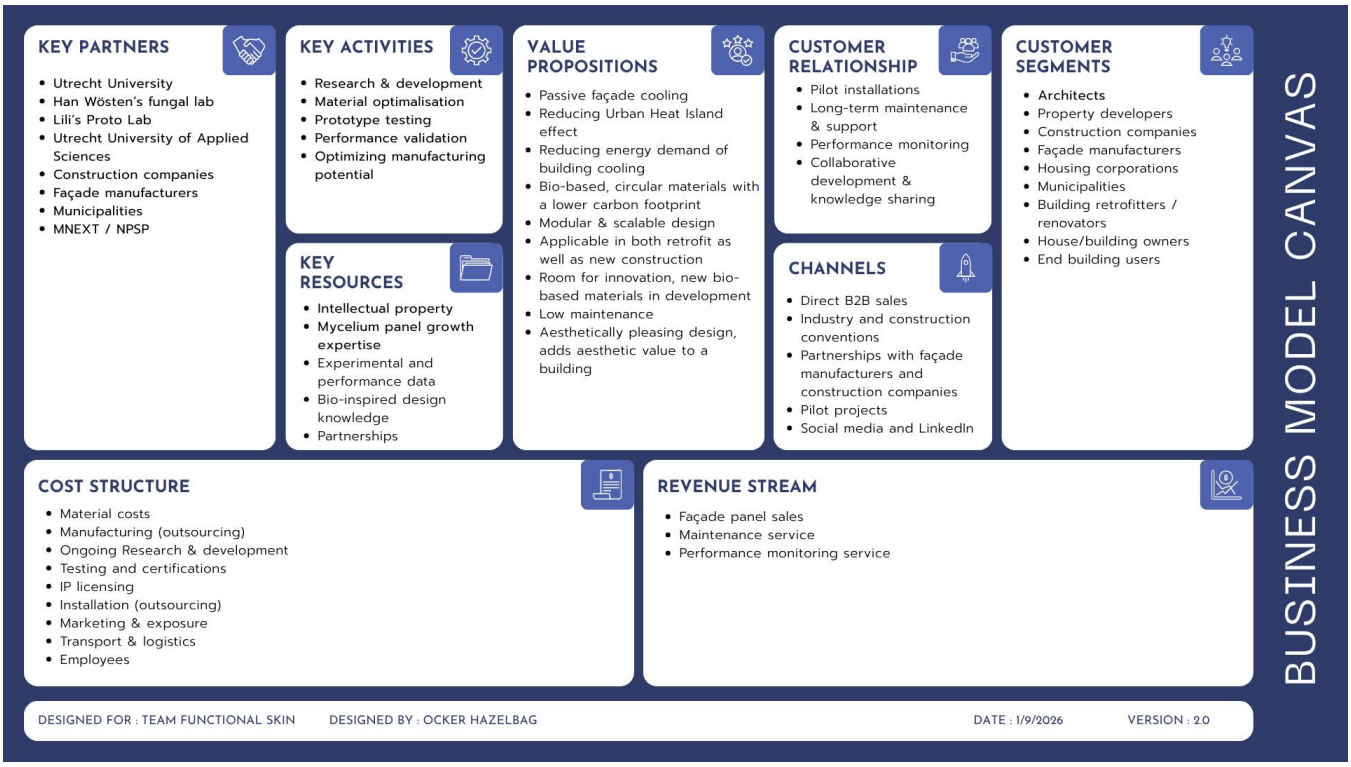


Figure 32: *Business Model Canvas*

C.2. Intellectual Property Analysis

One potential resource that was paid particular attention to is Intellectual Property (IP). An IP analysis serves the purpose of both identifying risks of IP infringement and chances of obtaining IP of our own. A preliminary search has not revealed IP infringements from the *FunctionalSkin*. However, this preliminary search is not equal to a full freedom to operate analysis, which could be used in communication with investors.

The relevance of our product components within the patent landscape are underlined in the quantitative patent analysis findings. Over the observed time of twenty years almost all categories which apply to the *FunctionalSkin* have seen a continuous increase in patent activity - with the sole exception of wood and wood based cladding systems, which had slightly less patents published in from 2022 to 2025 than during the previous four-year period (Appendix G). The highest increase in patent publications over the observed periods were patents on mycelium materials and forms of passive cooling, each having a three to fourfold increase in patents filed from 2022 to 2025 compared to the baseline of 2006 to 2009. Nonetheless, especially in the application of façade panels, wood-based inventions number almost 500 times higher than patent publications using mycelium materials. Most passive cooling façades focused on cooling via convection, while reducing solar heat gain via reflection and radiative cooling (also included in our prototype) were sparsely represented in new patent publications (Appendix G).

From a business model perspective these gaps can be interpreted as unoccupied niches in the market giving us opportunity to set our product apart and obtain an advantage over competitors. On the other hand, using wood as a material and utilising natural convection for

passive cooling offers more established alternatives, while maintaining the sustainability advantages of using biomaterials and energy free cooling mechanisms (Appendix G).

For our own IP strategy, we decided not to invest money and time into obtaining a patent of our own. This decision is based on consultancy with an expert in IP, Maud Kerstholt and a representative of a company active in the field, Naud van der Padt from Leafy. Both experts suggested that obtaining and enforcing a patent is in many cases not a viable option for small startups (Appendix A). Furthermore, the developed product likely would not meet the requirements needed for a patent, as the design is not new (Pujadas-Gispert et al., 2020). Instead, the focus will be to obtain forms of IP that can be obtained by default. While copyright does not apply for our technologically defined product, information we do not publish can constitute a trade secret if we take some protective measures (Appendix G). Nonetheless we consider our commitment to open science and scientific transparency as more important than an optimal IP strategy.

C.3. Implementation Plan

On a technological level there are also multiple hurdles that need to be addressed to convert the prototype into a finished, commercial product. One of the steps required is an attachment mechanism to the façade. The attachment mechanism needs to support the structural load of the panel, which is primarily influenced by wind pressure and secondarily influenced by the weight of the panel. It also needs to be flexible to the dimensional changes of wooden panels which occur through moisture and temperature changes. It should leave the outer portion of the panel uninterrupted to ensure optimal functioning of the cooling mechanisms, resulting in a continuous reflective surface and unobstructed airflow. Lastly for maintenance purposes, the panel should be easily (de-) mountable.

Façade panels which leave the outer portion of the panel uninterrupted are classified as concealed systems. Point fixing systems are generally suitable for thin wall panels, as they do not create areas at which pressure on the panels becomes too great and support movement of the panel, without risk of breaking. The NV3 system by NVELOPE (SFS Benelux) is such a point fixed concealed system and fulfils all given requirements and can therefore be treated as a general recommendation. However, depending on local conditions like higher wind speeds, different mechanisms might be needed.

Another hurdle is the longevity of the product. The current prototype is made only of indoor plywood, which would have a lifespan of around one year in outdoor conditions (Van den Bulcke et al., 2010). For a commercially viable product a longer lifespan is needed, which can be achieved by using more resistant wood and treating the wood. For the *FunctionalSkin* douglas fir will be used (Appendices C, E). Viable treatment options for increased longevity include thermal modification and acetylation (Aro et al., 2014; Candan et al., 2012; Lovrić et al., 2017; Slabohm et al., 2022). The most ecofriendly alternative is impregnation with natural oils or waxes; however, this protective layer would need to be reapplied regularly (Jebrane et al., 2017). In general, to make the product attractive for customers, some form of maintenance should be offered as a service (Appendices C, F, H). This can also serve as an

additional revenue stream for us. In the case of a wooden panel, gentle washing instead of high-pressure cleaning is required (Gustavsen et al., 2026).

A key hurdle that must be addressed is ensuring fire safety, otherwise, the entire panel would be unusable. For application in a high rise building the panel needs to adhere to the Euro-fire class B and in any case at the minimum fire class D has to be reached (EOTA, 2012). This poses a challenge, as the *FunctionalSkin* has multiple features which make it susceptible to fire. Both materials of wood and mycelium are flammable (M. Jones, Bhat, Kandare, et al., 2018; X. Li et al., 2024) and the chimney-like structure that utilizes upward convection of warm air to remove heat from the façade, could serve to supply a fire with more oxygen and transport it upward the façade quickly (Appendix H). Therefore, fireproofing mechanisms like fireproof coatings and fire barriers will be necessary (Fire Protection Association, 2023), as the in the end the whole façade needs to adhere to the fire class, even if components of the façade would have different individual classifications (Kaukanen, 2026).

Other considerations to be measured for the final panel is degree of thermal and acoustic insulations, which according to the *Besluit Bouwwerken Leefomgeving* (BBL), should insulate at least 20 dB of sound and have an Rc value of at least 4.7 (Rijkswaterstaat, n.d.). Lastly the manufacturing process needs to be scaled up to reduce the cost of a single panel, and an eco-friendly end of life plan should be provided (Appendix H).

Each technological challenge requires additional time and money to develop and produce a solution. Furthermore, for each technology introduced there is an additional risk of patent infringement. From a business perspective it might therefore be an attractive option to either license another company's technology or collaborate with another company to arrange for parts of the product preproduced by them.

C.4. Technological Readiness Level and Development Roadmap

At the beginning of this project the client expressed the wish of bringing the product to a Technology Readiness Level (TRL) 5. This following section will investigate to which extent we succeeded in this endeavour, and which steps would be necessary to advance the product further on the technological readiness scale.

The TRLs are a scale originally developed by NASA used to define the status of technological products. There are multiple iterations of the TRL scales, the definitions that will be used for this paper are taken from the European Commission's criteria to evaluate EU funded projects (Bruno et al., 2020). It ranges from the observation of basic principles (TRL 1) all the way to a fully established product on the market, proven in an operational environment (TRL 9).

This project continued a previous project initiated in 2025 by another group of students, who established an experimental proof of concept for the technology (Boer et al., 2025), corresponding to a TRL 3 product. Due to differences in testing methodology, it was not possible to continue working on the same prototype and advance it to higher TRL stages. Instead, building on their work, the design was improved further, effectively returning the

project to TRL 2. Subsequently, an experimental proof of concept, using temperature and wind-speed measurements in simplified chimney models was established, thereby advancing the project to TRL 3. Finally, a functioning prototype was developed and tested under laboratory conditions, advancing the project to TRL 4.

The goal of a TRL 5 product, according to the European Commission's definition, is a functioning prototype of a product component that has been tested in a relevant environment. What this translates to in the context of the *FunctionalSkin* would be a panel installed to an outside wall of a building. Testing in outdoor conditions can be a realistic goal for future research groups continuing the *FunctionalSkin* project. It would not require the covering of an entire façade, as this would already correspond to TRL 6. Finally, successfully carrying out all of the propositions of the implementation plan and integrating the product in the proposed business model is what would allow the product to reach the final TRLs of 7-9 (Bruno et al., 2020).

General Discussion and Conclusion

This Major Innovation Project investigated how a façade panel can passively cool buildings and the surrounding urban environment, using nature as a source of inspiration. Several literature studies have been carried out, finding bio-models for passive cooling, analysing material properties and choosing materials, and finding the most suitable shape and design. Also, experiments were carried out to address the research question and validate the performance of the panel, consisting of modular 15 x 15 cm tiles made of 5.5 mm thick plywood, coated in CryoPaint®. Lastly, the business case was strengthened by developing a business model, assessing intellectual property options, and drafting an implementation plan

The panel demonstrated passive cooling properties in the experiments, demonstrating that the design, and material functioned as hypothesized under experimental conditions. Analysing the results, convective airflow was observed within the test setup with the simplified shaft, as well as the final design, being substantially higher than the negative controls. Both a greater panel thickness (5.5 – 9 mm vs 12 mm) and a radiative cooling coating lowered the temperature of the backwall. Thus, subject to conditions, indicating the highest passive cooling properties.

When testing the final design in the climate chamber under a 30 °C summer baseline, the CryoPaint®-coated prototype lowered the STBI to 39.2 °C under natural convection. This is a 24.8 °C reduction compared to the unventilated control (64.0 °C) and 7.0 °C lower than the flat CryoPaint® reference (46.2 °C), which substantiates the effect of the added geometry. When introducing an external wind field (2–4 m s^{-1}), the upper cavity exhaust velocity accelerated nearly fourfold, from 0.47 m s^{-1} to 1.77 m s^{-1}, suppressing *STBI* further to 34.2 °C and backwall backside temperature (*STBB*) to 29.0 °C, below the 30 °C ambient chamber baseline. This demonstrated the feasibility of integrating multiple complementary passive cooling strategies into a single façade panel. Although these strategies contribute to UHI mitigation, the effect of the panel on building cooling demand and local urban air temperature was not quantified in this project and requires further real-world assessment.

Currently, heat-pressed MBCs as a façade material are only at TRL 3, as its main value lies in its potential as a future material. The MBC blocks can be heat-pressed into a rounded, three-dimensional shape using a mould, but the current mechanical and moisture-related properties prevent the material from being considered for façade application. The heat-pressed blocks were tested for white value, water contact angle, water absorption and flexural strength. These tests revealed the need for a surface coating for use in wet weather and to optimise surface reflectivity. An added coating could also improve the flexural strength of the material, which showed improvements in comparison with its flat-pressed counterparts, potentially due to the rounded three-dimensional shape.

Thermal performance

In previous research, the thermal performance of façades with the same geometry was assessed using different materials (Nabasco® 8010 and aluminium), under both experimental and real-world conditions (Boer et al., 2025; Pujadas-Gispert et al., 2020).

Pujadas-Gispert et al. (2020) evaluated the thermal performance of their ventilated façade in Dubai, finding that the temperature and air velocity in real-world measurements showed upward airflows that contributed to heat dissipation along the façade during the daylight hours. Upward airflow velocities of up to 8 m s^{-1} were achieved within the façade during daytime hours, with a difference of about 4 m s^{-1} between the top and bottom. In addition, a relation between outdoor wind speeds and cavity air velocity was found. This relation was tested under Dutch climate conditions in the climate chamber. The addition of a wind stream (2–4 m s^{-1}) increased cavity air velocity from 0.47 m s^{-1} to 1.77 m s^{-1}, moving accumulated thermal energy out of the cavity, reducing the vertical cavity temperature differential from 5.5 °C to 0.8 °C, and lowering STBI from 39.2 °C to 34.2 °C. This agrees with the results of Boer et al. (2025) and Pujadas-Gispert et al. (2020): both indicate that ambient wind adds to buoyancy forces to drive upward convective airflow within the façade cavity, rather than causing aerodynamic blockage or stagnation.

Pujadas-Gispert et al. (2020) found a temperature difference of up to 15 °C between the outer panel and apartment outside wall in 40 °C weather. This was attributed to the shading, insulation and airflow provided by the façade layers. But the airflow claim was not evaluated using negative controls, leaving room for further research. In this Major Innovation Project (MIP), the effect of convective airflow on the panel's cooling properties was evaluated using a simplified shaft, showing that sealing the lower intake in the negative control resulted in stagnant recirculation, a vertical air gradient of only 0.60 °C, and an elevated backwall temperature (*STBI*) of 44.3 °C. By contrast, opening the bottom inlet to allow natural buoyancy-driven airflow established a vertical cavity gradient of 7.2 °C and reduced the backwall temperature by 4.54 °C (to 39.80 °C) in the 8 cm deep panel, and by 10.48 °C (to 33.86 °C) in the 16 cm deep panel.

However, Pujadas-Gispert et al. (2020) showed a variation in air velocity between lower and upper parts of the façade, claiming that the airflows within the façade were disjointed and non-continuous. This is likely because the individual tiles were mounted directly onto a backboard, closing off an extra cavity behind that wall. In the final prototype in the MIP, a framework was constructed that ensured that a continuous flow of air could be generated along the façade, in the cavity between the panel and the backwall, leading to larger air velocities throughout the whole cavity. Boer et al. (2025) support this choice, as they showed experimentally that air velocity, and thus the passive cooling capacity, was highest when lamellae were placed and shafts were formed. This suggests that continuous vertical shafts reduce flow disruption within the panel and sustain higher air velocities than segmented cavities.

Although these previous experiments by other universities were done under different conditions, they support the panel's expected thermal performance, even in the Middle Eastern climate of Dubai (Pujadas-Gispert et al., 2020). However, the findings in this project in the first setup, as well as the climate chamber, cannot directly be translated to performance on an existing façade.

First, different temperatures, humidity levels, wind speeds, rain, and UV exposure should all be tested in real-world installations as well as controlled environments. It can then be determined what effect the application of such a façade panel actually has over the course of a whole year, and whether it only offers benefits in summer under ideal conditions. Furthermore, the weather resistance of the final chosen material should be evaluated over longer periods of time, finding out if the performance of the panel deteriorates as the panel wears down. The panel has so far only been tested under conditions simulating a hot summer day, so its performance in other seasons or weather conditions remains unknown.

Compared to Boer et al. (2025), who reported a 3.4 °C reduction in the *STBI* temperature comparing a configuration with the panel to one without it, the prototype tested in this MIP achieved a reduction of up to 24.8 °C under natural convection alone, rising to almost 30 °C with added wind. While a direct comparison with Pujadas-Gispert et al. (2020), who reported a 15 °C difference between the panel's outer layer and inner wall in Dubai, is limited by differing metrics and climates, the scale of reduction in this MIP indicates that the design's passive cooling capacities are at least competitive with, or even exceed, those of previously studied panels.

Material requirements

Based on the panel's requirements and the *Material selection report* findings, plywood was selected as the prototyping material (Appendix C). Its possible use was also included in the LCA (Appendix E). However, the *Material selection report* concluded that Douglas fir would be the most suitable wood type for the project. Further analysis and expert interviews showed that Douglas fir would be too thick for the design of the prototype, which is why plywood was chosen (Appendices A, C). Increasing the panel dimensions from 15 x 15 cm to 30 x 30 cm (following the tile size used by Pujadas-Gispert et al., 2020), could already enable the use of thicker, solid timber replacing the plywood. Plywood should therefore be regarded as a prototyping material, rather than the final optimal material for the panel, as it is not suitable for outdoor façade applications without an extra durable coating. At higher TRLs, the required thickness, durability and manufacturing process may instead favour solid timber or a biobased composite.

In addition, the results of the LCA showed a GWP of 4.02 kg CO_2e/FU for Douglas fir, while outdoor plywood had a GWP of 15.52 kg CO_2e/FU (Appendix E). When corrected for annual lifespan, the difference increased further. This suggests that the environmental impact of the panel made from plywood could potentially be reduced significantly by replacing it with raw Douglas fir timber, if the required panel thickness increases. The potential of a thicker panel was substantiated in the simplified shaft experiments, where the 12 mm thick panel showed substantially higher thermal insulation and delayed thermal breakthrough, reducing the inner backwall temperature to 35.0 °C (compared to 39.80 °C for the 5.5 mm standard panel) while extending the time to reach thermal equilibrium ($t_{plateau}$ = *1.35 vs. 0.96 h*) and dissipating 123 W of convective heat.

Additionally, heat-pressed MBCs were chosen as a possible future biomaterial for the panel. Literature review in the *Material selection report* showed that heat-pressed MBCs have not yet been extensively tested for outdoor durability, which presented a research gap that could be investigated (Appendix C). The literature review also showed that a cubical MBC had not yet been heat-pressed in a three-dimensional shape, using a mould. Through testing and consultation of mycelium experts dr. Jordi Pelkmans and Fran Ortega Exposito (Appendix A), it was shown that heat-pressing into a desired shape was possible using only a heated aluminium mould and 115 kN of force for 5 min. However, the cubical blocks repeatedly showed signs of tearing at the middle bulge, implying that the flexural strength of the MBC was too low in this area. Future development could involve growing the MBCs directly into the required shape, decreasing the flexural tension in these locations. Another option could be to repair the torn MBCs after heat-pressing, by regrowing that part. However, this option seems less viable at larger manufacturing scale.

To test the properties needed for possible panel application, the whiteness, flexural strength, water contact angle and water absorption were tested. From these tests, the materials were observed to darken in colour and show inadequate water resistance indicating the need for a coating for outdoor weather exposure due to the material deforming and swelling with water. Upon flexural strength testing, the 3D-pressed MBCs had similar or higher strengths than their flat counterparts (and broader literature (Appels, et al., 2019; Liu, et al., 2019; Shakir, et al., 2021)) and gave insight into the behaviour of the three-dimensional shape under bending stress. With the 3D-samples showing similar flexural strengths with larger material deformations, this could indicate a more durable material under bending stress thanks to the geometry of the round design (due to distribution of pressure). The prototypes show potential as a façade panel, however further testing and iterations are necessary to determine further applicability especially in an outdoor context, as well as more standardised methods in accordance with the highly variable, broader literature (Andrade, et al., 2026) especially in terms of opening a new avenue of research into novel heat-pressing into three-dimensional shapes.

MBCs have also been included in the LCA, with a GWP of 11.34 kg CO_2e/FU, which is slightly lower than indoor plywood and about three times as high as Douglas fir. The GWP of the MBC was not corrected per year of the lifespan, as the outdoor durability is currently unknown. Recent research has shown that MBCs can be grown over diurnal temperature cycles, eliminating the need for an incubator (Livne et al., 2024). Livne et al. (2024) found that the incubator accounts for 73% of the embodied energy and more than 40% of the CO_2 emissions, this type of production could drastically reduce the GWP of MBCs.

Scalability and manufacturability

The current production process is developed primarily for prototype production and therefore does not reflect the requirements for large-scale manufacturing. Currently, the prototypes are cut into shape using a small-scale CNC mill, and must be steamed, curved and glued one by one. Realistically, this process is not suitable for large-scale production, as building one 120 x 60 cm panel already took more than a full day of labour.

At larger production volumes, the fabrication time per panel could be decreased by processing in larger batches, reducing average setup time per panel, and increasing process refinement. However, the steps to be carried out for each individual panel would still take the same amount of time even when upscaling. By identifying the bottlenecks, the process could be streamlined for large-scale production. For example, the process of CNC milling is well suited to the current stage of prototyping, as it allows the plywood to be cut with high precision. However, the use of a CNC mill for processing a high number of panels becomes a limiting factor. As CNC milling is a subtractive process, each panel must be cut individually. The steaming, curving and gluing are likewise done one at a time, which means that this part of the processing time will increase linearly with the production volume.

Pujadas-Gispert et al. (2020) used an alternative manufacturing approach, heat-pressing the raw material (Nabasco® 8010) directly into a mould. This type of manufacturing would be more suited for large-scale production, as the wanted geometry of the panel can be made in one step, instead of several. In addition, the number of panels that can be made at once is limited by the size of the press and the number of moulds that are available. Within that constraint, however, several panels can be pressed simultaneously. This can both reduce the processing time per panel and make the manufacturing process easier to scale. Naturally, future research should investigate the manufacturing process together with the final material, rather than optimising these independently.

Bio-models / bio-inspiration

At the start of the project an analysis was carried out, where nature's passive cooling strategies were listed (Appendix B). Next, suitable bio-models were analysed and evaluated for application in the façade panel. The bio-models of the *Archaeoprepona demophon* butterfly, silkworm and Saharan silver ant seemed most promising because of the nanostructures that can perform radiative cooling (Lee et al., 2022; Ling et al., 2023; Shi et al., 2015). Moreover, a combination of the simplified *Insubark* corrugated prototype and the cactus bio-model inspired the implementation of ridges onto the design, but it was decided that this fell outside the scope of the project (Henrion & Tributsch, 2009; Lewis & Nobel, 1977).

Looking back, the bio-models served mostly as a source of inspiration rather than as direct blueprints for the final solution. This is an important distinction, as the resulting cooling strategies may have been developed through conventional engineering solutions. This shows the risk of forced or backwards bio-inspiration, where biological analogies are applied to solutions that are already there, or could have been identified without the use of bio-models.

The bio-models for a bio-inspired radiative cooling coating showed most potential for integration in the product. Although it can be hypothesized that this coating could be used in the final product, CryoPaint®, a commercial radiative cooling coating was purchased for the experiments, as the bio-model could not be recreated within the scope of the project. This allowed the same passive cooling principle to be tested, but without implementing the bioinspired solution, leaving open whether bio-inspiration would offer any advantage over the conventional coating. It would require evaluating whether the biological mechanism could

then be translated into a scalable, durable and weather-resistant material, instead of simply sourcing the non-bioinspired coating from another company.

Sustainability

At the start of the project, a number of sustainability objectives were set out for the panel: the panel must passively cool building façades without any operational energy required, the panel must be made from recyclable, biobased and eventually circular materials, the panel must be made out of modular, easily replaceable components, and the panel must have reduced environmental impact in comparison to its (in)direct competitors. Although design choices were made with the intention to improve the sustainability of the product, meeting these objectives does not necessarily result in lower environmental impact. This leads to the central question: To what extent does the proposed panel represent an environmentally sustainable solution when considering both its environmental impact and its potential cooling benefits?

In the LCA, it is addressed that the cooling power, or reduction in cooling energy demand of the product, is not included in the analysis. This represents a significant limitation, as this would provide a direct way to justify the environmental impact of its production. Future research should try to calculate under what conditions the operational benefits of the façade panel could outweigh its environmental impacts. Susca & Pomponi (2020) propose two approaches to incorporate the interactions between the built environment and the local climate, such as the passive cooling properties of a façade panel, into an LCA. The paper proposes site specific climate modelling (Approach A) or simplified physical relationship calculations (Approach B). Approach A seems promising, but would require the use of site-specific climate data, which would mean that the product's LCA needs to be assessed for each specific location of use. However, Approach B would be less demanding, but also less accurate, because of the simplified nature of the calculation. Approach A would be more suitable, as the passive cooling effect of the panel is extremely dependent on local environmental conditions, such as solar radiation, surrounding buildings, and local wind speeds. Future services offered by the company could therefore include modelling the panel's performance for a specific location and conducting an environmental cost-benefit analysis. This enables the company to communicate the sustainable nature and benefits of the panel to future customers in a transparent manner.

Consequently, the central question asked at the start of this paragraph cannot yet be answered conclusively. Although the LCA quantifies the environmental costs, it remains unclear if the cooling benefit can offset these costs. Whether the panel is truly an environmentally sustainable option, should be determined in future research through the ways described above.

Recommendations for future product development

Future product development should primarily focus on further optimalisation of the panel under realistic real-world conditions. Outdoor testing across different weather conditions and seasons would provide more insight into the long-term thermal performance and the weather resistance. Future research could use urban-scale UHI mitigation modelling to generate more data on the panel's performance in different scenarios. Afterwards, including the urban cooling benefits into the LCA would allow the environmental impacts of the panel to be weighed against the potential operational and urban cooling benefits.

Simultaneously, long-term weathering tests of the different wood species and MBCs, together with further investigation in heat-press and mould manufacturing of eventual other biobased materials, and investigation of larger panel dimensions to enable the use of a thicker solid timber, are needed to assess the durability, manufacturability and scalability. And, to make a final choice on the optimal material of the panel.

At last, future research should investigate whether the panel geometry could be simplified while maintaining the convective cooling capacities, and if a bio-inspired alternative to CryoPaint® can provide comparable radiative cooling performance.

AI Statement

This paper used GenAI (ChatGPT, Claude & Gemini) responsibly for the following components: brainstorming, grammar correction, sentence reforming, calculations, feedback. DeepL was used for translation purposes.

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Statement of Contribution

This report was co-authored and as such both their contributions here and throughout the project can be described and attributed below.

- **Alexandra Cooke-Kasyanto:**
  research and reporting on convection-based cooling mechanisms and their bio-models research and writing of design report; digital modelling of heat-pressing mould and previous prototype iterations; CNC assistance for aluminium mould fabrication at LPL (with Felipe); mycelium experimental planning and design; conducting fungal experiments at the UU fungal laboratories with Ruben (and consulting with expert dr. Jordi Pelkmans when needed); mycelium experimental results analysis and writing of mycelium section; research and writing parts of implementation plan (with Ruben); interviews with experts Arne Hendriks and Naud van der Padt (together with Felipe).
- **Felipe Krebs:**
  Research and reporting on conduction and evaporation based cooling mechanisms; data collection, analysis and reporting on the Life cycle assessment for mycelium materials; data collection and reporting on the Intellectual property analysis; writing of report section C, abstract and Lay summary; expert interviews with Maud Kerstholt, Naud van der Padt; 3D-modelling in Autodesk Fusion, for models such as the basic façade panel, the ridge design, the source file for CNC milling, and other designs featured in the design report, such as a simplified block design; manufacturing of the aluminum mould used for heat-pressing the mycelium panel, milling and folding, gluing of the individual wooden tiles needed to produce the wooden panel.
- **Jochem ten Pas:**

Literature research on radiative cooling and its biomimetic models, and writing the corresponding mini-report (WP1); developing, constructing and carrying out simplified shaft and climate chamber experimental setups together with Ocker (WP4); writing the complete experimental report on the simplified shaft and climate chamber experiments (WP6), including developing the theoretical framework for data analysis and performing the data analysis in R; booking and coordinating workspace for the duration of the MIP; 3D-modelling in Fusion 360 of jigs to aid prototype construction; learning to program Arduino-based sensors, including system setup and soldering, for the experimental setup, although it was ultimately not used; assisting with mycelium-based composite laboratory work under Alex's lead; contributing to the construction of both the wooden and mycelium-based prototypes, including constructing the wooden batten framework and assembling the prototype onto it, designing and laser-cutting mycelium moulds, laser-cutting components for the first proof-of-concept wooden panel, and heat-pressing mycelium into a corrugated shape for the mycelium panel; preparing and conducting an interview with Peter Oei on MBCs; preparing and conducting interviews with Willem Böttger and Fran Ortega Exposito together with Ocker.

- **Ocker Hazelbag:**
  Literature research on and writing about natural and conventional reflective cooling mechanisms, the possible integration section and the conclusion (WP1); literature research on and writing the section about mycelium-based composites (WP2); thorough analysis of all sections and writing a discussion and the final recommendations for the final report (WP6); writing most of the expert interview section in final report; final formatting and layout, title page, Zotero check final report; developing, constructing and carrying out simplified shaft & climate chamber experimental setups together with Jochem (WP4); finishing final prototype for climate chamber testing together with Jochem; thinking out an Arduino-based sensor setup, learning (simple) Arduino code, system design, troubleshooting and soldering of the final setup, although it was ultimately not used; conducting a business model analysis, including a value proposition and business model canvas (WP5); preparing and delivering final presentation together with Ruben; preparing and conducting interviews with Willem Böttger together with Jochem and Fran Ortega Exposito.
- **Ruben Langendoen:**
  Literature research on and writing the material report, besides the MBC part (WP2); Conducting and writing the four types of wood part of the LCA (WP2); Researching and writing of parts of the implementation plan (Attachment requirements, attachment methods and recommendations, weathering of timber, timber treatment, maintenance) (WP5); Writing the theoretical background and part A of the final report; do terminology check of the final report(WP6); preparing and delivering all three presentations during the project (with Felipe, Alex and Ocker respectively); Assembling and constructing the final panel together with Felipe and Jochem; involved in the process of growing the fungal blocks and heat-pressing them into a three-dimensional shape; helped conducting the fungal experiments (Alex had the lead).

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Appendix

Appendix A. Expert Meetings and their Outcomes

Interviews and informal discussions with experts from many different fields have continually shaped and guided the direction of the project. They helped to define the question behind the question, informed the final material choices and progressively refined the design. This has proven to be a continuously evolving process, moving the emphasis from trying to reinvent the wheel, to going back to the fundamental principles behind the panel's passive cooling and its shape and trying to understand the material behaviour. During the expert talks, a few reoccurring themes came back across the different conversations: refining the scope of the project, material orientation, outdoor applicability of MBCs, coatings, shape and prototyping, and the experimental testing setup.

Name	Profession	Discussed topic
Dr. Joris Remmers	Associate professor of Composite Materials – TU Eindhoven	Mechanical Engineering
Peter Oei	Program manager Biobased materials & Circular economy – Ministry of agriculture	Constructing and shaping MBCs
Willem Böttger	Director - NPSP	Possibility of using Nabasco 8010
Dr. Pieter Kooijman	Lab Manager – Lili's Proto Lab	Prototyping
Nico van Hijningen	Project Support – Lili's Proto Lab	Prototyping
Dr. Jordi Pelkmans	Lab Manager and Researcher – Microbiology Utrecht University	Growing and heat-pressing MBCs
Steve Szell	Architect & Interior designer – Freelance Guitar builder – Szephyr guitars	Prototyping, Woodworking
Fran Ortega Exposito	Junior researcher biobased construction – Avans Hogeschool / MNEXT	MBC limitations and opportunities
Naud van der Padt	Founder - Leafy	Implementation plan for the panel
Maud Kerstholt	Advisor Intellectual Property – Netherlands Patent Office	Intellectual Property Analysis
Arne Hendriks	Artist with existing MBC sculptures outdoors	Building/sculpting with MBC, long term material behaviour, social perception with MBCs

Scoping of the project

At the start of the Major Innovation Project, there were several key areas of the panel that could be reinvented and explored. This made it more difficult to define and formulate a clear scope for the project and led to several possible approaches for the project: redesigning the original shape, changing the material, test different coatings for durability and reflection, getting the product to TLR5, and growing mycelium panels and try to get the heat-pressed mycelium into the original shape. This led to a lack in clarity on where to focus the limited time and resources of the project.

This was also reflected in the feedback that multiple experts gave throughout the first months of the project. Joris Remmers, associate professor at the department of mechanical engineering at TU Eindhoven, was the first expert to point out that the combination of figuring out different shapes, configurations and materials at once was too much, and told us to focus on one of those and really dive into that one. However, this advice was not immediately acted upon, as the scope was not refined immediately. Willem Böttger of NPSP repeated this statement, questioning if the team would be able to act out two projects simultaneously, and suggested that growing and heat-pressing the mycelium should be more of a side project and that we should focus more on currently feasible façade panel materials. This led to the mycelium project being put more on the back burner in the beginning. Lastly, Nico van Hijningen also added to this idea, emphasising that we should first shift the focus to fully understand the method of heat transfer and convection in a simplified tube, before testing on the more complex final shape.

Material orientation

At the start of the project, the MaterialDistrict convention showed how many different companies are currently developing new sustainable biobased materials for façade panels. Also notable was that mycelium materials are currently hot in the field of material innovation but are still a luxury product and mostly for indoor use, like furniture or acoustic panels. This opens up a potential niche for the development of mycelium panels for outdoor use. During the many talks with experts, there were broad explorations of many different biobased materials that could be used in façade panel context. Peter Oei introduced the broadness of biobased materials that could be used in the project, with additions of biobased binders and even a C5 sugar resin that could be a sustainable alternative to the petroleum resins currently in use. However, Nico casted doubt about the use of Douglas fir wood in 5 mm thick panels, indicating that the wood would warp, while 16 mm thick panels would probably become too thick and heavy for the wall attachment. Steve, in contrast, was very enthusiastic about the use of wood in the specific shape and context. He gave options on what types of plywood could be used and information on how individual grains of wood interact with each other in plywood.

Outdoor applicability of MBCs

There is a great deal of scepticism surrounding the use of MBCs in outdoor environments, yet there were also experts who had a much more positive view on the topic. Experts on biobased materials like Willem Böttger and Peter Oei both expressed their doubts on the topic, claiming that the panels do not handle humidity well. Willem talked about a solution, using epoxy coatings to improve the lifespan and mechanical properties of the panels. But they would still not last long. Fran Ortega, expert on façade materials and MBCs at MNEXT and the Avans Hogeschool, sees the potential of MBCs in the construction sector and façade panel usage. Fran explained that while lifespan is the most difficult property to manage in outdoor context, as the panel is highly susceptible to microbial attack and will mould when it

cannot dry out anymore, that these challenges can be overcome with the right approach. He explained that heat-pressing improves the properties, and makes it less susceptible to moulding, while he strongly advised against cold-pressing for outdoor use. In addition, he pointed at UV degradation to be a concern, as prolonged exposure causes the heat-pressed to delaminate and change colour.

Despite these problems, Fran remained optimistic: “There are always ways to overcome the downsides, whether they're feasible or not is a separate question, but there are ways to protect mycelium and make it more durable. It's done with wood, and mycelium composites aren't fundamentally that different. It really depends on the production process and whether there can be more investigation into substrate combinations that would improve water resistance.”

Artist Arne Hendriks remarked that when working with living mycelium as a construction material, it is incredibly unpredictable and highly prone to changes in outdoor settings. However, it can act as a great “bio reactor” as it can attract life and increase biodiversity, as well as sparking debate and narrative amongst local community in regard to what it means to have nature as part of the urban environment. In our context, the mycelium would no longer be living, but it is an interesting point about the social aspects.

Coatings

Another topic that came up many times during interviews was the use of coatings to improve the outdoor applicability of MBCs and wooden planks. Fran Ortega explained that certain coatings can make heat-pressed and non-pressed MBCs sustain outdoor conditions, but these are often epoxy and not biobased. However, there are a few up and coming biobased coatings that come close to matching the performance of epoxy. He gave the example of the Impershield coating that is used on the Growing Pavilion. However, Fran explained that the biggest current knowledge gap in the field of MBCs is in biobased coatings that actually perform as well as non-biobased epoxies. Xyhlo was another coating that came up several times during the meetings, with Fran, Willem and Peter all mentioning the promising biobased coating. Willem indicated that the coating doesn’t work on Nabasco, as it cannot penetrate the material for the fungus to live in. Fran explained that it works on heat-pressed MBC boards and that they’re currently in advanced testing, needing to be tested still in the outdoor environment.

Another interesting option that Fran suggested is the use of pure mycelium itself as a coating. At MNEXT they are developing mycelium-based coatings using white rot fungi. If successful, this could offer a fully biobased solution that mimics the natural skin that grows on mycelium.

Prototyping

There were also tips for the prototyping part of the project, coming from multiple experts. Steve mostly helped with understanding different types of wood and their properties, explaining that grain direction is extremely important while using thin plates of wood. As non-

processed woods warp because of the grain direction, plywood does not, as it's thin slices of wood with the grains working against each other. He explained that you can CNC plywood not all the way through, so the thin outer layer acts as a membrane keeping the pieces together and it aids the convection leaving no crevices.

Fran helped with mycelium prototyping, indicating that using a positive, as well as a negative mould during heat-pressing will yield the best results, and recommended using borders in the mould for better results. Additionally, he suggested trying CNC'ing heat-pressed boards, as he had never tried it. He also noted that that heat-pressed boards can easily be stapled or nailed together.

Artist Arne Hendriks also advised getting familiar with the mycelium materials, as the there is usually diversity in the biomaterial naturally (it's difficult to make every sample exactly the same) which was also mentioned by dr. Jordi Pelkmans. Getting more familiar would increase embodied knowledge, problem solving and foresight. Therefore, replicates would be ideal, and experiential knowledge will enhance the process along the way.

Implementation plan

To get an idea of how to get from a simple prototype to a fully developed product that is commercially viable we talked to an expert who already achieved this step: Naud van der Padt, Cofounder of the company "Leafy" based in Amsterdam.

Naud stressed the importance of collaborating with other companies, in particular other startups, when trying to establish your product on the market. For a product as complex as a building façade doing every step by yourself is in his opinion not feasible.

He gave examples of collaborations with companies focused on materials and suggested that aspects like mounting mechanisms or fire safety mechanisms, could both be solved more easily by collaborating with another company. In his experience many startups would be eager to collaborate on joint projects as it also gives them the opportunities to get their product out on the market. Façade specific challenges he told us to look out for were lifespan of the product and the end-of-life scenario, which in his opinion is best solved by making the panels modular, as a whole façade is harder to dispose of.

Intellectual property

To understand more about how Intellectual property matters work in the Netherlands, we consulted with Maud Kerstholt, an expert from the Dutch Patent Office. She gave us a breakdown of the different types of Intellectual properties there are in the Netherlands, and how we could use them for our project. She also informed us about the possibility of a freedom to operate (FTO) analysis and how we could perform a shortened version thereof to have some indication of which types of patents we need to look out for. This recommendation was given because a full FTO analysis would not be possible for us as according to her they would need to be conducted by professionals and would cost between 10000 and 50000€ depending on the scope. She urged us to pay attention to patents with general claims about

mycelium materials. Sometimes patent claims can cover the use of a biological organism as a material entirely, or a common mode of growing mycelium. Even if that process used is common practice in the fungal lab of UU, it might be protected from commercial use.

A piece of advice Naud van der Padt gave us on IP is to not focus on obtaining patents in the starting phase of developing a company, as realistically a start up in that stage does not possess the resources to successfully enforce a patent, as the legal cost required to prevent competitors to use patent technology is too great. Instead, we should focus on distinguishing our product through its quality. The talk with Maud Kerstholt added the insight that with the prior published literature, obtaining a patent for our design would not be possible, which confirmed our IP strategy of not trying to obtain patents at the current stage of the project.

Appendix B. Cooling Mechanisms Report

Biological Cooling Mechanisms

Mini-Report for MIP 2026 Team X – Functional Skin for Passive Cooling



Joshua Tan, Alexander Scott Kalyan, Felipe Krebs, Ruben Langendoen, Ozker Hazelbag

Inhoudsopgave

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Introduction

Rising global temperatures and ongoing urbanization have led to an increase in the demand for artificial cooling inside of buildings, as well as urban environments. The global population has been facing an increasing demand for cooling energy, as traditional cooling technologies, like air-conditioning, use large amounts of electricity (Auffhammer et al., 2017). Cooling systems account for nearly 20% of the global building energy use (Falchetta et al., 2024). In addition, global warming in combination with the Urban Heat Island (UHI) effect further increases the need for sustainable cooling strategies that minimize energy use (Hong et al., 2022).

To understand how cooling can be achieved more sustainably and passively, it is essential to understand how heat is transferred, dissipated and lost. Heat transfer can occur through several mechanisms in nature: reflection, radiation, convection, conduction and evaporation. The balance between these mechanisms ultimately determines the thermal state of a system (Carey et al., 2008). In urban environments, this balance is often disrupted. Urban materials, like concrete, asphalt and steel absorb and retain heat, while the dense urban environment limits airflow and thus reduces heat dissipation (Hong et al., 2022). In addition, the limited presence of vegetation in urban areas reduces the passive cooling through evapotranspiration and decreases the availability of shading. This contributes to the UHI, where heat becomes trapped in these so-called urban canyons, and accumulates there due to a lack of effective thermoregulation. One way to tackle this problem is by using façade panels on buildings that can contribute to the passive cooling of the building and its urban environment, while also reducing the amount of heat that is absorbed in the first place. Existing designs (see Appendix A) could be optimised through the exploration of mechanisms that exist in nature.

Through evolution, nature has evolved many ways to keep organisms cool, even in harsh environments. For example, in the Sahara desert, termites construct their mounds in such a way, that convective airflows cool the inside, even in the extremely hot and arid climate (Korb, 2003). Secondly, the Saharan silver ant, *Cataglyphis bombycine,* is able to forage during the middle of the day at harsh temperatures, due to their triangular hairs that reflect solar radiation and emit infrared light. In addition, plants cool their surrounding environment through the mechanism of evapotranspiration, which is the process where plants lose water vapor to the atmosphere though their stomata (Y. Yang et al., 2023). These biological models highlight the potential of bio-inspired or biomimetic approaches for sustainable cooling.

This mini report explores how the key heat transfer strategies, reflection, radiative cooling, convection, conduction and evaporation are utilized in nature as strategies to passively cool organisms and the environment. Additionally, the underlying mechanisms of these strategies are examined to gain a deeper understanding of both the strategies and the biological models, enabling their application in the façade panel.

Reflection

Reflection is one of the primary passive cooling mechanisms within the façade panel. Solar reflection aims to minimize solar heat gain, by reducing the fraction of incoming radiation[1] that is absorbed by a material (Hernández-Pérez et al., 2014). **Solar reflectance** can be expressed as **ρ**, or ρ_{solar}. The ability of a surface to release the heat that is absorbed in the form of thermal radiation is called **thermal emittance**, or **ε**. Solar radiation is the primary source of heat input for solar reflection, so the ability to emit absorbed energy back to space as thermal radiation is crucial for its net cooling performance (Hossain & Gu, 2016). High values of ρ and ε enable surfaces to remain cooler under prolonged solar exposure, because less energy is absorbed and more is dissipated (Hernández-Pérez et al., 2014). These parameters play critical roles in the overall performance of the material of the façade panel.

Spectral selectivity

In addition to overall reflectance, the **spectral selectivity** of reflection plays a crucial role in the overall passive cooling performance of a material (Y. Li et al., 2020). Solar radiation consists of radiation in the ultraviolet (UV), visible light spectrum (Vis) and near-infrared (NIR) spectra (Sang et al., 2015). Out of all the solar energy, UV light accounts for 5% of the energy, visible light for 45% of the energy, and NIR for 50% of the energy. So, for the most efficient reflection over the full spectrum of solar radiation, the emphasis should be placed on the NIR range, as this account for the largest portion of the total solar energy. While visible light accounts for the perception of colour, high reflection in the NIR region is most important for thermal management, as it directly influences the heat gain without affecting the aesthetics of the material (Levinson et al., 2007) This is also the reason why radiative cooling has proven to be such an efficient passive cooling strategy in nature (Didari & Mengüç, 2018).

Therefore, the application of spectrally selective materials is extremely important in the field of reflection and passive cooling in facades. These materials are designed to exhibit the highest reflection in the NIR region, while also maintaining the optical properties that are desired to preserve the colour and aesthetics (Y. Li et al., 2020).

Energy balance of a surface

The thermal behaviour of a façade surface can be understood through a simplified surface energy balance (Kiehl & Trenberth, 1997; Trenberth et al., 2009). The absorbed solar energy can be expressed as:

$$(1 - \rho) \cdot I_{solar} \quad (1)$$

In this equation **(1 − ρ)** is the solar absorption, as **ρ** is the fraction of the incoming energy that is reflected. The remainder is what is absorbed. The average reflection on earth is about 0.3, meaning that 70% of the solar energy is absorbed (Trenberth et al., 2009). I_{solar} is the incident solar irradiance, most often expressed in W/m².

[1] Radiation generally refers to the energy that is transmitted through space in the form of particles and waves (Cameron Pimperl, 2009).



While the emitted thermal radiation follows the Stefan-Boltzmann law:

$$\boldsymbol{\varepsilon \cdot \sigma \cdot T^4} \qquad (2)$$

In this equation, **ε** is the thermal emittance of the surface, as mentioned before. In addition, **σ** is the Stefan-Boltzmann constant (5,67 × 10^{-8} W·m^{-2}·K^{-4}), while **T** is the temperature in Kelvin of the surface that is emitting the radiation.

Cool materials

Reflective materials, also referred to as **cool materials**, are designed to reflect a large amount of the incoming solar radiation, while also emitting thermal radiation (Hernández-Pérez et al., 2014). These materials have been shown to reduce surface temperatures by 10 – 25 °C, in comparison to other conventional building materials. The application of these materials is therefore essential in the passive cooling of our façade panel. For these reflective materials, two main categories can be distinguished. **Cool white materials** exhibit high reflectance across both the visible light spectrum, as well as the near-infrared spectrum. The solar reflectance value (ρ) of these materials can reach up to 0.9, resulting in large surface temperature reductions compared to non-coated materials. By contrast, **cool coloured materials** might offer a more aesthetically appealing option, as they can provide colour to the façade panel. However, they reflect a smaller fraction of the visible light while reflecting a larger fraction of the NIR spectrum, when compared to a white panel. These materials can reduce the surface temp by 5 – 13 °C, in comparison to the 10 - 25 °C of cool white materials.

Reflection and albedo

Another term is often used in the concept of reflectivity is **albedo**. Albedo, often expressed as **α,** can be defined as the fraction of solar radiation that is reflected by a particular surface (Stephens et al., 2015). In theory, albedo and solar reflectance are equivalent concepts, but albedo is often used on larger scale (e.g., earth, urban surfaces), while solar reflectance is often used to describe the solar spectral reflectance of materials in construction and material science (Hernández-Pérez et al., 2014; Veloso et al., 2021). However, reflection is not solely determined by the material composition or its colour. While the average conventional material rely mostly on pigments, more recent research has shown that reflection can also be caused by **micro- and nanoscale structures** (Mundekkad & Mallya, 2025; Veloso et al., 2021). These materials, often inspired by biology and nature, can scatter the light that would otherwise be reflected. These mechanisms are particularly relevant for the bio-inspired design, as natural models can achieve high reflectance through structure rather than solely colour.

Retro-reflectivity

In addition to conventional reflective materials, **retro-reflective materials** also present another approach to improving solar reflection in the urban environment. These retro-reflective materials redirect incoming solar radiation back directly towards the original source, instead of scattering the light back in several directions, like conventional reflective materials

(**Fig. 1)** (J. Wang et al., 2021). This difference in reflection trajectory has large consequences for UHI mitigation, as the conventional counterparts can cause secondary heating, which is the mechanism where the reflected solar radiation is absorbed by the surrounding urban materials. This merely shifts the problem, rather than solving it.

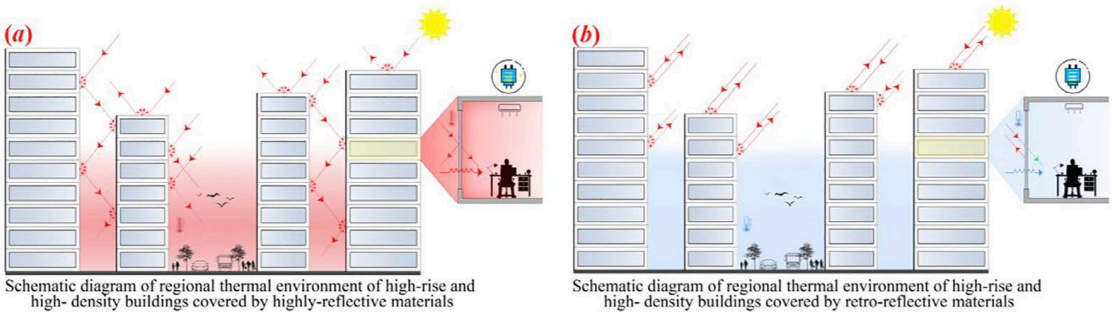


Figure 3: Schematic drawing of the thermal environment of high rise and high density buildings. A. Buildings covered in conventional reflective materials. B. Buildings covered in retro-reflective materials (Wang et al., 2021).

Retro-reflectivity in materials and surfaces is most often achieved through structural designs, like the application of glass beads, prism geometry or saw-tooth reflective films (**Fig. 2**) (J. Wang et al., 2021). When these are applied to building facades and roofs, they increase the upward solar reflection to 60-70%, in comparison to 45-55% for conventional glass. This results in reduced solar energy being trapped in the urban canyons that cause UHI. In addition, they increase the overall urban albedo by up to 36% by reducing the uptake of solar heat in these urban environments. However, these materials still have some limitations. The effectiveness depends heavily on the solar angle, the structure of the surface its is applied to and the material weathering and degradation over time (J. Wang et al., 2021). In addition, the performance of the material is hard to assess, as the reflected light is directed in the same direction as the light source.

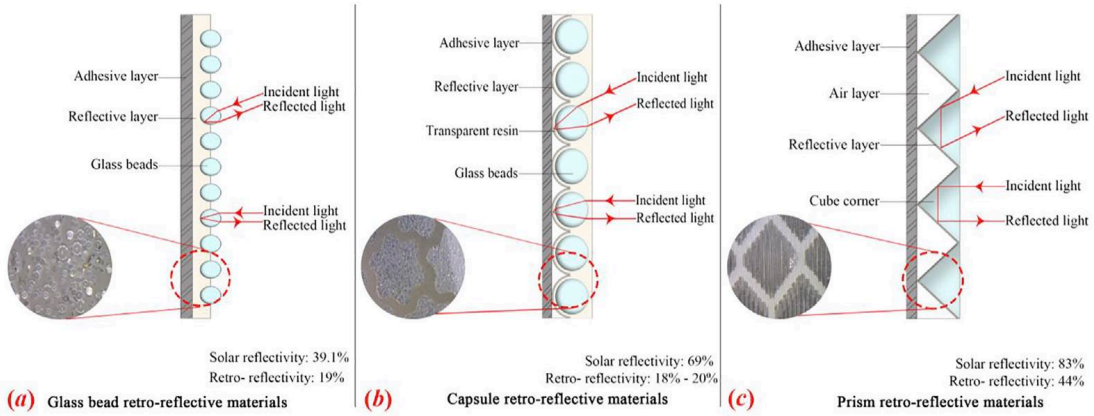


*Figure 4: Optical mechanism and structure of the different retro-reflective materials. **A.** Glass beads. **B**. Capsulated glass beads. **C.** Prism-shaped saw-tooth shaped (Wang et al., 2021).*

Cardinali et al. (2022) investigated the optimum design of retro-reflective materials as building envelopes, to enhance optical performance. This paper provides a framework on how to construct retro-reflective materials manually, using a combination of glass beads and a highly reflective (high albedo) paint. In this method, white pain is applied as a coating for an exterior building plaster sample (**Fig. 3**). The glass beads are deposited manually over the wet paint,

dried and then the excess beads are removed. This would prove to be a relatively easy way to construct these retro-reflective materials ourselves. Results further show that both the size and surface density of the glass beads strongly influence the optical performance of the coating. Among the configurations that were tested, beads with a diameter of 200 µm – 300 µm proved to have the best balance between maintaining solar reflection, as well as providing the retro-reflectivity. This size range showed the most favourable angular distribution[2] of the reflected radiation, making it the optimal choice.

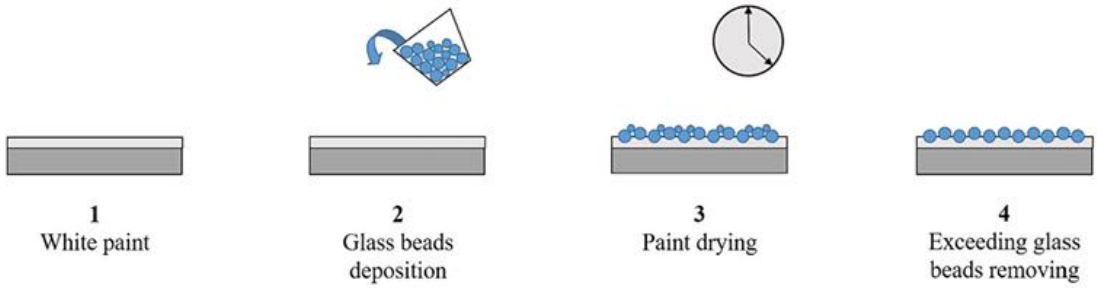


Figure 5: Protocol made by Cardinali et al. (2022) for the creation of a retro-reflective coating.

Bio-models

In addition to the current conventional solutions for reflective materials, biological systems provide other interesting design principles for passive cooling, mostly through micro- and nanoscale structures that were developed through natures intricate ways in 3.8 billion years of evolution.

Archaeoprepona Demophon

One of those bio-models is the butterfly *Archaeoprepona Demophon,* which provides a clear example on how structures at nanoscale can achieve radiative cooling, as well as provide colour through structural colouration (Lee et al., 2022). The dorsal side of the wing of this butterfly consists of a hierarchical structure, consisting of a nanoporous matrix, combined with a periodically structured nanograting structural colouration ridge on the surface (**Fig. 4**). This combination of nanostructures provides the material with high visible light reflectance, reducing the heat gain by the solar rays, as well as high emissivity on the infrared spectrum, which enhances heat dissipation through the mechanism of radiative cooling. Additionally, the nanograting ridge structure provides iridescent colouring through **Bragg diffraction**[3] (T.-D. Nguyen et al., 2018). Because this type of colouration is structure-based rather than pigment-based, the colours do not degrade over time, and contribute to the camouflage of the butterfly in natural environments (Lee et al., 2022). By controlling the spacing of this structure, specific wavelengths can be constructively reflected, which enables structural

[2] This describes how the reflected light is spread over different directions, and shows how much light is actually sent back in the direction of the source (Cardinali et al., 2022).
[3] This describes the principle where the incident light scatter of a material in specific angles, which determines the reflected wavelength (T.-D. Nguyen et al., 2018).

colouration that is tuneable to desire. The porous matrix is what gives the dorsal wing its radiative cooling abilities. This matrix contains nanopores that are randomly distributed and range in sizes from 200 to 800 nm, which includes the wavelengths of light in the visible spectrum, which range from 400 to 700 nm (Austin et al., 2021). These pores scatter the incoming solar radiation, which increases the overall reflection of the butterfly wings, and reduce the solar heat absorption (Lee et al., 2022).

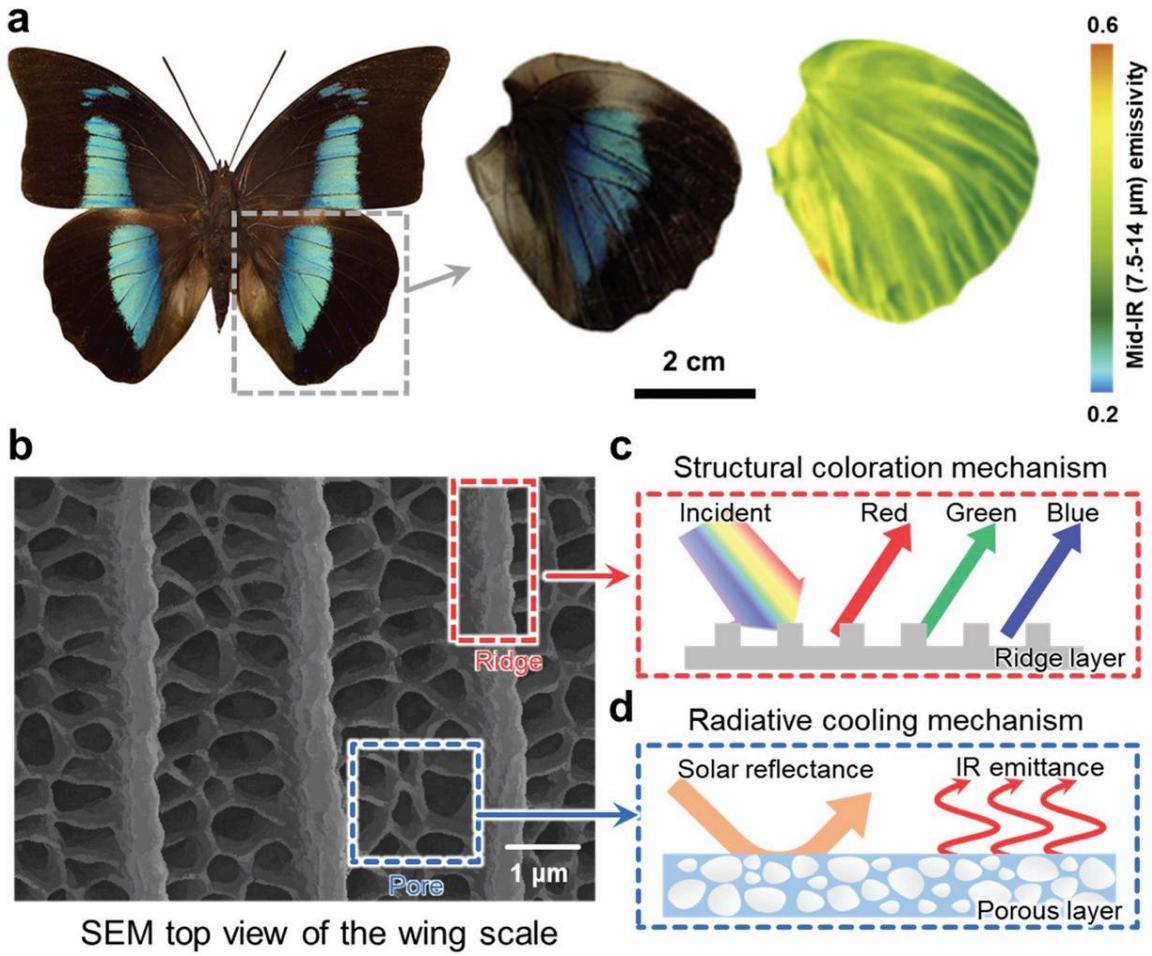


Figure 6: ***A.*** *Photograph of the Archaeoprepona Demophon and cut-out of the dorsal wing. Showing the mid-IR emissivity of the wing structure.* ***B.*** *Scanning Electron Microscopy image of the Archaeoprepona demophon wing, consisting of the pores and ridges.* ***C.*** *Schematic figure of the structural colouration mechanism.* ***D.*** *Schematic image of the radiative cooling properties of the dorsal wing, caused by the nanoporous properties of the wing (Lee et al., 2022).*

To replicate these functionalities, a biomimetic film was constructed by Lee et al. (2022). By drop-casting, a solution from SiO2 was transformed into a random porous structure, mimicking the pores of the dorsal wing of the *A. demophon* (**Fig. 5**). Laser interference lithography was applied to produce the nanograting structural colouring ridge. The resulting material, which was placed onto a PVDF-HFP polymer, exhibits both extremely high solar reflectance ($\rho \approx 0.94$), as well as high infrared emissivity ($\varepsilon \approx 0.96$). This biomimetic nanofilm achieves up to 8.45 °C reduction in temperature, compared to conventional materials, under full solar exposure. Along with the radiative cooling capability, the biomimetic film produces

the structural colouring of a large colour gamut (91.8% of sRGB[4]). This shows the potential of this material in combining radiative cooling with functional colouration and aesthetics in one multifunctional film for the passive cooling façade panel.

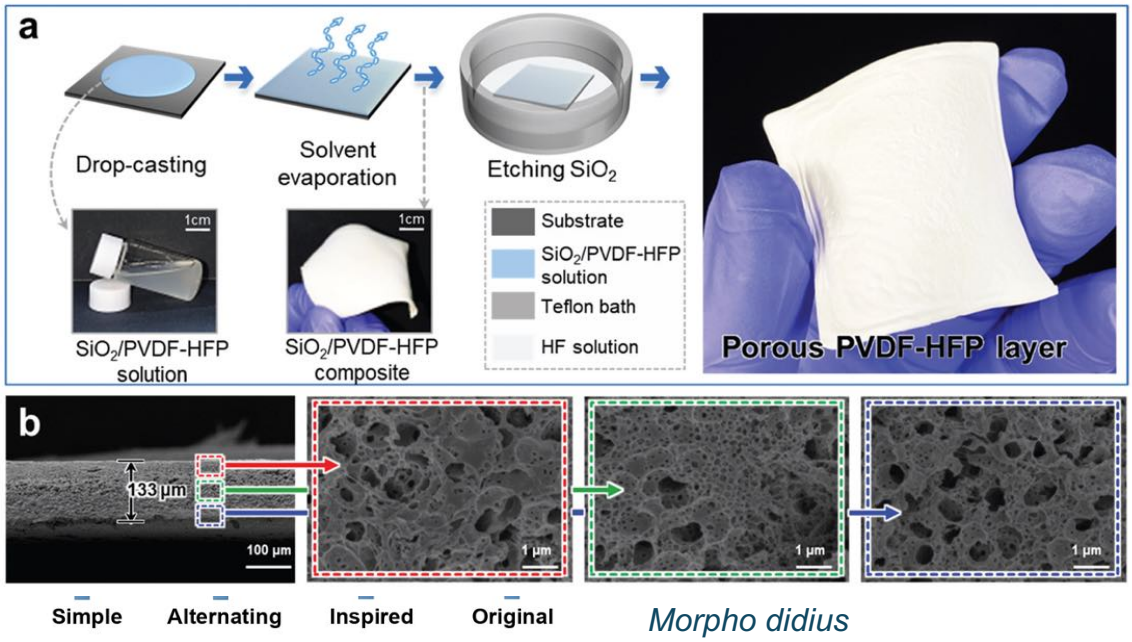


*Figure 7: **A**. The formation of the biomimetic film. **B**. The nanoporous structure of the man-made butterfly-inspired optical film.*

Morpho didius is another case of biomimicry for structural colouring. This butterfly is widely known for its bright iridescent colouration. Similarly to the *A. Demophon*, the butterfly does not rely on pigments for its bright colour, but on the structural colouration nanostructures that manipulate and reflect the incoming light (Didari & Mengüç, 2018; Siddique et al., 2013). The wing scales of *M. didius* consist of these tree-like ridge structures, with a periodic arrangement of lamellae (**Fig. 6)**. These structures act like multilayered reflectors, where variations in the refractive index between the layers cause the reflection of certain wavelengths, mostly in the blue region of the visible colour spectrum (450 - 495 nm) (Vukusic et al., 1999).

[4] International standardized colour space created for consistent colour representation.

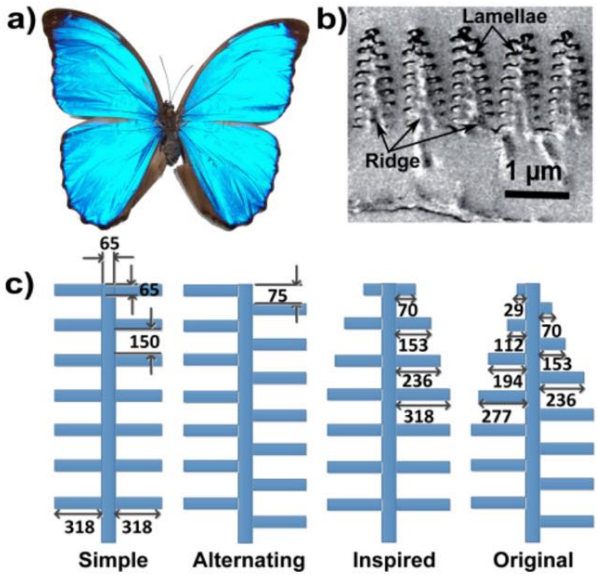


*Figure 8: **A**. Morpho didius butterfly. **B**. Electron microscopy image of the ridges and lamellae that provide the structural colouring. **C**. Simplified figure of the different configurations of the structural colouring that were tested in the paper (Siddique et al., 2013).*

Next to the ordered structure of the lamellae, small irregularities within the wing and the tree-like structure diffuse reflection, which allows the colour to be visible from different angles (Didari & Mengüç, 2018). This combination of order and disorder in the wings provides both colour stability and selectivity for specific broadbands of the spectrum. In addition to the *A. demophon* model, this model demonstrates how these nanostructures can be optimized for specific light reflection, thus for specific colours and how to recreate them.

Silk worm fibre

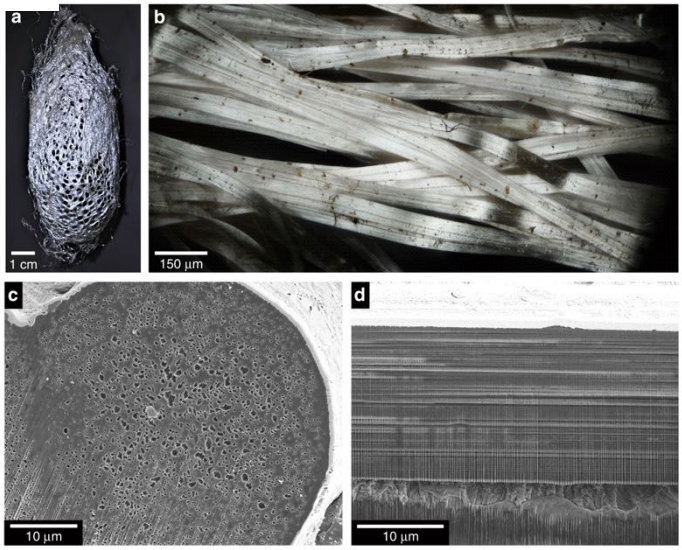


*Figure 9: **A**. The silk worm cocoon, constructed out of silk fibres. **B**. Image of individual silk fibres. **C**. Image of the air voids inside of the silk fibres that cause the solar rays to be reflected. **D**. Cross-section of silk fibre (Xiang et al., 2023).*

In nature, many other nano- and microstructures have evolved to enhance reflective properties. The silk worm, for example, constructs a cocoon made out of silk fibres to protect its pupae from the rapid temperature fluctuations and solar radiation from the environment, allowing them to develop safely in the protective cocoon (J. Zhang et al., 2017). These cocoons can be 6-10 cm in length, and conduct optical properties that are unique from any other optical materials constructed in nature (Shi et al., 2018). The optical properties from this material are a result of the air voids inside of the separate silk fibres, which have diameters similar to the wavelengths of visible and NIR light, thereby scattering incoming radiation and enhancing solar reflectance **(Fig. 7**). In addition, the open, randomly stacked nanofibrous network within the silk fibres, combined with the specific molecular bonds (N-H, C=O, C-O and C-O-C), maximize both the reflection of solar heat and the emission of excess thermal energy to the surrounding environment (Xiang et al., 2023). These chemical bonds exhibit strong vibrational absorption in the mid-infrared region: incident infrared radiation excites quantized stretching and bending modes of the bonds allowing them to efficiently absorb thermal energy at characteristic wavelengths. According

to Kirchhoff's Law[5], the absorbed energy is subsequently re-emitted as IR radiation, resulting in high MIR emissivity.

In Xiang et al. (2023), this randomly ordered nanostructure was recreated, and filaments of several nano- and microstructures were compared. In the paper, a series of procedures, including degumming, dialysis and electrostatic spinning create silk fibres which are also biodegradable. The average fibre diameter averaged from 320 nm to 2850 nm in these tests, and the silk fibres were randomly distributed and stacked on a membrane. The tests showed that loosely stacked nano fibres exhibited better daytime radiative cooling than more densely stacked micro-sized ones. Additionally, nano-sized fibres showed solar reflectance up to 96% (**$\rho = 0.96$**) and infrared immittance of up to 97% (**$\varepsilon_{IR} = 0.97$**). These optical properties led to a 6 °C decrease in surface temperature, even under high solar irradiance.

[5] Kirchhoff's Law states that a material's emissivity (ε_λ) equals its (vibrational) absorptivity (α_λ) at thermal equilibrium. Experimentally, emissivity can therefore be obtained by accounting for all incident solar radiated light. These contain the portions that are reflected, scattered (Mason et al., 2011)

Radiative cooling

Radiative cooling is a mechanism that essentially uses the cold vacuum of outer space (3 Kelvin) as a massive heat sink. In the context of the Urban Heat Island (UHI) effect, where cities become significantly hotter than their rural surroundings due to the inability of heat to dissipate, employment of this mechanism shows potential. This cooling mechanism relies on two key principles of thermodynamics. First, Planck's Law[6] is relevant as it describes that every object at terrestrial temperatures (around 300 Kelvin) emit heat in the mid-infrared (MIR) spectrum (**Fig. 8**). The second principle is the atmospheric window, which is a wavelength interval (8-13 µm), where radiation emitted within this interval passes through the atmosphere into the cold vacuum of space. The atmospheric gases, like CO_2 and water vapor, do not absorb this infrared radiation at this interval, which reduces the amount of heat that is trapped in the atmosphere.

The main challenge in obtaining passive radiative cooling is that it can only happen during the day, when sunlight is present. This is where two functions come into play: solar reflectivity and selective emissivity. In this section selective emissivity will be discussed.

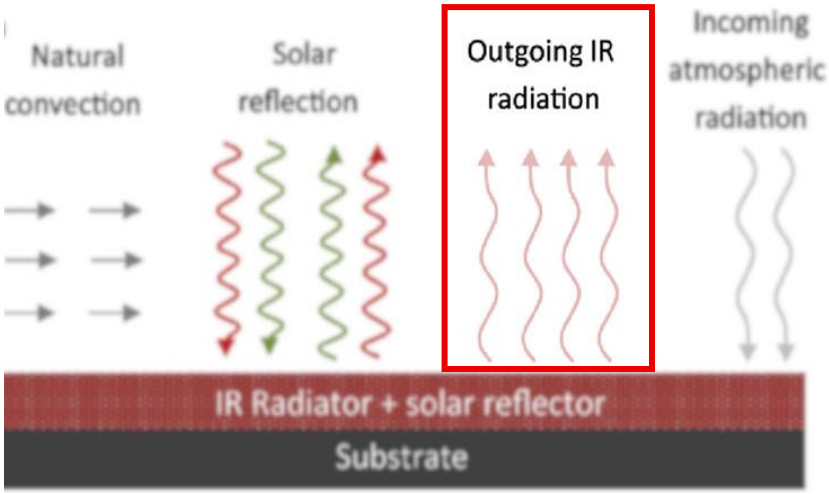


Figure 10: Schematic image of outgoing infrared radiation of a solar reflector.

Selective and broadband radiators

A broadband radiator, like a blackbody, emits within the entire emission band of the atmosphere (**Fig. 9**), except for the main solar spectral band (AM1.5; from 300 nm to 2.5 µm) (Hossain & Gu, 2016). Broadband radiators are highly beneficial for above-ambient cooling (removing heat from objects that are already hotter than their environment), as their high emissivity across the entire infrared spectrum facilitates efficient radiative heat exchange with the ambient environment (Pirvaram et al., 2022).

To achieve temperatures below the surrounding air (sub-ambient cooling), a radiator must release heat as infrared (IR) radiation specifically through the atmospheric window, where

[6] Planck's law defines the spectral distribution of radiation emitted by a blackbody as a function of temperature, showing that objects around 300k predominantly emit energy in the mid-infrared MIR range (Bose, 1994).

the atmosphere allows it to escape into space. However, broadband radiators also emit heat outside of this window, where the atmosphere is opaque. In these wavelength regions, CO_2 and water vapor in the atmosphere absorb the radiator's heat and partially reflects it right back down (Hossain & Gu, 2016) This counteracts the cooling effect and prevents the broadband radiator from reaching temperatures significantly below the ambient temperature.

By contrast, a selective radiator features high emissivity specifically within the 8-13 μm wavelength range and low emissivity everywhere else (Pirvaram et al., 2022). This selective property is critical for below-ambient cooling, as it maximizes the heat radiation into the cold vacuum of space, simultaneously minimizing the amount of radiation absorption and reflectance by the atmosphere.

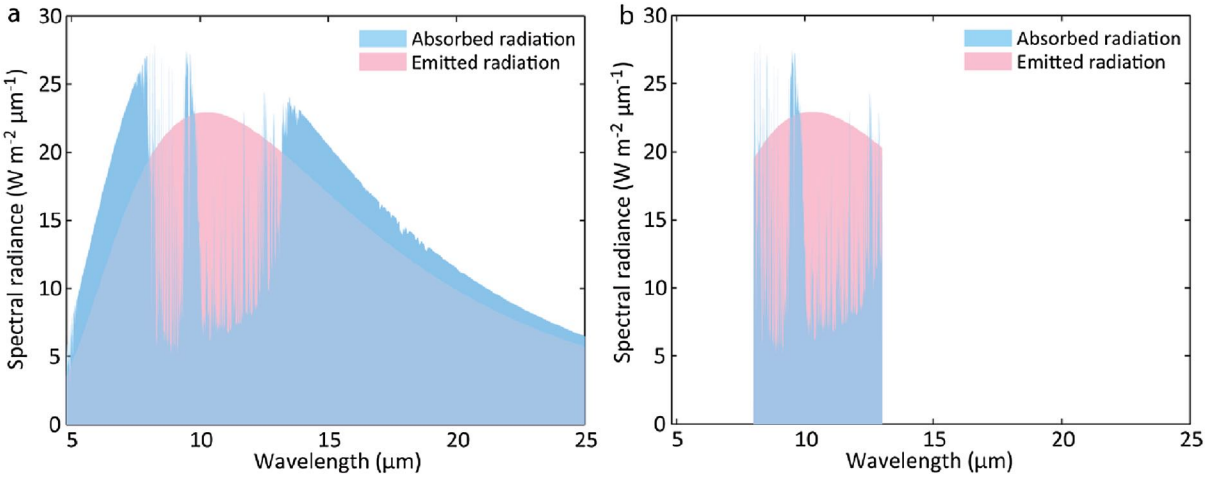


Figure 11: Comparison of absorbed atmospheric radiation and emitted thermal radiation for (a) broadband and (b) spectrally selective radiators, both operating at a temperature 16 °C below the ambient temperature (T_a) (Hossain & Gu, 2016).

Commercial products utilizing radiative cooling (RC)

Below different commercial products will be discussed that employ radiative cooling on a surface.

Inorganic pigment paints

Traditional cooling paints have heavily relied on titanium dioxide (TiO_2) due its high refractive index[7] (n > 2.5). However, TiO_2 inherently absorbs ultraviolet (UV) radiation, which limits its maximum solar reflectance to approximately 95% (**ρ = 0.95)** (Mandal et al., 2020b; Mishra et al., 2024). To surpass this limit, researchers have shifted to "wide-bandgap" dielectric materials (materials that become polarized when exposed to an electric field) like barium sulphate ($BaSO_4$), aluminium oxide (Al_2O_3) and calcium carbonate ($CaCO_3$). (Atiganyanun & Kumnorkaew, 2023; Dong et al., 2022). For instance, $BaSO_4$-based acrylic paints can achieve an ultrahigh solar reflectance of 98.1% (**ρ = 0.981**) and high atmospheric window

[7] A refractive index (n) is a dimensionless quantity that describes how strongly a material slows down and bends (refracts) light compared to its speed in a vacuum. It is defined as the ratio of the speed of light in vacuum to that in the material, with high values indicating stronger light scattering and reflection at interfaces (Born et al., 1999).

emissivity (**ε = 0.89**) yielding remarkable sub-ambient cooling even under direct sunlight, where a theoretical radiative cooling power at nighttime and daytime could reach 119.3 W m^{-2} and 94.3 W m^{-2} (Dong et al., 2022). In textbox 1, the experimental setup for testing the cooling performance of a self-manufactured radiative cooling by Dong et al. (2022) is explained.

Textbox 1: Protocol Dong et al. (2022)
The outdoor cooling performance of the manufactured PDRC coating was evaluated using an experimental test system established on the rooftop of a laboratory building in Weihai, China. The characterization was based on maintaining an overall thermal balance across four distinct components: solar radiation absorption, atmospheric radiation absorption, heat dissipation to deep space via electromagnetic waves within the atmospheric transmission window, and parasitic heat losses caused by the temperature differential between the coating and the environment. The experimental apparatus consisted of a data logger, a laptop, a custom-designed test cavity, and an outdoor weather station. To minimize extraneous heat dissipation, the main body of the test cavity was constructed from polystyrene foam, with the exterior surfaces covered in aluminium foil to reflect incident solar radiation and the top aperture sealed with a transparent polyethylene (PE) film to suppress convective heat transfer. The PDRC coating was applied to an aluminium substrate with dimensions of 10 x 10 cm. Given the high thermal conductivity of the aluminium plate (237 W m^{-1} K^{-1}), a thermocouple was secured to the rear surface of the plate to record representative temperature variations. The cooling efficacy of the PDRC coating was demonstrated by comparing its temperature profile against both a commercial white coating and the air temperature within an empty control cavity over a continuous multi-day testing period. Furthermore, environmental variables including solar radiation intensity, relative humidity, and wind speed were monitored in real-time via the integrated weather station to correlate thermal performance with localized atmospheric conditions.

Porous polymer coatings

Polymers such as polyvinylidene fluoride (PVDF), poly(methyl methacrylate) (PMMA), and polydimethylsiloxane (PDMS) contain functional groups (e.g., C-O, C-F, Si-O) that naturally emit strongly in the mid-infrared (MIR) region (Guo et al., 2023). By introducing micro- and nano-scale air voids these polymers form sponge-like porous structures that strongly scatter and reflect sunlight without the need for metallic reflectors. These porous structures can achieve a solar reflectance of 96% (**ρ = 0.96**) and an IR emissivity of 97% (**ε_{IR} = 0.97**), thereby delivering a radiative cooling power of 93 W m^{-2} (J. Liu et al., 2023a).

Bilayer and multilayer stacking

Coating performance can be further optimized by stacking layers with distinct optical properties. For example, a bouble-layer system consisting of a PVDF upper layer and a PDMS lower layer exploits the refractive index difference between the layers to enhance the backscattering of the solar spectrum, reaching 98% solar reflectance (**ρ = 0.98**) (F. Wang et

al., 2024). Other bilayer configurations use a thick TiO_2-PMMA base paint combined with a thin UV-reflective topcoat (such as $BaSO_4$ with polyvinylideenfluoride (PVDF)) to overcome TIO_2's inherent UV absorption limitations (Altamimi et al., 2023).

Colored and fluorescent paints

Because brilliant white or silver coatings can cause severe glare and limit aesthetic architectural choices, coloured passive daytime radiative cooling (PDRC) paints have been developed. Bilayer coloured designs use a thin visible-absorbing top layer for coloration and a highly reflective underlayer to maximize NIR to SWIR (short wave infrared) reflection, significantly mitigating the heat typically absorbed by dark or coloured surfaces (Altamimi et al., 2023). Furthermore, embedding photoluminescent materials allows the coating to convert absorbed high-energy UV (visual light) into lower-energy fluorescence, thereby actively shedding heat as light rather than thermal energy (Liu et al., 2023).

Thermochromic coatings

To prevent the "winter penalty" of overcooling buildings during cold seasons, temperature-adaptive coatings utilize thermochromic materials; most notably vanadium dioxide (VO_2) (Guo et al., 2023). VO_2 undergoes a metal-insulator phase transition that changes its emission and absorption profiles based on the ambient temperature, passively switching the surface from a radiative cooling mode in the summer to a solar heating mode in the winter.

Superhydrophobic and self-cleaning coatings

In real-world environments, dirt, dust and rainwater can quickly degrade a coatings' solar reflectance and PDRC's performance (Fan et al., 2022). Applying superhydrophobic surface treatments (e.g. using PTFE composites, modified silica) imparts self-cleaning capabilities, allowing water droplets to easily roll off and carry away contaminants, which ensures long-lasting cooling efficiency of the coating (Fan et al., 2022; J. Liu et al., 2023b). This inspired by the lotus effect. This effect refers to the natural water-repellent properties of lotus leaves which arise from microscopic structures on their surface. These structures cause water to form droplets that roll off rather than adhere, preventing accumulation. As the droplets move across the surface, they pick up and remove dirt and other contaminants, resulting in a self-cleaning effect that maintains an optimal surface condition (Collins & Safiuddin, 2022).

Using structure to employ radiative cooling

This section examines the role of nano- and microstructures in facilitating radiative cooling, drawing on two biological models that are briefly introduced here (with further detail provided in the dedicated section on biological models). It then explores the biomimetic mechanisms inspired by these models.

Saharan silver ant

The Saharan silver ant forages in extreme desert environments, enduring surface temperatures up to 70 °C (Shi et al., 2015). In these extreme conditions the ant can cool its body temperature by utilizing a dense array of specialized, triangularly shaped hairs that cover the top and sides of its body. These hairs feature two corrugated top facets and a flat bottom facet facing the ant's cuticle (**Fig. 10**). In the visible and NIR spectrum, where solar radiation peaks, this geometric shape significantly enhances solar reflectivity through *Mie scattering* and total internal reflection occurring at the flat bottom facets of the hairs. Concurrently, in the MIR range, the hair layer acts as a broadband antireflection coating that enhances the ant's thermal emissivity. This dual-effect allows the ant to reflect intense solar heat while efficiently unloading its own excess body heat through the atmospheric window via radiative cooling (Shi et al., 2015).

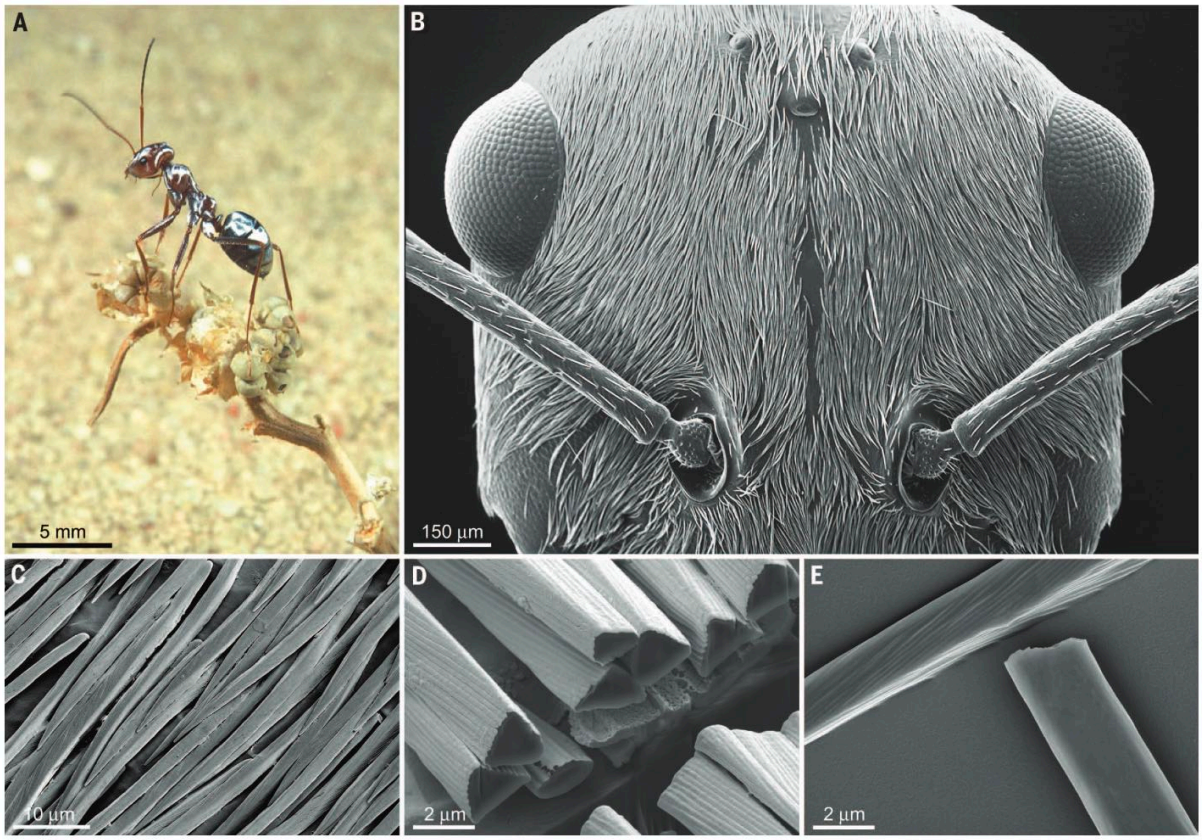


Figure 12: Image of the Saharan silver ant in its natural habitat. ***B.*** *Scanning electron microscopy image of the head.* ***C.*** *SEM image of zoomed in hairs.* ***D.*** *FIB image where the triangular shape of the hairs is clearly visible.* ***E.*** *SEM image of two separate hairs (Shi et al., 2015).*

Goliathus goliatus

The white beetle (*Goliathus goliatus*) survives high temperatures in equatorial forests by employing a unique shell/hollow cylinder structure within its dense epidermal scales (**fig. 11**) (Xie et al., 2019). The scales consist of an outer chitin cortex that encloses an interior packed with randomly distributed hollow cylinders and major air voids. This complex architecture achieves brilliant whiteness and broadband omnidirectional reflection in the visible spectrum.

The outer shell induces thin-film interferences for shorter wavelengths, while the internal hollow optimizes *Mie resonance* and total internal reflections for longer wavelengths. To facilitate heat dissipation, the scale arrays possess a sub microscopic nipple-like contour that provides a continuous refractive index gradient, acting as an antireflective layer that significantly enhances MIR emissivity and enables the beetle to passively radiate heat into the atmosphere (Xie et al., 2019).

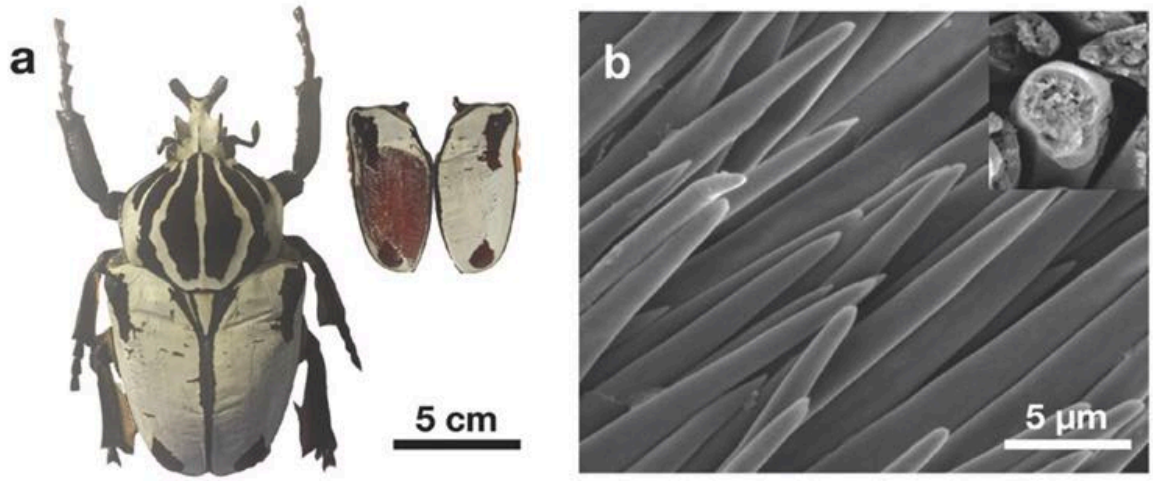


Figure 13: ***A.*** *Photograph of a Goliathus goliatus specimen.* ***B.*** *SEM image of the white scales on the exoskeleton of the Goliathus goliatus (Xie et al., 2019).*

Convection

The phenomenon of convection is found throughout nature in either a forced or passive capacity. Particularly of interest is the way in which convection is utilised for cooling purposes that does not require the expenditure of energy so that it may inspire a similar mechanism in an urban setting.

Mechanism

Convection is broadly defined as the **transfer of heat by movement**, this occurs in a fluid such as water or air (Britannica, 2026). It is the combination of two key mechanisms: **passive (diffusive) or active (advective)**. Passive convection refers to the natural behaviour of molecules at different temperatures. Most fluids[8] expand when heated and thus become less dense, this means they become more buoyant and can rise in elevation. This can also happen in reverse, where the cooler molecules sink towards the ground. In the context of air, this is seen in hot air ballons, or the rising of moist air to form clouds in the sky. In an active convection context, the movement is forced by more than the buoyancy of the molecules themselves, but rather the pressure differences (or viscous entrainment (see Appendix C)) caused by factors such as wind in air, the running water of a river (imagine silt being pulled by the current of the water), or a powered mechanism such as an electric fan.

In the context of air and built structures, there are a few common effects seen when it comes to convection, one of them is the Venturi effect, which can be explained by Bernoulli's principal (See Appendix C). The Venturi effect is the observed circulation of air in a system based on the air pressure differences; whereby warm air will move from an area of high pressure to an area of low pressure by diffusion (Design Buildings, 2022). We see examples of this when fast wind speeds up high can pull the air upwards from lower areas where the air is slow moving. In relation to this, the stack (or chimney) effect, is the passive convection of air in or around a structure that happens based only on the density (buoyancy) of the air molecules themselves (Design Buildings, 2022). Circulation of heat in air occurs due to this principle - the warm air travels up while the cool air sinks, often this is cyclical because the air cools as it gets higher with increase in density and a resultant sinking is then observed.

Measuring convection

Convection is measured by determining the difference in temperature between two points of a system such as between a fluid and a solid (e.g. a material and the air) (*Newton's Law of Cooling,* 2026). The **rate of heat transfer (Q)** can be calculated by determining the temperature difference between the surface and the surrounding fluid (air) $(T_2\text{-}T_1)$ multiplied

[8] Gases in this context also behave like a fluid in terms of how heat moves around, as such the air around the material is also considered a fluid.

by the surface area (A) through which the heat is transferred and multiplied by the convective heat transfer coefficient (*h*). This is described in equation 3:

$$\boldsymbol{Q = hA(T2 - T1)} \qquad \boldsymbol{(3)}$$

Looking at equation 3, we can deduce that there is a positive correlation between the rate of heat transfer and the other factors, whereby the higher the temperature difference, or the convection coefficient, or the surface area, the higher the rate of heat transfer. The convective heat transfer coefficient for air is usually around 1-10W/m^2.K for passive convection, or about 20-100 W/m^2.K for forced convection (Xiao et al., 2018). There are numerous factors that can affect convective heat transfer, such as geometry of the surface, properties of the fluid or solid (such as their thermal conductivity), as well as the fluid's viscosity and velocity.

Methods of experimental testing

The rate of heat flux can be calculated using equation 3. Additionally, heat mapping of the façade can be performed using a thermal camera or thermometer to record temperature at different locations to locate hot and cold spots of the design, and changes over time influenced by convective heat movement. In a 1977 study on convection in cacti (Lewis & Nobel, 1977) the local convection coefficient was determined for each nodal level using a regression equation involving the measured change in wind speed with height. Additionally, the behaviour of airflow over designs can be observed using either an air flow mapper (clean room fogger), or by utilising computational fluid dynamics when a physical model is not possible.

Bio-Models

Looking at nature for examples of convection can be quite broad, as it is a mechanism that exists almost everywhere. When researching ways to optimise or utilise the mechanism for the benefit of cooling, there were some key examples that show promise. These are highlighted below.

Cycling air in wetland plants

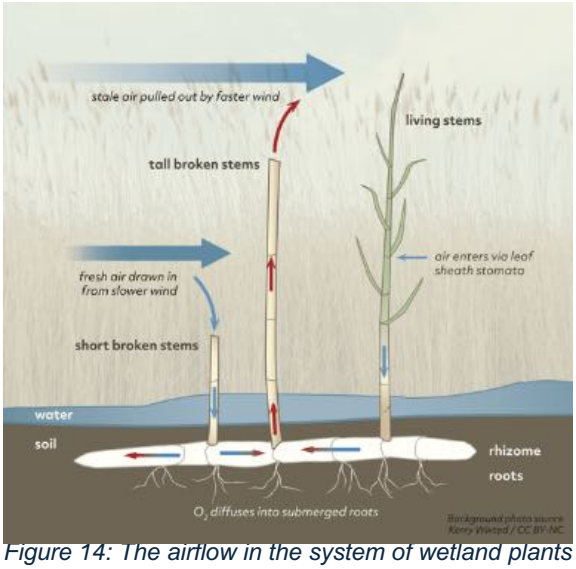


Figure 14: The airflow in the system of wetland plants (How Wetland Plants Get Air to Waterlogged Roots, 2022)

Wetlands plants that are tall and have hollow spaces, can utilise the flow of air and differences in pressure to passively cycle fresh and stagnant air throughout its system (Colmer, 2003). This occurs by the tall and broken stalks of the plant providing openings to the air up high, where the air is traveling at faster speeds. Meanwhile there are also broken stems down low where the air is not moving very fast. This pressure difference draws air into the lower open stalks whereby also forcing air out of the taller stalks. This process does not require any energy expenditure by the plant and is completely passive, all the while providing fresh and oxygen rich air to the lower areas of the plant such as the roots and living stems. (Colmer, 2003) (**Fig. 12**).

Asymmetric burrow openings

Black tailed prairie dogs build burrows that can take advantage of wind speed gradients to ensure there is optimal airflow through the internal spaces of the underground burrows. The animals build one or more openings upwind at higher elevation, and one or more openings downwind at lower elevation (**Figure 15**) (Vogel et al., 1973). Thanks to the fact that wind moves slower closer to the surface of the ground, and Bernoulli's principle, this results in a one-way air flow through the underground tunnels. The air can rush out of the higher opening while being pulled into the lower opening, much like the wetland plants seen above.

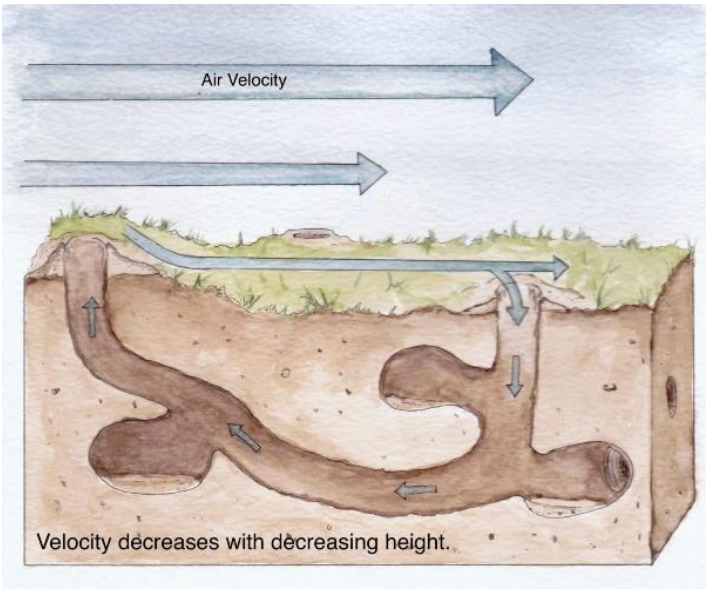


Figure 15: Burrow openings and airflow (Asymmetric Burrow Openings Create Passive Ventilation, 2020)

Termite mounds

One of the more well-known bio models for passive cooling are termite mounds that are seen in warm environments around the world, such as Africa and Australia. Large cathedral termite mounds found in the savanna house the best examples of convective cooling (Korb & Linsenmair, 2000). Convection occurs thanks to contrasting temperature differences (between the outside ambient air, the internal spaces, and the thinner walls of the mound) and the dissipation tendencies of the warmer air molecules as described earlier. The tendency of this mechanism to work better on larger mounds is also due to the role the sun plays in heating the outer faces of the mound, providing a larger temperature gradient, thus increasing the rate of convection (Korb & Linsenmair, 2000).

In terms of design, termite mounds can either be open at the top with openings at the base of the mound (an "open chimney mound" where mounds can allow warm air to naturally rise to the top of the mound, while cool air near the base is drawn inside) or completely closed at the top, but with porous walls ("capped chimney mounds") that allow the air exchange to occur all over the mound's surface via small inlets (**Fig. 14**) (Turner & Soar, 2008). They can also be likened to a human lung, in that the convection methods used are a mix of small-scale diffusion and larger scale, forced (wind) convection as well (see appendix C). Termite mounds can also reduce unwanted convection in areas of the mound thanks to bumps and dimples to trap air on the surface, acting essentially as insulation. With all these factors in mind, termite mounds are a complicated thermoregulatory system, with a multitude of interacting mechanisms at play.

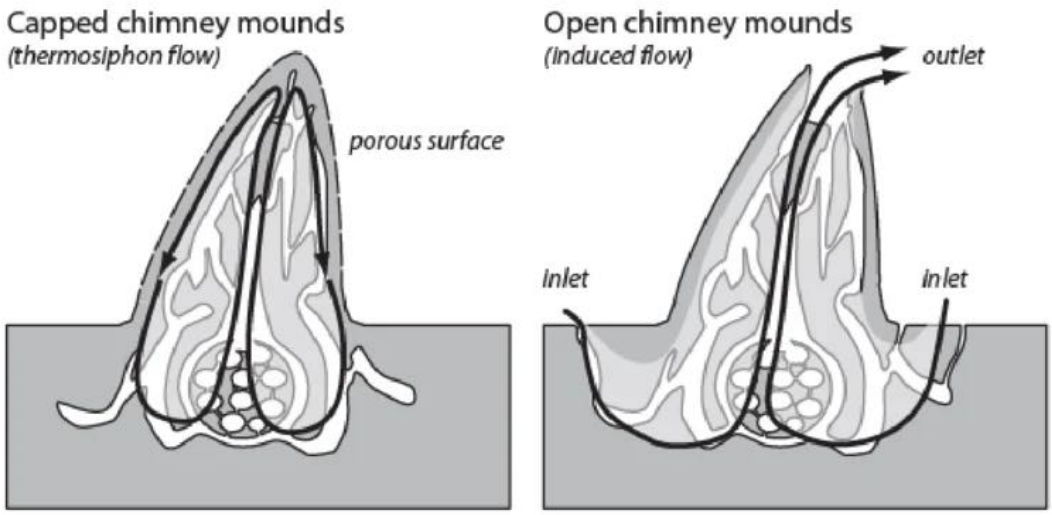


Figure 16: Different types of termite mounds (Turner & Soar, 2008).

From what has been discussed, we can say that for a large structure, convection certainly plays a role but will not be the only driver of heat dissipation because both the solar and wind driven enhancing factors can be variable throughout the day and seasons. Therefore, the diversity of mechanisms in fact works in favour of thermoregulation in this instance. Additionally, thanks to the prevalence of these structures, easily available research and

recent history in human-made structures, we can see current realisations of this bio-model in the construction industry in such buildings as the well-known Eastgate building in Zimbabwe by the architect Mick Pearce in 1996, and the common stack effect seen in architecture.

As a fascinating, additional mechanism, Turner and Soar (2008) also highlight the ability of some termite mounds to make use of resonant frequencies (a mound is usually between 20-30Hz) that are attuned to that of turbulent winds (10-100Hz) which, if achieved, can activate the mixing of air inside the mound, promoting gas exchange at a small scale.

Cacti

The mascot for warm environments is often the cactus, with its sharp spines and ribs. These surface qualities allow the cactus to maintain optimal temperature despite the harsh environment. Vertical ribs on the body of a cactus allow for surface shading, increased surface area to dissipate heat, and increase turbulence to enhance convective heat removal (Lewis & Nobel, 1977). The troughs of the ribs are shaded and thus cooler, allowing for more heat to be drawn out from the body of the cactus, which is turn rises towards the peaks where it can waft away (**Fig. 15**). Lewis & Nobel (1977) also note that with the ribs present, the convection coefficient is 44% higher than its theoretical smooth counterpart. Noting as well, that the warmer tip of a rib is 3-fold higher in convective coefficient than the cooler trough. In this assessment, the ribs can be morphologically described: rib depth ~1.9 cm; diameter of cactus at mid-height 26 cm; number of ribs 26 (see additional information in Appendix C). Additionally, they can be spaced differently on different sides of the cactus (e.g. closer together on the southwest side) for different temperature regulations based on the orientation to the sun (Lewis & Nobel, 1977).



Figure 17: Thermal image of a cactus from Cactus Ribs Keep Them Cool, 2017; a barrel cactus photographed in Nevada (Marketa Mark, n.d.).

In species such as the barrel cactus, spines serve multiple functions: they hold back predators, collect moisture and reduce convective heat loss on the plant's surface. Without these spines,

the convection coefficient is 13% higher, meaning that the spined reduce convection along the surface of the plant (Lewis & Nobel, 1977). The presence of spines reduces *h* (heat transfer coefficient), increasing the thickness of the air boundary layer of the cactus. The barrel cactus has a higher density of spines at the apex to make for more shading and effectively eliminating convective air transport. However, lower on the cactus there is less density of the spines and wider spacing of the ribs, which does allow for convective heat transfer. In studies of barrel cacti with spines, reduced daytime temp by 2-6 °C were observed (compared to without). These spines were 1-2 cm long and shades ~62% of the surface of the cactus (for the *mammillaria* species) (Nobel, 1978). Therefore, it is generally concluded that the spines' role in cooling is primarily based on shading from shortwave irradiation and reducing the outgoing net longwave radiation during the night (Lewis & Nobel, 1977). There is, however, very little empirical research on the topic of cactus and spines for the purpose of thermal regulation. Many of the other papers on this topic refer to these two key papers mentioned so far, so it is important to keep this in mind. However, there have been buildings designed after this bio-model since then, which take advantage of increased surface area and jutting shade structures like spines, for example the well-known Eastgate building in Zimbabwe with concrete "teeth" aiding in cooling the external air by 3°C (*Eastgate Building Harare*, z.d.).

Conduction

Conduction is the transfer of heat within a material from one molecule to the other, without the movement of molecules themselves. In metals, which are generally among the most conductive materials, the energy is transferred mostly by free electrons in the material, but phonons are also efficient carriers of energy between the molecules of a material, especially in non-metals (Dincer & Siddiqui, 2018). Other material properties making a material highly conductive are an ordered crystal structure, tight packing off atoms and strong bonding between atoms (Dincer & Siddiqui, 2018). To reduce a building's energy consumption by limiting heat gain from warm outdoor air and solar radiation, an effective strategy is to minimize heat conduction by selecting a material with low thermal conductivity.

Air has a low thermal conductivity, making it essential for insulation in the construction sector. In fact, without the conductive properties of air, almost all conventional insulation materials would fail (Juanicó & González, 2017). Wool, rock/ glass wool, fiberglass, EPS, and even biobased materials, including straw or mycelium materials all rely on air pockets within to material to prevent heat being conducted through the material.

With the panel, the aim is to achieve more than just insulation and cooling down the building. Therefore, relying solely on low-conductivity materials is not sufficient. Unlike the other mechanisms, conduction does not remove heat from the façade, as the net thermal energy of the object stays the same. However, by adjusting the thermal conductivity of the materials, it is possible to influence how the heat is distributed through the façade, which in turn affects where and how it can be released. An example of the use of conductivity to increase infrared emissivity, is the usage of graphene on top, or directly under, the surface of a façade. Graphene is highly conductive along the plane of the coating, but less conductive through the plane (**Fig. 16**) (Song et al., 2025).

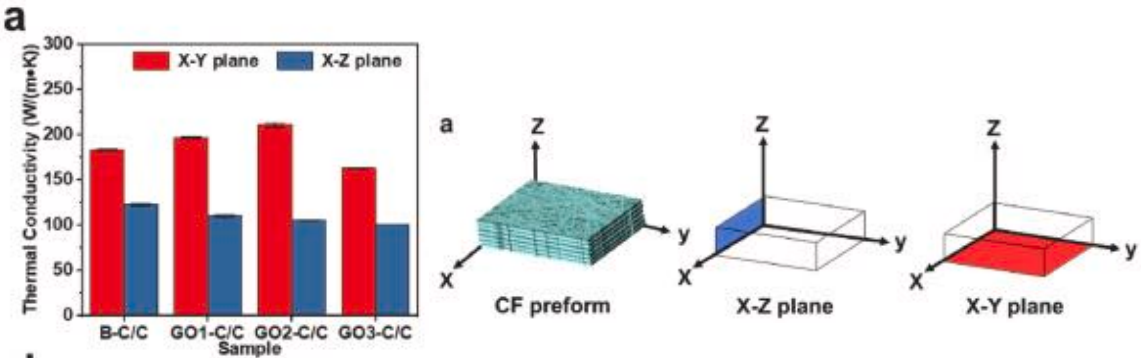


Figure 18: Demonstration of conductivity differences along and through a graphene sheet from (Song et al., 2025).

This means that heat can easily dissipate along the surface but not enter the core of the material. This results in the heat being concentrated at the surface, meaning the emissivity at the surface is increased. The biocompatibility and biodegradability of graphene based materials varies based on number and size of graphene layers. In many biomedical applications the material has been found to be non-hazardous, but that does not mean it is a biobased and easily degradable material (Chiticaru & Ionita, 2022).

Alternatively, to heat dissipation, it is also possible to concentrate heat at a few local hot spots, by guiding the heat flux within a material. This can be done by using so called metamaterials, usually composite materials consisting of a highly conductive material, usually a metal, and an insulative material. This allows heat to be conducted unevenly, creating local temperature differences and hotspots (Arepolage et al., 2024). Furthermore, the reliance on highly conductive materials, like metals, which need to be intricately integrated into the material, means that the entire material will not be biodegradable.

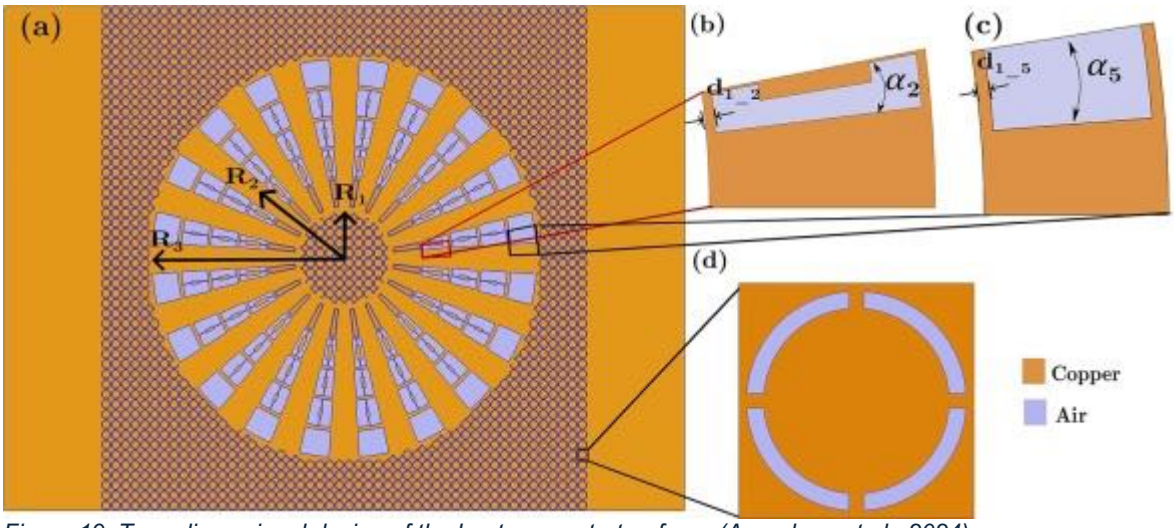


Figure 19: Two- dimensional design of the heat concentrator, from: (Arepolage et al., 2024).

Figures 17 and 18 demonstrate the design and resulting heat flow of a possible thermal concentrator. Because highly conductive metal components of the material point towards the centre, the heat also moves faster into the centre of the circle relatively to how it moves through the rest of the material (Arepolage et al., 2024). As the direction of heat flow cannot be changed, it also moves more easily away from the centre, which is the reason that the dark blue area is also distorted towards the centre. Nonetheless the arrangement leads to local temperature differences inside the circle even at equal distance to the heat source.

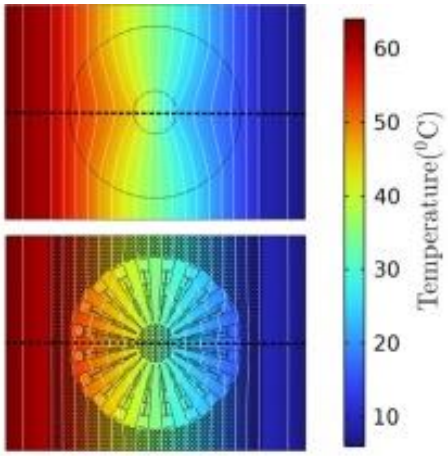


Figure 20: Heat distribution after heat source is applied on the left side, from: (Arepolage et al., 2024).

Next to the possible application in radiative cooling, heat concentrators could be useful in facilitating convective cooling. As in the example of the biological model, the barrel cactus, convective airflows are shaped largely by temperature differences in the air immediately surrounding the material (Laver et al., 2008). These differences could be created if parts of the material are naturally hotter than its surroundings, as these areas would naturally emit more heat to the surrounding air layers, which creates temperature differences along the material, facilitating airflow. In the example of the barrel cactus having enhanced the relative temperature difference between the ridges and groves, by steering conductive heat flow towards the already hotter ridges, could increase the speed of

convective airflow around the cactus and remove even more heat from the material. To do so a simple design of spacing vertically oriented metal rods on the ridge part separated by non-conductive groves can already achieve this function. If the metal rods absorb more solar energy than the groves the heat will be dissipated mostly via the conductive material, so mostly vertically, meaning the temperature difference between ridges and grove remains. This difference in absorption is achieved by the same self-shading principle as in the cactus and can be enhanced further by using a reflective coating in the groves and absorptive material for the ridges. As long as parts of the material are exposed to solar radiation this effect would work even on parts of the panel where sunlight does not directly reach as the heat will reach these parts via conduction through the metal rod.

Evaporative cooling

Evaporative cooling is a very effective cooling mechanism that can be used on the facades of buildings. At normal building temperatures evaporating 1 kg water results in the removal of 2,450kJ of energy, meaning that 2000 m³ air can be cooled by 1 degree from just the evaporation of one kg water (Santamouris, 2019). In real life simple evaporative cooling walls, such as walls using porous bricks can achieve temperature reductions of 1.45°C in hot humid regions (L. Li et al., 2025a). Evaporative cooling is also believed to be the main contributor to the cooling effect that green walls have (Kapilan et al., 2023). However, the dependence on water creates design problems which often jeopardize the feasibility of evaporative cooling facades. The first challenge is to obtain the water that can evaporate from the wall. In regions with high rainfall, this can be achieved by relying solely on rainwater, however that means the panels will not cool after extended periods without rain (L. Li et al., 2025b). Alternatively, water can be purposely added with irrigation mechanisms to the wall, however this method is not passive and consumes water and energy. A work around would be to use irrigation from sources that already leak water, like condensed water from air conditioning units, or other types of greywaters that form in the building. A third option to source water is to use groundwater, which can be transported upwards, without the use of external energy, thanks to the physical effect of capillary action. With the pull created from evaporation porous materials with very small pore sizes could transport water 10 meters above the ground, without energy input (Holbrook & Zwieniecki, 2008).

Another downside of evaporative cooling is that the inevitable exposure to water can drastically shorten the lifespan of the panel, especially if it is produced from a biodegradable material. Biodegradation would be more likely to already occur during use of the product if it is continuously exposed to water (Soh et al., 2025b). Coatings can be applied to mitigate this, however, they might then reduce the materials capability to take up water and would make the panel less biodegradable, if they are not also sourced from a biodegradable material. Hydrophobic coatings would also drastically reduce the water holding capacity of the material, which would then reduce evaporative cooling of the façade.

Existing bioinspired applications of evaporative cooling include elephant skin inspired tiles. Cracks on elephant skin increase the amount of water that can be held on a surface. Elephants bathe daily if they have access to water and if not usually cover themselves in cooling mud, which means evaporative cooling regulates their body temperature throughout the entire day (Martins et al., 2018). A research team from Nanyang Technological University, Singapore used this inspiration to create Mycelium based façade tiles, which, like the biological inspiration, can store five to tenfold more water than a flat surface could (**Fig. 19**) (Soh et al., 2025a). Nonetheless, despite applying a waterproof coating made from beeswax these biobased tiles still face the problem of biodegradation during exposure to outside weather conditions.



Figure 21: Elephant tiles by Nanyang Technological University to increase water holding capacity (Soh et al., 2025).

Another biological model that has not yet been applied in a bioinspired context is that of ants, which release bursts of water mist that evaporate almost instantly, since the tiny water droplets in air together have a much higher total surface area than just a flat surface could have. Mist driven evaporative cooling has already been tested and utilized as a possible strategy to mitigate urban heat. A study conducted in Italy showed a reduction of outside temperature surrounding the mist by as much as 7.4°C (Ulpiani et al., 2020). However, these systems are not passive as they require water to be squeezed through a small nozzle at high pressures to create the mist. While it is possible to generate pressure without additional energy supply, e.g. via air expanding when heated in a closed space, this would require a highly complex control mechanism, which would be too difficult to install in a simple façade panel. In general mist formation and droplet size are determined by the nozzle size and the pressure and speed at which water is ejected (Ulpiani et al., 2020).

Possible integrations

In practice for the façade panel, these passive cooling mechanisms will not operate on their own, but rather as one connected system where different heat transfer mechanisms will work together to passively cool the façades of buildings. The effectiveness of the panel will therefore depend on the intertwined function of reflection, radiative cooling, convection, conduction and optionally evaporation into a coherent design.

The most obvious interaction between the mechanisms lies in the link between **reflection** and **radiative cooling**. As explained in this report, radiative cooling consists of the reflective properties, as well as the infrared emissive properties of a surface. High solar reflectance (**ρ**) reduces the heat gain of a surface during initial contact with the solar rays, while the thermal emissivity (**ε**), mostly in the 8-13 μm atmospheric window, enables the heat dissipation to the sky. These two functions should be optimized simultaneously, and several coatings and materials were named in this mini report that could be used. The bio-inspired options, like the *A. demophon* inspired coating could prove to be efficient but might be hard to acquire or to recreate. Additionally, spectrally selective coatings could be used in this context as they reflect strongly over the whole solar spectrum, as well as emit in the mid-infrared range. Lastly, a retro-reflective coating could be applied, but its performance in the mid-infrared range must also be evaluated. A possible approach would be to test these three coatings against each other to determine whether the bio-inspired option performs the best.

Conduction can control heat distribution within the material and facade while it's heating up. This mechanism creates the necessary conditions to carry heat away from the panel and the building, by distributing thermal energy within the material. Moreover, if we were to increase the heat dissipation over the surface of the material and limit the dissipation through the material (with graphene for example) this could result in a concentration of heat at the surface, meaning that the heat emissivity at the surface is increased as well, allowing for the heat to leave the building. **Convection** can further add to this mechanism by moving the heat that is conducted through the material into the cavity of the panel and dispersing it into the surrounding air.

Key to utilizing the mechanism of convection is increasing the material surface area or increasing the temperature gradient - promoting the heat flux away from the material and the building. Airflow should also be encouraged rather than restricted to prevent the heat from being trapped on the surface of the building. The efficiency of this mechanism will be influenced strongly by the geometry of the panel. Textured and corrugated surfaces, inspired by the biological model of the cactus and the earlier prototypes from the client, can increase surface area, create local temperature gradients and enhance the airflow and turbulence and thus enhance the convective air removal when the inside of the panel gets hot by conduction. Additionally, the macro-structure of the panel, inspired by the design challenge years ago by the biological model of bark, promotes airflow through the chimney effect and additional shading. These channels and cavities function in the same way as they do in the biological models of the termite mound and the prairie dog burrows, where the natural, buoyancy-driven

convection can also be enhanced by wind. Geometry that allows for shading can also be helpful to reduce the building temperatures. In addition, the incorporation of wind to encourage more active transport of heat away from the building will be beneficial. Without this, there will still be a small amount of passive convection, based on the temperature gradients mentioned. Current designs seen in the built environments provide many possible examples and inspirations, such as the Eastgate building, solar chimneys in architecture, and the façade design seen in Appendix C by Laver et al. (2008).

Lastly, **evaporation** has been proven to be an effective strategy to cool buildings and the environment, both with green walls and roofs, as well as transpiring bricks and mist-driven evaporative cooling. However, all these strategies require some type of active mechanism, or active storage of water, which leaves evaporation out of scope for this project. Investigating how mist evaporative cooling can be integrated into a façade panel can nonetheless pose an interesting research question for future research. For our panel the technological restraints around water supply to the panel and the challenges around waterproofing the material, while purposely exposing it to water led us to choose against incorporating evaporative cooling in our design.

Conclusion

The heat transfer mechanisms are highly interdependent. From a design perspective, it will be important to focus on multi-scale integration of all the mechanisms into one panel. At the micro- and nanoscale, coating composition, (structural) colour and structure determine the solar reflectance and infrared emissivity. The biological models of the *Archaeoprepona demophon*, Silk worm and Saharan silver ant seem most promising for the application of possible bio-inspiration into our panel. At the macroscale, panel composition and geometry cause convective airflow, heat exchange with the environment, and insulation of the façade. For most efficient convection, the biological models of the prairie dog burrows and termite mounds should be further analysed. Additionally, the application of cacti-like spikes to the surface of the existing panel could be investigated to improve convective airflow and shading.

A successful passive cooling panel should therefore be designed using a holistic approach, in which material selection, structural design, and interaction with the environment are optimized harmoniously. The façade design should work together with existing relationships in urban context, such as using materials that are locally accessible and sustainable; make the city more liveable for humans, wildlife, and plant species; reduce the energy demand of the city by enabling passive thermoregulation; and enhance other cooling strategies such as increasing the airflow from green spaces thanks to increased convection capabilities, enabling the city to function as a resilient interconnected system.

For the façade panel in this project, these are the requirements for a holistic, hybrid design, combining all the heat exchange mechanisms:

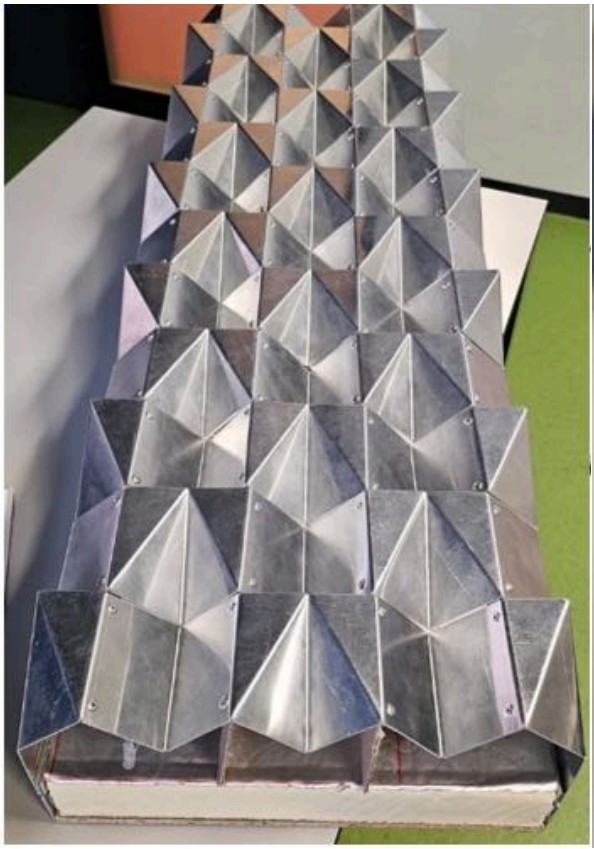
- High solar and NIR reflection, together with a MIR emissivity coating or outer layer (reflection + radiative cooling)

- Controlled (and steered) thermal conductivity to guide heat towards heat dissipative structure (conduction)
- Surface geometry structures that enhance the heat transporting airflow and shading of the façade (convection)

By integrating these principles, the façade can function as a passive, adapting, cooling interface, minimizing the heat gain, maximizing the reflection, and removing the trapped heat out of the urban canyons, aiding passive mitigation of the Urban Heat Island effect.

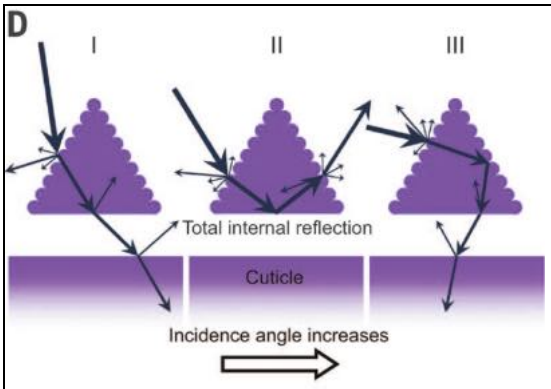
Appendix A.

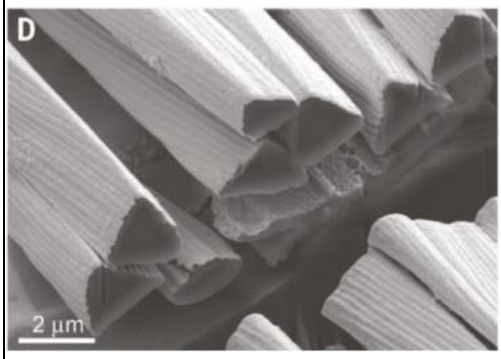
Photographs of the original façade panel designs which are to be improved upon based on this research.

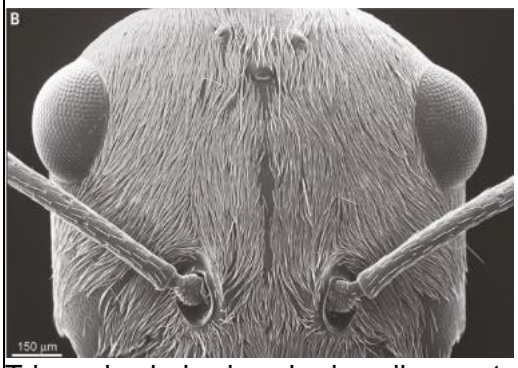




Appendix B.

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Triangular hairs in saharian silver ant and the solar reflection.
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Mechanism

The dorsal and lateral sides of the Saharan silver ant (*Cataglyphis bombycina*) are covered in a dense array of hairs that gradually taper off at the tips and are locally aligned in the same direction. These hairs possess a highly specialized triangular cross-section characterized by two corrugated top facets and a flat bottom facet facing the ant's body. This unique structural geometry protects the ant from overheating through three physical mechanisms:

- **Visible and Near-Infrared (NIR) reflectivity**: The triangular hairs act as a highly reflective coating against solar radiation. This is achieved through Mie scattering, where light gets trapped within the hairs and reradiates out in all directions, combined with total internal reflection occurring at the flat bottom facets of the hairs.
- **Mid-Infrared (MIR) emissivity**: The hair cover significantly enhances the ant's thermal emissivity in the MIR range. At wavelengths much larger than the hair dimensions, the hair structure acts as a gradient refractive index layer, giving the surface broadband, antireflective properties that allow the ant to efficiently offload excess heat back to its environment via blackbody radiation.
- **Reduced absorption from the ground**: The bottom surface of the ant is completely bare, which effectively reduces the radiative energy transfer between the hot desert sand and the ant, reflecting MIR radiation from the ground better than a hair-covered surface would.

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Function

Maintain a lower steady-state body temperature and prevent overheating beyond the ant's critical thermal maximum (53.6°C) while foraging in extreme desert temperatures.

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Strategy

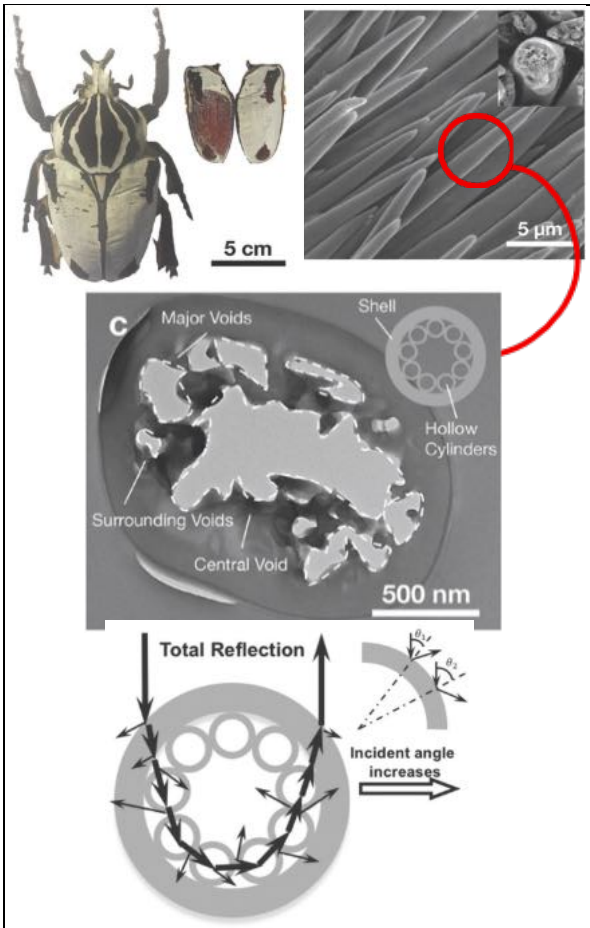
Use precisely shaped micro-structural features (triangular hairs) to simultaneously reduce solar heat absorption and maximize the dissipation of excess body heat to the surrounding environment.

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Design principles

A surface-integrated micro-structure composed of dense, triangularly shaped elements featuring corrugated upper facets and flat bottom facets can passively cool an object. This is achieved by maximizing visible and near-infrared optical reflection through Mie scattering and total internal reflection, while simultaneously acting as an antireflection layer that enhances mid-infrared thermal emissivity for highly efficient radiative heat dissipation.

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Mechanism

The elytra (wing covers) of the white beetle (*Goliathus goliatus*) are covered in dense scales that feature a shell/hollow cylinder structure, consisting of an exterior chitin shell and an interior of packed hollow cylinders (air pockets). This specialized micro-structure protects the beetle from overheating via two mechanisms:

- **Broadband omnidirectional light reflection**

The scales significantly enhance reflection and minimize the absorption of solar energy (of wavelengths in the visible spectrum). High reflection of short wavelengths is achieved by the exterior shell through **thin-film interference**. The interior system of hollow cylinders enhances the reflection of longer wavelengths through **Mie scattering** (resonance).

- **Mid-infrared (MIR) emissivity**

The scale array significantly increases the thermal emissivity of the elytra in the MIR range. The exterior layer of shell arrays consist of a submicroscopic nipple-like contour that acts as a continuous refractive index that allows the beetle to dissipate excess heat to the cooler outer space via **passive radiative cooling.**

Function

Maintain a lower equilibrium body temperature and prevent overheating (reducing temperature by up to 7.8 °C in air) while resting exposed to direct sunlight in high-temperature equatorial forests.

Strategy

Use a specifically optimized shell/hollow cylinder micro-structure to simultaneously minimize solar energy absorption and maximize the radiation of excess body heat to the surrounding environment (passive radiative cooling).

Design principles

A surface-integrated dual functional microstructure composed of an outer thin-film shell and an interior array of hollow cylinders that can passively cool an object. This is achieved by combining the mechanisms of thin-film interference, Mie resonance and total internal reflection to maximize visible light reflection. This while simultaneously acting as an antireflective gradient-index layer that enhances MIR emissivity for highly efficient radiative heat dissipation.

Appendix C.

Relevant terms & principles (and other points of interest)

- Advection/advective transport: transfer of heat or matter by the flow of a fluid (esp. horizontally). E.g. flow of water in a river taking silt along the riverbed. Essentially another term for viscous entrainment.
- Diffusion (molecular): movement of particles from high concentration to low concentration.
- Viscous entrainment: the process where a moving viscous fluid or solid surface drags, traps, and transports surrounding fluid (usually gas) along with it. Used in leaf cutter ant nests (air moving outside pulls the still air of the nest along with it = ventilation) (Design Buildings, 2022).
- Bernoulli's principle is a general principle of fluid dynamics based on the theory of conservation of energy, whereby an increase in pressure will mean a decrease in speed of a fluid (and vice versa) → it is that increased pressure = decreased speed and vice versa. Bernoulli's equation

$$\Delta P = -\frac{1}{2}pV^2 \quad (4)$$

(Armstrong et al., 1992)

- Venturi effect is the demonstration of bernoulli's principle physically.
- The stack effect (aka chimney effect) the use of convection in a building setting; making use of the differences in temperature and density of air in/outside of buildings, whereby hot air wants to rise and escape (Design Buildings, 2022).
- The Nusselt number (Nu) is defined as the ratio between convective and conduction heat transfer, indicating the efficiency of heat transfer in a fluid.
- Bonus mechanism: Resonant Acoustic Mixing (biomodel = termite mounds). The resonant frequency of a space can cause the air particles to mix quite thoroughly. Termites use this with the resonant frequency of their mound, to mix stagnant air. Could we make a panel like a flute that makes the frequency when the wind goes through it?
- Phase change materials (PCMs). Substances that absorb and release high amounts of latent heat during phase transition (usually freezing and melting) to provide thermal management. E.g. when it's hot the material melts and stores energy, when it's cold it then freezes and releases that energy to the microenvironment (material example is wax).

Other convection considerations

Barrel cactus morphological details from Lewis and Nobel, 1977:

Table I. Summary of Morphological Characteristics of F. acanthodes used in Model

Height above ground	41 cm
Depth below ground	4 cm
Diameter at midheight	26 cm
Number of ribs	26
Rib depth	1.9 ± 0.4 (26) cm
Total aboveground surface area	0.47 m^2
Spine length	5.4 ± 2.1 (30) cm
Spine diameter	0.12 ± 0.031 (30) cm
Spine shading of stem	
levels 2 to 5	36 ± 5 (16) %
level 6	76 ± 3 (16) %
level 8	84 ± 3 (16) %
Apical pubescence thickness	1.2 cm
Thermal conductivity	
stem tissue	0.57 ± 0.03 (10) W m^{-1} C^{-1}
apical pubescence	0.042 ± 0.015 (10) W m^{-1} C^{-1}
Shortwave absorptance	
stem	0.65 ± 0.02 (10)
apical pubescence (level 8)	0.61 ± 0.03 (10)
spine	
levels 2 to 5	0.69 ± 0.03 (12)
level 6	0.64 ± 0.03 (12)
level 8	0.58 ± 0.03 (12)

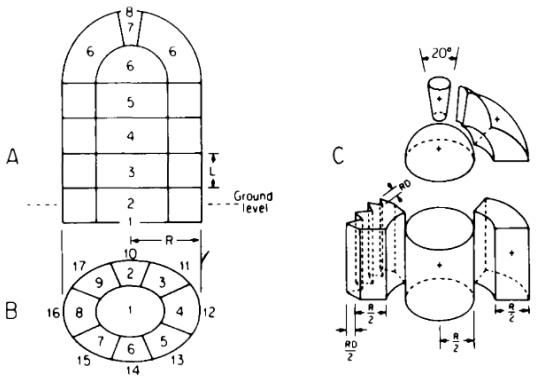
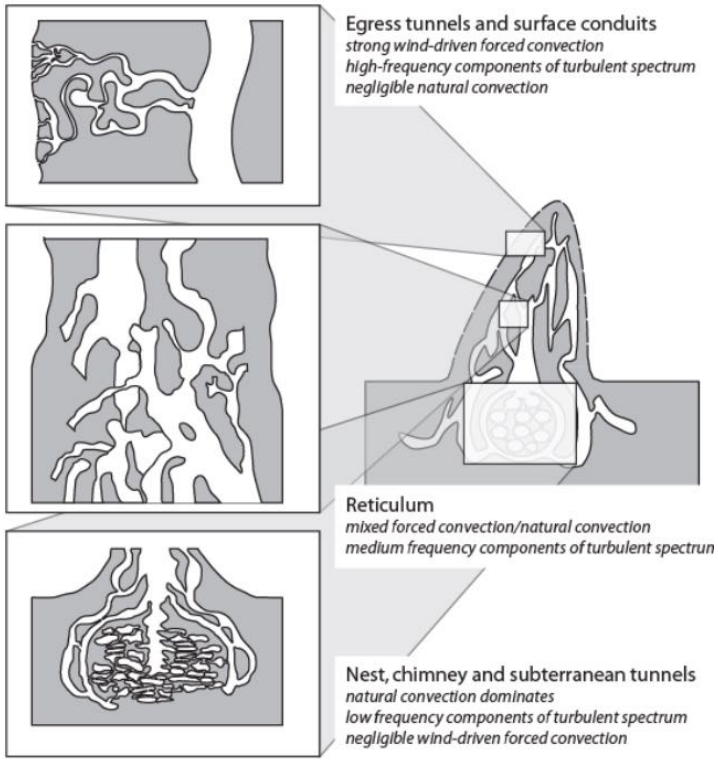
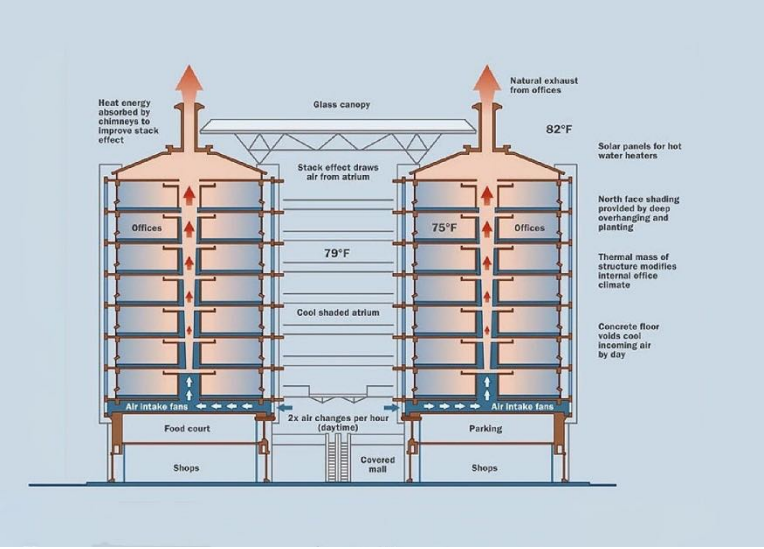


FIG. 1. Nodal system for *F. acanthodes*. A: Vertical section indicating levels; B: horizontal section indicating positions; C: nodal volume elements indicating centers of mass (+). L: level separation; R: corrected plant radius; RD: rib depth.

Termite mound movement of air from Turner & Soar (2008) that shows the relationship of forced and natural convection at different levels of the mound:

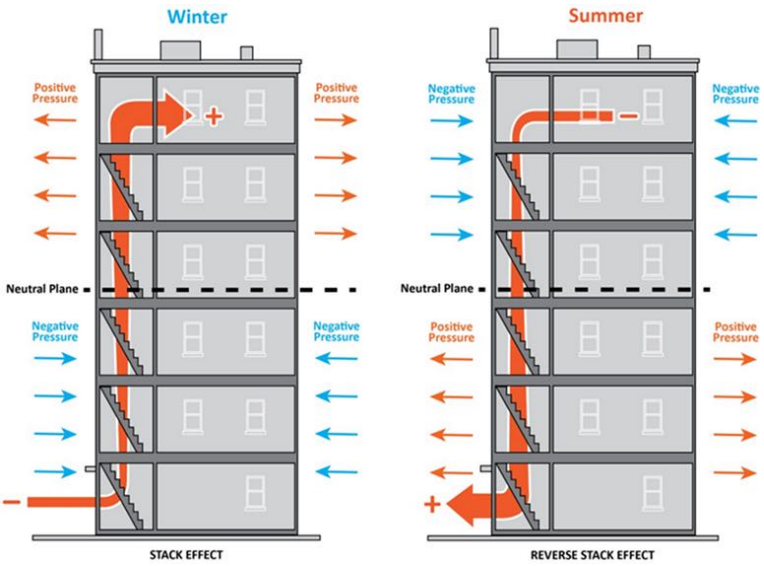


Eastgate building (‘The Eastgate Centre’, n.d.) which is a well-known architectural example of bio mimicry. It increases the existing natural convection with added fans at the lower intake areas of the building. It’s also designed in the image of a cactus, with high surface area and many shading “spine” like structures:





Stack effect can also be reversed in summer due to difference in pressure (based on ventilation design of the whole building, not just the facade) (Kocaman, 2025). Interesting detail to note is the neutral pressure plane of the building which is an especially relevant characteristic of tall buildings. Above and below this plane is the different pressure (positive or negative) whereby air always wants to travel to the low pressure side due to diffusion. Generally speaking, the area of high pressure is the warmer area because of the accumulation of air from the mentioned diffusion:



Other interesting designs

Modular and biophilic design from Laver et al., (2008) which incorporates both shading and air inlets:

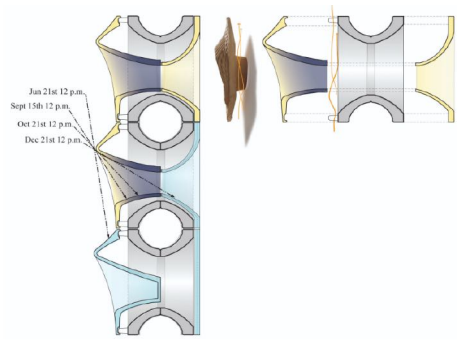


Figure 8: Wall system diagram, with notes on solar angles and surface textures.

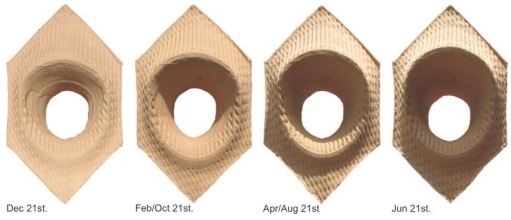


Figure 9: Prototype of face tile. Over the course of the year the face tile either shades itself or receives full sun. This enables us to tune the geometry of the face tile to a specific location for the heating and cooling months. From Arial to August the face is in 50% shade reducing the radiant thermal gains, while from November to February face tile receives less than 10% shading enabling radiant heating of the wall.

Façade designs from Turner & Soar, (2008) that are inspired by termite mounds:

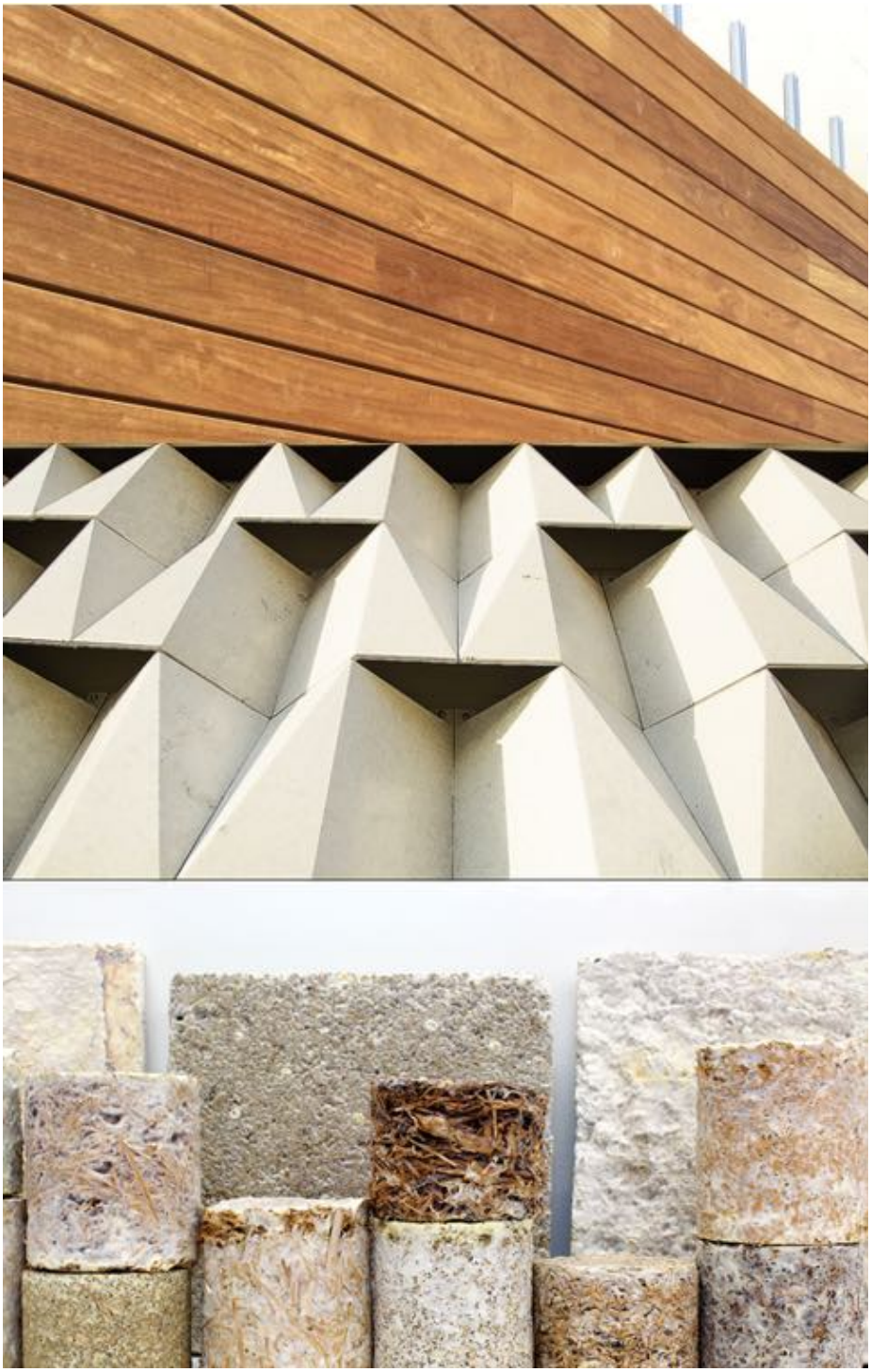


Fig. 9. Some imagined biomimetic designs. Top left. The surface conduit-egress tunnel complex. Top right. A rendering of a building enveloped by porous "surface conduits." Bottom. The block elements for an artificial surface conduit.

Appendix C. Material Selection Report

Material selection: façade cooling panel

Report for MIP 2026 Team X – Functional skin for Passive Cooling



Jochem ten Pas, Alexandra Cooke-Kasyanto, Felipe Krebs, Ruben Langendoen, Ocker Hazelbag

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Introduction

Façades are the barrier between the interior and exterior of a building, like a skin separating the internal body to the outside environment (Sandak et al., 2019). In the past, a façade just consisted of one layer and therefore only one material, as the inside was the same material as the outside. However, over time, façades have become more refined with multiple layers that each have their own function, showing analogy with the human skin **(Fig. 1).**

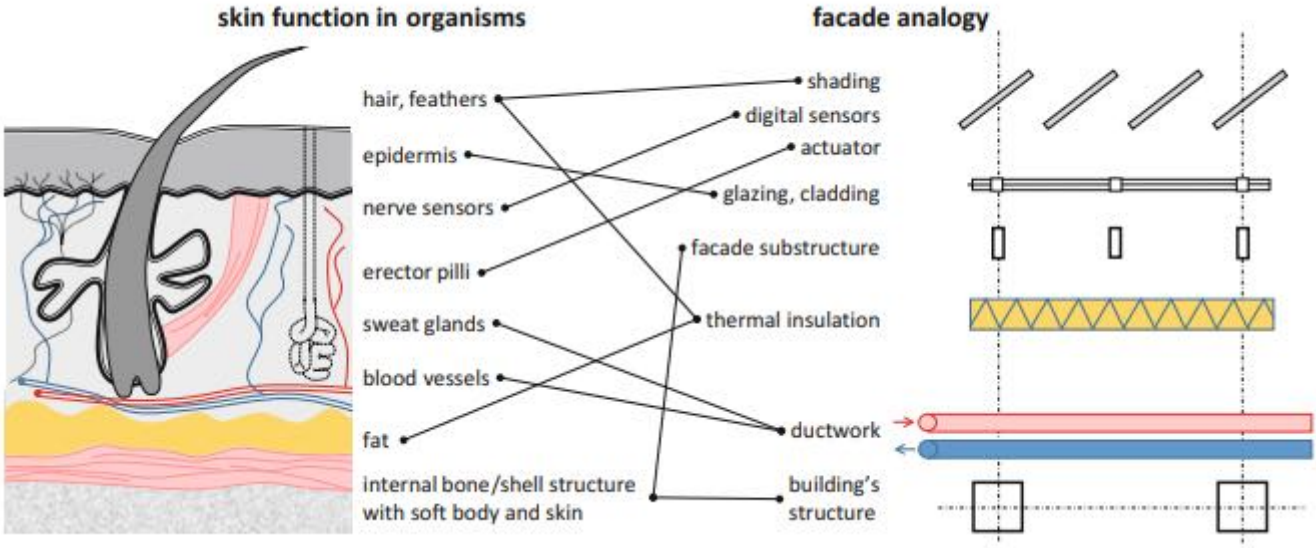


Figure 22: A comparison between the different layers of an animal skin and a building façade (Sandak et al., 2019).

Façade cladding is the outer, non-load-bearing layer of a building skin (**Fig. 2**). Consequently, it is exposed to all types of weather and one of its main purposes is to protect the control and structural layers against these environmental elements. Besides its protective role, façade cladding serves aesthetic purposes, as it determines the building's appearance. Next to those

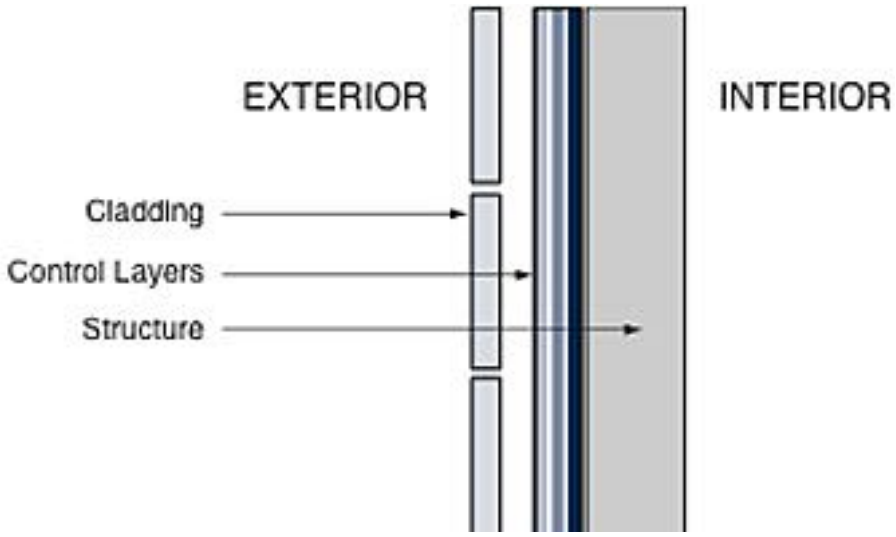


Figure 23: A sketch of the composition of a façade (from Whole Building Design Guide, z.d.).

two main functions, it can improve thermal or sound insulation and fire resistance (Thevega et al., 2022).

Façade cladding comes in all different type of forms, shapes and materials, depending on the façade system and its requirements. Traditionally, walls were constructed with locally resourced materials, such as stone and mud bricks. In more northern areas of Europe they started to use wood which had higher availability and offered better insulating properties (Sandak et al., 2019). In the Netherlands this was also the case and façades were a load-bearing part of a building. However, in the 1930s, concrete frame structures started to become the building standard and façades were no longer part of the load-bearing structure. In addition, other requirements started to arise for the exterior of a building (De Jonge, 2024). Aesthetics became more important and with the development of new building techniques and materials, façade systems evolved to become more advanced.

The material selection of façade cladding is vital as it impacts the aesthetic look of a building as well as its functions (Shrivastava & Sahu, 2024). The most common materials used for façade cladding are stone/brick(slips), metals, aluminium, fibre-reinforced composites (FRPs), ceramics and wood. However, there is increasing interest in biobased materials as sustainability is gaining importance in the building industry. However, most biobased materials do not yet withstand the environmental conditions these are exposed to when used as a façade cladding. Therefore, it is important to assess the different types of conventional and biobased cladding materials that are commonly used for façades. In the following section, these materials will be discussed, and their material properties will be assessed against the requirements established for these materials.

Material requirements

Choosing the right material for façade cladding is a multi-criteria decision problem (Ferreira, Dias, et al., 2021). The façade cladding should not only have an aesthetic appeal and protect the underlying layers (Shrivastava & Sahu, 2024). Designers also consider durability and cost, need for maintenance, thermal and acoustic properties, weight, formability (the ability of a material to be formed into a desired three-dimensional shape), fire safety, and sustainability of the material (Ferreira, Dias, et al., 2021; Thevega et al., 2022). The number of criteria that need to be considered shows that material selection depends on the project context and which functionalities are the most important.

Thermal conductivity

In this project it is therefore important to narrow down the number of material properties for our cooling panel to make an informed decision on the material selection. First, the panel must have a passive cooling effect on the building as well as the environment. However, as assessed in work package 1 (WP1), the cooling properties of the panel will mainly result from the geometric design of the panel enabling convective cooling. Moreover, we aim to apply a reflective and/or radiative coating/paint. Additionally, the thickness of the panel will likely not

exceed 20 mm, which limits the thermal mass contribution regardless of the specific heat capacity. Thermal mass and heat capacity will therefore be excluded from the material requirements for this project (**Table 1**).

However, it is still important that the material has low thermal conductivity, ideally below 0.5 W/(m·K), as materials with high thermal conductivity would counteract the convective cooling mechanism. Thermal insulation materials typically have thermal conductivity values below 0.07 W/(m·K) (Petcu et al., 2023), whereas the thermal conductivity values of wood, concrete and plastics, which are considered poor thermal conductors, range from 0.1-0.57 W/(m·K) (Cabral & Blanchet, 2021; Kijo-Kleczkowska & Gnatowski, 2022). Materials that are considered thermally conductive, such as steels or metals, typically have values ranging from 50 to over 200 W/(m·K). Together, a material with a thermal conductivity < 0.5 W/(m·K) is preferred.

Adhesive properties

The material should have adhesive properties for applying reflective or radiative paints and/or coatings. These requirements are difficult to quantify with a specific threshold value and therefore will be considered as qualitative screening criteria: the material should be compatible with paints and/or coatings or rather, the material should not have properties that inherently prevent compatibility with paints and/or coatings.

Durability

Exterior façade cladding has an average estimated service life of 12 to 60 years, which requires the material of these panels to degrade minimally over the duration of this use phase (Silva & de Brito, 2021). As it is exposed to all the weather elements, several material properties are important for the durability. Solar radiation is one of the factors impacting materials, specifically from UV exposure, resulting in colour degradation, material brittleness or biodegradation (Andrady et al., 2023).

Additionally, moisture, especially in the Dutch climate, impacts the durability of materials (Ferreira, Barrelas, et al., 2021). Water absorption is one of the metrics to assess the moisture resistance of materials. Although there is no defined standard threshold for water absorption at material level, water resistance of façade constructions is regulated at systems level in the Dutch Besluit bouwwerken leefomgeving (Bbl), which is crafted by the Dutch government (article 4.118) (Koninkrijksrelaties, 2026). Testing procedures for this are defined in NEN 2778. As there is no standard threshold for water absorption rates for materials, a material-level threshold can not be constructed in the requirements table. Therefore, this metric will be assessed as a ranking criterion, where lower water absorption rate is preferred. However, compliance with NEN 2778 should ultimately be demonstrated in later developmental stages of the panel.

Freeze-thaw resistance

The Dutch climate also undergoes freeze-thaw cycles during winter, which can impact the mechanical properties of a material. Water can infiltrate the pores of materials and expand when it freezes, putting internal pressures on the material which can result in cracks or deformation. Upon thawing, the water leaves the material that is now deformed and has a

reduced strength (Asare et al., 2024; Bogas et al., 2016; H. Li et al., 2012). Cycles of this process can lead to significant strength loss in materials, depending on their porosity and water adsorption and content (Bogas et al., 2016).

Thermal expansion

Materials also expand during exposure to high temperatures. The degree of expansion of a material is called the thermal expansion coefficient. If a material has a high thermal expansion coefficient, it can affect the appearance and structural integrity, which can lead to safety hazards in high-rise buildings (Yang et al., 2024). Ideally, a façade cladding material has a low thermal expansion to minimize the chance of deformation or degradation, however, there are ways to circumvent deformation as a result of thermal expansion in the form of fixing and jointing techniques or reinforcement (Baker, 1964). Therefore, it will not be included in the list of requirements.

Weight

Façade cladding elements are non load-bearing but should generally still be able to hold their own weight. Lightweight materials are easier to install and transport, resulting in reduced costs and environmental impact. Therefore, a lightweight material for the façade panel in this project is preferred. The weight of façade cladding panels is assessed by their density (kg/m^3). However, in the construction industry, surface density (kg/m^2) is used as comparing metric, as it incorporates the thickness of a panel into the weight. Some materials have a high density but can be thin, thereby reducing the overall weight of a panel. Therefore, surface density will be used as comparative metric regarding the weight.

By comparing different panels that are made from materials that are considered lightweight, such as aluminium (3 mm thick, 8.1 kg/m^2), Rockpanel (stone, 8 mm thick, 9.4 kg/m^2), plastic (Keralit, 17 mm thick, 7.7 kg/m^2) or Thermowood (18 mm thick, 7.2 kg/m^2), a surface density < 10 kg/m^2 can be considered a realistic target for our material.

Fire safety

According to the Bbl, new buildings or renovations should comply with EN 13501-1 that describes the fire safety of facades (Koninkrijksrelaties, 2026). Façades of a single-family home should be fire class B (very limited contribution to fire), whereas high-rise buildings have more complicated regulations. Façades between the height of 2.5 and 13 meter of a building should be fire class D (medium contribution to fire), whereas facades below 2.5m and above 13 meters should fire class B (**Fig. 3**). For smoke production, all materials should comply with smoke class 2 (s2, limited contribution to smoke production), as well as burning droplet production class d0 (does not produce flaming droplets). For our project, ideally a material that is inherently fire resistant is selected. However, materials that do not comply with these

standards can be modified with coatings to improve their fire safety. Therefore, fire safety is considered a ranking criterion in this list of requirements.

Sustainability

Recyclability, biobased content and biodegradability are important properties that can impact the material choice. A biodegradable material would be preferred for our project, however, biodegradability often significantly impacts the durability of a material, therefore recyclability is the preferred baseline. As by ISO 140121, a material is considered recyclable if it can be recovered from a waste stream through existing processing, separating and reprocessing methods, and then converted into new products. Choosing a material that is recyclable therefore improves circularity. Biobased content is another metric of sustainability. Choosing

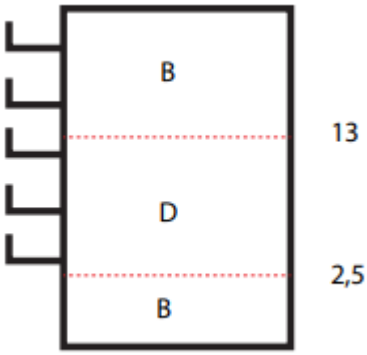


Figure 24: Visualisation of the fire resistance regulations (Bbl) for high-rise buildings.

a material that is at least 100% biobased is preferable, however, it is considered a ranking criterion where higher biobased content is preferred.

Formability

The panel that we are planning to make has a certain 3D-geometry including cm-scale structures incorporated into it. Therefore, it is important that the material can be formed in this shape or that these structures can be easily introduced into the material.

Cost and availability

Some materials are easier to obtain given the costs, availability and contacts that we have. Apart from this, availability of equipment to shape a certain material will influence the choice of a material.



Table 1: Overview of requirements on material properties, describing the type of requirement (threshold or qualitative) and whether it is a hard constraint or ranking criterion.

Criterion	**Threshold or qualitative**	**Hard constraint or ranking criterion**	**Does the material comply**
Cooling function			
Thermal conductivity	Below 0.5 W/(m·K)	Hard constraint	
Paint/coating			
Paint/coating compatibility	Qualitative	Hard constraint	
Weather resistance			
UV resistance	Qualitative	Ranking criterion	
Water absorption rate	Qualitative	Ranking criterion	
Freeze-thaw resistance	Qualitative	Ranking criterion	
Weight			
Surface density	Below 10 kg/m^2	Minimum requirement – Ranking criterion	
Fire resistance			
Fire resistance	Fireclass B, s2, d0	Ranking criterion	
Sustainability			
Recyclability	Qualitative	Hard constraint	
Biodegradability	Qualitative	Ranking criterion	

Biobased content	Qualitative	Ranking criterion	
Formability & availability			
Material availability	Qualitative	Hard constraint	
Material formability	Qualitative	Hard constraint	
Instrumentation availability	Qualitative	Hard constraint	
Prototyping feasibility	Qualitative	Hard constraint	

Comparison of different materials

Natural stone, concrete, ceramics, glass, metals, plastics, fibre reinforced composites (FRCs) and wood are some of the most used façade cladding materials. With the use of Table 1, materials can be assessed and eliminated or ranked, depending on their assessment. If a material does not comply with a property that is a hard constraint, it can be immediately eliminated as it is not suitable for our project. For this reason, metals and in particular aluminium was eliminated as a potential material as it has a thermal conductivity that far exceeds the threshold (237 W/(m·K)). Similarly, ceramics have a slightly higher thermal conductivity than the threshold (1.55 W/(m·K)), as well as a far too high surface density (12mm, 27.8 kg/m^2) and was therefore also eliminated.

Different types of plastics are used as cladding (PE, PP, PVC) as they possess low thermal conductivity, high flexibility, corrosion resistance and are low density and cheap (Riahinezhad et al., 2021). However, there are concerns about the weather durability of plastics. Depending on the type of plastic, different type of environmental conditions can significantly affect the durability of plastic (Riahinezhad et al., 2021). For example, UV radiation can cause photochemical degradation in several organic-based polymers, leading to micro-cracking, colour degradation, embrittlement, increasing water absorption rates and loss of strength (Andrady et al., 2023; Riahinezhad et al., 2021). In addition to the concerns about weather durability, plastics are eliminated based on their recyclability. Most plastics are technically recyclable, but require complex, multistage techniques and therefore often end up being landfilled or degraded into microplastics, which impacts the health of the soil, oceans, plants and animals (Lalrinfela et al., 2024; Naderi Kalali et al., 2023). Lastly, most plastics are considered combustible and therefore should be restricted in their application in the building industry (Sandak et al., 2019).

Glass is a common material used for the exterior layer of buildings, especially in curtain wall systems (Sandak et al., 2019). However, the thermal conductivity of different types of glass exceeds the threshold for thermal conductivity for this project. The three types of glass often used for curtain walls, float glass, heat-strengthened glass and tempered glass, all have thermal conductivity values > 0.5 W/(m·K) (~1 W/(m·K)) (Geotti-Bianchini & Lohrengel, 1993; Malinarič, 2025). Additionally, although it is often considered lightweight, glass actually has a high density (2200-3000 kg/m^3) (İlerisoy & Çolak, 2018). A glass panel with a thickness of 10 mm would then have a surface density of 22-30 kg/m^2, which would exceed the threshold of 10 kg/m^2. As a final constraint, glass is difficult to form in the 3D-geometrical shape that is needed for our project. Additionally, we lack the instrumentation to create complicated shapes with glass and because of all these reasons, glass is also eliminated as a potential material for our project.

A wide range of materials can be categorized under fibre reinforced polymer composites (FRPCs), of which 25% is produced for the construction industry (Effing, 2018; Q. Nguyen et al., 2013). The combination of matrix of a polymer and a fibre as reinforcement offers increased mechanical, thermal and insulating properties (Bedon, 2016). Fibers can be natural,

organic, metal or inorganic. As for polymers, in the building industry, three main groups can be identified, thermoplastics, elastomers and thermosets (Bedon, 2016). However, there is increasing interest towards bio-composites, which consists of natural fibres and a biological polymer (Manu et al., 2022). 99% of the composites produced are glass-fibre reinforced polymers (GFRPs) (Effing, 2018). FRPCs contain, next to the previously named properties, other properties which makes them appealing for the construction industry. They are mouldable and have excellent corrosion resistance (Qureshi, 2022). However, similarly to glass, they are often considered lightweight, while in reality GFRPs have a density ranging from 2270 to 2500 kg/m^3 and carbon fibre 1700-1800 kg/m^3 (Shakir Abbood et al., 2021). This would make their surface density 1.7-2.5 kg/m^2 per mm thickness. With a standard 6-8 mm thick façade panel, this implies that the weight is exceeding the threshold. Additionally, FRPCs are difficult to recycle, especially ones with thermoset and rubber-based matrices that are widely used, which makes it an unsuitable material for our project (Ghosh, 2023; Radu et al., 2026) .

FRPCs can also be composed of biopolymers, natural fibres and bio-fillers. When made from natural materials, they can be categorized as: fully green bio-composites and partially green composites (Shelly et al., 2025). Fully green composites use a biobased matrix in addition to natural fibres and bio-fillers. Partially green bio composites are composed of natural fibres and a petrochemical matrix. Partially green bio composites can face difficulties with recycling, depending on the type of petrochemical matrix that is used (thermoset or thermoplastic), but are still considered recyclable four to six times until a degradation of thermomechanical properties is observed (Zhao et al., 2022). Fully green bio composites however can be composted or biodegraded, offering inherent benefits sustainability-wise (Faheem & Khan, 2025; Fartash Naeimi et al., 2025).

Students at Eindhoven University of Technology developed a façade panel with a similar 3D geometry as the design of our project, using a bio-composite from the Dutch company NPSP. Nabasco® 8010, the bio composite that was used, is one of the leading biobased materials in the Netherlands that is used for façade cladding and is therefore an interesting material option for our project. It consists of flax, reed, cotton, sisal or jute, lime and a biobased polyester resin (Pujadas-Gispert, Alsailani, van Dijk (Koen), et al., 2020a; Vancso et al., 2024; Verspeek & Burgh, 2019). The product has been used in numerous projects in collaboration with many companies, such as Staatsbosbeheer, Rijkswaterstaat and Province Noord-Brabant, and has built a reputation for being a reliable bio composite for façade cladding (Ortega Exposito et al., 2023). It is reported to have similar mechanical properties and density to glass and certified with fire safety class B-s1, d0 (NPSP, 2025a; Pujadas-Gispert, Alsailani, van Dijk (Koen), et al., 2020a). Additionally, it is reported by the manufacturer to have an estimated lifespan of 50-100 years (Romano, 2023; Vancso et al., 2024), can be mechanically recycled by crushing it and reusing the residual as raw material for new panels (NPSP, 2025b) and can be formed into advanced three-dimensional shapes by using moulding techniques and hot pressing (Romano, 2023).

However, besides the observed benefits of Nabasco®, some material properties do not fulfil the requirements of our project. First, the surface density of Nabasco® 8012, a following iteration on Nabasco® 8010, has a surface density of 10.7 kg/m^2, which exceeds the threshold of 10 kg/m^2 (Vancso et al., 2024). Additionally, Nabasco® 8010/8012 is produced by a private company, which makes the material difficult to obtain within our project, and requires a steel or aluminium mould to form complex 3D-shapes, which is costly. Lastly, the paper from Pujadas-Gispert et al. (2020) was conducted in Dubai, a hot and dry climate, whereas the panel for our project is meant to be developed for use in the Netherlands. In that paper, the dimensions of the panels had decreased because of the hygrothermal conditions of Dubai. The Netherlands has a more demanding climate regarding moisture, which increases insecurity about the dimensional stability of Nabasco®. Additionally, some parts of the panel, such as the small flaps at the tile edges were prone for breaking, which impacted the functionality of the tiles. All these considerations decided us to not pursue Nabasco® as a material for our project.

Historically, wood is a commonly used material for façade cladding in Europe. Buildings from as far back as the Stone Age have been discovered which used wood for external walls (Hill et al., 2022). When brick became a cheaper material, popularity of wood for building façades declined. Additionally, for multi-story buildings, there were concerns regarding the durability and fire safety of wood. However, over the last 20 years, wood started a revival. Sustainability is gaining importance in the building industry and wood modification techniques are becoming more advanced (Hill et al., 2022). Wood complies with most of the requirements of our project. The thermal conductivity of wood depends on the type, as density, structure and moisture content affect the value (Pelaz et al., 2017; Sonderegger & Niemz, 2009; Suleiman et al., 1999). However, the thermal conductivity of soft- and hardwood is below the threshold of 0.5 W/(m·K) with a safe margin (Cabral & Blanchet, 2021). Additionally, the density of most wood species and types is below 1000 kg/m^3, which would make the surface density of a 10 mm thick panel below the threshold of 10 kg/m^2 (Gérard et al., 2011).

Wood is a material that is readily available (depending on the species), within budget and complex 3D shapes can be developed with instrumentation that is available for us, which makes it an appealing material. Additionally, wood is a sustainable material as it is inherently biodegradable, 100% biobased and recyclable. The two properties in which timber does not excel are durability and fire resistance. However, the durability depends on the type of wood that is used. Different natural durability classes of wood are provided through the EN 350 standard, whereas the natural durability is defined as the natural ability of the wood to protect itself against biological wood-destroying agents (fungi, beetles, termites and marine organisms (EN, 2016; Hill et al., 2022). Class 5 (non-durable) is the lowest, whereas 1 (very durable) is the highest (EN, 2016). Wood types considered class 3 or lower are suitable for outdoor purposes such as cladding.

Lignin within wood is susceptible to solar radiation, specifically UV light. This can result in colour change, depolymerization of lignin and cellulosic polymers, and eventually erosion within the wood (Jirouš-Rajković & Miklečić, 2021). Similarly, moisture is a common

degradation agent of wood. However, both these concerns can be resolved by applying moisture and UV resistant coatings to the wood, which are readily available, or by thermally or chemically modifying the wood which alters its properties.

Overall, wood meets most of the material requirements, which is why the next section assesses which type of wood would be most suitable for our project. Different treatments and coatings are also discussed, which can be used so wood complies with all the material requirements for our project.

Wood as a cladding material

In Europe, the tree species considered most suitable for timber cladding are larch (Siberian or European) (*Larix sibirica or Larix decidua*), fir (*Abies* spp.), Douglas fir (*Pseudotsuga menziesii*), spruce (*Picea* spp.), pine (*Pinus* spp.), European oak (*Quercus robur*) and western red cedar (*Thuja plicata*) (Hill et al., 2022; Ivanović-Šekularac et al., 2016; Svensson Meulmann & Sjökvist, 2023). These species can be categorized in several ways, such as natural durability classes, hard- or softwood and European native or non-native species. It is notable that some of the species (spruce, fir and pine) listed by Hill et al. (2022) are not necessarily durability class 3, as previously described needed for cladding applications. However, by treating them they are still often used as façade cladding materials. Some papers even investigated the possibility of using some of these species untreated for façade applications (Svensson Meulmann & Sjökvist, 2023).

Other notable species are European oak, as this is the only hardwood in the list, and western red cedar. Hardwoods are generally stronger and more durable than softwoods, but more difficult to process, heavier, pricier and more susceptible to shape deformation and crack formation (Hill et al., 2022; Štěrbová et al., 2021). However, because of its strength, durability, wide availability and aesthetic look, European oak is still used for façade cladding purposes despite its downsides. Western red cedar is a non-European wood type that is native to North America and therefore needs to be imported to Europe. It is however a very suitable wood type, as it is lightweight, considered durability class 2 and keeps good dimensional stability.

Plywood consists of sheets of plies (thin layers) of wood veneer, stacked and glued together. The wood veneers are stacked in such a way that the grain of each layer is perpendicular to its adjacent layers (cross laminated). This method is applied to mitigate the disadvantages of wood, such as poor dimensional stability or poor processing, and protect its advantages. Wood types often used for plywood are larch, pine, oak, fir, birch and poplar (Tangellamudi et al., 2025).

To assess which wood type is most suitable for our project, a systematic comparison is going to be made between each type by looking at the natural durability class per EN 350, density, price and the supplier (**Table 2**). Additionally, different types of modifications and treatments can be done to improve the properties of wood types, such as thermal modification, impregnation, acetylation, coating or surface modification. Therefore, wood types that might have worse properties than others, might be improved significantly with modification

techniques and should not be disregarded as easily. In the end, a final recommendation will be made for the most suitable wood type and modification for our project based on our assessment. Siberian larch is excluded from the comparison as it is unavailable since the invasion of Ukraine by Russia.

Table 2: Comparison of wood species properties used for species selection.

Durability class (EN 350)	
Density at 12% MC (kg/m^3)	
Price (low, medium or high)	
Supplier	

Comparison of different types of wood

The data on the durability class and densities of the different types of wood is sourced from houtinfo.nl by Centrum Hout, the online version of *Houtvademecum*, a wood catalogue of over 250 species used by architects and constructors for information about wood types (Centrum Hout, z.d.-a). The values from this database have been cross-referenced with the CIRAD Tropix database, a French agricultural research institute (Gérard et al., 2011). This provides ranges for the durability classes and densities, which is usual as wood as a natural material varies in terms of its properties. The price for the different types of wood is qualitatively assessed based on current supplier prices, rather than specific prices, as they constantly fluctuate depending on supplier, grade, dimensions and other factors. The supply of the wood will be categorized on three levels: national, regional and continental. National means that the wood type is grown and sold in the Netherlands, regional means that it is grown and sold in Europe, whereas continental means it needs to be imported from different continents. For most species, it will be a mix, as 94.8% of all the wood in the Netherlands used for material application is imported (Teeuwen et al., 2025).

Spruce

Table 3: Material properties of spruce (*Picea abies*).

Durability class (EN 350)	4
Density at 12% MC (kg/m^3)	450-460
Price	Low
Supplier	National / Regional

Spruce is a softwood which originates from a vast area in Europe, from Scandinavia to the Balkan and the alpine regions of Central Europe (Seim et al., 2022). It is low weight with

relatively good strength and although it is considered durability class 4 if not treated, it can be argued that due to its pit aspiration it can perform well as cladding material (**Table 3**)(Hill et al., 2022). Generally, spruce wood is widely available in hardware stores in the Netherlands and is considered an inexpensive type of wood. Most of the spruce wood is imported from Northern- and Central Europe, however, 4% of Dutch forests consist of Spruce trees (Assink et al., 2022). Spruce is considered a refractory wood species; a wood type difficult to penetrate with preservatives (Gérard et al., 2011). On one hand this means that it is difficult to impregnate, but also that it naturally limits moisture uptake, in the form of liquid water, when uncoated (Hill et al., 2022). It is still, however, susceptible to vapor sorption. Spruce has a moderate volumetric shrinkage coefficient (0.37% per %), which is not a problem with proper detailing. However, spruce has a 2.1 St/Sr ratio, which means it shrinks 2.1 times as much tangentially as radially, which can lead to warping (Gérard et al., 2011).

European larch

Table 4: Material properties of European larch (*Larix decidua*).

Durability class (EN 350)	3-4
Density at 12% MC (kg/m^3)	600-650
Price	Medium
Supplier	National / Regional

European larch is a tree species commonly found in the Alp mountains and Central Europe (Gérard et al., 2011). It is one of the heaviest and more durable European softwoods and commonly used for cladding (**Table 4**). European larch is generally slightly more expensive than spruce, pine or fir but cheaper than other hardwoods and mostly imported from other parts in Europe, although 5% of the trees in Dutch forests are larch trees (Assink et al., 2022). European larch is considered non-permeable, which makes it more difficult to impregnate but again also naturally more durable (Gérard et al., 2011). Additionally, wood from European larch is prone for warping due to its high St/Sr ratio (2.0) and relatively high coefficient of volumetric shrinkage (0.48% per %) (Gérard et al., 2011). One important remark is that with a 15 mm thick panel made from European Larch, the surface density of a panel would come close to 10 kg/m^2, which is undesirable. Therefore, the eventual chosen thickness is important for the final material decision.

Scots pine

Table 5: Material properties of Scots pine (*Pinus sylvestris*).

Durability class (EN 350)	3-4

Density at 12% MC (kg/m^3)	550
Price	Low
Supplier	National / Regional

Scots pine is a tree species native to large parts of North, North-East and Central Europe and introduced in large parts of Western Europe and Great Britain (Gérard et al., 2011). It is a relatively heavy softwood compared to spruce but less heavy than European larch or other hardwoods (**Table 5**). Scots pine is the most common tree species in Dutch forests (32% of all trees) (Assink et al., 2022), which partially contributes to the low price it has in retail. Scots pine has the same permeability as spruce, poorly to non-permeable (Gérard et al., 2011). Similar to European larch, Scots pine has a relatively high coefficient of volumetric shrinkage (0.45% per %), but a lower St/Sr ratio (1.6) (Gérard et al., 2011). However, despite this lower St/Sr ratio when compared to spruce, it has been found that Scots pine is more prone to cupping than spruce, presumably because of the low pit aspiration degree compared to spruce (Virta et al., 2005).

Douglas fir

Table 6: Material properties of Douglas fir (*Pseudotsuga menziesii*).

Durability class (EN 350)	3-4
Density at 12% MC (kg/m^3)	480-540
Price	Low - Medium
Supplier	National / Regional / Continental

Douglas fir is a tree species native to the west of North America, there referred to as Oregon pine, but introduced in large parts of Europe (Centrum Hout, z.d.-b). Douglas fir has a similar density to Scots Pine and is reported to have better compression strength than spruce and pine, but less than larch (**Table 6**)(Zeidler et al., 2022). It is considered moderate to poorly durable (class 3-4), similar to Scots pine and European larch. However, reports have suggested that the natural durability of Douglas fir in outdoor applications might be better than expected and that it can be used for outdoor applications without any preservative treatment. This might be due to the refractory nature of both the sapwood and heartwood of Douglas fir to water uptake (Kutnik et al., 2011).

In terms of impregnability it is non-permeable, similar to European larch (Gérard et al., 2011). Douglas fir is considered moderately prone to warping, with a low St/Sr ratio (1.6) but moderately high coefficient of volumetric shrinkage (0.46% per %) (Gérard et al., 2011). 6% of the trees in Dutch forest are Douglas firs, which means that most of the Douglas fir wood is imported from other European countries

Douglas fir from North American has different properties than Douglas fir from Europe. North American Douglas fir grow slower, which makes them dimensionally more stable and less resinous than European Douglas firs (Humar et al., 2022). The price of Douglas fir is considered to be low to medium, as in Western Europe it is approximately 25% more expensive than spruce, which is considered one of the most inexpensive types of wood (Pulkrab et al., 2014).

Fir

Table 7: Material properties of European silver fir (*Abies alba*)**.**

Durability class (EN 350)	4
Density at 12% MC (kg/m^3)	500-540
Price	Low
Supplier	National / Regional

The European silver fir (*Abies alba*), commonly referred to as fir, is a tree species that naturally grows in forests in the Alps, Pyrenees and Balkan and has been introduced in forests in Western- and Northern Europe (Mauri et al., 2016). The natural durability of fir wood is similar to spruce, class 4, but with a higher density, similar to Douglas fir and Scots pine (**Table 7**). Less than 4% of the trees in Dutch forests are firs, which makes them the least abundant coniferous tree species of the ones assessed in this report (Assink et al., 2022). It is considered an inexpensive type of wood with reasonable strength. A downside of fir is that, based on market research conducted in this project, it is not as widely available across the Dutch hardware stores as other types of softwood like spruce and Scots pine. Additionally it is often used for construction application, not for higher-end applications (Centrum hout, z.d.). Fir wood is moderately prone to warping with a coefficient of volumetric shrinkage of 0.44% per % and a St/Sr ratio of 2.2. It is however easier to impregnate fir wood, with an impregnability rating of 2-3, which translates to poorly to moderately permeable (Gérard et al., 2011).

European oak

Table 8: Material properties of European oak (*Quercus robur*).

Durability class (EN 350)	2
Density at 12% MC (kg/m^3)	670-760
Price	Medium
Supplier	National / Regional

European oak is the only hardwood listed in this comparison, which is also determined due to the density, which is the highest among all tree species (**Table 8**). This already raises weight concerns as the surface density would exceed the threshold of 10 kg/m^2 with a thickness over 15 mm. However, due to its density it is also one of the most durable tree type in this list, with natural durability class 2 (Gérard et al., 2011). Hardwoods are typically stronger than softwoods, which highlights the structural integrity of European oak. It is the most abundant deciduous tree species in the Netherlands (18%) but also gets imported from other parts of Europe (Assink et al., 2022; Oldenburger, 2009). European oak is considered difficult to impregnate (not permeable) (Gérard et al., 2011). Good quality oak wood is often more expensive than the other tree types in this comparison as it is often used for veneer or furniture production. Besides that, oak is particularly prone to greying as a result of UV exposure (Oldenburger, 2009).

Western red cedar

Table 9: Material properties of Western red cedar (*Thuja plicata*).

Durability class (EN 350)	2
Density at 12% MC (kg/m^3)	330-390
Price	High
Supplier	Regional / Continental

Western red cedar is a tree species that originates from North America and is still mostly imported from there, but it has also been introduced to Great Britain, Germany, Italy, Denmark, Norway and Finland (Aldana et al., 2023). As a softwood it is both the most naturally durable (class 2) and lightweight species in this comparison, which also translates to its price (**Table 9**)(Gérard et al., 2011; Oldenburger, 2009). Despite its relatively high price, it is still one of the most popular choices of wood for façade cladding in the Netherlands, as it has moderate to good dimensional stability, does not need any preservative treatment for outdoor application and is aesthetically wanted (Gérard et al., 2011; Oldenburger, 2009). However, the main downside of Western red cedar is that it needs to be imported from North America, as the Western red cedars grown in Great Britain have a lower natural durability (Centrum Hout, z.d.-c). This raises concerns about sustainability regarding the environmental output needed for import as well as the price.

Plywood

Plywood has similar or even increased mechanical properties compared to solid wood. However, due to the method of manufacturing, the variability of mechanical properties is reduced (Castanié et al., 2024). Additionally, it can have a lower density than solid wood,

depending on veneer species, and have higher resistance to cleaving and warping (Hrázský & Král, 2014).

The main problem of plywood is its durability, in particular their resistance to moisture. Alternating wetting and drying of plywood could cause reduction of mechanical properties of plywood (W. Li et al., 2016). Especially the sawn edges of plywood are prone for moisture and need primer finish when used outdoors. Additionally, most plywood types use urea-formaldehyde resin glue, which is not water-proof and therefore unsuitable for outdoor use (Liang et al., 2023). However, there are moisture-resistant plywood types on the Dutch market.

Table 10: Material properties of Okoume plywood at 12 mm thickness.

Use class (EN 636)	3 (Exterior)
Bonding quality (EN 314)	3 (Weather and Boil Proof (WBP))
Density at 12% MC (kg/m^3)	500
Price	Medium - High
Supplier	Continental

Plywood made from Okoume, a tree species native to West-Africa, is considered moisture resistant which makes it usable for outdoor applications. It is often used for boatbuilding, as it is lightweight, durable and marine resistant (Negro et al., 2011). It is, however, more expensive than plywood made from other tree species, depending on the thickness (**Table 10**). Plywood is excluded from EN 350 natural durability classification, as wood veneers are made from sapwood which is excluded from the norm. Therefore, in this table, EN 636 (use class) and adhesive bonding quality (EN 314) are included.

Okoume 12 mm plywood both has use class 3 and bonding quality class 3, which means it can be used outdoors and the glue is WBP (*Okoumé multiplex FSC 12 mm 250 x 122 cm*, z.d.). It has similar density to other softwood types. The price of Okoume plywood however is higher than other types of plywood and needs to be imported from West-Africa. Similar to western red cedar, this impacts the sustainability of this type of plywood. Additionally, the phenol-formaldehyde adhesive used for this type of plywood is petrochemical-based, which reduces it the biobased content of plywood and increases its environmental output (Kariuki et al., 2019).

Modification methods of wood

Commercially sold wood is often modified to improve its mechanical properties and dimensional stability (Hill et al., 2022). Thermal modification, impregnation and acetylation are commonly used techniques to improve wood properties. In this section, several modification methods will be discussed and assessed for our project. The chemical processes used for modification are reported in the literature but are not relevant for the scope of this report. Therefore, the focus will mainly be on the alteration of material properties as a result of the modifications.

Thermal modification

Thermal modification is a process in which wood is heated to a temperature between 160 °C and 240 °C in absence of oxygen, which leads to the degradation of the least heat-resistant components of cell wall polymers (mainly hemicelluloses), which are also the most prone to moisture induced deformation. This process can improve the natural durability, dimensional stability and reduce weight of wood (Jančíková & Jablonský, 2025). It is also reported that thermal modification can improve the fire resistance (Čekovská et al., 2016). In terms of produced volume of modified wood, thermally modified wood (TMW) has the biggest market share in Europe, also for cladding purposes (D. Jones et al., 2020a). The biggest downside of thermal modification is that it reduces the mechanical strength of wood, however, façade cladding is often not load-bearing, which reduces the severity of this problem (Hill et al., 2022; Jančíková & Jablonský, 2025).

Compared to other modified wood types, TMW is the most inexpensive and widely available type (Brelid, 2016). The wood types most commonly used for thermal modification are spruce and Scots pine (van Blokland et al., 2020). Their durability classes can be pushed up to 2 when thermally modified (Jebrane et al., 2018; van Blokland, 2020). Additionally, both species lose 7-9% of their mass due to thermal modification, which is directly linked to the improved durability and reduced mechanical properties (Jebrane et al., 2018). Furthermore, the equilibrium moisture content (EMC) is lower in TMW, which increases the dimensional stability. The EMC is the point at which moisture content of the wood is in balance with its environment (air humidity). When the EMC is reduced, it means wood can absorb less moisture at any given humidity, which decreases its degree of swelling and therefore improves its dimensional stability (Esteves & Pereira, 2008). The loss of EMC in TMW can go up to 50-60%, leading to increased dimensional stability (van Blokland et al., 2020).

Two important remarks need to be made regarding TMW for this project, which is about coating adhesivity and price. In our project, reflective or radiative coatings will be applied to the panel. The adhesion quality of coatings can be affected with TMW (Chu et al., 2023). For spruce and Scots pine specifically, the coating penetration was reported to be the same compared to untreated wood (Altgen & Militz, 2017). The adhesion strength however is coating-system dependent. Certain waterborne systems showed reduced adhesion while others performed sufficiently. Therefore, if TMW ends up being the final material for our project, compatibility with a radiative or reflective coating needs to be verified before application. Furthermore, TMW is more expensive than untreated wood, which also is an important factor in our project. It is however still less expensive than Western red cedar.

Acetylation

Wood treated with acetic anhydride has changed material properties. Accoya™ is the only commercially available acetylated wood product, which involves the reaction of acetic anhydride with the polymers in the wood cell wall. This reaction forms ester-bonded acetyl groups, which occupy space in the cell wall. This so-called bulking effect expands the cell

wall, which reduces the available space for water molecules to penetrate. As a result, less water can be taken up by the wood, making it more moisture resistant and dimensionally stable (Dunningham et al., 1992). Other benefits of acetylated wood are improved UV and erosion resistance, as well as natural durability class 1 (Sandak et al., 2021; Slabohm et al., 2023). Additionally, it has been reported that the acetylation of wood increases the performance of coatings as the interface between the wood and the coating is more stable due to the reduced swelling of acetylated wood (Rowell & Bongers, 2015).

A downside of acetylation is that it makes the wood less fire resistant, which needs to be accounted for when using it for cladding purposes (Rabe et al., 2020). Another downside is that the aesthetics of acetylated wood deteriorate in a similar way as untreated wood, which means it loses its appearance within the first few years (Sandak et al., 2021). But the biggest downside of acetylated wood is its price tag, as it is considered a premium type wood, sometimes more expensive than tropical hardwoods or high quality oak (Zelinka et al., 2022).

Impregnation

Impregnation is a method to modify the properties of wood by applying a monomer that polymerizes with wood cell wall polymers or fills cell cavities without any chemical reaction (Hill et al., 2022). When an impregnating agent reacts with the cell wall polymers and changes the wood properties, it is called active modification, whereas passive modification is where no chemical reaction is present (Augustina et al., 2023). Wood impregnation can increase the natural durability of wood and its dimensional stability.

Impregnating compounds can be divided into three categories: nonbonded-leachable (oils and waxes), nonbonded-nonleachable (thermosetting resins) and bonded-nonleachable (anhydrides and organic acids). One important remark needs to be made about thermosetting resins, which is that they are petrochemical based, which increases the environmental impact of wood when applied (J. Liu et al., 2021). Nonbonded-leachable impregnating agents however, like oils and waxes, are biobased and would therefore be an option to use for impregnation. Other benefits include that it is widely available, relatively inexpensive and non-toxic (Chau et al., 2015; J. Chen et al., 2020). The limitation of oils however, is that they can only improve water repellency, not dimensional stability, especially not in the long-term, which is an important reason for wood modification (Augustina et al., 2023).

Additionally, all impregnating compounds do significantly increase the weight of the wood. This is an important remark, as the weight gain can go up to 60%, which impacts the suitability of these woods for our project (Augustina et al., 2023; Ulvcrona et al., 2006). Furthermore, some species are less permeable than others, which limits their suitability for impregnation. Like noted before in the wood type comparison, species like spruce and Scots pine are less permeable than species like fir, which makes them less easy to impregnate than fir (Augustina et al., 2023; Hill et al., 2022). Therefore, if impregnation is chosen as a method to modify the wood, the type of wood needs to be carefully chosen.

Mycelium-Based Composites

Most conventional materials used in the construction sector are nonrecyclable, energy-consuming, costly and therefore greatly environmentally unfriendly (Alemu et al., 2022). In 2010 in developing countries, the production of construction materials like concrete, accounted for about 32% of the total energy consumption, as well as 19% of the greenhouse gas emissions (Intergovernmental Panel Climate Change et al., 2014). If the use of cement and other high-CO_2 emitting construction materials isn't reduced, it is expected that this will be doubled or even tripled by 2050. Therefore, the search for sustainable and durable biomaterials is seen as the future, with increasing research being done in the field (Alemu et al., 2022). Bio-based materials are seen as the ideal combination of low embodied energy and carbon, lower production costs, recyclability, often available as waste or able to grow on waste and byproducts and can be locally available, reducing the need for long distance transportation. One of the most promising upcoming biomaterial options is mycelium. Mycelium has already been utilized for some time now to produce bioactive molecules in the medicinal industry, as well as to produce dietary supplements. Evidence from Ötzi the iceman even demonstrates the early human use of fungi, as he carried *Fomes fomentarius* most probably to start fires and as a medicine (Peintner et al., 1998) In recent years this research has gone a lot further, with mycelium being used as a biomaterial in self-healing concrete, biocement, synthetic leathers, wall and ceiling panels, packaging items, and even furniture (Appels & Wösten, 2021; Dougoud, 2018; Joshi et al., 2020; Menon et al., 2019).

Mycelium as a Biomaterial

In nature, fungi gather the organic compounds needed as energy and carbon sources in several ways. Fungi can establish mutualistic interactions with plants, bacteria, and animals, where they exchange essential nutrients and compounds with the other mutualist, in exchange for necessary compounds like sugars (Appels & Wösten, 2021). In addition, fungi can acquire food by degrading dead organic matter. Their hyphae secrete enzymes that break down the substrate into molecules that can be taken up as nutrients. For instance, enzymes like laccase, lignin peroxidase, and manganese peroxidase degrade lignin, cellulose and hemicellulose in wood-like substrates. However, the breakdown process is relatively slow explaining why a large part of the particles in the wood substrate are bound together by the hyphae (Alemu et al., 2022). This binding activity explains why fungi can make mycelium-based composites (MBCs). These MBCs can be produced by several ways. The most conventional method is to stop the growth of the mycelium before full degradation of the substrate, thereby also preserving the structural integrity of the substrate (M. Jones, Bhat, Huynh, et al., 2018). Mycelium inactivation can be achieved by drying the mycelium, which places the fungus in a kind of dormant 'hibernation' state. This state can therefore be reactivated when the material is rehydrated, which would give MBCs regenerative properties (H. Wang et al., 2024). Mycelium inactivation can also be done by thermal treatment. This method kills the fungus and can therefore not be reversed (M. Jones, Bhat, Huynh, et al., 2018). This method addresses the public misconception that MBCs are alive (Bonenberg et al., 2023).

MBCs can also be produced by mixing pure mycelium[9] with other fibres to reinforce the material, which is a more controlled but less conventional process (Appels & Wösten, 2021). MBCs have also been explored using more than just bio-based materials, as research has also been done on mixing pure mycelium with materials like metals or ceramics. This could potentially shift and broaden the applications for these composite materials, also for the usage in façade panels, or other applications in the construction industry.

The final properties of a MBC are highly dependent on multiple varying factors, including the fungal species used, the substrate composition, the growth conditions and post-processing of the material (Rovers & Wösten, 2026). An analysis by Rovers & Wösten (2026) showed that the growth conditions of MBCs are the most influential variable in the compression strength of the material. As differences in substrate led to a maximum absolute increase of 4.03 MPa, second was the composition (3.74 MPa), followed by the fungal species (1.63 MPa), and post-processing (0.82 MPa) (**Fig 4a**). For tensile strength, the difference is even larger when changing conditions (**Fig 4b**). Here, Rovers & Wösten (2026) showed that difference in species had the most impact, and strains led to a maximum absolute increase of 42.5 MPa. This was followed by substrate (26.4 MPa), growth conditions (24.8 MPa), and post-processing (17.2 MPa). In the next section, the influence of each of these factors on MBC

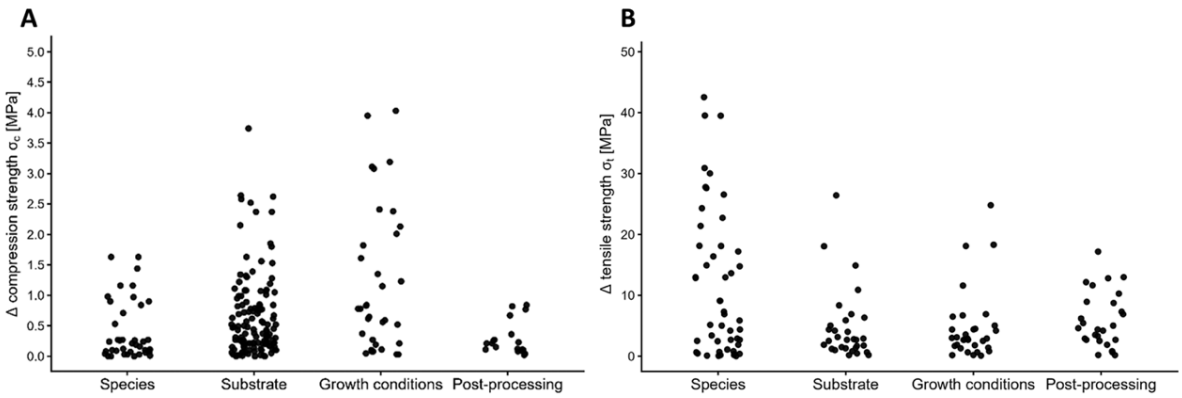


Figure 25: ***Effects of change in species, substrate, growth conditions and post-processing*** *on the* ***A.*** *Compression strength of MBCs and the* ***B.*** *Tensile strength of MBCs. All data points represent findings from a particular study. Analysis by Rovers & Wösten (2026).*

properties will be evaluated.

Species Selection

Many criteria must be considered for selection of a fungal species to produce MBCs for façade panels. Properties like mycelium density and growth-rate, cost of growth, toxicity or harmfulness to humans or other organisms, ease of cultivation, and most importantly the mycelium structure, must be considered (Alemu et al., 2022). Although research on many different strains has been done for mycelium materials in general, 96% of the species that have been used so far are of the Basidiomycota species (Rovers & Wösten, 2026). This

[9] Pure mycelium is the fungal network (the hyphae) of a fungus, grown without added substrate (Appels & Wösten, 2021).

emphasis is justified, as these fungi can grow large mycelial networks and can excrete lignolytic enzymes, and are therefore able to use lignocellulosic agricultural waste streams as substrate.

Appels et al. (2019) demonstrates how the choice of a fungi strain can influence the formation of MBCs. In a comparison between *Pleurotus ostreatus* and *Trametes multicolor,* it was shown that the largest differences were observed in characteristics like fungal skin thickness, substrate colonization behaviour, and water resistance. *T. multicolor* consistently formed a thicker, softer, more velvety fungal skin[10], which led to better water resistance of the MBC, together with a better surface integrity of the layer. This thicker skin also led to improved mechanical properties, as this layer of the composite is more resistant to bending than the part around the substrate. *P. ostreatus* demonstrated more brittle, but dense material properties. However, (type of) pressing impacts the tensile strength and elasticity of the materials significantly more than the type of fungus.

Another widely used mycelium strain is *Fomes fomentarius*, which is commonly known as the tinder fungus and has only recently been introduced into research on composite materials (Pohl et al., 2022). The fruiting body of the fungus can grow a dense outer mycelium layer and a hierarchical structure that combines low weight with great mechanical performance (Pylkkänen et al., 2023). The outer layer of the mycelium consists of tightly packed hyphae that provide protection of the inside, as well as hydrophobicity and mechanical strength. These characteristics make it a suitable choice for MBC usage. *Fomes inzengae* was recently reclassified as a new species distinct from *Fomes fomentarius*. *F. inzengae* will be used in the project, because of availability and good experiences in the fungal lab of Han Wösten over the last couple of years.

Substrate

The properties of MBCs are influenced by the choice of the organic substrate, as this is what the composite consists of largely (Rovers & Wösten, 2026). The analysis by Rovers & Wösten (2026) showed that the composition of the substrate of an MBC is the second most influential variable in the compression strength, as well as the tensile strength of the material. As differences in substrate led to a maximum absolute increate of 3.74 MPa and 26.4 MPa respectively (**Fig. 4**). Appels et al. (2019) demonstrated this by growing *Pleurotus ostreatus* and *Trametes versicolor* on beech sawdust, rapeseed straw and cotton waste. They found that substrate type had significant influence on the material properties. *T. multicolor* grown on rapeseed straw showed the highest tensile strength, while *T. multicolor* grown on beech sawdust showed the highest Young's module[11]. Elsacker et al. (2019) compared *T. versicolor* MBCs resulting from the lignocellulosic substrates hemp, flax, flax waste, straw and softwood, while using different types of fibre processing (loose, chopped, dust, pre-compressed, and

[10] 'Fungal skin' is a term that is often used in literature to describe the thin surface layer of mycelium growth (Appels & Wösten, 2021). Usually this is the soft, white outer layer of MBCs, when they've been incubated again after being taken outside of the mold.
[11] The Young's module is the stiffness of a material, and how much it resists elastic deformation when force is applied.

tow). While differences in mechanical properties were observed between the different fibre types, the conducted tests revealed that the mechanical performance of MBCs depends more on the processing of said fibres. Overall, dust was found to be a poor substrate for mycelium growth, most likely due to the availability of nutrients and absence of air voids for the mycelium to grow into. Wood and straw-based substrates resulted in varying growth of the fungus, making them less reliable choices for the project. On the contrary, hemp, flax, and flax waste as substrates resulted in reliable results and well-developed composites.

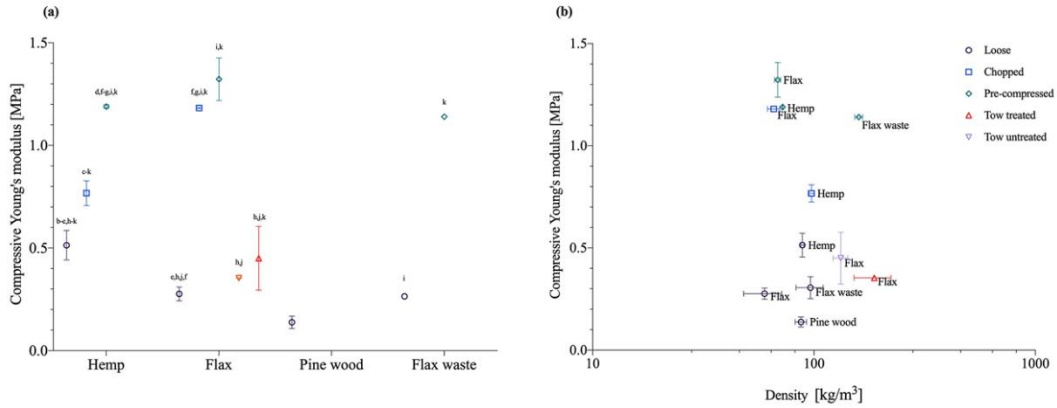


Figure 5: ***The compressive Young's Modulus of MBCs. A.*** *The modulus of elasticity of the MBCs with different types of substrates.* ***B.*** *Young's modulus vs density of MBCs with different types of substrates (Elsacker et al., 2019).*

In loose substrate condition, hemp achieved the highest compressive stiffness of 0.51 MPa, followed by flax waste that scored 0.31 MPa (**Fig 5a**) (Elsacker et al., 2019). When these fibres were cut < 5 mm, the stiffness significantly improved across all substrate types, with flax reaching 1.18 MPa and hemp reaching 0.77 MPa. Elsacker et al. (2019) attribute this to better packing of the substrate and a higher number of contact points for mycelium binding. Pre-compression of the substrate scored highest in compressive stiffness, which resulted in the Young's modules of 1.32 MPa for flax, 1.19 MPa for hemp, and 1,14 MPa for flax waste (**Fig 5b**). For the water absorption of the separate substrates, hemp-based MBCs demonstrated the most favourable behaviour, absorbing only 24.45% of the substrate weight after 24 hours, while straw absorbed 26.78% and flax 30.28% (Elsacker et al., 2019). This is attributed to the denser hydrophobic outer skin that the *T. versicolor* mycelium develops when grown on hemp, as this slows the absorption of water into the MBC. Together, pre-compressed hemp seems like the most favourable substrate, as it has the most favourable combination of stiffness and water resistance among the substrates that were tested.

Post-processing of Mycelium-Based Composites

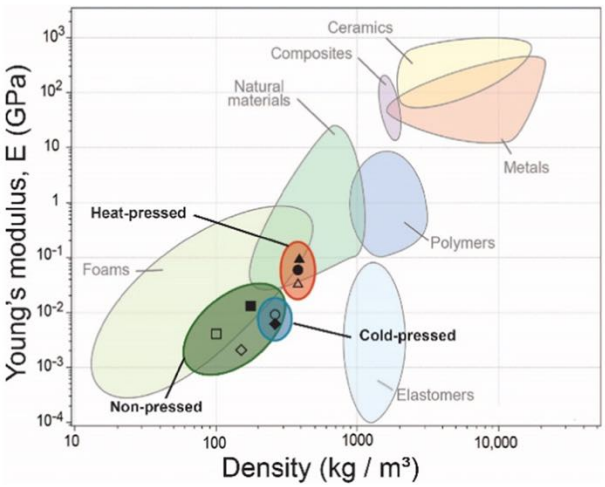


*Figure 6: **Material family chart of the Young's modulus vs the density** of pressed, cold-pressed, and heat-pressed MBCs (Appels et al., 2019).*

As **figure 4** and the analysis by Rovers & Wösten (2026) has shown, the least research has been conducted on the effects of post-processing on the material properties out of the four variables. However, the analysis showed that difference in post-processing was the least influential variable for tensile and compression strength. Appels et al. (2019) also conducted tests on the effects of pressing the MBCs into board-like materials. In these experiments, the non-pressed material was compared to cold-pressed and heat-pressed equivalents. This turned out to be the most dominant factor in determining the material performance, in comparison to substrate or even fungal strain.

As seen in **figure 6,** heat-pressed MBCs have a significantly higher density and Young's module than cold-pressed and non-pressed equivalents. Cold pressing the material increases the density about two times (from 0.10–0.17 g/cm³), while heat-pressing increased the density three or more times (up to 0.35–0.39 g/cm³). This results in significantly higher tensile strength for the heat-pressed material, moving from 0.01 – 0.05 MPa in the non-pressed material to 0.13 – 0.24 MPa in the heat-pressed material. Lastly, the Young's module increases from 2 – 13 MPa in the non-pressed material to 35 – 97 MPa in the heat-pressed material.

Pressing of MBCs thus significantly enhances the material properties and performance, moving from a foamy material-like performance more to a natural material-like performance (**Fig 6**). Heat-pressing the material improves the homogeneity and reduces the variation in between different individual units (**Fig 7**). This provides more homogenous and predictable performance of the material. In addition, post-processing of the MBCs makes the material much more suited for usage as a façade material slowly shifting the applicability from indoor



*Figure 7: **MBCs from P. ostreatus** grown on cotton substrate with heat-pressing (PCH), cold-pressing (PCC) and without pressing (PCN) (Appels et al., 2019).*

to outdoor usage.

Additionally, coatings or resins could be applied to the (pressed) mycelium material to make the material more durable and the material properties more desirable for façade panel usage. In nature, mycelium can grow an outer fungal skin that protects the inner core of the mycelium from environmental elements like moisture and heavy rainfall (M. Zhang et al., 2023). Research has shown that this skin contributes heavily to the hydrophobicity of the material, as the absence of the fungal skin led to poor hydrophobicity of the outer layer. This is underlined by the fact that the filamentous fungi contain hydrophobic proteins and lipids, attributing to the hydrophobicity of the material.

But in the case of heat-pressed MBC's, this hydrophobic outer skin and its function is mostly diminished and/or demolished. Hence, to improve hydrophobicity bio-based coatings could be explored to improve the lifespan of the material in outdoor application. For example, Farrahnoor et al. (2024) demonstrated that the usage of a beeswax-coconut oil mixture as a coating onto non-pressed MBCs led to a significant decrease of water absorption after submerging. While the non-coated MBC control absorbed about 261.65% of its own weight in water after five days, the MBC with 100% beeswax content only absorbed about 18.02% of its own weight. In addition, the coated units also showed a lower contamination rate by other fungi, as well as significantly slower material degradation. The 100% beeswax coating led to 20.57% weight loss after being buried in soil for four weeks, while the non-coated unit lost 88.72% of its weight. Although this research has been done on non-pressed MBCs, it could prove to be transferable when applied onto heat-pressed MBCs, pending further research in the Major Innovation Project.

For the application of façade panels, post-processing of MBCs, heat-pressing and coatings show significantly improved material properties and suitability compared to non-pressed and non-treated materials. The increase in density, stiffness, and tensile strength results in a more homogenous and reproducible material, which is essential for structural integrity and weather durability in the outdoor application of a façade panel. Additionally, the transition from a material in the foam category to more natural wood-like properties enhance the resistance to (permanent) deformation and stress. Applying bio-based coatings like beeswax could further improve the material properties needed for façade panels. However, despite these improvements, MBCs still need a lot of improving to be used in real construction purposes and façade panels.

Material properties needed for façade materials

A list of criteria has been drawn up to ensure that the selected materials meet the requirements for the installation of façade panels. Although great advances have been made in the field of mycelium materials in the last decennium, there are still significant shortcomings based on these criteria. This chapter will list and critically evaluate heat-pressed MBCs for the usage in façade panels and will show where the further research possibilities lie.

As mentioned before, the material properties of MBCs are dependent of type of species, substrate composition, growth conditions, and post-processing (Rovers & Wösten, 2026). The right combination of fungal strain and substrate can lead to thermal conductivity of as low as

0.026 W/mK. MBCs are one of the pioneering biomaterials in the field of insulation materials. MBCs have a thermal conductivity between 0.03 – 0.07 W/mK, which is comparable to the conventional insulation materials EPS with ~0.035 W/mK, and mineral wool with 0.04 W/mK (Elsacker et al., 2019; Pohl et al., 2022). This performance is attributed to the porosity of the hyphae network, which traps the pockets of air. *F. fomentarius* composites offer potential, combining good insulation, improved fire resistance, and a sustainable alternative to conventional wooden particle boards when pressed (H. Chen, 2026). However, the air bubbles inside provide the insulation properties, which would mean that heat-pressed mycelium would have less favourable insulation properties. Yet, more research needs to be done on the thermal conductivity of heat-pressed MBCs.

Furthermore, criteria were set for the weather resistance needed for a material used for façade panels. No research has currently been done on the UV resistance of MBCs, which is explained by the fact that MBCs have been mainly developed for indoor applications out of direct sunlight. This presents an interesting new avenue of research to be done on heat-pressed and non-pressed MBCs to move towards outdoor application. Water absorption rate and hydrophobicity however have been researched thoroughly. While MBCs with a thick velvety fungal skin show greater hydrophobicity, because of the hydrophobic proteins and lipids, this is diminished when the skin is taken away, as well as when the material is heat-pressed (M. Zhang et al., 2023). When pressed and non-pressed mycelium materials were placed in contact with water, all samples increased in mass, indicating significant moisture uptake (Appels et al., 2019). The sample with skin did, again, show significantly lower moisture uptake. In addition, the stability of the materials was affected by the environmental humidity. All materials showed significant expansion at 60% and 80% relative humidity for 192 hours. But the heat-pressed samples showed most significant gain at 80%, while staying more stable at 60%. The authors reason this could be because the increased moisture content from 60% to 80% leads to structural changes in the material, potentially leading to collapse of the inner porous structure. It was suggested that MBCs should be used in a layered system, sandwiched between two water-resistant layers to reduce expansion or deformation of the material while still retaining its mechanical qualities (Schultz et al., 2024). Schultz et al. (2024) has described how MBCs behave under simulated, accelerated weathering conditions. The MBCs showed limited structural degradation after weathering in the mycelium fibres, but cracking and damage to the fibres of the substrate. The combined application of increased heat and humidity reduced the water absorption of the material; this suggests that temperature plays a larger role in the degradation of the material than humidity. The experiments also showed that water absorption by the fibrous substrate led to a decrease in hardness of the material, but not in the overall durability of the composite. Based on the findings, it can be expected that freeze-thaw cycles mostly imitate moisture-related damage, such as internal substrate cracking and softening, rather than structural failure of the composite. It should be noted that a hydrophobic outer layer can prevent moisture from getting into the composite, which avoids these outcomes. Again, pointing at the importance of a much-needed bio-based coating for the pressed-MBCs used for façade panels.



Façade materials, to be applied onto high rise buildings have to be fireclass B, s2, d0. Lingam et al. (2023) showed that the best-performing MBC took more than 16 minutes before the heat fully penetrated the composite, which adheres to the "Ignitability with a small flame (EN ISO 11925-2): flame spread ≤ 150 mm within 20 seconds, following a 15-second exposure" of fireclass B (Measurlabs, 2026). S2 means that there is limited contribution to smoke production, which would need to be further investigated. Additionally, d0 means that no flaming droplets need to be produced in a fire. This seems highly unlikely, as the substrate inside the mycelium would most likely lead to separate flaming droplets falling. Once again, the application of a bio-based coating needed for fire resistance should be investigated. Usage of beeswax for example seems unlikely in that sense, as the material reaches its flash point and ignites and combusts at 204 °C (Huanbutta et al., 2026)

As mentioned before, the usage of MBCs in façade materials would most definitely become one of the more sustainable material options in the construction sector. As a comparative LCA showed, usage of MBCs have better climate change impact, compared to the conventional insulation materials EPC, sandwich panels, foam concrete and rockwool (Volk et al., 2024a). In addition, the usage of waste materials from the agriculture sector reduces the environmental impact even further. By growing on waste materials, that would otherwise be burned and emit CO_2 in the process, MBCs act as a CO_2 sink (Livne et al., 2022). Further analysis by Livne et al. (2022) showed that the incubation stage of MBCs makes up about 73% of the overall energy used to produce the material. Further advances in energy usage of the incubator could reduce the environmental impact of MBCs even further. It should be noted that end-of-life and usage was not considered in this LCA, but we will conduct an own comparative LCA on this matter soon (Volk et al., 2024a).

Table 11: Table of completed criteria for the Mycelium-Based Composites.

Criterion	**Threshold or qualitative**	**Hard constraint or ranking criterion**	**Does the MBC comply**
Cooling function			
Thermal conductivity	Below 0.5 W/(m·K)	Hard constraint	Yes, non-pressed MBCs: 0.03 – 0.07 W/mK (Elsacker et al., 2019; Pohl et al., 2022) Pressed MBCs: to be examined
Paint/coating			
Paint/coating compatibility	Qualitative	Hard constraint	Yes, (bio-based) coatings and/or paint can be applied.



Weather resistance			
UV resistance	Qualitative	Ranking criterion	**To be examined in further research**
Water absorption rate	Qualitative	Ranking criterion	At 60% relative humidity: 7% bodyweight increase At 80% relative humidity: 11% bodyweight increase (Appels et al., 2019)
Freeze-thaw resistance	Qualitative	Ranking criterion	**To be examined in further research**
Weight			
Surface density	Below 10 kg/m^2	Minimum requirement – Ranking criterion	Heat-pressed MBCs: up to 0.35–0.39 g/cm^3 (Appels et al., 2019)
Fire resistance			
Fire resistance	Fireclass B, s2, d0	Ranking criterion	Fireclass B seems likely, but s2 and d0 less so
Sustainability			
Recyclability	Qualitative	Hard constraint	100% recyclable and renewable (Alemu et al., 2022)
Biodegradability	Qualitative	Ranking criterion	100% biodegradable
Biobased content	Qualitative	Ranking criterion	100% biobased
Formability & availability			
Material availability	Qualitative	Hard constraint	Available in the fungal lab of Han Wösten
Material formability	Qualitative	Hard constraint	Further research should be done on the possibility of heat-pressing MBCs

			into a mold. Possibly to be investigated in this project.
Instrumentation availability	Qualitative	Hard constraint	Available in the fungal lab of Han Wösten
Prototyping feasibility	Qualitative	Hard constraint	To be further investigated

Recommendations

The aim of our project is to design a façade panel that reduces the urban heat island effect. The design of the façade panel is built upon previous projects, where a specific three-dimensional shape was implemented to create a panel which exhibits convective cooling. In our project, we want to try and add radiative and/or reflective cooling properties to the panel by using a reflective or radiative coating or paint. Combining these two cooling strategies could potentially enhance the cooling properties of the panel. In this report, a literature review has been conducted to give recommendations which is the most suitable material for our project.

A specific set of requirements was used to systematically assess whether a material would be suitable for our project. This consisted of requirements on the thermal conductivity, weather resistance, weight, fire resistance, sustainability, formability, availability and paint or coating compatibility. It was concluded that wood would be the most suitable material for our project. Next, we assessed which type of wood would be the most suitable material.

Looking at the scope of our project, which excludes testing the durability of wood for an extended period of time but mainly focuses on the cooling properties of the panel, it can be concluded that the natural durability is of lesser concern, at least for the prototyping. Therefore, heavier, more durable wood species and types, such as modified wood (thermally or acetylated), oak, western red cedar and European larch can be eliminated for the final wood type choice. The main reason for this is that these types of wood are more expensive than the other types of wood and it is unnecessary to buy more expensive types of wood just because they are more durable, as durability testing will be outside of the scope of our project.

This leaves spruce, Scots pine, fir and Douglas fir as final potential wood types for the prototyping part of our project. As we want to apply specific radiative or reflective coatings on the wood, the coating/painting compatibility of the different types of wood is important to assess. Scots pine is a highly resinous tree species and resins can bleed through the surface with UV exposure, affecting the coating adhesion (Hill et al., 2022). The same can be said about spruce, which has a slightly lower resin content but still experiences the same problem, which is why both wood types can be eliminated.

Douglas fir has the benefit that it is naturally more durable, however, it is less permeable, which makes it more difficult to impregnate. Fir on the other hand is less durable but more permeable than Douglas fir. Fir, however, has one big downside, which is that it is not widely available in Dutch hardware stores. For this reason, in addition to the lower durability, it can be concluded that Douglas fir would be the most suitable wood type for our project.

However, after talking with CNC-milling and prototyping experts about our choice of material, we came to the conclusion that Douglas fir would be too thick for the design of the panel. This is why we chose to use plywood for the construction of our first prototype. Plywood can be as thin as 3 mm, can be easily milled with a CNC and is available in the facilities in which we are constructing our first prototype. The concern about the durability of plywood is less important, as we are first testing other properties of the panel, like convection and the ideal

thickness of the material. If the design of the panel is optimized and needs to be tested for durability, specific moisture resistant and suitable outdoor plywood such as Okoume plywood is recommended.

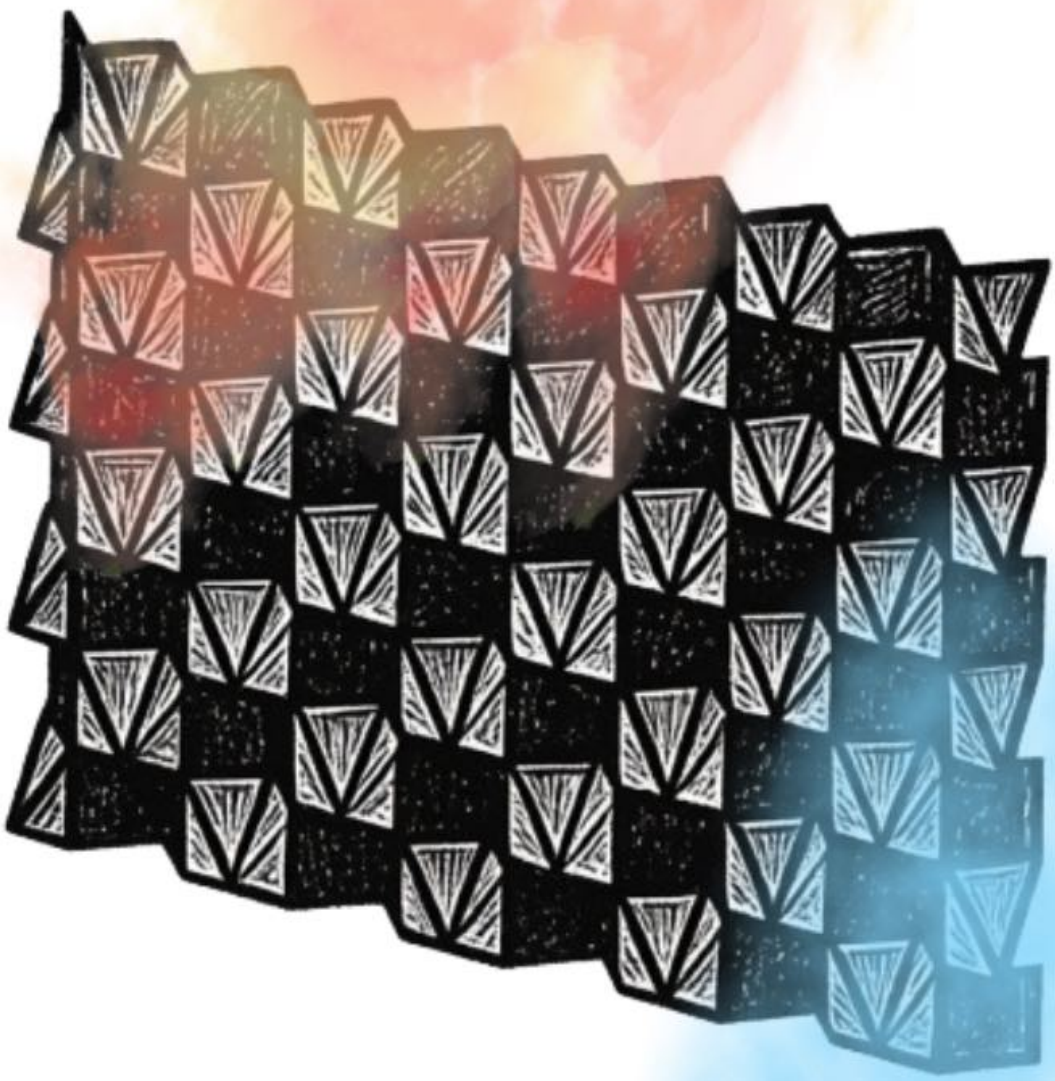
For a design with thicker materials and a higher TRL, thermally modified spruce or pine or western red cedar would be the best type of wood. Both types have high natural durability, are lightweight and do not need extra treatment (apart from the prior thermal modification) for outdoor applications. However, for the scope of this project, where durability is not tested, price, sustainability and available thicknesses are more important, plywood is the recommended material, and the coating system must be selected and verified to be compatible with the wood prior to application.

MBCs are also a highly promising biomaterial for construction applications. One of the main advantages of MBCs is their low environmental impact and footprint, as they are produced from agricultural waste streams and require relatively low energy input in comparison to conventional materials like concrete, EPS, or stone wool (Volk et al., 2024a). Additionally, MBCs are fully circular, being 100% recyclable and renewable, biodegradable and are therefore extremely future-proof biomaterials. The material's lightweight nature (with heat pressed MBCs at 0.35–0.39 g/cm³) makes them the ideal weight for façade panel materials and handle structural load requirements. Furthermore, their low thermal conductivity (0.03 – 0.07 W/mK for non-pressed MBCs) is comparable to that of conventional insulation materials and highlights their potential in that field, however, the performance of the heat-pressed MBCs needs more evaluation. This could be something that we investigate in this project.

However, MBCs have several limitations when applied as façade panel materials. One of the main drawbacks is the high water absorption of the material, losing its hydrophobic outer skin when heat pressed. Even when coatings are applied, high humidity or absorption can lead to swelling, deformation or degradation of the material. The mechanical strength also remains low in comparison to conventional materials. Even when heat-pressed, the material still performs like the least strong natural materials, not being near the Young's modules of synthetic composites, ceramics or metals (**Fig 5**). Another concern that needs further investigation is the fire resistance performance of the material. Although fireclass B seems likely, s2 and d0 would seem less likely and need further investigation. In addition, long-term weather durability, like UV resistance and freeze-thaw conditions remain unexplored. These limitations clearly define our scope for further research.

Despite these challenges, MBCs are still selected as a material for this project. This is due to their significant potential in the future as sustainable biomaterials, as well as material availability and prototyping potential. As shown in this report, methods such as heat-pressing and application of coatings already significantly improve the material properties of MBCs, suggesting that current limitations could be diminished through further research. By investigating the formability of the heat-pressed MBCs in our façade panel shape, together with investigating weather resistance, the project aims to contribute to adding to the current knowledge gaps that are still in the field of MBCs in outdoor environment applicability.

Appendix D. Design Report



FUNCTIONAL SKIN FOR PASSIVE COOLING

Alexandra Cooke-Kasyanto, Felipe Krebs,
Ocker Hazelbag, Jochem ten Pas, Ruben Langendoen

July 2026

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This report is to follow the *Biological Cooling Mechanisms* report and will refer to it throughout. The contents of this report entail the design process following that research.

1. Introduction

As described in the *Biological Cooling Mechanisms* report, the urban heat island effect (UHIE) is a growing problem in the Netherlands and requires efficient solutions that can be implemented in new and existing urban environments. This begs the question: could city buildings passively cool urban environments by utilising nature-inspired façade designs?

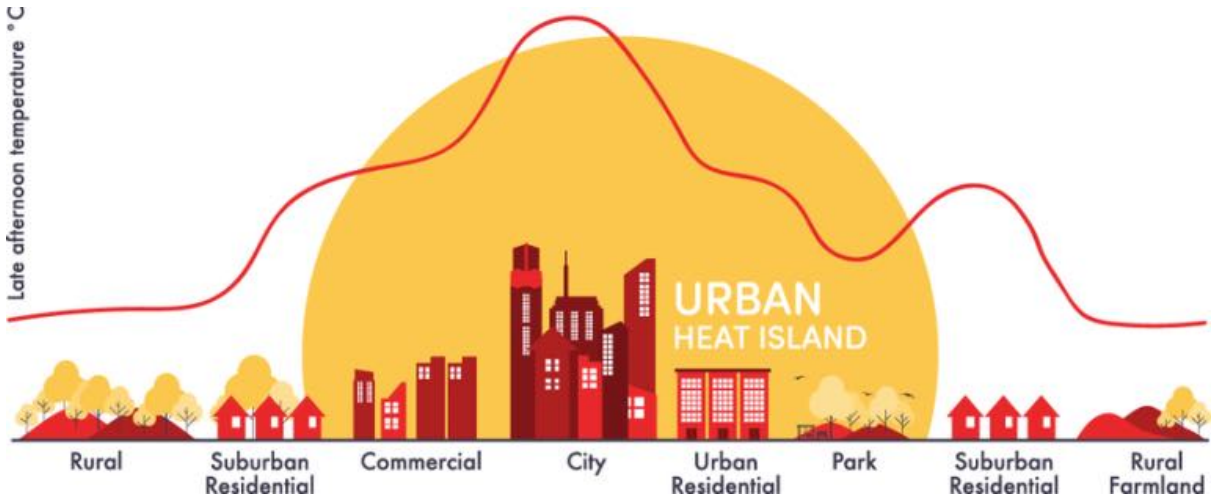


Figure 26: diagram of environments and the temperatures getting higher the closer to the city it is (Urban Heat Island, 2025)

This project aims to investigate this potential solution to combat the UHIE (Figure 26) in Dutch cities. Not only could the design of building facades be inspired by nature, but it could also utilise more sustainable materials - as the construction industry pollutes 6.2 billion tonnes of greenhouse gas emissions, meaning it is the 4th biggest polluting industry globally (Ritchie et al., 2020). To harmonise the construction, and architectural design industries with the scientific, regenerative method of biologists, our approach aims to bring elements of each into the design process. This will be through biological modelling, objective mapping and detailed methods of approach. The designs explored in this project will also be for the implementation of biomaterials as a means of optimising the culture and applicability of bio-design for a more sustainable future in our heating cities. The thinking behind the design process will be heavily influenced by nature and bio models that were uncovered during initial research. By using a nature centric approach, the design can better integrate the urban landscape with the natural environment while achieving the key function of cooling.

2. Background and context

Two prototypes were presented to our team (Figure 27) that are façade panels in the form of modular cassettes, designed to enhance the passive cooling ability of buildings and combat UHIE. They utilise geometric and ribbed designs to enhance convective air flow and shading of a building exterior (de Boer et al., 2025). The goal now is to improve these designs to enhance their passive cooling capabilities in a bio inspired way, prioritising the use of biomaterials. Our team has taken this previous research on board with the aim of developing them further with nature as our guide.

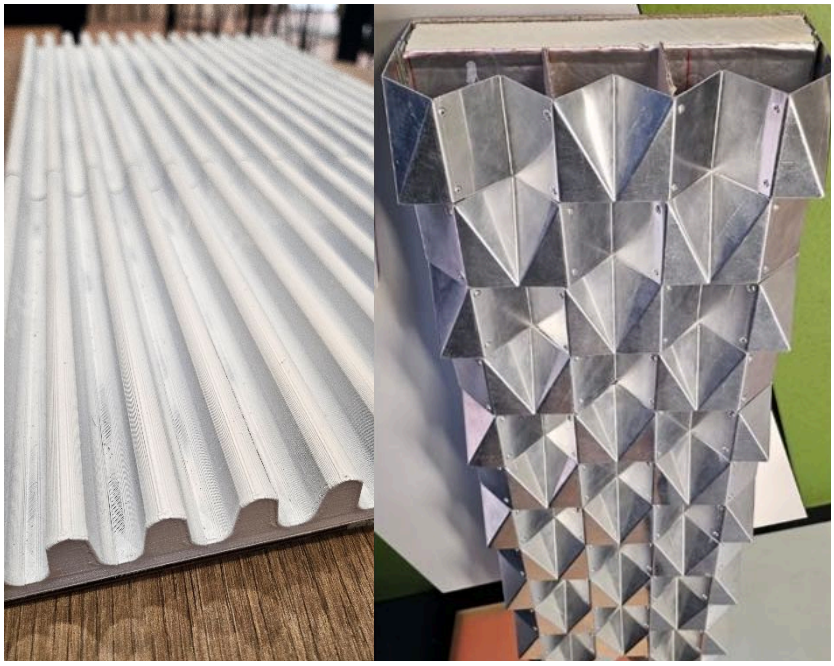


Figure 27: The two prototypes presented by the problem owner, upon which this project will build on (Left: "Insubark" panel (Pernice et al., 2022); right: "Stag Façade" (de Boer et al., 2025)).

2.1 Stakeholders

This project of creating a façade panel design for buildings in an urban context involves numerous stakeholders, each with differing values and goals. To understand who these stakeholders are, their relationships, and their involvement in the project, a salience model was performed (Figure 28*: salience model of the relevant stakeholders for this project.*Figure 28 below). The detail of this was discussed in the *Project Plan* and will again be discussed in the *Final Report*. For the purposes of design, the key stakeholders are identified below with their overarching desires for the project.

- Governing bodies of the city and the Netherlands - primarily regarding safety regulations. They require the product to be safe for citizens and the environment.
- Nature - since the façade must ultimately cool and fit within the sustainable bounds of practice as a material and functional object.

- The Utrecht University, and Lili's Protolab within it – as the available equipment will dictate the limitations of prototyping during the iterative design process.
- The problem owner - their desire for the product to eventually be a functional and pleasing façade panel. They value function overall, as well as a sustainable approach that uses biomaterials. While they are not seen in the salience model, they are the one who brought this project to the table and as such have provided the direction for the project and will receive the finished project at the end.

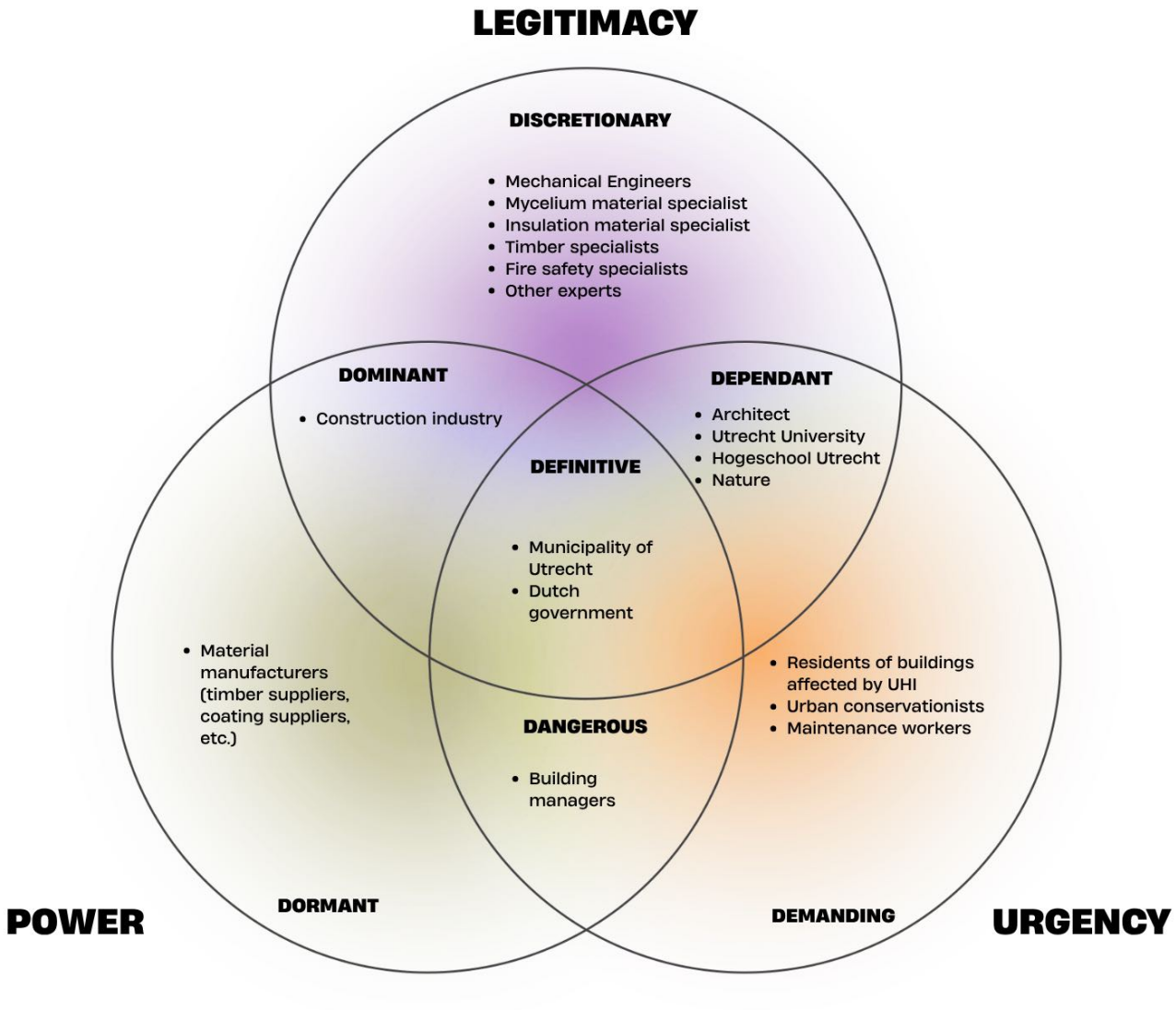


Figure 28: salience model of the relevant stakeholders for this project.

2.2 Five Why's

To identify the underlying causes behind the problem we want to tackle the 5 Why's tool was used to ask questions behind the questions.

1. **Why is the urban heat island effect a problem in the Netherlands?**
   Because buildings heat up over the day from the sun (and from human activity) which is not a problem in nature normally.

2. **Why do buildings heat up so much and heat up the surrounding urban area?**
   The materials that buildings and their façades are made from (concretes, bricks and metals) act essentially as heat batteries, soaking in solar heat and releasing it at night to the neighbouring buildings and city air.

3. **Why do facades of these materials take up heat and limit opportunity for cooling?**
   The materials often have too high of a heat capacity and not usually designed for cooling, only insulation. Additionally, there is also not enough airflow to transport heat away, so it often gets trapped in the urban environment which could be improved with better aerodynamic surface designs as seen in nature.

4. **Why don't building facades integrate passive mechanisms to stay cool and cool down the urban environment?**
   Intensive urban environments are facing warming unlike what's been seen before due to increased human activity, population and climate warming. Therefore, there needs to be a shift in the kind of designs seen in cities that are able to mitigate these factors, especially with nature as inspiration, as there have not been sufficient human invention to solve this problem yet.

5. **Why aren't buildings and their facades implementing the same mechanisms found in nature to achieve energy free cooling?**
   In the context of the Netherlands, this is a relatively new problem at its current (and future) intensity and nature-based solutions for this are in their infancy. Hence, there needs to be a push for integration of nature into the built environment to achieve the energy free cooling mechanisms that are needed to combat the growing problem of urban heat islands.

2.3 Fishbone diagram

Understanding the true nature of the problem means understanding the nuances of each component. To achieve this, a fishbone diagram (Figure 29) exercise was utilised to break down the context of the problem and look at each aspect as it connects to people, process, equipment, management, environment and materials involved in the problem. This resulted in the defining of the problem that current buildings **lack a functional skin that can cool buildings and their surrounding environment using bio-inspired passive cooling strategies**.

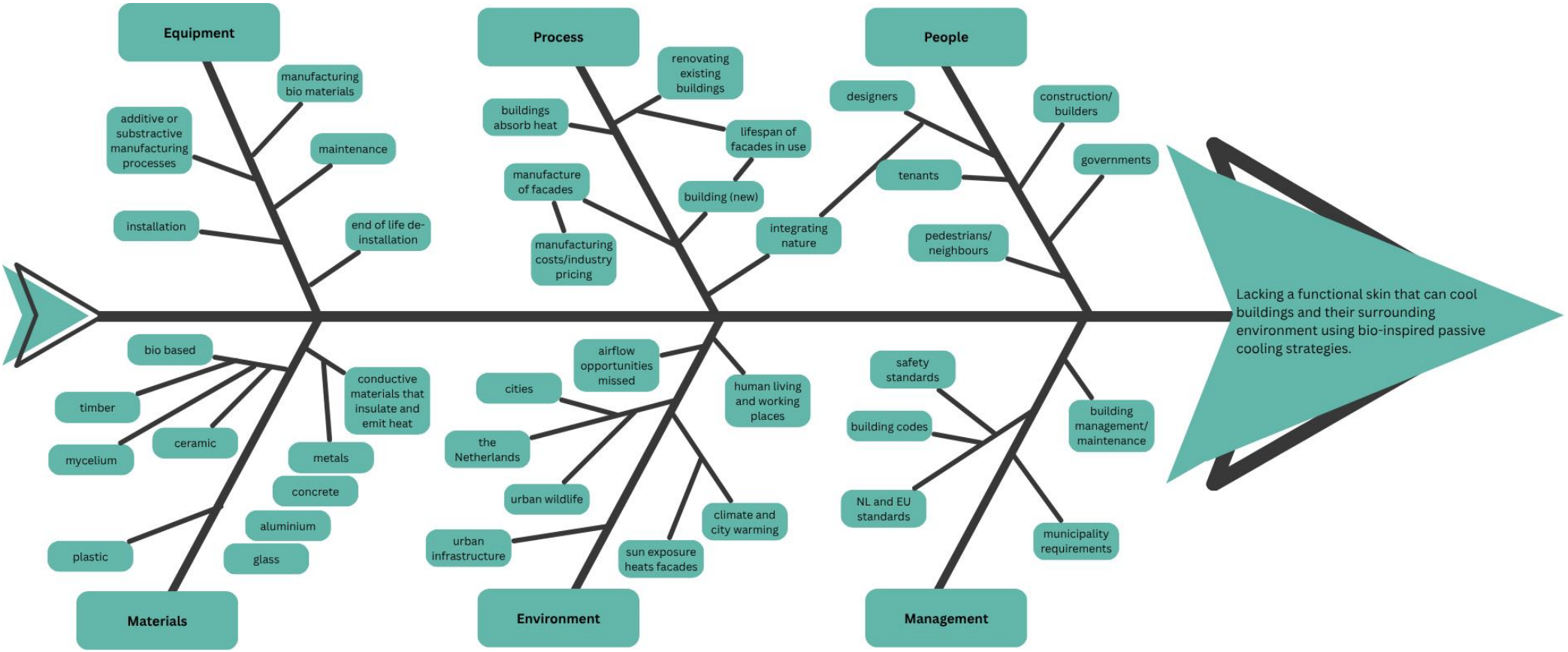


Figure 29: a fishbone diagram detailing the breakdown of components of the project context to gain deeper insight into the underlying problem.

2.4 Problem definition

The problem with current buildings was identified to be the lack of a functional skin that can cool buildings and their surrounding environment using bio-inspired passive cooling strategies. Therefore, the aim of this project will be to create a bio inspired "skin" (the front layer of the cassette) of a façade panel that enables passive cooling for a building and the adjacent urban environment. It must be bio inspired in its design, material and function. That is, it will be inspired by nature in its design (incorporating life's principles, bio models, and ecosystem services); will be made from a biomaterial; and will functionally cool using passive mechanisms that are found in nature and enhanced by modern fabrication technology.

2.5 Design objectives and requirements

Given the problem outlined above, the next step was to determine the objectives of the project, and requirements of the stakeholders. Throughout the first months of this project, various interviews with experts, the problem owner and other stakeholders were undertaken as a means of understanding their wants and wishes for such a project as well as their advice. University based stakeholders had strong desire for new and innovative material research, some engineering professionals were interested in function over sustainability, other experts had strong interest in novel biomaterials as the focus, and the problem owner outlined the desire for the end product to cool as the principal goal of the project. A summary of these can be seen in the table below, outlining the wants and wishes of the project as demonstrated by various interviews (summary of which will be included in the *Final Report*).

Table 1: the wants and wishes of relevant stakeholders for this project.

Wants	**Wishes**
Has passive cooling effect on building	Façade design will cool the urban environment around the building
Enhances convective airflow	The design will create a more biophilic urban environment
Provides opportunity for enhanced reflection and heat emissivity	The lifespan of the design will require little maintenance to ensure longevity
Made from biomaterials	The new design will be an improvement of the old design
Material has a sustainable or regenerative life cycle	It will offer enhanced ecosystem services such as providing nesting opportunities for wildlife
Design is inspired by nature and incorporates bio models	The design will be aesthetically pleasing and a selling point for the final product

Based on the requirements and wishes for this project, a set of objectives and their categorical reduction can be outlined as follows and explored in Figure 30 below:

1. Passively cools (function)
2. Inspired by nature (design)
3. Sustainable and/or regenerative (material)

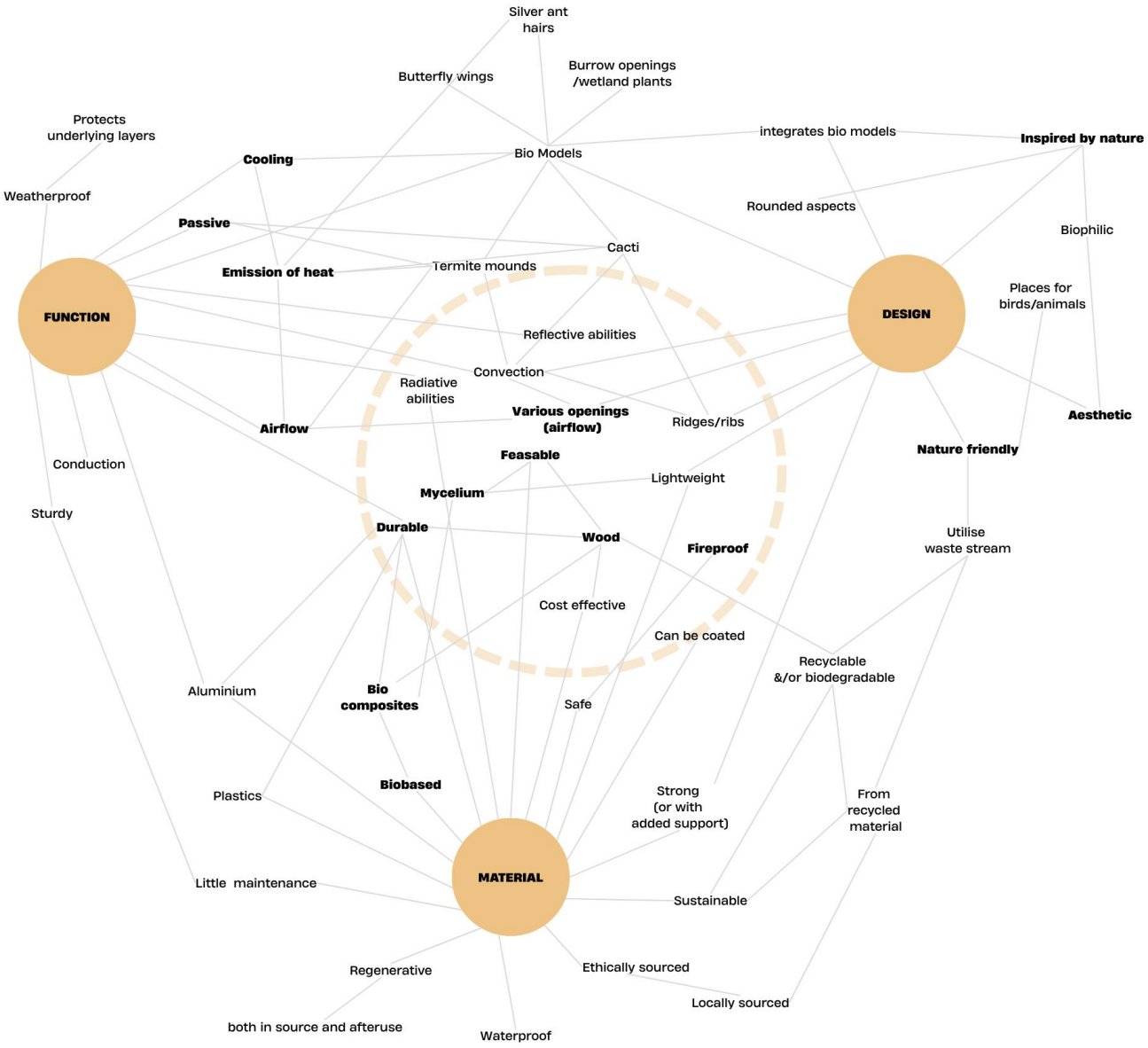


Figure 30: diagram showing each aspect of the design (function, design, material). Each aspect includes optimal attributes and where they intersect most (in the centre circle), there lies the best options for the design as they could potentially achieve all three aspect goals. Bolded words are aspects that are known to be important to the project based on wants and wishes mentioned earlier.

Based on the above and other research by the team so far (see *Material Selection Report* and *Biological Cooling Mechanisms Report*), the choice to make the design out of timber (and in parallel, pressed mycelium based composite (MBC)) has been made. Timber and mycelium were highlighted as the best available materials (*Material Selection Report*) with good

potential for incorporating the biological models. As seen in the above Figure 5, the three focal areas of the design process (function, material, and design) converge with these materials and their possibility to integrate various design options (such as ridges, openings and convective shapes), while providing options for different coatings that may enhance the reflectivity and emissivity of the design. Their low cost and workability as a material also make them feasible for this project.

2.6 Sustainability

In keeping the final product as sustainable as possible, the team also makes use of Biomimicry 3.8's "Life's Principles" which entails key characteristics that promote the creation of conditions conducive to life (see Appendix A). The key principles that will be integrated into the design are:

- Integrating development with growth
    - Combining modular and nested components:
      The design will be modular so that it may adapt to cover facades of many types of buildings. The smaller openings of the panels should work well with the larger undulating pattern of the façade to enhance the cooling effect.
- Locally attuned and responsive
    - Use feedback loops:
      The design aims to enhance the stack effect – this is a feedback loop whereby warm air transports heat away from the area. Therefore, designing the façade to take in and direct warm air in a passive way should enhance this feedback loop to act in a cooling way.
- Evolve to survive
    - Replicate strategies that work:
      Various biological models (detailed in the *Cooling Mechanisms* report) have informed strategies that will directly inspire the design of the façade. By replicating some of these (i.e. termite mounds, butterfly wings and cacti), the function of the façade will be effective (further discussed in section 3.2).
- Use life friendly chemistry
    - Breakdown into benign and useful constituents:
      By utilising timber and MBC as the primary materials, the aim of the design will be that it can contribute to a healthy environment at the end of its life and should not leach harmful chemicals into the area during its use.

During the design process these key principles will be part of the focus when brainstorming and considering options. At the end of the project, these will be re-evaluated in the *Final Report* to determine if the design and project as a whole have maintained these principles.

2.7 Scope

The redesigned functional skin must fit the context of buildings and climate of the Netherlands. The design will also be influenced by the resources available for the MIP course and a total of 7 months scheduled for the entire project (which includes building, experimental testing and adjacent research such as business model analysis, IP, etc).

3. Design Process

Improving the design of existing models must begin with understanding the current design, and what aspects should be brought along in the design process, and what should not. This is determined by the methods of passive cooling that work best. From there, we can begin an iterative process of improvement based on additional theoretical knowledge.

3.1 Background research and design considerations

The existing deigns have some experimental backing based on the reports presented by the problem owner. While these are still quite preliminary and require a more thorough testing approach, it can give some insight into what aspects of the design are most important.

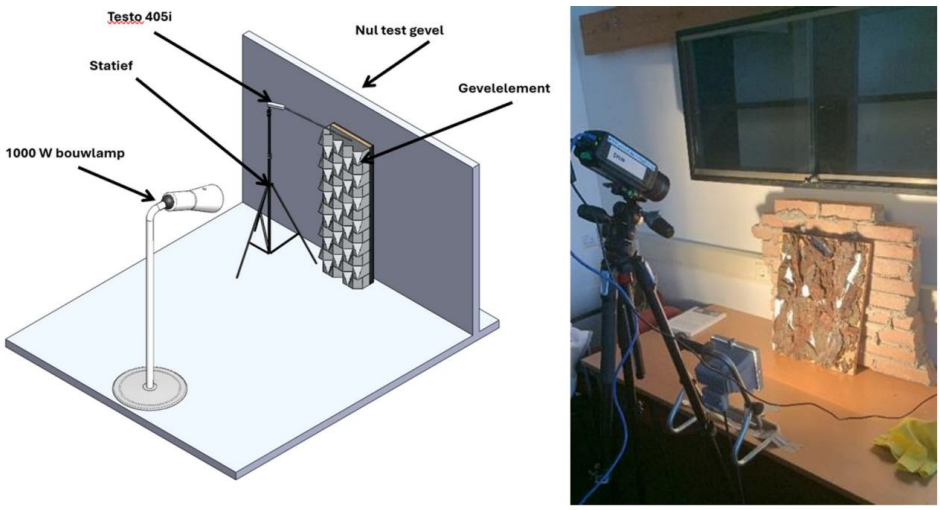


Figure 31: Experimental set up of the Stag Façade (de Boer et al., 2025) on the left, and the Insubark (Pernice et al., 2022) proof of concept on the right.

The Stag Facade had best results with an “Orgago” skin design and small “louvres” on the internal cavities of the cassette, which created smaller shafts with improved air velocity. This meant that there was an observed air speed of up to 0.3m/s, and a change in temperature of ~0.25°C (de Boer et al., 2025). Their experimental set up can be seen in the above figure. Additionally, the Insubark report highlights the rougher surface as more effective at maintaining lower temperatures than a flat surface (Pernice et al., 2022). From here, we have a better understanding of how the facades work based on their designs. The sketches (Figure

32) below depict the basic ideas – passing air currents are caught by the design and enhances the natural convection of the panel. This enables heat transport via optimal air velocity inside the cavity while the rough texture of the skin also aids in self-shading and maintaining an overall cool temperature for the building.

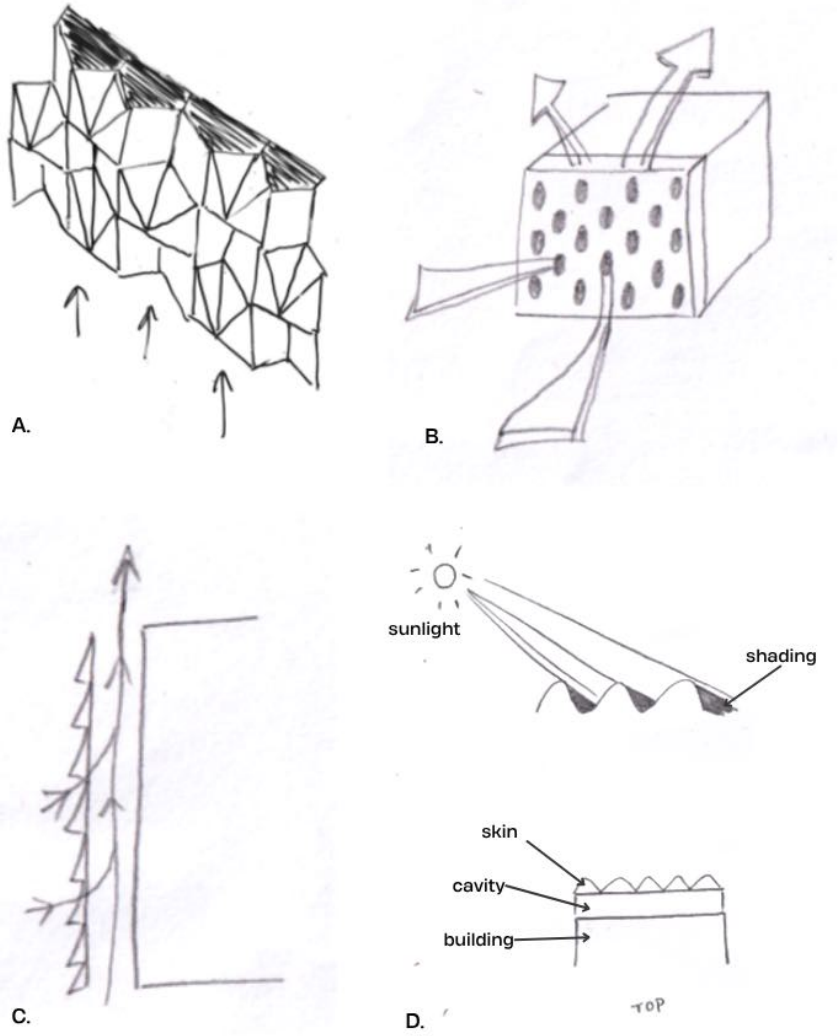


Figure 32: sketches depicting the basic function of the façade panel. Arrows show air moving through the façade. A: angled front view or Orgago skin; B: angled front view of building with façade (arrows indicating air movement); C: side view including building behind façade' D: the self-shading ability of the façade shown from top view.

Re-designing the current prototype designs involve only the "skin" portion of the cassette. In doing so, there are some considerations that will need to be incorporated. The depth of the cavity of the current cassette is determined to be about 4cm to the back face of the skin with another 4cm of protrusion from that point (see Figure 33 below). This is the best model of the initial design, as the Insubark panel has no cavity for convective movement of air at all. It instead relies on surface convection which is much harder to measure.

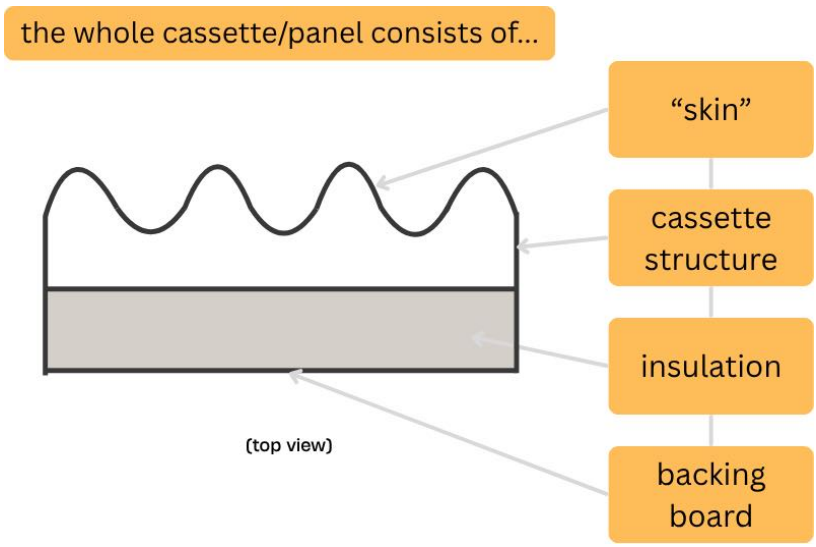


Figure 33: showing the façade from the top view and its components. This project will focus on the skin portion.

From previous research, the recommended **depth of the cavity** is quite variable. For a DSF with a glazed front, the recommendation is between 0.7 and 1.2m which ensures a balance of effective stack effect and reduced heat transfer to the building (Barbosa & Ip, 2014; Radhi et al., 2013). Other similar designs such as a "solar chimney" utilise a cavity depth of 0.2-0.3m when there is no wind, or 0.4-0.5m when there is external wind accounted for (Q. Wang et al., 2020). These designs are larger as they are built into the deeper walls of the building and utilise glass fronts. Since the façade panels in this project aim to be applicable retro-actively, the panels must not be overly bulky in order to ensure they can be feasibly implemented. The Stag Façade has a cavity depth of 4-8cm, so this project will aim to work with a similar depth as a means of continuing their research while simultaneously striving for a façade that can be easily implemented. Additionally, it seems there is no other background literature or industry examples of a façade panel for passive cooling, therefore this is an opportunity to test if a scaled down version with a new skin design works

The **angle** of the geometric protrusions of the current model shown above, are about 72°. This is an adequate angle, as according to a study which looked at venetian style blinds within a glass DSF (Ji et al., 2007), they found that an angle of 80° did demonstrate a 35% improvement of natural ventilation due to the turbulent airflow around the slats forming large upwards buoyancy momentum. While these blinds were behind glass, they still provide some indication of the buoyancy behaviour of air molecules in a windless state.

The **height** of the DSF is also very important, our cassette panels can fit in the category of shaft box, or if they were also connected horizontally, they could be classified as the multi-story design (see Figure 34). Both of these are the better performing categories due to their optimisation of the buoyancy effect over the extended heights (Barbosa & Ip, 2014). Barbosa and Ip recommend that the DSF should at least be 2 stories high (and even extend beyond

the roof), to achieve effective pressure differences that enable the optimal stack effect. The design of the cassette itself allows for continuous vertical movement of air through the façade thanks to its modularity, which will be maintained in our design as this is an effective design aspect. Horizontal movement of air will not be prioritised this had a less effective movement of air for the cooling function (Barbosa & Ip, 2014). This also gives more opportunity to maintaining structural integrity and modularity. Since it's estimated that in Netherlands that one story is about 3 m (Roy et al., 2023), the design would want to be effective at a minimum of 6 m in height for the best results.

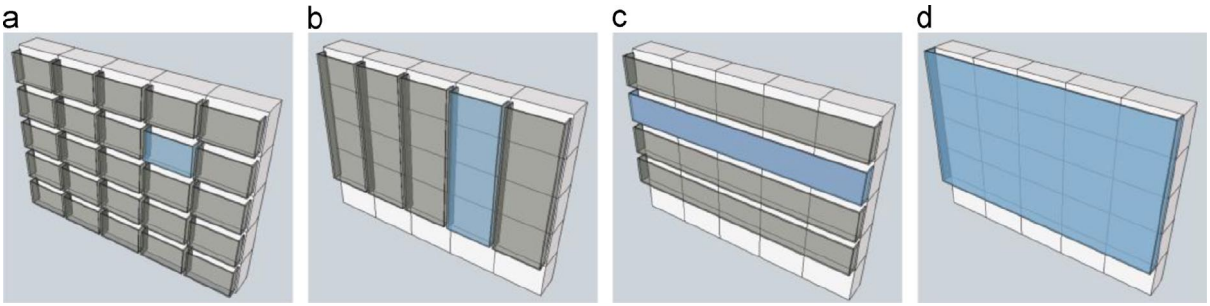


Figure 34 : Double skin facade classification: (a) Box Window, (b) Shaft-Box, (c) Corridor and (d) Multi-Storey double skin façade (Barbosa & Ip, 2014).

Visual summary of dimensions required:

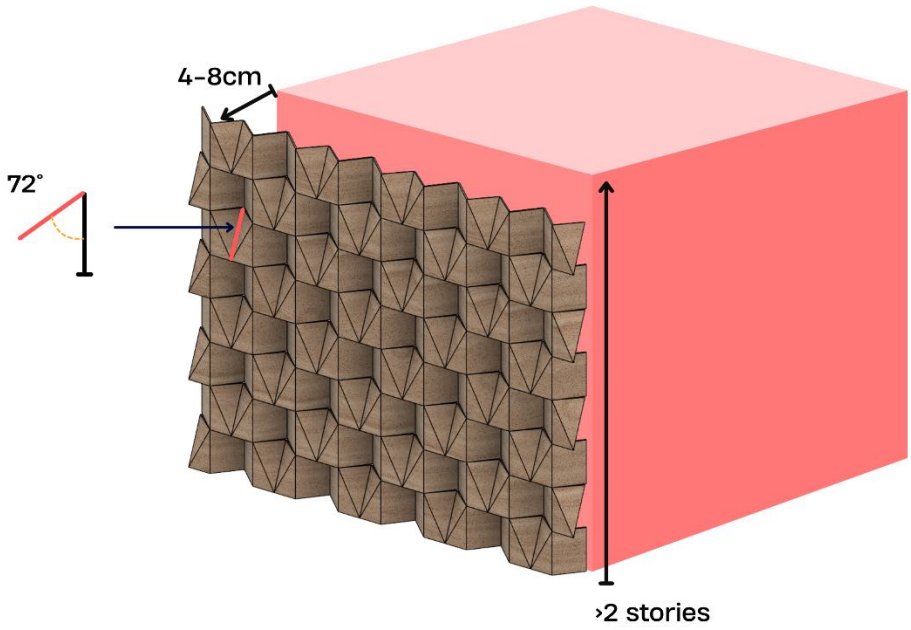


Figure 35: Summary of dimensions needed for design of the façade based on preliminary research (façade not to scale with building (red cube)).

Choice of **material** is also tied to the choices made by the team based on sustainability, performance, design potential and availability. The 2 materials that will be used are wood and MBC. With these in mind, the design will lend itself to shapes and details that are realistic (both for the material and the context). As well, the methods of production outline some other design considerations due to **manufacturability** (i.e. availability of equipment, machine constraints, etc). However, these will largely be outside of this report and will instead be discussed towards the end of the project.

The design must also be visually pleasing as it will eventually become a product that will be sold to the market. Therefore, the **aesthetics** of the design must be imbued with a sense of style that can fit both the urban context, and the natural environment as well. In this way it must pay homage to the bio models and original bio inspiration of the project and its broader goal of reducing the urban heat island effect and improving the environment we live in. When designing for an architectural context, there is a wide array of techniques depending on the design style, existing context of the building/environment and the clients. To determine what suits this project best we have compared the 5 principles of visual design in UX (user experience) (Gordon, 2020) as well as Kellert's 6 biophilic design principles (Kellert et al., 2008) in the table below.

Table 2: Visual design and Biophilic design principles.

Visual Design in UX	Kellert's Biophilic design principles
1. **Scale**: using relative sizes to illustrate composition. 2. **Visual Hierarchy**: guiding the eye over the design in order of importance. 3. **Balance**: evenly distributed visual signals on both sides of an imaginary axis. 4. **Contrast**: juxtaposition of visually dissimilar elements. 5. **Gestalt principles**: capture the tendency of people to perceive the whole rather than the individual elements.	1. **Environmental features:** characteristics of the natural world. 2. **Natural shapes and forms:** representations and simulations of the natural world. 3. **Natural patterns and processes:** incorporation of properties found in nature (rather than simulations). 4. **Light and space:** types and positioning of light and spaces. 5. **Place-based relationships:** connection of people, culture and ecology in a geological context. 6. **Evolved human-nature relationships:** fundamental aspects of the inherent human relationship to nature.

Taking cues from both above approaches, our design will follow these key principles as they suit the context of the project and the scale of the design in its function as both a retrofitting product and a new material for future builds:

1. Scale
2. Balance
3. Gestalt principles
4. Natural shapes and forms
5. Natural patterns and processes

For the final design, we utilised a poll to 10-15 people to assess the adherence of the design to these principles. The poll and its details can be found in Appendix B. The results of which will be discussed in section 4.

The design must, more than anything, **function** well. Previous studies by the University of Applied Science and Utrecht University provide proof of functionality for the existing design, and this must not be lost, rather it will be improved if possible. Other similar designs by TU/e student team called VIRTUe (Pujadas-Gispert, Alsailani, van Dijk (Koen), et al., 2020b) (Figure 36) and the Polytechnic University of Marche, Italy (Summa et al., 2025) (Figure 37) also provide evidence of the design achieving cooling abilities. The VIRTUe student team design was designed in collaboration with Nabasco, and as such was make from a biobased composite material. It was also tested in Dubai as part of the Energy Solar Decathlon there, where it had successful results. More recently at the Polytechnic University of Marche, their design used a similar geometrical pattern which explored the self-shading abilities and heat reflecting coatings on the façade. Their study found that the folded shape and its corresponding back wall were on average 10°C and 1.3°C lower than the flat module (tile) (Summa et al., 2025). They also found that the reflective coatings worked well to cool the flat wall, but had negligible benefits when added to their folded, self-shading design.

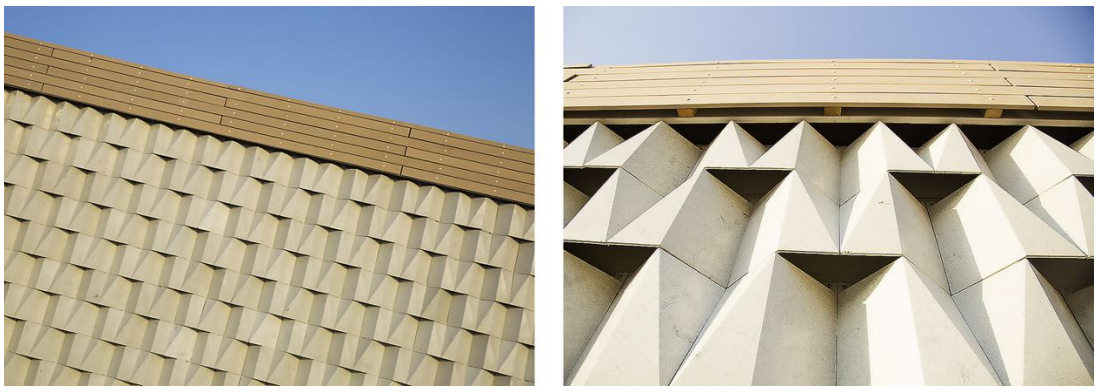


Figure 36: the façade design from TU/e that was successfully tested in Dubai and made from biobased composite in collaboration with Nabasco (Pujadas-Gispert, Alsailani, van Dijk (Koen), et al., 2020b).

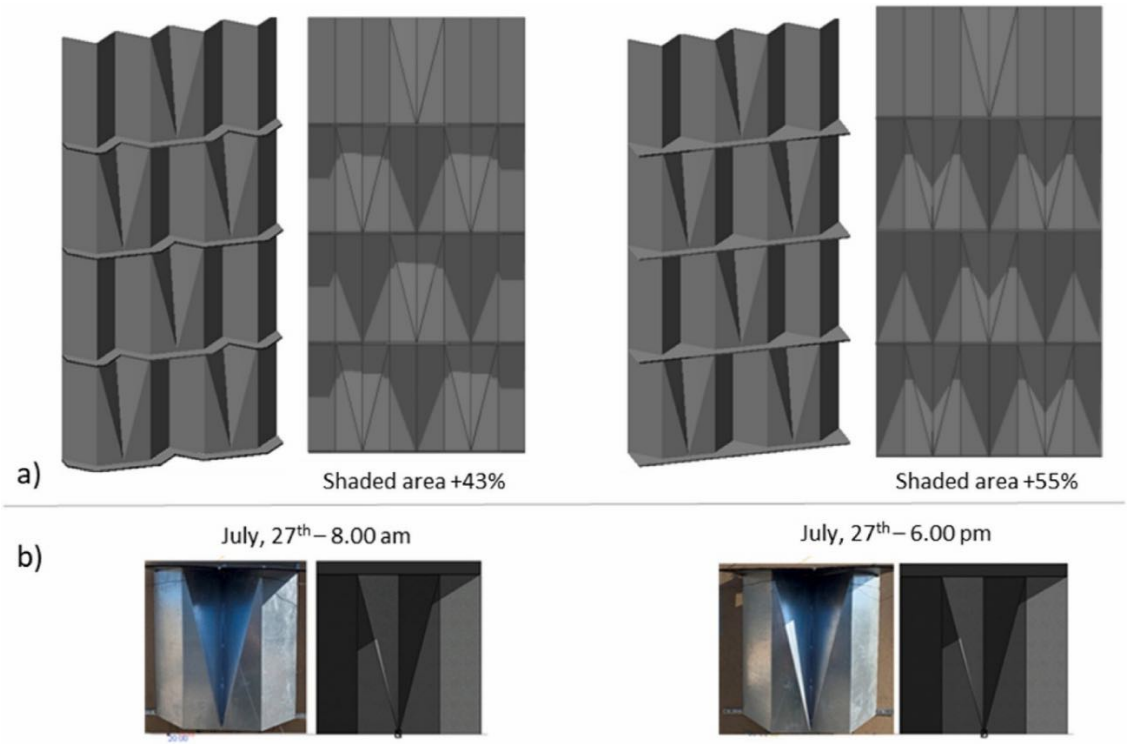
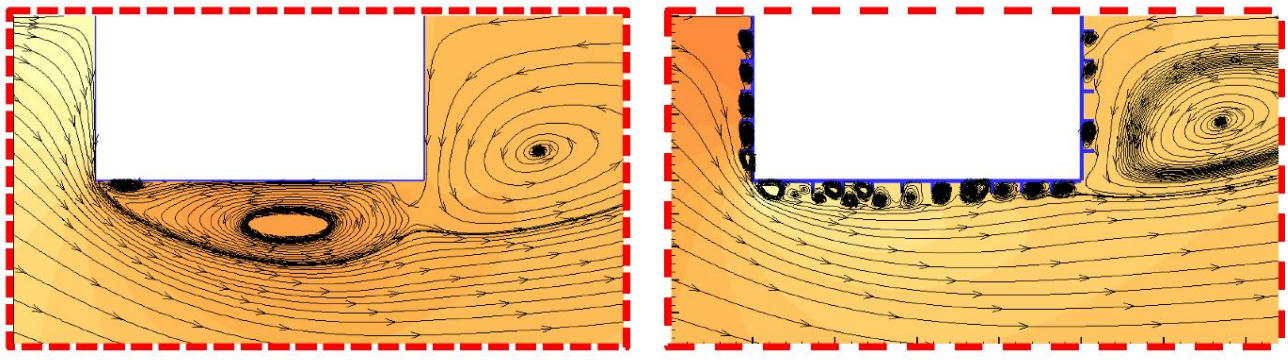


Figure 37: designs from Polytechnic University of Marche, Italy, that are similar to the Stag Façade and show the self-shading ability of the façade (Summa et al., 2025).

The use of "**ridges**" similar to the cacti bio-model (*Biological Cooling Mechanisms Report* and summarised in next section) (or "pleats" as they are also referred to in architectural settings) can influence the turbulence and movement of air at the façade (see Figure 38). The design will therefore make use of these aerodynamic behaviours in a way that enhances the passive cooling opportunities. Not only could the protruding design keep the flow of air close to the building, but the creating of small vortices could be beneficial in directing and catching air at each inlet of the façade.



Mean velocity streamlines of one-half of the horizontal section; (a) no ridges, (b) with ridges.

Figure 38: the use of vertical façade ridges influences how air interacts with a building (Cheng et al., 2025). From these diagrams, it allows more air interaction with the façade while reducing force on the building (the goal of the study). The corner of the building also becoming more chamfered with the façade was a positive factor. These aspects could be beneficial for our design in terms of enhancing uptake into the inlets of the façade based on the small vortices created along the sides of the building.

The design must also consider the **user experience.** In this sense, it should be simple to apply to buildings and must be able to withstand weathering for at least 5 years. In addition, it will likely be used by other-than-human lifeforms, such as birds who may make nests, or spiders and insects. The design will therefore allow this to occur, while maintaining functionality and lifespan. These aspects should also allow the positive contribution to **ecosystem services** (such as providing nesting spaces for insects and birds, assisting in thermoregulation of the environment, and enhancing cultural enrichment through aesthetic design) which will be discussed in the *Final Report*.

3.2 Biological inspiration

The *Cooling Mechanisms Report* outlined the key biological models that have been used for inspiration and background research for this project. Some of the more influential design principles are extracted and summarised below.

Bio model 1: *Termite mounds*

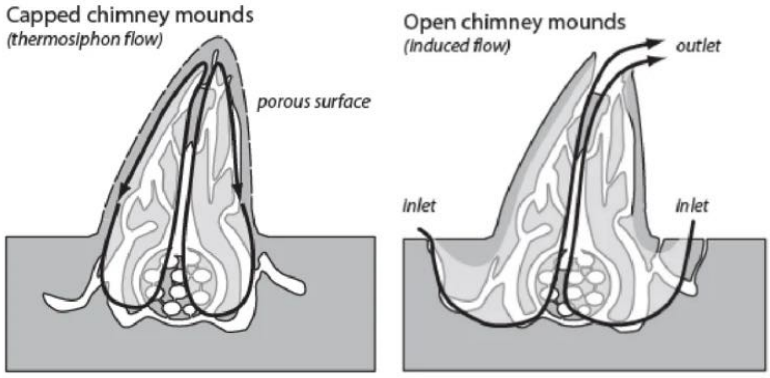


Figure 39: termite mound schematic (Turner & Soar, 2008).

Design principles

Materials that can warm up while the internal structure remains cool can effectively increase convective cooling. This, in combination with openings that allow for the movement of air and transport of heat, can enhance a desired the cooling effect.

Bio model 2: *Natural cycling of air in wetland plants and prairie dog burrows*

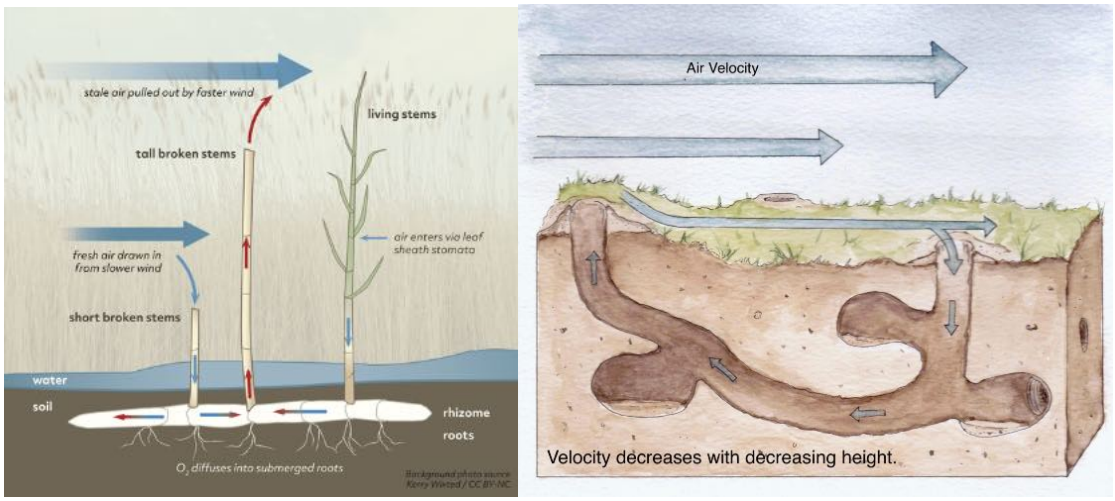


Figure 40: air cycling via openings of a system (Asymmetric Burrow Openings Create Passive Ventilation, 2020) (How Wetland Plants Get Air to Waterlogged Roots, 2022).

Design principles
Strategic openings in a system can utilise the pressure differences that occur thanks to contrasting wind speeds which result in efficient movement of air within the system.

Bio model 3: *Cacti cooling ridges*



Figure 41: thermal image of a cactus (Cactus Ribs Keep Them Cool, 2017); and a barrel cactus photographed in Nevada (Marketa Mark, z.d.).

Design principles
Surface geometries can enhance air mixing and boundary-layer disruption, increasing convective heat transfer.

Bio model 4: *Saharan silver ant's triangular hairs*

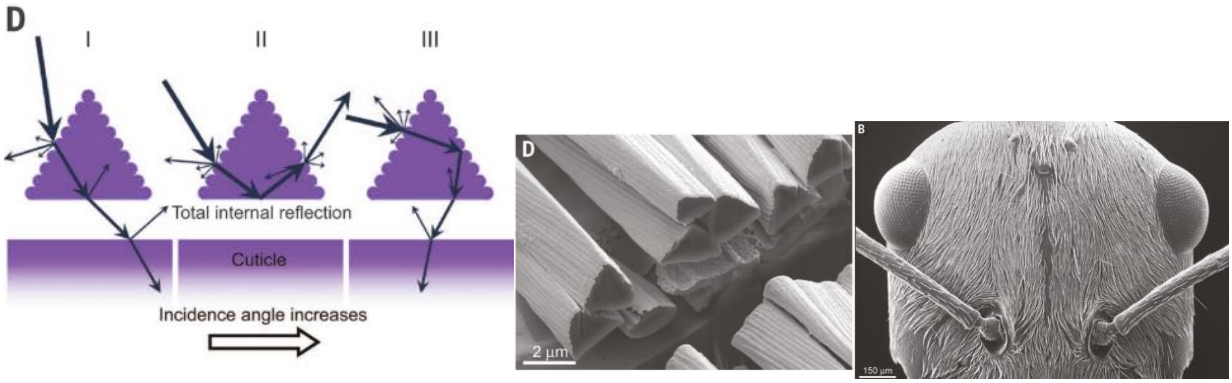


Figure 42: detailed images of Saharan Silver Ant hairs and function (Shi et al., 2015).

Design Principles
A surface-integrated micro-structure composed of dense, triangularly shaped elements featuring corrugated upper facets and flat bottom facets can passively cool an object. This is achieved by maximizing visible and near-infrared optical reflection through Mie scattering and total internal reflection, while simultaneously acting as an antireflection layer that enhances mid-infrared thermal emissivity for highly efficient radiative heat dissipation.

Bio model 5: *Butterfly wings of Archaeoprepona Demophon*

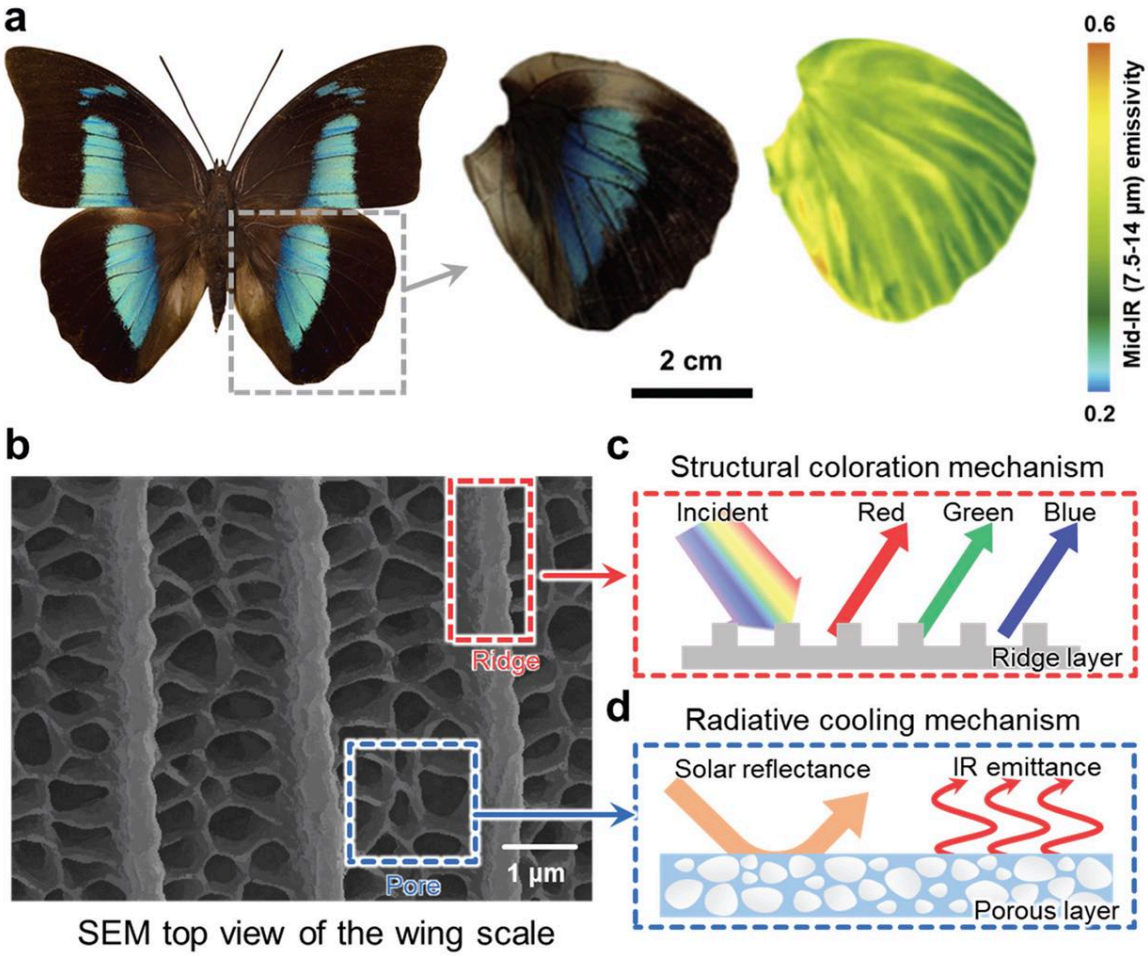


Figure 43: A. Photograph of the Archaeoprepona Demophon and cut-out of the dorsal wing. Showing the mid-IR emissivity of the wing structure. B. Scanning Electron Microscopy image of the Archaeoprepona demophon wing, consisting of the pores and ridges. C. Schematic figure of the structural colouration mechanism. D. Schematic image of the radiative cooling properties of the dorsal wing, caused by the nanoporous properties of the wing (Lee et al., 2022).

Design principles

Nanoscale structural elements of a surface give both colour and functional properties. Porous structures can reflect light and solar radiation, enhance cooling and maintain colour over time.

3.3 Incorporation of bio models into our design

The overall function of the skin will utilise the passive cooling mechanism similar to bio models 1 and 2 – enhancing the convective flow of air into, though, and out of the façade as one whole system. This is possible thanks to an array of small openings on the surface of the design and one central cavity that can transport warm air through the internal aspect of the cassette. This heat will be transported upwards and out of the top of the cavity where it can exit the urban environment. For an even more effective result that mimics these cooling systems, an integrated airflow system that incorporated the internal areas of the building could be investigated - however this is beyond the scope of this project.

Ridges such as those seen in bio model 3 will be incorporated through the addition of small ridges on the skin, this should increase the surface area of the design (increasing potential for convection), provide extra surface shading and also increase air turbulence leading to more movement into the openings. Additionally, the hierarchical design (small components making up a larger façade) will result in a larger ridged pattern overall – with the aim of further enhancing this effect.

Bio models 4 and 5 also inspire the enhancing of reflection and emissivity at the surface of the façade skin. The scale of these bio models is too small to replicate in this project (although bio model 5 has been replicated in previous studies, further discussed in *Cooling Mechanisms* report); therefore, the use of coatings will be the best way to incorporate them into the design. For this, we will test various coatings for heat reflectance, high albedo and IR emissivity to determine the best option post manufacture of the design form. These will be mechanical in their function (e.g. using physics to redirect heat/light), as we would like to avoid the use of harsh chemicals for the sustainability of the project and in keeping with the Life's Principle of using life friendly chemistry (Appendix A).

4. Design process

The process of designing a new façade panel that meets the desires of the stakeholders, while being informed by nature was iterative and spanned over multiple months. The beginning was marked by a deep dive into the physics and cooling mechanisms found in nature, and this is what gave the design process its backbone. The basic pathway the design process took is depicted below (Figure 44), branching off into multiple areas of focus in parallel with each other. Ultimately, we have chosen timber as a focus to investigate the potential of the Orgago shape (seen in the original Stag Façade), and an MBC focus to explore a rounded, biophilic design and a potential future material (pressed MBC).

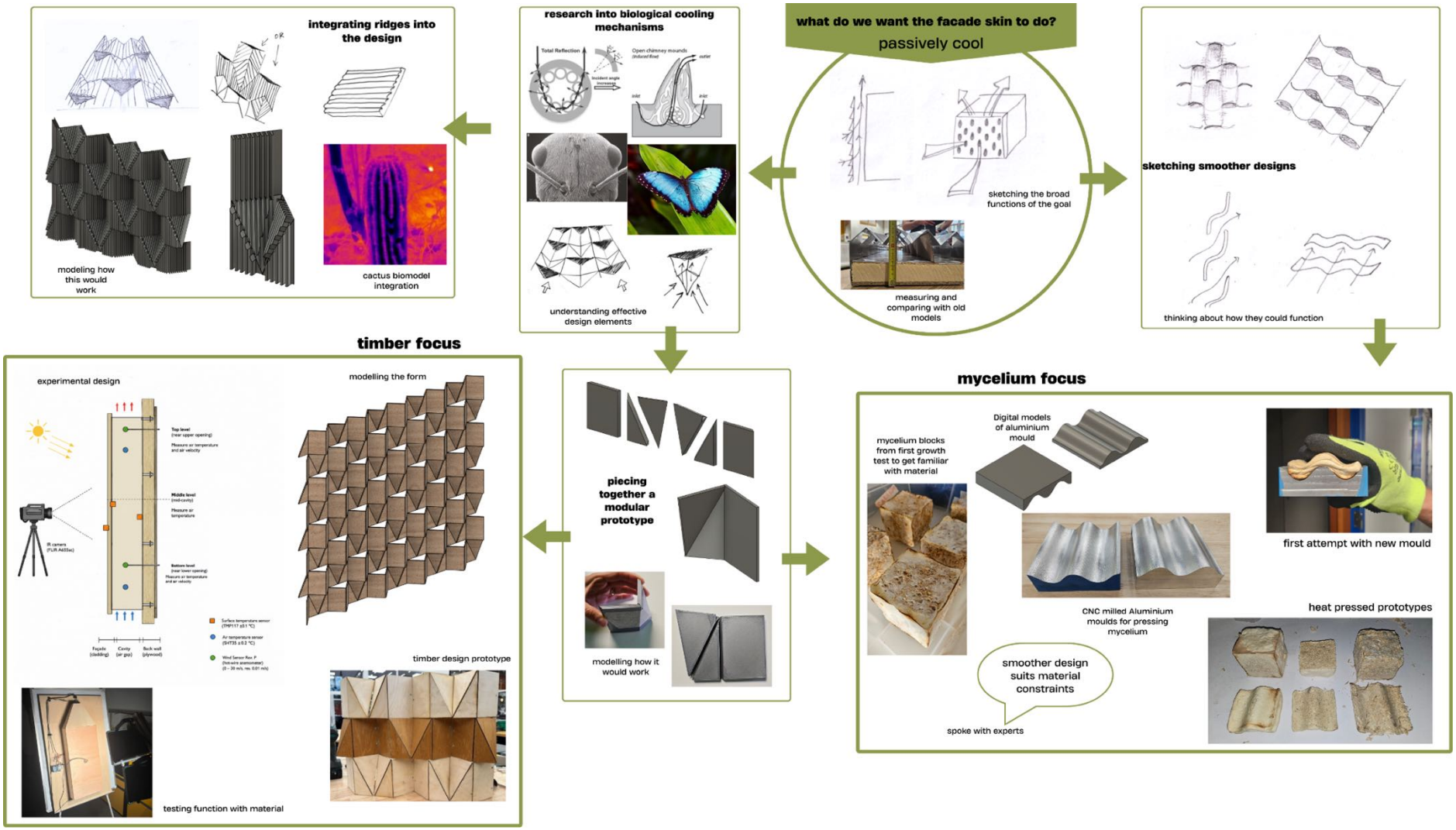
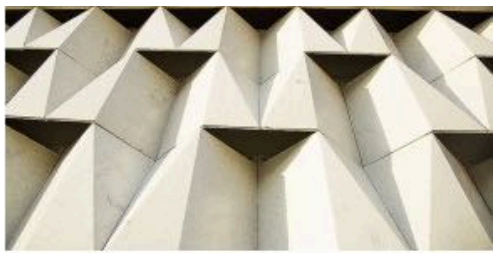


Figure 44: Design map showing the beginning of the project in the round circle, and the different directions the design process took. Ultimately, a parallel timber and mycelium focus was chosen.

Some initial sketches to understand the existing function of the designs and what opportunities exist. The project will look at replacing aluminium and plastic with a biomaterial such as wood or mycelium; therefore, it is prudent to create a version of the existing prototype that is the same design but with a new material to assess potential functional improvements.



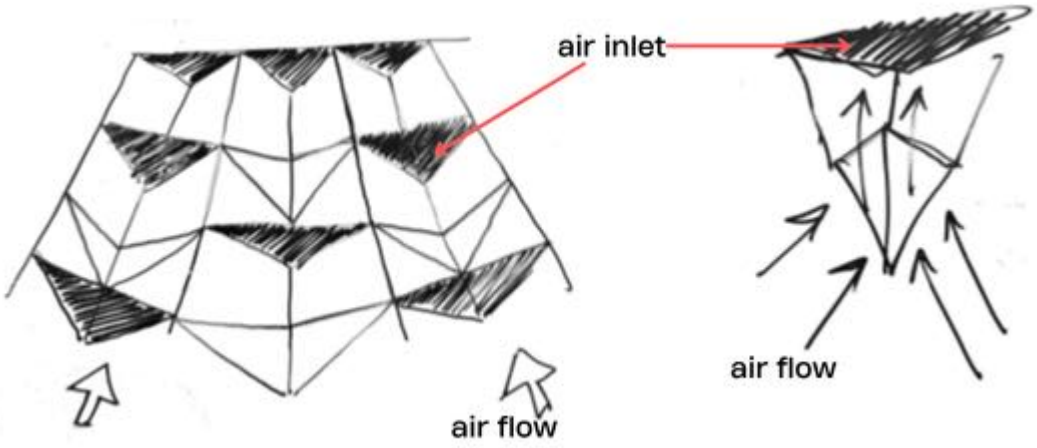


Figure 45

Considering how the design can be integrated with cacti-like ridges to achieve the same benefits. Integrating cactus inspired ridges into the design comes with many decisions to be made. For instance, choosing which faces and directions the ridges should go. We opted for external ridges only, as there didn't seem like much reason to place it on the inside for this project's scope, but perhaps beyond this project it could be explored. We also opted for ridges in the direction of potential airflow in the hopes that this will help direct more air into the inlets. This is something that needs to be tested in practice, however.

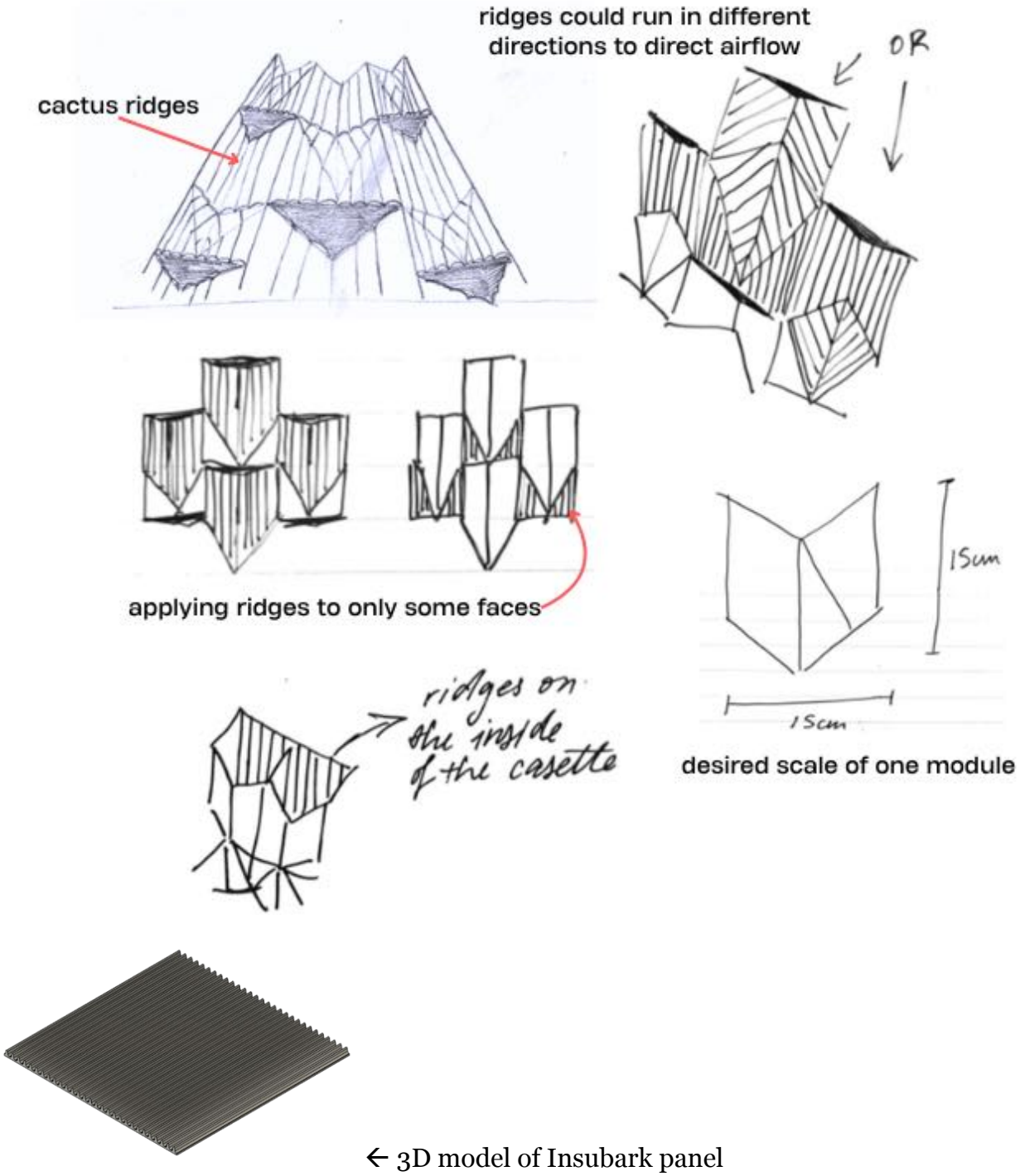


Figure 46

Considering a more 'biophilic' design that is rounder. This is also visually more similar to a succulent. Perhaps it could function better than the harsh angles of the previous design as well. There still needs to be research conducted into the functionality of an undulating surface like this, in terms of its possible better or worse ability to capture passing air currents and direct them into the cavity of the façade. If it is at least about the same effectiveness, then there is good reason to pursue this design further as it does lend more of a biophilic aesthetic which the problem owner or future clients might enjoy, and even perhaps the local wildlife.

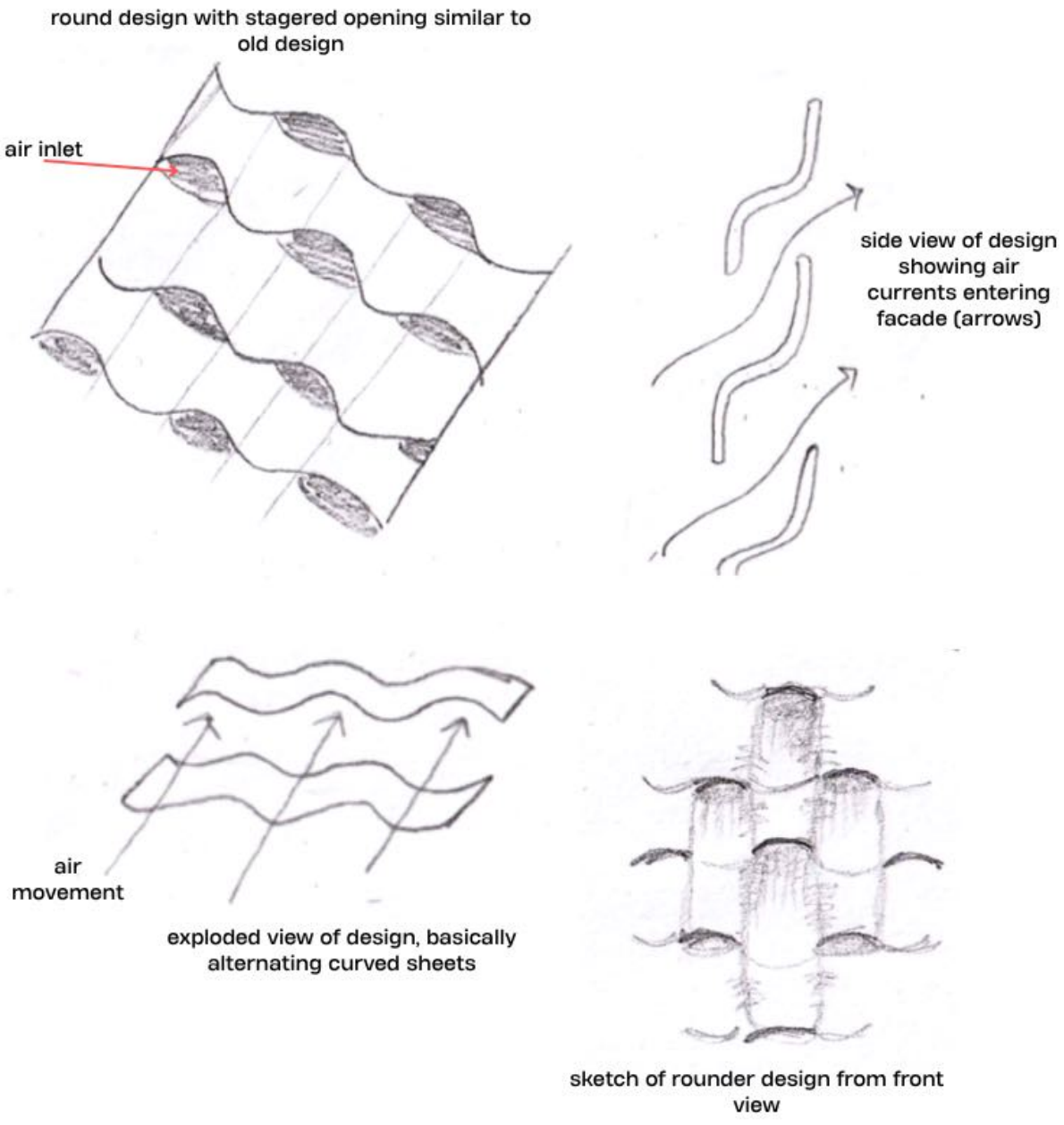


Figure 47

Results from the design aesthetics survey also provided some indication of what others found interesting and desirable. When it came to what aspect did best in each design, the timber geometric design performed best with a sense of balance and worst with natural shapes and forms; whereas the round mycelium design had a better sense of natural pattern and process but both scale and Gestalt principles were less evident to the viewers. Finally, when comparing the 2 designs/materials, the round mycelium design scored better, but only by 0.3 of a rating out of 5 (Appendix C, summarised in Table 3 below). Those surveyed also highlighted the importance of localised design (fitting local context and considering neighbourhood characteristics) which is in line with one of the sustainability principles of being locally attuned. It was understood that the timber geometric pattern suited the existing urban contexts, yet there was considerable interest in the rounder, mycelium design – indicating that both have their benefits.

Table 3: summary of design aesthetic survey results

Aspect in question	Average rating out of 5 (1 low, 5 high) (10 x responses)	
	Timber geometric design	Mycelium round design
Scale	3.7	3.6
Balance	4	4.1
Gestalt principles	3.9	3.6
Natural shapes and forms	2.8	3.8
Natural patterns and processes	3.4	4.2
Overall score	3.56	3.86

5. Final design selection

The final designs chosen for the project comes in 3 forms: an Orgago design (based on the Stag Façade) that will be made from timber; a hybrid design that incorporates the Orgago pattern as well as cactus style ridges; and finally, a rounder design envisioned as MBC that can inspire a more natural form in the urban environment.

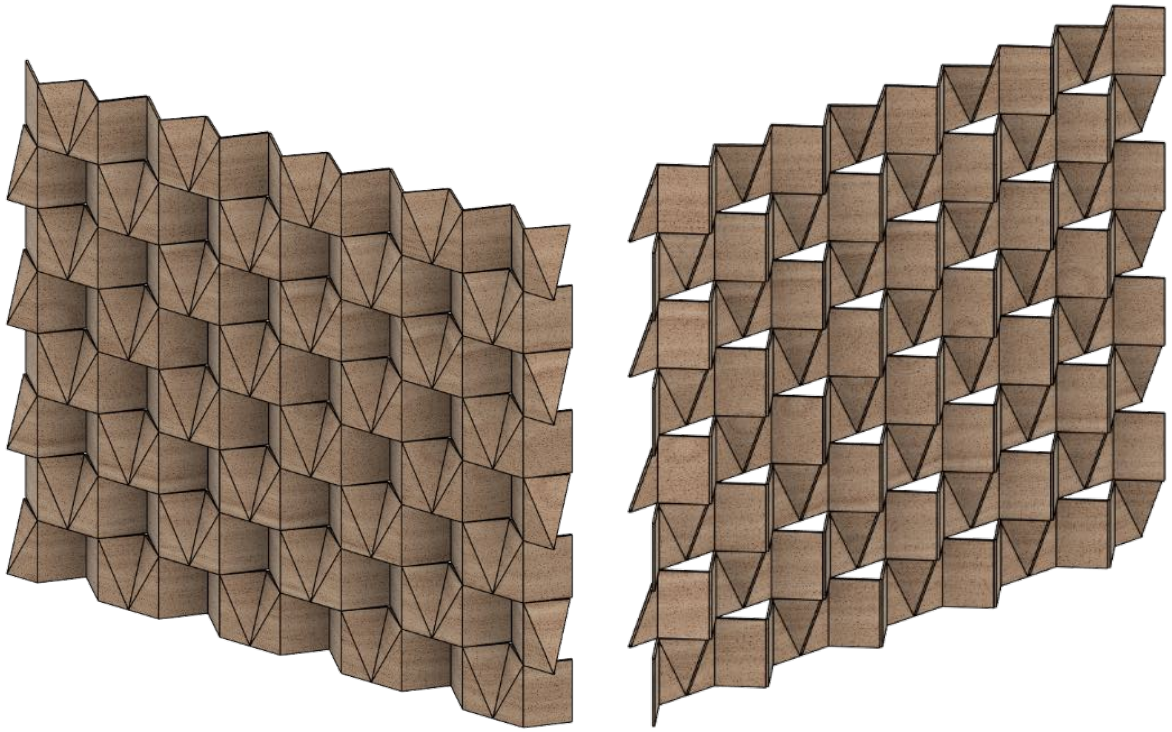


Figure 48: Fusion model of the Orgago design mocked up with timber as the material. On the left is the front, and on the right is the back face.

The gaps which allow for air to enter the façade and enhance the stack effect. The scale of this model is 120 cm x 135 cm. Each module (see below) will be 15 cm x 15 cm which can then be assembled into any desired scale to meet the requirements of the cassette or wall.

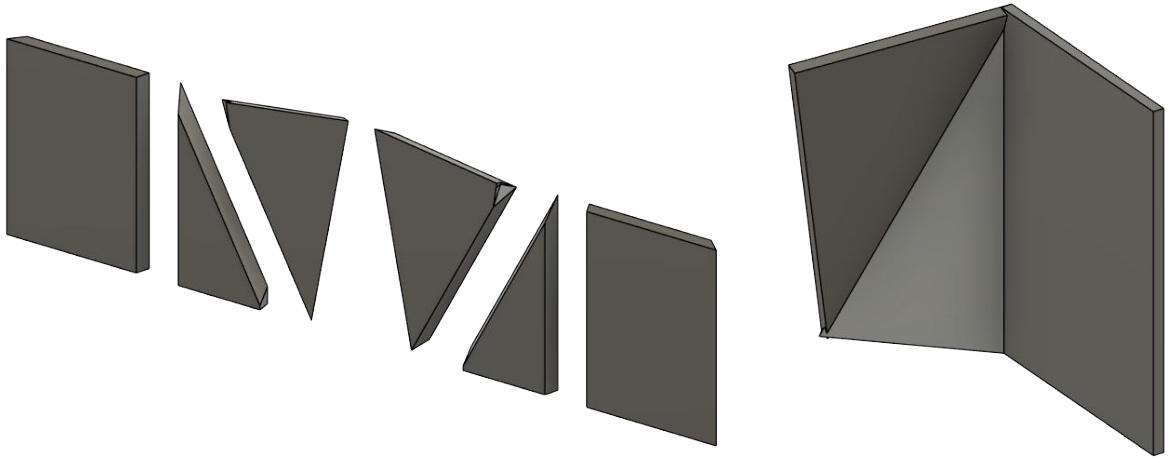


Figure 49: Fusion model of the individual modules that make up one "tile".

On the left are the individual pieces to make 2 tiles. There is a left and right version as this allows for the formation of those indents that lead air into the inlets. On the right is a left-side tile, see from behind so that you can observe how the pieces fit together.

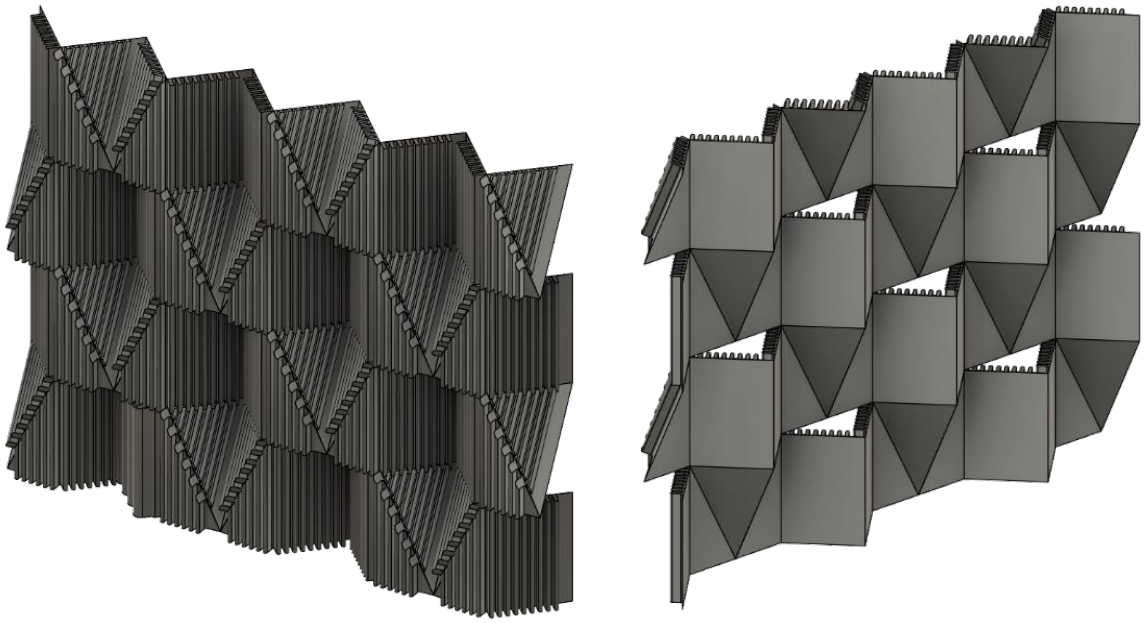


Figure 50: hybrid design of Orgago geometry with incorporated ridges.

The Orgago backing on the back view on the right. On the left are the ridges which follow the direction of each face towards the inlets. There are still some alterations to be made before it can be considered ready as the ridges are proving to require further level of design detail to ensure they run smoothly over the angled surfaces.

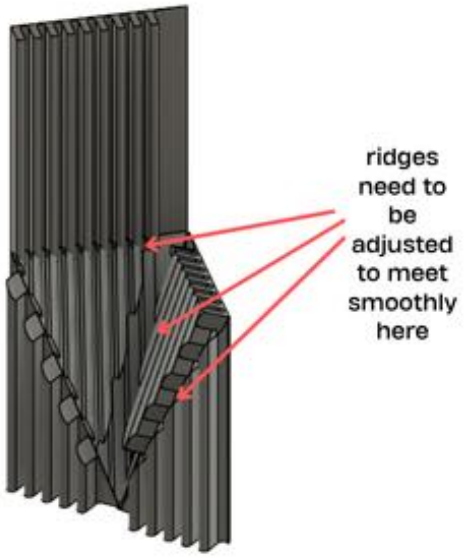


Figure 51: close-up view of a portion of the hybrid design.

There is still work to be done to best fit the ridges over the lip between faces and in the valley of the indent as well. Each of these ridges needs to be better fit to the shape before it can be CNC cut into a usable prototype.



Figure 52: Fusion model of smooth, mycelium based design.

The smooth design was modelled in Fusion 360 with the material envisioned to be mycelium. The smoother, wavy shape matches the material qualities - as the manufacturing process would involve pressing mycelium composite blocks into the desired shape. This design is intended to be a proof of concept as a future material, rather than a project focus since it would require far more research to ensure it also achieves the desired passive cooling function of the façade.

6. Implementation and testing

6.1 Prototyping/Manufacturing

The digital models in fusion 360 enabled initial and rapid pre-prototyping iterations. These iterations determined the necessary physical parameters of the design such as the functional angles, thickness of the material, and feasibility for scaling and testing.

6.1.1 Timber

A pre-prototype was made using laser cutting which allowed us to explore iterations early on. However, the machine's parameters are limited when it comes to material thickness, which is why CNC was chosen to for the final design. Additionally, the CNC allows for more intricate iterations of the design, and more closely resembles the method of construction when it comes to the final, scaled design for industry.

The final design will be made of Douglas Fir (refer to *Material Report*). However, for our first prototypes we used multiplex timber as a proof of concept since it is easier to construct iterations, and scrap material was available – in keeping with our aim to maintain sustainability wherever possible. Moreover, multiplex is not inherently unsustainable. It is largely determined by the glue used, while the combination of smaller sheets left over from other processes can be considered a good use of material since they are being diverted from landfill.

The following three figures show the prototyping process, starting with a rudimentary version of the design from scrap multiplex (Figure 53). From there we realised what details needed to be improved – mainly how the pieces within the tiles fit together to achieve a tight seam (Figure 54), which involved many 3D prints in small scale (Figure 55) to achieve this. Following this whole process, a more sophisticated version of the design can be made using the CNC machine to achieve smooth and aesthetically pleasing finishes and edges.

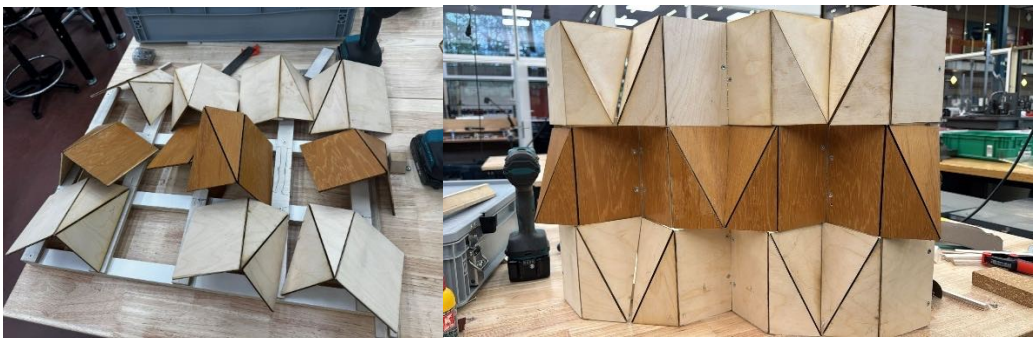


Figure 53: Timber pre-prototype made from laser cut scrap multiplex (hence the different types seen in photos) that was assembled at LPL. Note the gaps between the pieces which indicated the need for improving design so that the pieces can fit together flush.

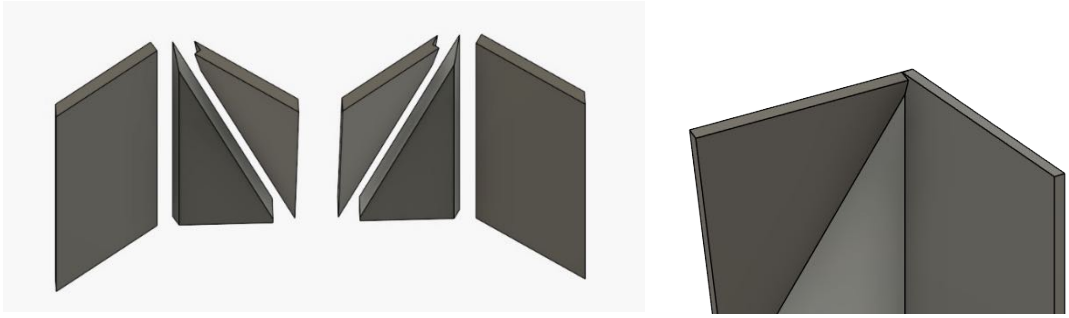


Figure 54: Improvement on the digital model in response to above pre-prototype results. Notice the joining top corner, this design ensures no gaps between pieces.

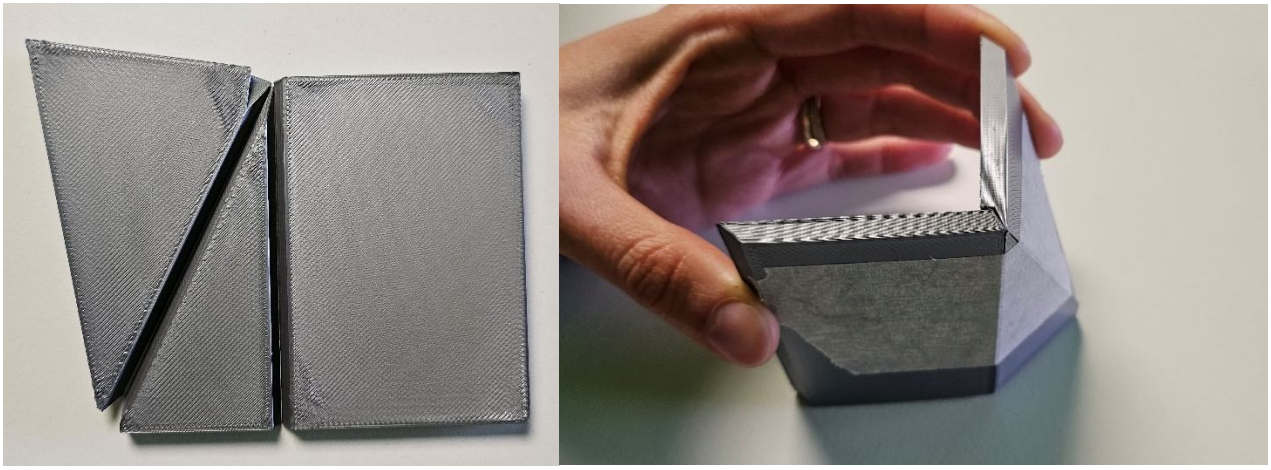


Figure 55: Iteration of the design using 3D printed PLA to determine the best method to fit three pieces together to form one tile of the design.

6.1.2 Mycelium

For this project, the MBC prototype will be in the form of a 10x10 cm tile only, used for experimental testing of the MBC surface, material properties and conceptual possibilities as a future material. Fungal experts (Dr Han Wösten, Dr Jordi Pelkmans, and Dr Pauline Krijgsheld) suggested pressed MBC, which can be achieved in the available fungal lab. This could be hot or cold pressed, possibly achieving different results in reflectivity and coating interactions. To our knowledge, mycelium/MBC has never been pressed into a 3D shape before, nor has it been used as a façade for a building. Pressing it into a complex design such as the Orgago shape will be difficult, therefore a simple, wavy design will be tested (also in line with expert advice).

For this project, we will grow the mycelium in cubic blocks and attempt pressing them into flat and 3D shapes. This is a functional first step in determining if the biomaterial can achieve a sturdy form, and its possibility as a material. Mycelium (*Fomes inzengae*) will be grown in a **10x10x3cm mould** on a substrate of hemp shives. They will be grown for a period of about 19-21 days (including time to grow skin) and will be dried at room temperature over 2 weeks. For the pressing of these blocks, heat (150°C) pressing will be tested, at differing lengths of time (5-20 minutes) to assess any variability in the resulting material. The shape of the press

will be flat, and also 3D thanks to an aluminium positive and negative mould we have designed and manufactured (Figure 56).

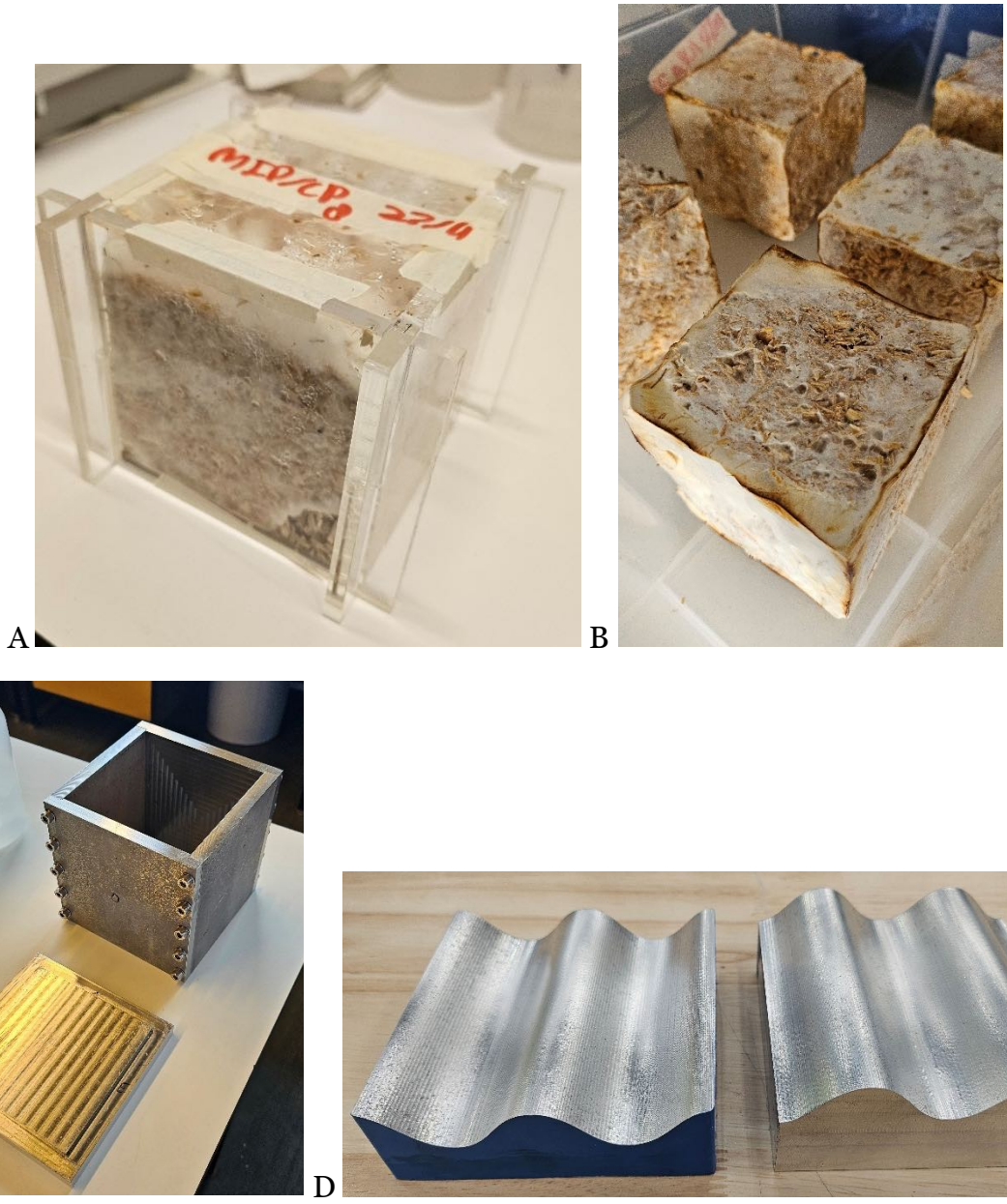


Figure 56: A: mycelium blocks in the acrylic mould; B: those blocks were later dried; C: the square, flat press that will be used to press blocks; D: the wavy, 3D moulds to press the mycelium blocks (left: top, right: bottom).

The smoother, wavy shape was chosen as it aligns with our interest in creating a more biophilic, rounded shape for the façade; as well as providing the best chance for success on this never before done process. The harsh angles of the Orgago shape were hypothesised to cause major cracking in the mycelium composite material, whereas the large angles of the smooth design allow for more movement to compensate for new form induced by the press.

In the beginning of the design process, it was hypothesised how the mycelium could be grown into and then pressed into the desired form, see below for a schematic method (Figure 57). This could make for interesting further research beyond this project. It will not be attempted in this project due to time constraints.

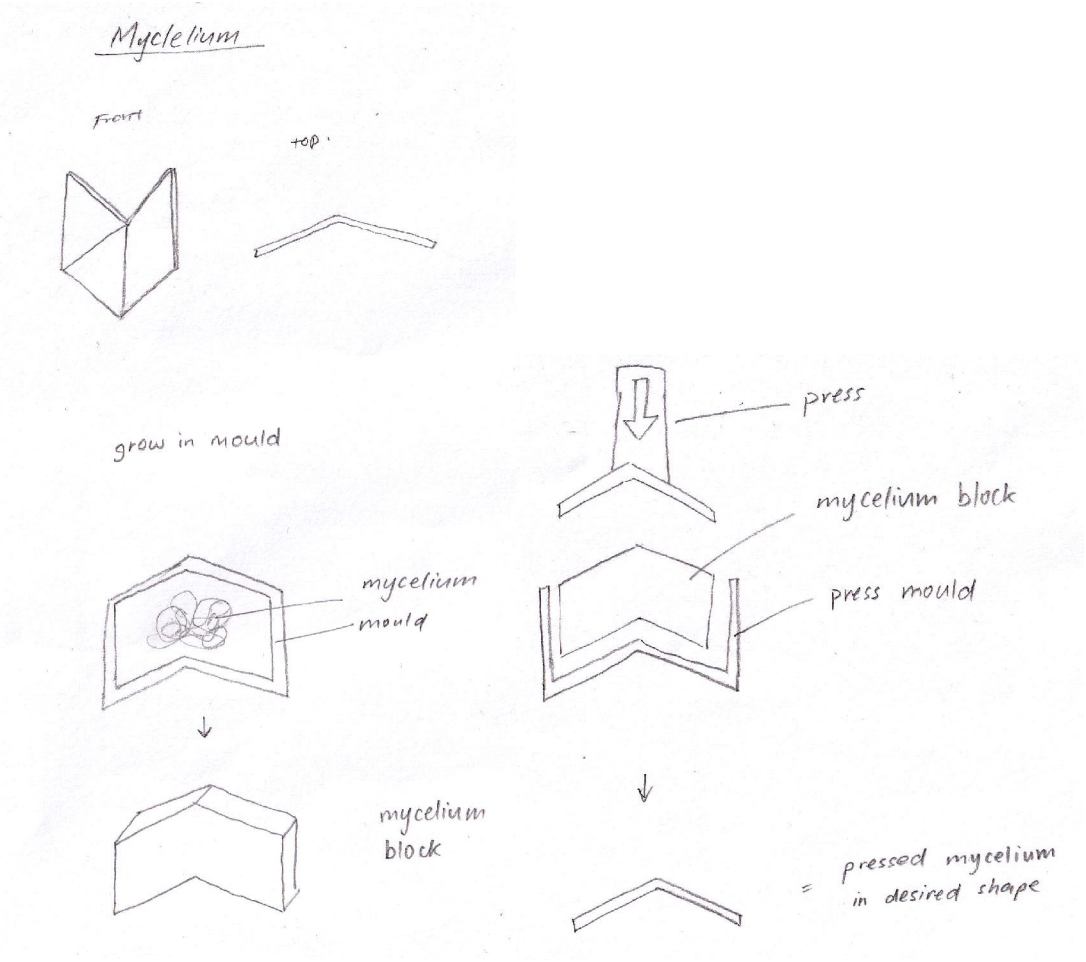


Figure 57: an idea of how mycelium could potentially be grown and pressed into a desired shape. Firstly, by growing it in a specially designed 3D mould, and then by using a press mould with a positive and negative side to press the mycelium block into that shape permanently.

7. Conclusion

The final designs chosen (Figure 58) have been informed by models found in nature such as desert cacti, termite mounds, ants, and butterflies. The design process has been guided by further literature research, expert consultation and material testing which has culminated in the below designs. The goal is that the façade can achieve passive cooling whilst maintaining a sustainable and nature focussed approach. The final prototype and experimental results will be discussed in the *Final Report*.

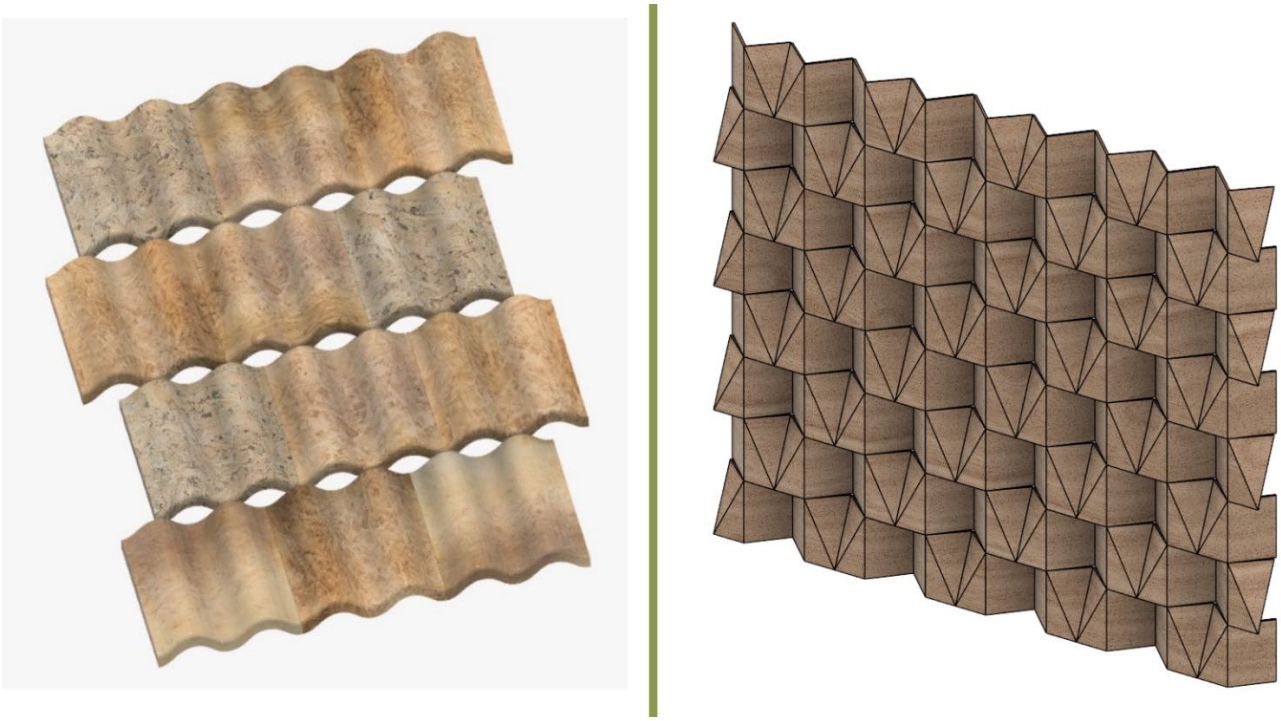
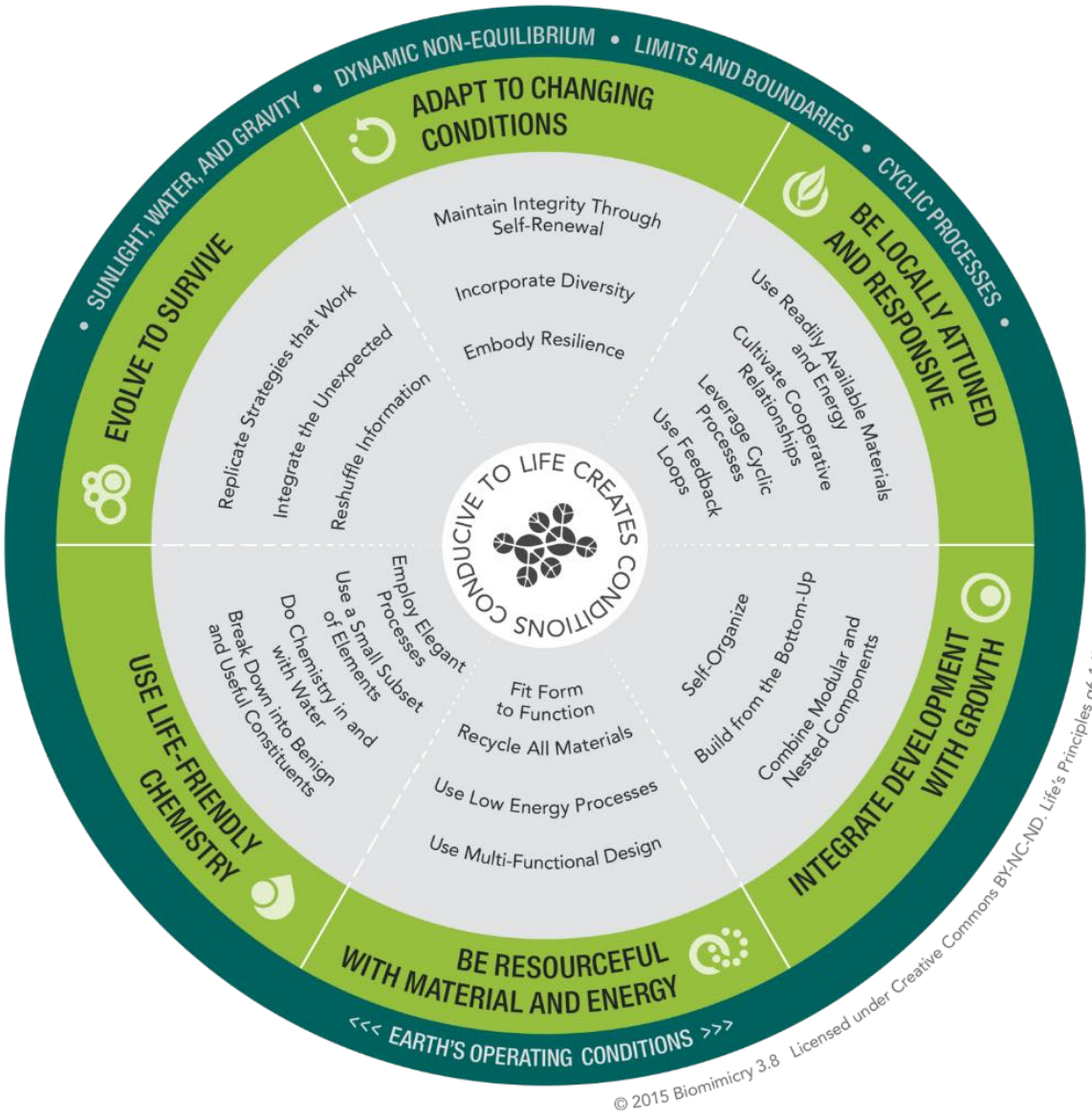


Figure 58: The final designs. Left: mycelium in a rounded design; right: timber in the Orgago design.

11. Appendix

A. Life's Principles

Life's Principles, which are used as a guide to ensure that the design and end product can contribute to a sustainable and/or regenerative future.



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B. Design Aesthetics Survey



This survey is to assess the aesthetic of the designs for building façade panels. These designs will be placed on external walls of a building to have a passive cooling effect. Based on visual design principles and styles, we have come up with 5 key principles to assess:

1. Scale
2. Balance
3. Gestalt principles
4. Natural shapes and forms
5. Natural patterns and processes



We will ask for a rating out of 5 to assess each of these principles for 2 different designs. This form should only take 5 minutes.

1. Firstly, how would you describe your aesthetic preference?
   - Minimalist ("less is more")
   - Bohemian (carefree, eclectic, breaks traditional rules)
   - Art Deco (glamorous, luxe, geometric, 1920's)
   - Brutalist, Bauhaus (bold, monumental, unapologetic)
   - Biophilic (incorporates nature, is like nature)
   - Contemporary green (experimental, hyper-current, eco-friendly, sustainable)
   - Modern (sleek, structural)
   - Neoclassical (grand, institutional, symmetrical)
   - Gothic (ornate, decorative, arches, vaults)
   - Other.
2. What do you value most, aesthetically, in the urban context, based on your preferred style? (e.g. material qualities, shapes, colours, textures, finishes, natural references etc.)

Looking at the images, please **rate design A and B on a scale from 1-5 (1 =low, 5= high)** on **how well the design ...**

3. Achieves a pleasing sense of scale (uses relative sizes to illustrate a pleasing composition).
   The imagined facade feels welcoming rather than intimidating; the design scale makes sense for the urban context; the scale communicates its purpose as both a facade and passive function of the building and environment; it's not too big or too small (pictured prototype is tiles of 10x10 cm).

4. Achieves a pleasing visual balance (evenly distributed visual signals).
   The design doesn't feel lopsided; you can imagine light and shade feels evenly distributed on the building wall; it looks stable and efficient; it balances form and function in a stylish way.

5. Captures a pleasing whole that is made up of individual elements.
   People tend to perceive the whole design rather than its individual parts. How well does the design unify the small, complex parts into a coherent, whole facade? the design should portray a unified purpose and rhythm.

6. Depicts natural shapes and forms that simulate the natural world.
   Biological patterns are translated into shapes and patterns of the building facade. Shapes and textures similar to what's found in cacti, termite mounds, burrows or plants are clearly referenced in the design.

7. Incorporates properties found in nature (not just simulates them).

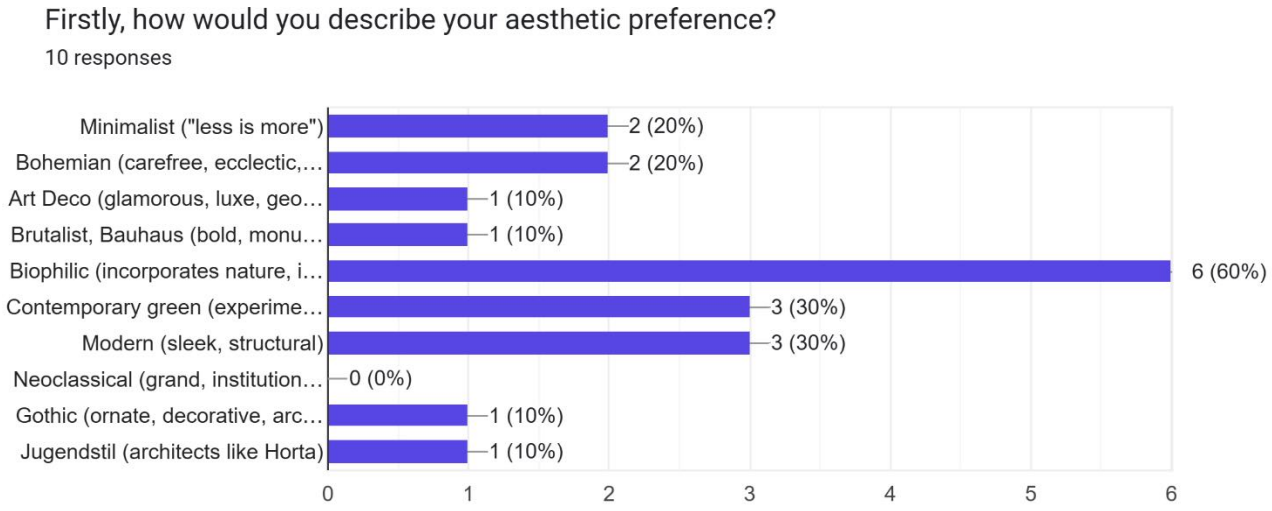
The design feels natural, you can identify natural materials used, you imagine the manufacturing process could be sustainable, it adds a natural feel to urban contexts.

Which design would you rather see in your city?

- Design A: Mycelium in rounded shapes
- Design B: Timber in geometric shapes
- Combination 1 - mycelium in geometric shapes
- Combination 2 - timber in rounded shapes

Optional: Any final thoughts you'd like to add? (what you like/dislike, how you imagine it in context, would you put this on your house, etc.)

C. Design Aesthetics Survey Results



What do you value most, aesthetically, in the urban context, based on your preferred style? (e.g. material qualities, shapes, colours, textures, finishes, natural references etc.)

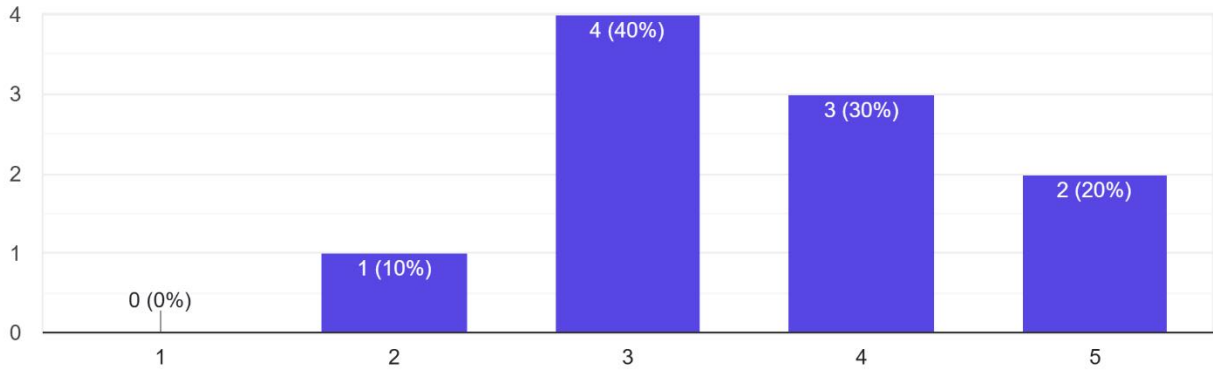
9 responses:

- (Material) quality, colour
- Calming flow and texture
- Colorful architecture combined with lots of nature
- Natural references
- Shapes and textures.
- colour, shape
- A working prototype we'll need in the future probably a range of designs that can be integrated in different styles of architecture
- colours, natural references

- textures

Design A: Mycelium in a rounded shape 1. Achieves a pleasing sense of scale (uses relative sizes to illustrate a pleasing composition). The imagin...small (pictured prototype is modules of 10x10 cm).

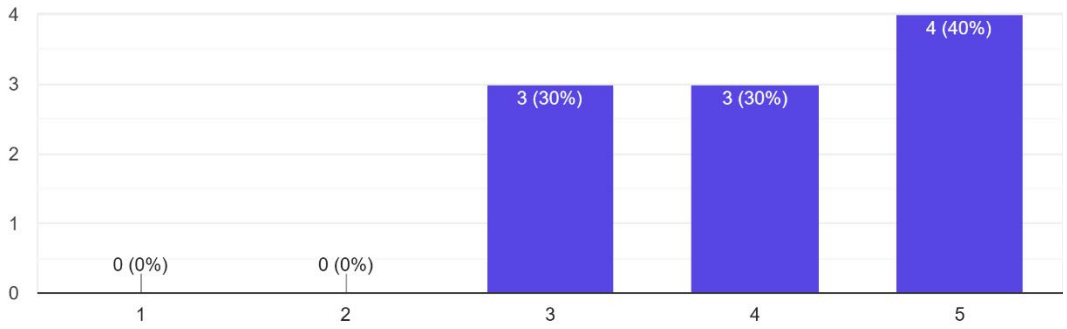
10 responses



Average rating = 3.6/5

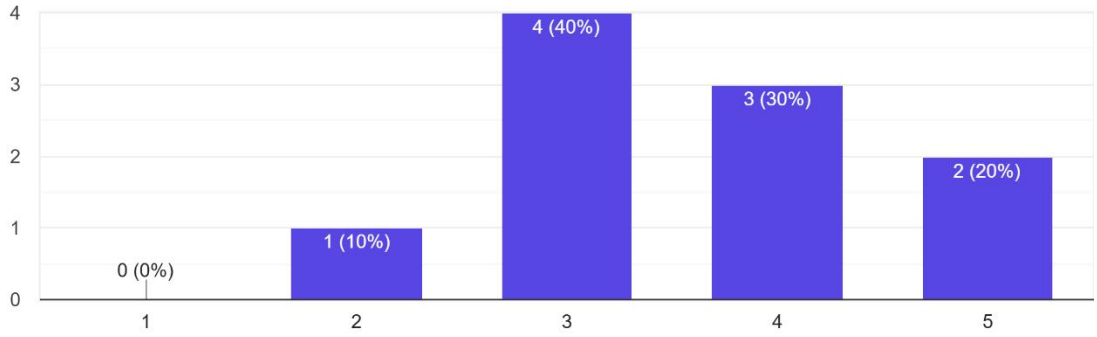
2. Achieves a pleasing visual balance (evenly distributed visual signals). The design doesn't feel lopsided; you can imagine light and shade feels even...t, it balances form and function in a stylish way.

10 responses



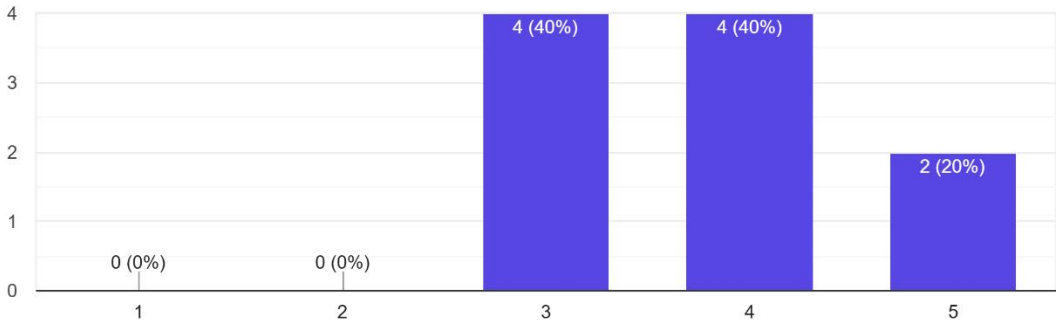
Average rating = 4.1/5

3. Captures a pleasing whole that is made up of individual elements. People tend to perccieve the whole design rather than its individual parts. How ...esign should portray a unified purpose and rhythm.
10 responses



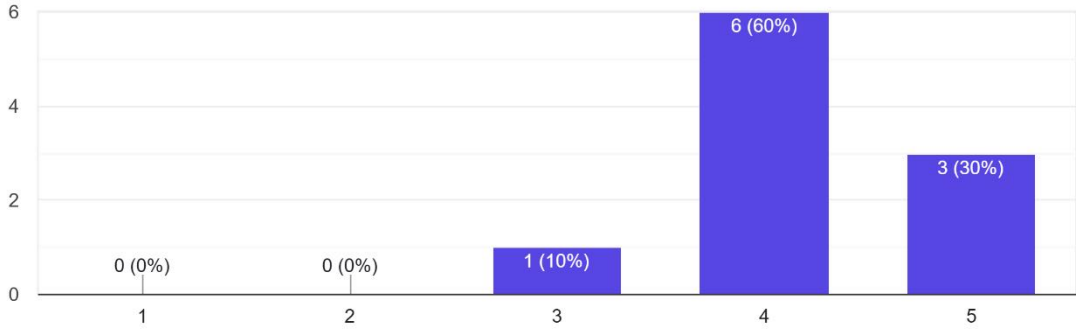
Average rating = 3.6/5

4. Depicts natural shapes and forms that simulate the natural world. Biological patterns are translated into shapes and patterns of the building ...rows or plants are clearly referenced in the design.
10 responses



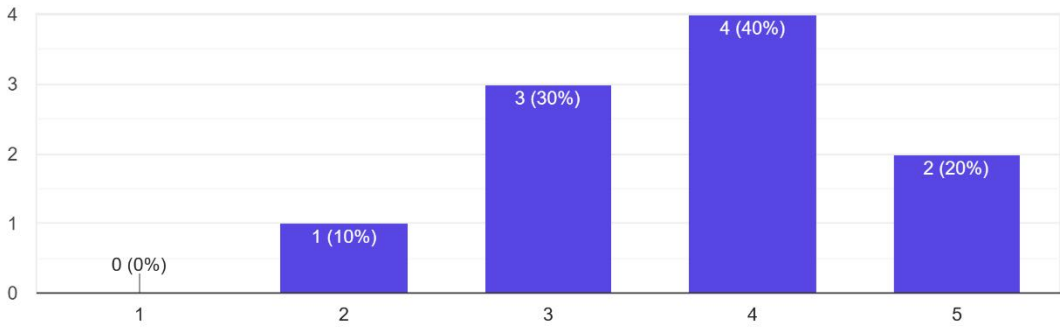
Average rating = 3.8/5

5. Incorporates properties found in nature (not just simulates them). The design feels natural, you can identify natural materials used, you imagine the ...tainable, it adds a natural feel to urban contexts.
10 responses



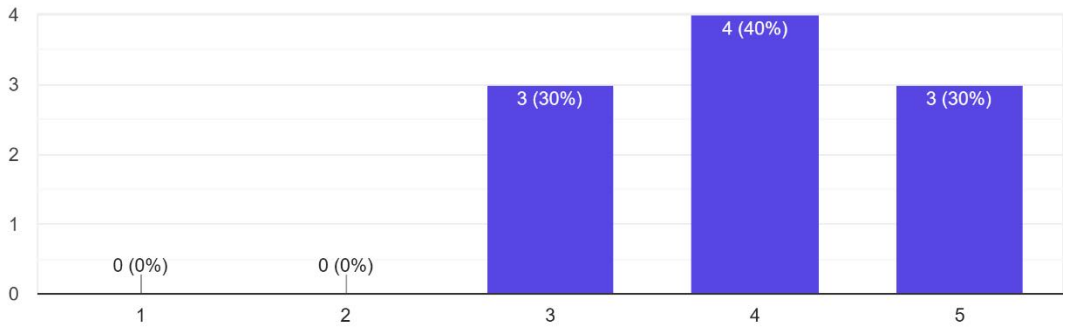
Average rating = 4.2/5

Design B: timber in geometric shapes 1. Achieves a pleasing sense of scale (uses relative sizes to illustrate a pleasing composition). The imagined ...small (pictured prototype is modules of 15x15 cm).
10 responses



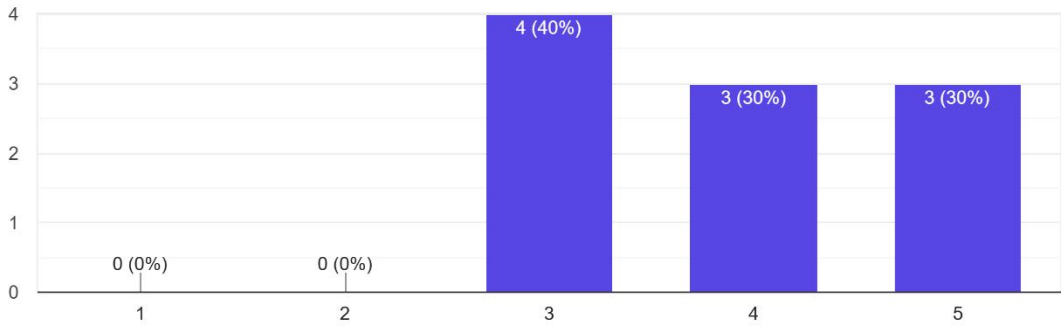
Average rating = 3.7/5

2. Achieves a pleasing visual balance (evenly distibuted visual signals). The design doesn't feel lopsided; you can imagine light and shade feels even...t, it balances form and function in a stylish way.
10 responses



Average rating = 4/5

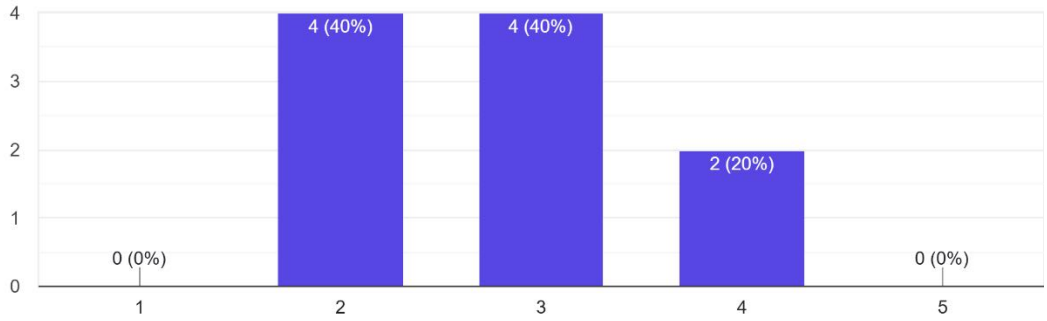
3. Captures a pleasing whole that is made up of individual elements. People tend to perccieve the whole design rather than its individual parts. How ...esign should portray a unified purpose and rhythm.
10 responses



Average rating = 3.9/5

4. Depicts natural shapes and forms that simulate the natural world. Biological patterns are translated into shapes and patterns of the building ...rows or plants are clearly referenced in the design.

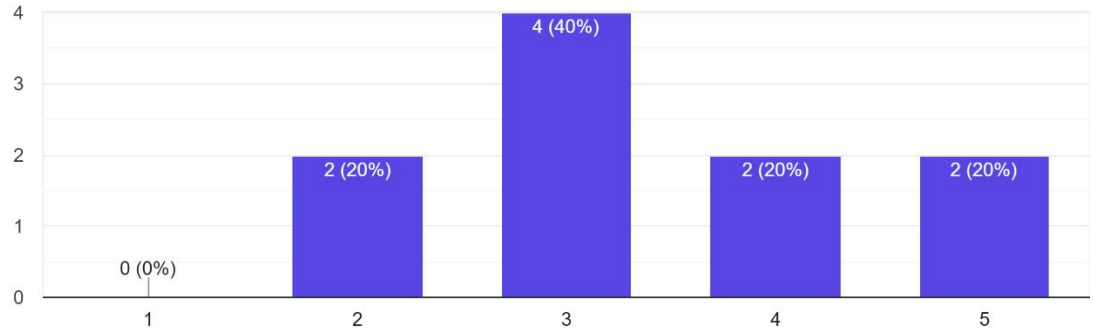
10 responses



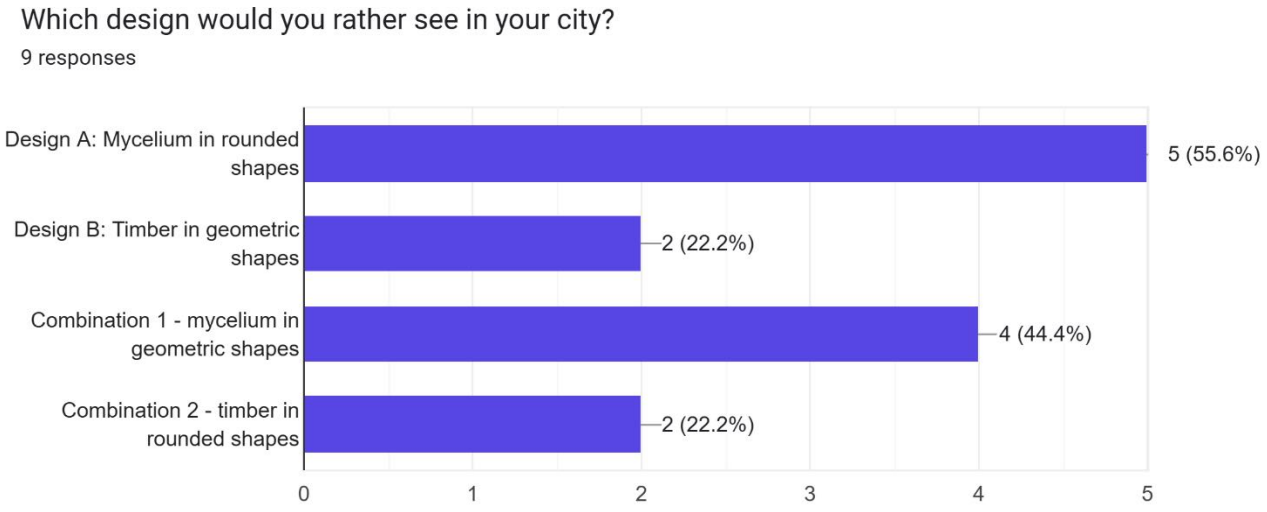
Average rating = 2.8/5

5. Incorporates properties found in nature (not just simulates them). The design feels natural, you can identify natural materials used, you imagine the ...tainable, it adds a natural feel to urban contexts.

10 responses



Average rating = 3.4/5



Optional: Any final thoughts you'd like to add? (what you like/dislike, how you imagine it in context, would you put this on your house, etc.)

5 responses:

- The geometric lines are not really my style but fit the modern urban context.
- I would put design A in my home
- I think you are working on two nice alternatives, I would not choose because it depends very much on te particular building and its surroundings what would be a 'good' design.
- Liked the one without holes for better functionality and noise proofing but shape of A
- I'd be very interested to see how the facades develop over time and how this affects the physical form and resulting aesthetic - how do they age & patina, how do they incorporate this patina and can this be an opportunity ?

Appendix E. LCA: Cooling panel made from wood and mycelium

LCA: Cooling panel made from wood and mycelium

Goal and scope

This study presents a comparative Life Cycle Assessment (LCA) of two alternative materials, pressed mycelium-based composite (MBC) and different types of wood, for the production and assembly of a modular façade tile to mitigate the urban heat island effect. The façade tile is specifically designed to enhance air convection along the building envelope, thereby contributing to passive cooling and improving overall building energy efficiency.

Façade cladding panels or tiles are the outermost layer of a building, the so-called skin. Wood has historically been used as a material for façade cladding and recently its use in construction has become increasingly popular, as it is sustainable and has become more durable and weather resistant due to advanced wood modification techniques (Hill et al., 2022). MBCs are novel bio-based materials, whose application in construction is currently being explored in literature. It has low embodied energy and carbon, can be locally produced and is biodegradable. Additionally, it has low density, inherent fire resistance and acoustic absorption (George et al., 2026; Lewandowska et al., 2025). However, there are concerns regarding the durability and mechanical properties of MBCs for outdoor application, which has limited application in the construction industry thus far (Lewandowska et al., 2025).

The primary objective of this LCA is to evaluate and compare the environmental impacts associated with the material choices used in the tile's production and assembly. By focusing on the material stage, the study seeks to quantify the differences in environmental performance between pressed MBC and wood, providing insight into their applicability as sustainable construction materials for passive cooling façade panels. The environmental impact should be considered in relation to the material's lifespan. A material that is more energy-intensive to produce will have a higher environmental impact per unit, but it may also last considerably longer. When these factors are evaluated together, a higher production impact can be justified if the increased durability means the material needs to be replaced less frequently over the building's lifetime.

Systems in scope

Research has been done to choose the most suitable type of wood for this project. Plywood has been chosen as the type of wood to prototype, while Douglas fir and thermally modified wood have been suggested as suitable wood types for later iterations of the panel. To make a complete assessment, all wood types will be part of this LCA. Many studies have been conducted on the environmental impact of the production of 1 m^3 of different types of wood (cradle-to-gate) (Adhikari & Ozarska, 2018; González-García et al., 2013, 2014; Hill et al., 2021) and will therefore not be discussed in this study but simply used as input for the LCA calculations.

For the production of a 15x15 cm modular piece of a panel, a rectangular piece of 17 x 21.33 cm is cut from a bigger sheet with a circular saw (Figure 1). One piece will then be milled to the correct shapes and edges using computer numerical control (CNC), after which it is glued together and one piece is formed. The production process for a prototype using Douglas fir or

thermally modified wood is similar, besides that these types of wood are not available in sheet material but in planks.

The process of making MBCs is described in a protocol provided by dr. Pelkmans (Microbiology, Utrecht University) and visualized in Figure 2. In short, *Fomes inzengae* is grown for five days on potato dextrose agar (PDA) at 30˚C. The resulting mycelium is sliced into 5 mm cubes, transferred to a sterilized aeriation bag filled with reclaimed cellulose spawn at 80% water saturation, and incubated for four days at 30°C. The resulting spawn is mixed at a 1:10 ratio with sterilized hemp shives at 80% water saturation, placed inside a mould (10x10 cm, aluminium), incubated for 7-10 days at 30°C, and left to dry on a table at room temperature Pressing of the MBC was done at 11.5 MPa and 150°C using a Carver 3889 (hydraulic, heated press).

Functional unit

For this study, 1 m^2 of façade coverage is the functional unit as frequently adopted in other LCA studies (Jandaghian et al., 2025). The size of one modular façade tile is 15 x 15 cm, which would mean that covering 1 m^2 of façade equals to 44.4 tiles. The per-tile inventory can therefore be multiplied by 44.4 to calculate the impact of covering 1 m^2 of façade by a 20 mm thick panel. For the mycelium material a basic element of 15 x 15 cm at an unpressed thickness of 3 cm weighs 136 g, meaning a 1m² façade section covered by these panels would weigh 6.0 kg. After heat-pressing the 3 cm thick panel a thickness of 15 mm is expected.

System flowcharts

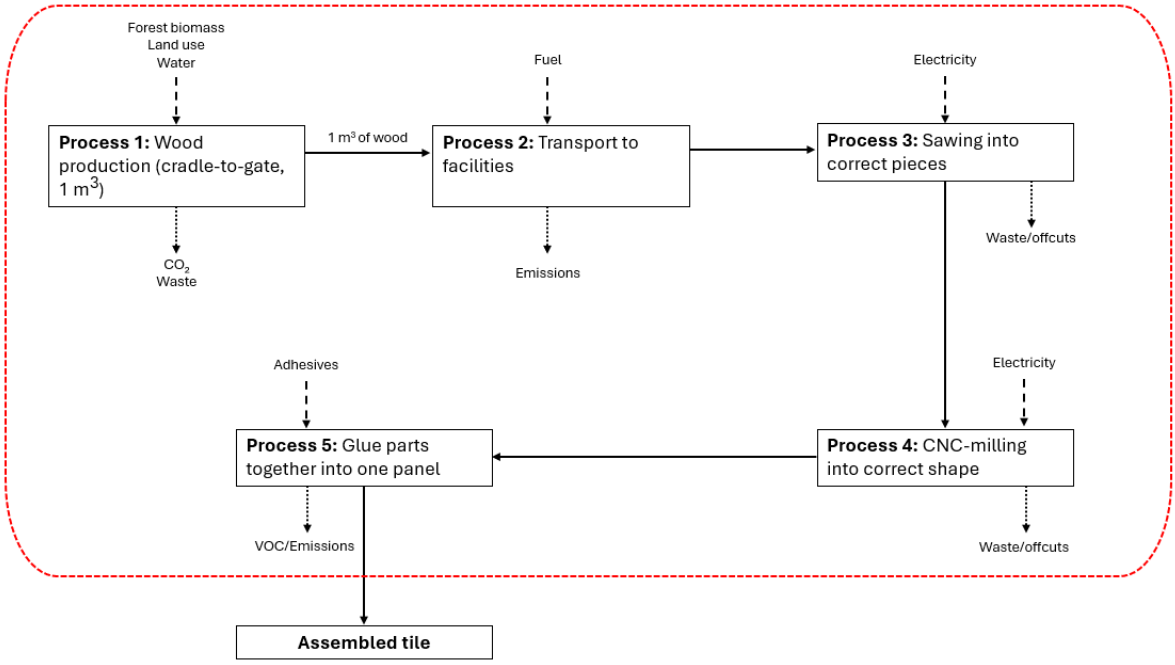


Figure 1. Cradle to gate system flowchart of production of one 15x15 cm wooden façade tile. Red marked line is the system boundary.

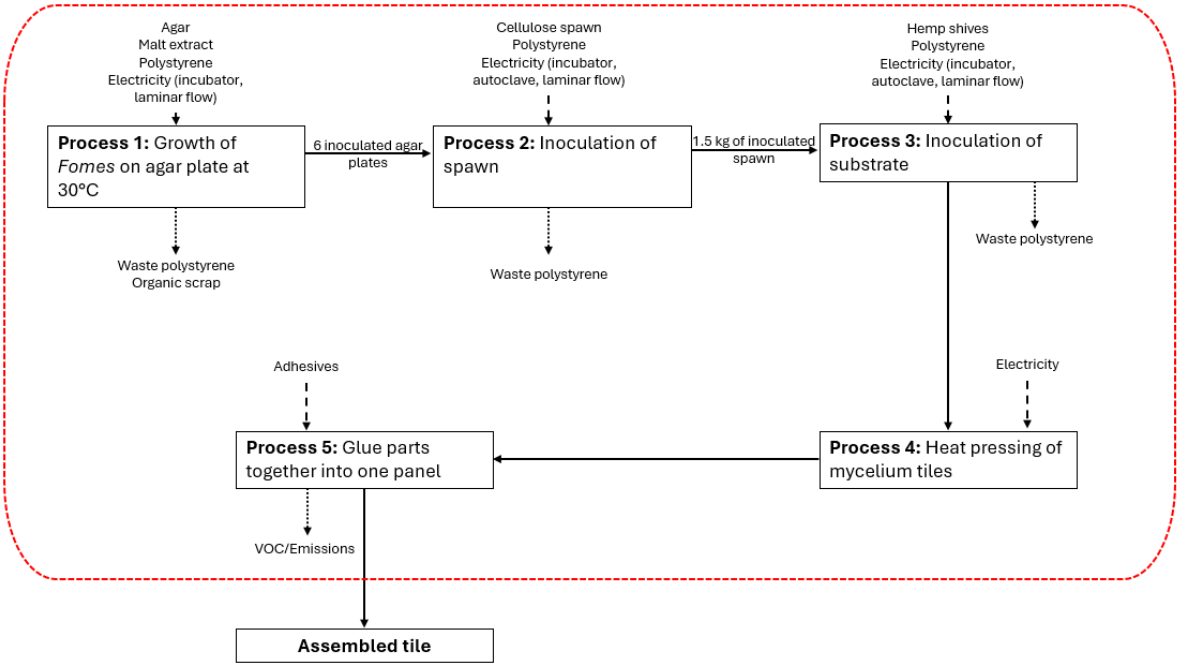


Figure 2. Cradle to gate system flowchart of production of one 15x15 cm mycelium composite façade tile, cradle to gate. Red marked line is the system boundary.

Cut-off criteria

Inputs and outputs contributing < 1% of the calculated global warming potential (GWP) were excluded, provided that the total sum of the excluded flows remains < 5% of the total calculated GWP. The energy use of the circular saw was estimated at ~0.05 kg CO_2e/FU (<1% of the total) and therefore excluded from the Life Cycle Inventory (LCI). Additionally, the cellulose contributed < 1% of the total GWP of the mycelium panel (0.065 kg CO_2e/FU) and was therefore excluded from the study. The Rotterdam-Utrecht transport for both types of plywood also fall within < 1% criterion and could therefore be excluded. However, it was retained as it was already quantified and its inclusion improves completeness without affecting the result.

System boundaries

The conducted LCA is cradle-to-gate. For the wood, besides the material manufacturing, the transport to the facilities, processing of the material (sawing and CNC machining) and panel assembly is included in the system boundary. For pressed MBC, the system boundary includes substrate preparation, inoculation, incubation, heat-pressing and panel assembly. The use-phase and end-of-life scenario are excluded for both the wood and the pressed MBC. The output of this LCA thus is the environmental impact of the production of 1 m^2 of façade coverage with the panels. With this output, combined with the reported lifespan, we can assess whether wood or pressed MBC is the most sustainable option to use. The durability of both wood and pressed MBCs can be increased with the use of coatings or paints. However, this is excluded from this analysis. This decision was made on the grounds of the variability of the coatings to be used for the panel and by the assumption that the same coatings could be applied on both the mycelium and wood cladding.

Assumptions

1. TMS is assumed to be produced and imported from Estonia by Thermory, as it is the largest Estonian exporter of TMS (360 Research Reports, 2025; Hill et al., 2021).
2. "Short cycle" (< 100 years), i.e. biogenic CO_2 is not counted in the LCA. Most LCAs on wood exclude short cycle CO_2, following IPCC's approach (SSIM, 2022).
3. It was not possible to confirm whether the indoor and outdoor plywood entries in Idemat had the same system boundaries as sawn-wood entries (cradle-to-gate production including transport from the country of origin to Rotterdam). Therefore, it is assumed that Idemat uses consistent system boundaries and both plywood entries follow the same methodology as sawn-wood entries. As the transport flows represent < 5% of the total impact, this assumption does not majorly impact the final result. It is however noted as a limitation.
4. Both wood and mycelium tiles are assumed to be glued using the same type and quantity of adhesive. This would therefore not affect the comparison and is therefore excluded from the inventory analysis.
5. The aluminium mould is not expected to deform in a realistically achievable number of cycles (see Calculations). At a theoretical use for 1000 functional units the inputs and processes to create the mould already fall below the cut-off criterium, therefore the production of the mould is excluded in the final result.
6. Due to the restriction posed by the pressing the material, the moulds made for growing the mycelium blocks were also made in the dimensions of 100 x 100 x 30 mm. Calculations are already made based on mass of material used and therefore the climate change impact per FU also remains the same, since the thickness of the material does not change.

Inventory analysis (LCI)

Data collection procedures & key numbers needed

Data was collected from a combination of the Ecoinvent and Idemat databases, literature and own measurements. For each input and output, available data most closely representing the production process (as described in the systems flowchart) was used. For the Ecoinvent data entries, the closest to the "NL" region were chosen in selecting a production site. In the case of insufficient data availability for the exact input/output, the most similar entry available was used instead. In cases where specifically measured data for production inputs was available outside of the Ecoinvent or Idemat databases, such as from literature or own measurements, this data was used instead.

Wood

Table 1: Source of the entries used for the calculation of the impact of wooden panels. The category and entry name refer to the classification and name used in the respective database. For Ecoinvent entries, system models is given in the entry name (APOS: Allocation at the Point of Substitution; S: system process), and the year refers to the start of the data collection. For Idemat, the year refers to the publication year of the source.

Input/ Output	Database	Category	Entry name	Region	Year
Douglas fir	Idemat	Woods, Class III, 25-40 years (m3)	Oregon pine/Douglas (FSC, PEFC)	Vancouver, Canada	2017
Plywood indoor	Idemat	Woods, Wood products (m3)	Plywood indoor	Unknown	2017

Plywood outdoor	Idemat	Woods, Wood products (m3)	Plywood outdoor	Libreville, Gabon	2017
Thermally modified spruce	(Hill et al., 2021)	-	TMT spruce	EST	2014
28t truck transport (t.km)	Idemat	Other, transport	Truck, container, 28t (ton.km)	Unknown	Unknown
Electricity	Ecoinvent	3510:Electric power generation, transmission and	Market for electricity, low voltage | electricity, low voltage | APOS, S	NL	2020

Many LCAs and Environmental Product Declarations (EPDs) have been conducted calculating the environmental impact of wood production (González-García et al., 2013; Hill et al., 2021). Idemat provides the numbers on different types of wood (Table 1). The numbers for the production of 1 m^3 wood that were available in Idemat were used as input for wood. For thermally modified wood, no data was available in Idemat and data was sourced from scientific literature (Hill et al., 2021).

Idemat draws their system boundary at the harbour of Rotterdam, therefore only the transport from Rotterdam to hardware stores Utrecht (61 km) by truck is included in the calculations (SSIM, 2022).

Data gathered in Hill et al. (2021) regarding thermally modified spruce is from Estonia. Thermory is the largest Estonian exporter of thermally modified wood, with factories close to Tallinn (360 Research Reports, 2025). Therefore, imported thermally modified spruce is assumed to be transported via truck from Tallinn to Utrecht (2134 km).

In 45 minutes of continuous operation, 349 Wh were used by the CNC machine. This was converted to an hourly power usage of 465.3 Wh or 1.674 MJ.

Mycelium based composite

Table 2: Source of the entries used for the calculation of the impact of the mycelium panel. The category and entry name refer to the classification and name used in the respective database. For Ecoinvent entries, system models is given in the entry name (APOS: Allocation at the Point of Substitution; Cutoff: cut-off by classification; S: system process; U: unit process), and the year refers to the start of the data collection. For Idemat, the year refers to the publication year of the source.

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Input/ Output	Database	Category	Entry name	Region	Year
Hempshives	Ecoinvent	0116: Growing of fibre crops	Sunn hemp production | sunn hemp plant, harvested | Cutoff, U	RoW	2016
Cellulose	Ecoinvent	1709: Manufacture of other articles of paper and paperboard	Cellulose fibre production | cellulose fibre | APOS, S	RoW	2012
Demineralized water	Ecoinvent	3600: Water collection, treatment and supply	Water production, deionised | water deionised | APOS, S	RER	2012
Potato	Ecoinvent	0112: Growing of vegetables and melons, roots and tubers	Potato production | potato | APOS, S	RoW	2001
Dextrose	Ecoinvent	1062: Manufacture of starches and starch products	Glucose production | glucose | APOS, S	RER	2015

Agar	Ecoinvent	0145: Raising of swine/pigs	Swine production | swine for slaughtering, live weight | APOS, S	RoW	2024
Polystyrene plate	Idemat	Plastics	PS (GPPS)	Unknown	2020
Waste polystyrene	Ecoinvent	3822: Treatment and disposal of non- hazardous waste	Treatment of waste polystyrene, municipal incineration | electricity, for reuse in municipal waste incineration only | APOS, S	GLO	2006
Electricity	Ecoinvent	3510: Electric power generation	Market for electricity, low voltage | electricity, low voltage | APOS, S	NL	2020
Aluminium	Idemat	Common Metals	Aluminium, secondary	unknown	2016

The data for the hemp shives that were used as substrate for the mycelium production was not available in the databases used. Volk et al. (2024), who conducted an LCA on a similar mycelium-based composite, used the Ecoinvent data for Sunn hemp. Therefore, the same dataset was adopted as a substitute for the hemp shives façade (Table 2).

The PDA plate was weighed on a scale two times, one time filled with the medium and one time as the empty plastic container, to determine the weight of the polystyrene as well as the medium used in a plate. A standard recipe for PDA uses four g of potato extract, and 20 g of dextrose and Agar to obtain one Liter of medium (Smit et al., 2011). To get 4 g of potato extract an infusion of 200 g of potatoes is needed (Aryal, 2022).

In comparable LCAs, a common replacement for the dataset for agar is gelatine (Stelzer et al., 2021; Volk et al., 2024b). However, within the Ecoinvent database, no data for gelatine production is available. Therefore, the best proxy available and therefore used was that of swine raising for slaughtering, as gelatine is typically extracted from swine skin (Rather et al., 2022). For the energy use of the laboratory equipment, the Ecoinvent entry for end user electricity was taken.

Saluena Martin et al. (2026) conducted a cradle-to-gate analysis of reclaimed cellulose used for the production of mycelium insulating materials. As confirmed with one of the co-authors of this study, the equipment used was the same as for the production of our mycelium-based façade panel. Therefore, the measurements for the energy use of the autoclave, laminar flow chamber and incubator, as well as the holding capacity of the incubator, was taken from this paper. Over a period of 1 h and 28 min of continuous heat pressing at 150°C and 115KN the heat press consumed a total 0.594 KWh energy. This implies an hourly energy consumption of 0.405 KWh or 1.458 MJ for standard operation without breaks. While Saluena Martin et al. (2026) used the same reclaimed cellulose as we used as spawn for the mycelium, this data was not taken from the paper, as they used a different form of carbon allocation that included biogenic carbon changes. Instead, a more general data entry from Ecoinvent was taken.

Standardized unit

The standardized unit chosen for this LCA is CO_2 equivalent (CO_2e). This metric measures the effect of a production process or product on the greenhouse effect by normalizing all emissions generated during production to the reference gas CO_2. It therefore represents the

amount of CO_2 that would cause the same global warming effect as all the greenhouse gases emitted during the product's production (Myhre et al., 2013). Using this metric as a standardized unit is in line with industry trends of LCAs in the building sector (Barbhuiya & Das, 2023), as well as being a frequently used metric for assessing biomaterials (Pawelzik et al., 2013). Comparable LCAs on mycelium composite materials and wood cladding use CO_2e. as standardized unit (Cordier et al., 2022; Stelzer et al., 2021; Volk et al., 2024b). Furthermore, the emphasis on climate change as the primary environmental performance indicator follows from the product's intended function. The façade panel is intended to mitigate the UHI effect, which is worsened by rising temperatures (Lizana et al., 2026). Therefore, a panel with a large climate impact would undermine the very problem it aims to combat.

Input/output table

Below are the simplified input/output tables for the production of 1 m^2 of façade coverage with panels from different material types (Table 3). Complete inventory tables with all inputs and calculations are available in the Appendix**.**

Table 3: Input/output table of 1 m^2 of façade panels made with different material types (Douglas fir, indoor plywood, outdoor plywood, thermally modified spruce and heat-pressed mycelium-based composites).

Douglas fir		
Input	**CO2 eq/FU (kg)**	**Impact contribution to total (%)**
Douglas fir (Idemat)	3,02017	75,34%
Transport		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,078189251	1,95%
Processes		
CNC milling	0,910473449	22,71%
Total	**4,0088327**	**100,00%**
Indoor plywood		
Input		
Indoor plywood (Idemat)	12,524988	92,58%
Transport		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,09289812	0,69%
Processes		
CNC milling	0,910473449	6,73%
Total	**13,52835957**	**100,00%**
Outdoor plywood		
Input		
Outdoor plywood (Idemat)	14,529823	93,63%
Transport		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,0774151	0,50%
Processes		
CNC milling	0,910473449	5,87%
Total	**15,51771155**	**100,00%**
Thermally modified spruce		
Input		

Thermally modified spruce (Hill, 2021)	20,303	91,87%
Transport		
Tallinn to Utrecht, 2134km, 28t truck, m3km	0,8850765	4,01%
Processes		
CNC milling	0,910473449	4,12%
Total	**22,09854995**	**100,00%**

Mycelium based composite		
Input		
Hemp Shives	0,24819328	2,19%
Agar Plate (total)	0,199473256	1,76%
Processes		
Autoclave	2,209838	19,49%
Incubation (spawn)	0,261578	2,31%
Incubation (block)	5,231568	46,15%
Heat pressing	2,248935	19,84%
Working under Laminar flow hood	0,935981	8,26%
Total	**11,335566**	**100,00%**

Data allocation

The inventory represents the actual prototype-scale process rather than an optimized, upscaled, industrial production process. During the construction of one modular 15 x 15 cm panel, waste material is produced, such as sawdust and wood scraps. All the waste material and carbon emissions related to those within this study is allocated towards the product.

Similarly, the energy use of all the equipment used is fully allocated to the product. For example, the autoclave that was used had a capacity of 18L and could therefore have sterilized more than the 600 g of spawn needed for one FU. However, during the experimental runs, the autoclave was not used for additional material, which is why the energy use is fully allocated to the product. The only exception to this is the use of the incubator, where only the fraction of the total volume occupied by a single FU was allocated to the mycelium product. This allocation approach was chosen to better reflect the reality of the production process, in which the incubator was shared among multiple samples simultaneously.

Calculations

The calculations to obtain the amount of CO_2e. for each in- and output were calculated in the OpenLCA software using the “IPCC 2021” Impact assessment method.

Wood

To find out the total volume of wood needed per FU, it needed to be calculated how many panels could be obtained from a plank or sheet of wood from Dutch hardware stores (Figure 3).

Douglas fir and thermally modified spruce are mostly available in planks. For these two types of wood, a 20x400 cm plank was chosen to calculate the required volume of wood needed per FU. Out of this size plank, 18 panels can be made, which equals to 2.47 planks of Douglas fir

or TMS per FU. As the thickness is 2 cm, the total volume of Douglas fir or TMS per FU is 0.0395 m^3, including the waste of offcuts.

Plywood is available in bigger sheets, which changes the total volume needed per FU. Out of one plywood sheet of 122 x 244 cm, 77 panels can be made, which means that 0.577 sheet of plywood is needed per FU. Using the same thickness of 2 cm, this means that 0.0343 m^3 of plywood is needed per FU. These volumes were used to obtain the kg CO_2e. per FU for the different types of wood. The transport of Douglas fir, indoor plywood and TMS is assumed to be by truck. In the Idemat database there are several options available for truck transport. The largest truck (28 tonne) was chosen. In Idemat, two options with different CO_2e. values as output are available for this type of truck transport, depending on the weight/volume ratio of the load. If the weight/volume ratio of wood is higher than 0.41 t/m^3 (410 kg/m^3), the CO_2e. was calculated per ton x km. This was the case for Douglas fir and indoor plywood. The density of TMS is 350 kg/m^3, which means that the maximum transport of this type of wood is limited by the volume of the wood instead of the weight (Hill et al., 2021). Therefore, the impact of transporting the TMS is calculated by multiplying the volume with the distance (m^3 · km).

CNC energy use was calculated by multiplying the measured hourly power usage by the milling time required per FU (1.67508 MJ/hour · 5.18 hour).

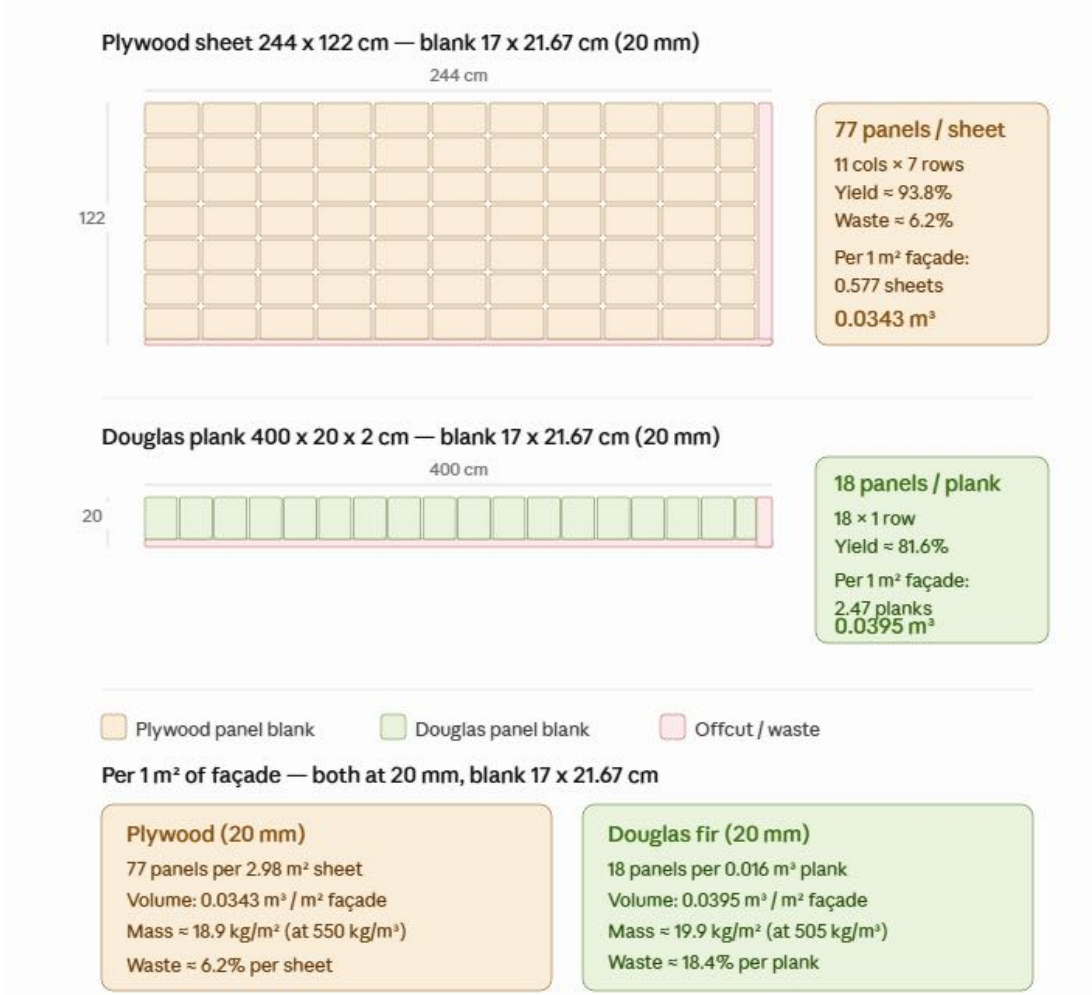


Figure 3: The number of panels that can be obtained from one 244 x 122 x 2 cm plywood sheet and 400 x 20 x 2 cm Douglas fir plank. Included is the volume and mass per m^2 of façade, as well as waste per sheet/plank. Figure partially generated with AI.

Mycelium

Specific calculations

During use, the mould experiences a force of 115kN divided over an area of 10,000 mm². This means the average pressure on the mould is 11.5 MPa. While exact pressures might vary locally, the pressure it is subjected to are far below the compression strength of aluminium alloys (Hopperstad et al., 1999). Even the high cycle fatigue strength for some alloys at 10^7 cycles, or the pressure a material can withstand over a duration of 10 million cycles, can be as high as 125 MPa, much higher than the pressures used to press the mycelium (Wu et al., 2021). Even at elevated temperatures of 180°C fatigue strength is reduced only by a factor of 1.2 to 1.4 (Bahaideen et al., 2009). It is therefore unlikely that any realistically reachable number of cycles would significantly deform the aluminium mould.

The total volume of the incubator is 0.491m³. The volume of wet mycelium material needed for the FU is: 1m² · 3cm= 0.03m³. This is also roughly the volume taken up by the mycelium material in the incubator therefore 0.03m³ / 0.491m³ ≈ 6.1% of the incubator is occupied per FU. For the spawn, assuming the density is equal to that of the hemp shives, the volume is 9.1% of the total mycelium. Therefore approximately 0.55% of the incubator is filled by the spawn.

The total mass of one FU is 6.0 kg. Using a pre-established formula developed by the fungal lab, it is possible to calculate the mass of hemp shives needed for this amount of mycelium block at 80% water saturation: (0.376 / 0.918) · 6.0kg = 2.3 kg (the coefficient 0.376 has been determined as the coefficient to obtain a mycelium material at 80% water saturation, while 0.918 adjusts for the incubation temperature, with the baseline set at room temperature). The density of hemp shives ranges between 87-122 kg/m³ (Nazmul et al., 2023), therefore 2.3 kg of hemp shives have a volume ranging between 2.3 kg / 87 kg/m³ = 26 l and 2.3 kg / 122 kg/m³ = 19 l. The autoclave used for the production has a maximum capacity of 18 l, meaning that to autoclave the hemp shives for one FU two repetitions are needed.

End result

The total environmental impact of the production of 1 m^2 of façade cladding was calculated for different types of wood, as well as mycelium composites (Figure 4). Douglas fir, which is the least modified type of wood, had the smallest environmental impact of the wood types (~ 4.02 kg CO_2e/FU). Indoor plywood had the second least environmental impact of the different wood types, with more than three times the amount CO_2(~13.53 kg CO_2e/FU), closely followed by outdoor plywood (~15.52 kg CO_2e/FU). The impact of TMS was the highest of the different types of wood (~22.10 kg CO_2e/FU). Compared to the different types of wood, mycelium-based composite has the second smallest carbon output (~11.34 kg CO_2e/FU).

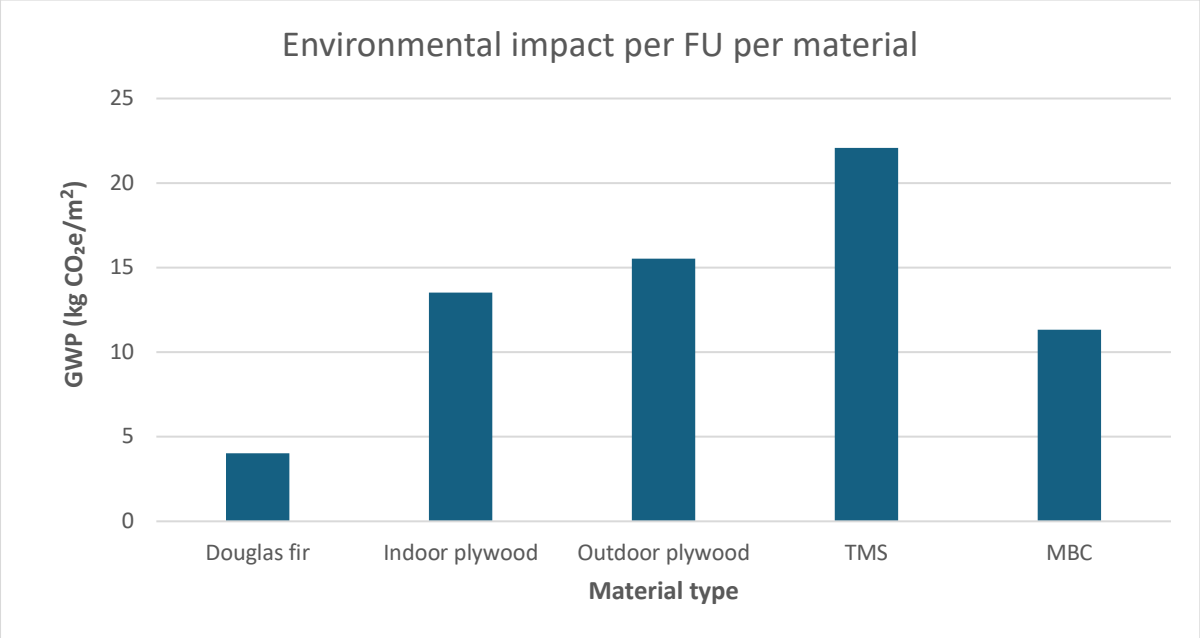


Figure 4: Global warming potential (CO_2e/m^2) per functional unit (1 m² of façade coverage) for each material, cradle-to-gate.

Impact analysis

In this study, the global warming potential over 100 years (GWP100), expressed in kg CO_2e, was determined. This choice is partly determined because three of the four inputs for wood were taken directly from Idemat, which provides pre-characterized carbon footprint values (GWP100, CO_2e.), rather than multiple impact categories. Other potentially relevant categories could therefore not consistently be quantified across all compared materials, which is why no separate LCIA method was applied. The limitation of only a single impact category is that the underlying effects of the production of the panel are unknown. For instance, a material that wins on carbon output, might lose on acidification or vice versa.

The numbers sourced from Ecoinvent used IPCC 2021 as LCIA method, which also gives GWP in kg CO_2e. Both the values from Ecoinvent and Idemat exclude captured biogenic carbon, which aligns the different methodologies. However, Ecoinvent and Idemat are different datasets which limits the comparability between the values drawn from either dataset. Nevertheless, the impact indicator and biogenic convention is similar for both datasets, which justifies the comparison despite the database difference.

For all the different types of wood, the actual material input contributed most to the total environmental impact. The trifold difference between Douglas fir and indoor plywood (4.02 vs 13.53) is likely due to the extra manufacturing steps in the production process of plywood, which are energy intensive.

Indoor and outdoor plywood had a similar contribution profile, where the material production contributed to over 90% of the total the environmental impact (92.58% and 93.63% respectively). Because Idemat embeds the transport to Rotterdam within the material value, the main difference between the CO_2e values likely is caused by the transport distance. Outdoor plywood is imported from Gabon, while indoor plywood is either made locally or imported from other European countries (Mbang, 2013; Teeuwen et al., 2023).

The production of TMS is energy-intensive, as it requires a lot of heat for a considerable amount of time. Therefore, the largest contributor to the total environmental impact for a panel made with TMS is the material itself (91.87%).

For the production of a mycelium panel, incubation of the block, heat-pressing, and autoclaving were the most impactful steps (46.15%, 19.84% and 19.49%, respectively). The reason for the high impact of incubating the block is that the incubator requires a constant energy flow to maintain consistent conditions. As the blocks need to incubate for more than two weeks, this results in the second largest contribution to the total environmental impact. Similarly, the heat-press and autoclave require significant energy to heat up and maintain that temperature (~1.5 MJ/h and ~4.7 MJ/h, respectively).

To reach a lower total impact, the incubation time could be shortened, as this is one of the most impactful steps of the production of a mycelium panel. Alternatively, growing the mycelium using ambient temperature is suggested to reduce the environmental impact of the growing stage (Livne et al., 2024). However, this also limits the structural stability of the panel and therefore is not preferred.

Discussion, conclusion & recommendations

This study compared the environmental impact of different façade-cladding materials and related it to their lifespan, to assess how durable a material must be to justify its production impact. The GWP of each material was therefore divided by their expected lifespan in outdoor conditions, which gives the GWP of the material per year of expected use. Of all materials assessed, the production of a panel using Douglas fir has the lowest GWP per FU (Table 8).

Table 8: Comparison of GWP for each material in relation to the estimated lifespan in literature.

Material type	**GWP (kg CO_2e/FU)**	**Compared to DF**	**Lifespan (in outdoor conditions)**	**GWP/year**
Douglas fir	4.02	1	10-15 years	0.27-0.40
Indoor plywood	13.53	3.36	1 year	13.5
Outdoor plywood	15.52	3.86	5-10 years	1.55-3.10
TMS	22.10	5.49	20-25 years	0.88-1.11
Mycelium based composite	11.34	2.82	Unknown	Unknown

Compared to Douglas fir, indoor plywood had ~3.36 times the GWP, outdoor plywood had ~3.86 times the GWP and TMS had ~5.49 times the GWP. Comparing the GWP in relation to the expected lifespan in outdoor conditions, Douglas fir has the lowest GWP per year of expected use (0.27-0.40 GWP/year), followed by TMS and outdoor plywood (0.88-1.11 and 1.55-3.10 GWP/year, respectively). Indoor plywood has a high GWP per year (13.5 versus 3.10 or lower) compared to the other types of wood. This is expected, as indoor plywood is, as the name implies, not manufactured to be used outdoor.

This means that Douglas fir would be the preferred type of wood, sustainability wise, to construct the final prototype with. There is no determined lifespan of heat-pressed MBCs in literature, as it is a new field of research with lots of uncertainty regarding its biodegradability and response to moisture. However, based on the GWP/FU, it should last at least 28.5 years

to match the worst-case GWP/year of Douglas fir (11.34 / 28.5 = 0.4 kg CO_2e / year) and around 42 years to match its best case (0.27).

At this stage, it does not seem feasible for MBCs to maintain in outdoor conditions for that period, as it is biodegradable and is not resistant for moisture without any treatment. Therefore, under the assumptions and system boundaries of this study, heat-pressed MBC is not environmentally preferred to Douglas fir regarding façade cladding application. However, the GWP of heat-pressed MBC is lower than every wood option besides Douglas fir, which means it could become a viable alternative to wood if the durability gap is reduced. In the future, the impact of the production steps of mycelium-based composites might be reduced, for example by using more efficient incubators or even directing excess heat from surrounding laboratory equipment into the incubators, which operate at relatively low temperatures (30°C). When this is the case, mycelium based could become a more sustainable option than Douglas fir.

Modified types of wood do not come close to the environmental performance of raw timber. Even though, for example, TMS already lasts roughly twice as long as Douglas fir (Table 8), its far higher production impact (5.49 times more compared to Douglas fir) results in a GWP/year which is roughly two to three times higher than Douglas fir. It would need to last several times longer than Douglas to match it.

One of the limitations of this study is that it was not possible to test how long each panel would last in outdoor conditions and therefore the numbers in Table 8 are estimations based on literature (Thermory, z.d.; Van den Bulcke et al., 2010; Wong & Singh, 2005). Moreover, the lifespan of wood cladding depends on the local climate, detailing and moisture dynamics. Therefore, the ranges given in this estimation are not definitive and if one of the panels would be found to last twice as long or short, it could alter the results significantly. Because the gaps between the different types are large, the ranking is robust to these differences. The precise GWP/year values should be considered as indicative rather than exact. If testing the lifespan of the panels could have been included, an improved FU for this LCA could have been “1m² of façade covered for 5 years”.

Another limitation of the FU chosen for this study, is that it does not include the cooling efficiency of the panels. A panel that cools buildings more efficiently but is more outputting during the production phase would not be favoured in this LCA, even though the energy savings from cooling the building might be greater than the difference in embodied energy in the material. A potential way to adjust for this would be to conduct a cradle-to-grave analysis of the entire life cycle and allocate the energy savings from cooling demands, compared to a reference building without the façade, to the product. However, even then the contribution to the UHI would be neglected, as no impact assessment method including the effects of a product to the UHI and subsequent risks to human health exists yet (Chessa & Susca, 2023).

Since the purpose of this study was to compare the impact of two different building materials for an exterior cladding, the analysis excluded additional coatings, such as a radiative cooling coating or reflective paint. However, to draw conclusions about the environmental impact of the panel it would be necessary to include additional coatings as well as the use and end of life of all materials.

Lastly future LCAs could improve their relevance by comparing a more upscaled production process of the wood and mycelium panels, as these will certainly differ from the production process at the prototyping stage. Upscaling may allow for an optimized use of materials and machinery, significantly affecting the embodied energy of the product.

Appendix

Complete input and output tables

Douglas fir:

Input	m3/FU	Density (kg/m3)	Mass (kg/FU)	Distance (km)	tkm/FU	Emission factor (kg CO2eq/m3)	CO2 eq/FU (kg)	Impact Contribution to Total (%)
Douglas Fir (Idemat)	0,03950					76,46000	3,02017	75,34%
Transport	**m3/FU**	**Density (kg/m3)**	**Mass (kg/FU)**	**Distance (km)**	**tkm/FU**	**Emission factor (kg CO2eq/tkm)**		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,03430	505,00000	17,32150	61,00000	1,05661	0,07400	0,078189251	1,95%
Processes	**Hours/FU**	**Energy use MJ/hour**	**Energy use/FU (MJ)**			**Emission factor (kg CO2eq/MJ)**	**CO2 eq/FU (kg)**	
CNC milling	5,18000	1,67508	8,67691			0,104931	0,910473449	22,71%
Total							**4,0088327**	**100,00%**

Indoor plywood:

Input	m3/FU	Density (kg/m3)	Mass (kg/FU)	Distance (km)	tkm/FU	Emission factor (kg CO2eq/m3)	CO2 eq/FU (kg)	Impact Contribution to Total (%)
Indoor plywood (Idemat)	0,03430					365,16000	12,524988	92,58%
Transport	**m3/FU**	**Density (kg/m3)**	**Mass (kg/FU)**	**Distance (km)**	**tkm/FU**	**Emission factor (kg CO2eq/tkm)**		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,03430	600,00000	20,58000	61,00000	1,25538	0,07400	0,09289812	0,69%
Processes	**Hours/FU**	**Energy use(MJ)/ hour**	**Energy use(MJ)/FU**			**Emission factor (kg CO2eq/MJ)**		
CNC milling	5,18000	1,67508	8,67691			0,104931	0,910473449	6,73%
Total							**13,52835957**	**100,00%**

Outdoor plywood:

Input	m3/FU	Density (kg/m3)	Mass (kg/FU)	Distance (km)	tkm/FU	Emission factor (kg CO2eq/m3)	CO2 eq/FU (kg)	Impact Contribution to Total (%)
Outdoor plywood (Idemat)	0,03430					423,61000	14,529823	93,63%
Transport	**m3/FU**	**Density (kg/m3)**	**Mass (kg/FU)**	**Distance (km)**	**tkm/FU**	**Emission factor (kg CO2eq/tkm)**		
Rotterdam to Utrecht, 61 km,28t truck, tonkm	0,03430	500,00000	17,15000	61,00000	1,04615	0,07400	0,0774151	0,50%
Processes	**Hours/FU**	**Energy use(MJ)/ hour**	**Energy use(MJ)/FU**			**Emission factor (kg CO2eq/MJ)**	**CO2 eq/FU (kg)**	
CNC milling	5,18000	1,67508	8,67691			0,104931	0,910473449	5,87%
Total							**15,51771155**	**100,00%**

Thermally modified spruce:

Input	m3/FU	Density (kg/m3)	Mass (kg/FU)	Distance (km)	tkm/FU	Emission factor (kg CO2eq/m3)	CO2 eq/FU (kg)	Impact Contribution to Total (%)
Thermally modified spruce (Hill, 2021)	0,03950					514,00000	20,303	91,87%
Transport	**m3/FU**	**Density (kg/m3)**	**Mass (kg/FU)**	**Distance (km)**	**tkm/FU**	**Emission factor (kg CO2eq/tkm)**		
Tallinn to Utrecht, 2134km, 28t truck, m3km	0,03950	350,00000	13,82500	2134,00000	29,50255	0,03000	0,8850765	4,01%
Processes	**Hours/FU**	**Energy use(MJ)/ hour**	**Energy use(MJ)/FU**			**Emission factor (kg CO2eq/MJ)**	**CO2 eq/FU (kg)**	
CNC milling	5,18000	1,67508	8,67691			0,104931	0,910473449	4,12%
Total							**22,09854995**	**100,00%**

Mycelium based composite:



Input		Weight of FU (I	Kg/FU (for exar	CO2 eq/kg (kg)	CO2 eq/FU (kg)	Impact Contribution to Total (%)
Cellulose		6	0,26400	0,245	0,06460872	0,56%
Hemp Shives		6	2,27	0,109	0,24819328	2,14%
Water Deionized		6	3,91200	0,00027	0,00105624	0,01%
Agar Plate (total)		6	0,12480		0,199473256	1,72%
Potato			0,02496	0,146	0,003648403	0,03%
Glucose		6	0,002496	2,065	0,005155288	0,04%
Agar/Gelatine		6	0,002496	8,890	0,022189565	0,19%
Polystyrene		6	0,07800	2,160	0,16848	1,45%
Aluminium (Mold)		6,000000	0,001000	0,750000	0,000750	0,01%
					0,000000	0,00%
Total Inputs			6,681752		0,713555	6,15%
Output						
Waste Polystyrene		6,000000	0,126000	0,007700	0,000970	0,01%
Processes	**Hours used**	**Energy use(MJ**	**MJ/FU (for exar**	**CO2 eq/MJ (kg)**	**CO2 eq/FU (kg)**	**Impact Contribution to Total (%)**
Autoclave	4,500000	4,680000	21,060000	0,104931	2,209838	19,05%
Incubation (spawn)	120,000000	0,020774	2,492872	0,104931	0,261578	2,25%
Incubation (block)	240,000000	0,207739	49,857434	0,104931	5,231568	45,09%
Heat pressing	14,700000	1,458000	21,432600	0,104931	2,248935	19,38%
Working under Laminar flow hood	4,000000	2,230000	8,920000	0,104931	0,935981	8,07%
CNC	0,004000	1,674000	0,006696	0,104931	0,000703	0,01%
Total processes					10,887899	93,85%
Total					**11,601454**	**100,01%**

Appendix F. Business Model Analysis

Business Model Analysis: Functional Skin That Can Passively Cool Façades

Major Innovation Project Functional Skin / Façade Cooling Panel

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1. Value Proposition

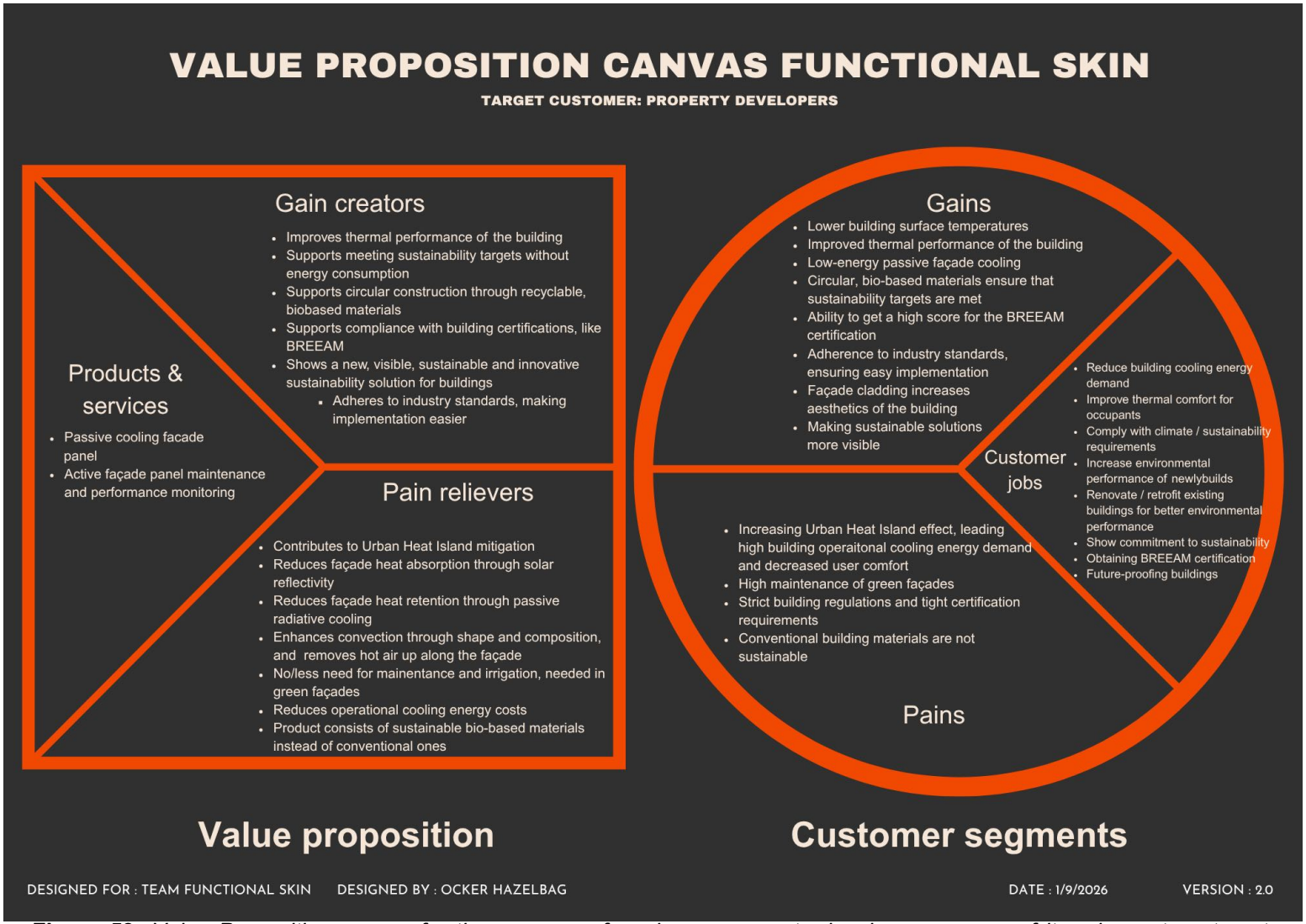


Figure 59: *Value Proposition canvas for the company, focusing on property developers as one of its primary target customer groups.* The Value Proposition is specifically tailored to address the pains, gains, requirements and challenges for this customer segment.

2. Customers

The FunctionalSkin façade panel passively cools buildings and their surrounding urban environment by reflecting solar heat, radiating into space, and channelling warm air up along the façade through a combination of natural and wind-induced convection. It can be retrofitted onto existing façades or built into new ones.

FunctionalSkin is intended to mostly be purchased by professional clients, rather dan private homeowners. In the different types of intended customers, we can distinguish between two different groups: primary customer audience, and secondary customer audience. There's also a third non-customer group, that is still very important to consider, which are the end users.

The primary customer audience consists of the companies and stakeholders that are expected to purchase and implement the product directly into their build or renovation. These include property developers, housing corporations, building owners, and building retrofitters/renovators. These stakeholders are responsible for the construction, renovation, and/or management of buildings that exist in cities where the UHI effect plays a major role, and energy demand for cooling continues to rise. These stakeholders can make the investment decision to innovate and implement passive façade cooling panels.

The secondary customer audience consists of the companies and stakeholders that do influence the decision of purchasing the product but are not direct buyers. These stakeholders implement the product in the building design, manufacture or must install the system, or support the implementation directly. These stakeholders include architects, façade manufacturers, construction companies, and municipalities. Although these stakeholders do not purchase the product directly, they play a crucial role in the adaptation and implementation of the façade panel within these construction project.

Lastly, another important group are the end users, who cannot purchase the product or influence the implementation of the product directly, but play a large role in the overall value that is generated by the panel. This group includes building residents and office workers, who benefit from the improved thermal comfort and reduced need for air-conditioning. This also creates additional value for the primary customers, as it makes the building more attractive and desirable for users. Therefore, the benefits experienced by end users may strengthen the business case for implementing the panel and help recoup the initial investment for this specific primary customer.

3. Revenue Streams

The revenue streams describe how the company generates revenue out of the proposed value proposition. The customers are expected to invest in a passive façade cooling system, as it contributes to a lower building cooling energy demand, reduces operational costs, improves the building sustainability and its compliance with strict environmental regulations

Table 1: *Revenue streams, sources, drivers and prices for the FunctionalSkin passive cooling façade panel.*

Revenue streams	Revenue source (Who pays?)	Revenue drivers (What drives the price?)	Pricing (assumption)
Façade panel sales (primary)	Primary customers: property developers, housing corporations, building owners, renovators	The area of the façade per project in m^2, times the number of projects	€150 - €200 per m^2, quoted per project
Maintenance subscription	The building owner who bought the panel	The area of façade panels installed, times the number of customers that subscribe (Minimum subscription of 5 years)	2-3% of the panel's price, per year (about €4 - €6 per m^2). Cleaned every 2-3 years. (*Facade Cleaning Costs*, z.d.)
Performance monitoring	Building owner or housing corporation	Number of buildings monitored (fixed annual price)	Set price of €2000 - €5000 per building per year, for integrated, analysis and annual performance report

Pricing strategy: The FunctionalSkin panel is priced as a premium product in the façade panel industry, compared to conventional ventilated façades, but will still be far below the pricing of a normal green wall. Green walls would cost €450 - €1500 per m^2 to install (*Cost Comparison of Different Types of Green Facades*, z.d.), while conventional ventilated façade cladding made from wood would cost between €50 - €120 to install (Kurian & Karthi, 2022). When looking at Douglas fir cladding, the price per m^2 is normally between €60 - €80 (*Wood Cladding Prices per Square Metre*, z.d.). Based on this, a target price of €150–€200 per m² is assumed. This more premium price can be justified by the cooling function, the complex geometry in which the wood needs to be reworked, the low embodied carbon and the fact that the panel needs no energy, irrigation or complex controlling to function.

Sales channel: The *FunctionalSkin* panel is marketed directly to professional clients through a business-to-business (B2B) model, and revenue is created mostly by selling to customers in the construction sector. Because of the B2B sales, the primary revenue stream will consist of project-based sales of the façade panels. Each building will need a specified solution that meets the local façade regulations, which is why project-based B2B sales will fit the business model better than retail sales, like DIY stores.

Services: In addition, the secondary revenue streams could come from inspection services, performance assessments and maintenance inspections, where the panels will be closely examined on their passive cooling performance. In combination with this, maintenance service subscriptions could increase revenue even more, ensuring the best performance of the façade panels through periodic inspection, cleaning and repair services. This would be offered to the buyer as an extended warranty for a minimum of 5 years.

4. Competitive Analysis

For façade cladding systems, the 2025 market size was about 270.5 billion USD, and is expected to grow to 489.1 billion USD in 2033 (Grand View Research, 2026). This demonstrates that façade cladding systems represent a rapidly expanding global market. Currently, the market for ventilated façades in this category is also quickly rising due to several underlying reasons (WiseGuyReports, 2026). Firstly, the demand for sustainable and energy-efficient façade options in new construction is rising. Secondly, there is a growing popularity for prefabricated methods in construction and ventilated façades are perfect for this type of application(?). In addition, there is an increasing demand for aesthetically pleasing buildings, and these façade cladding systems offer a wide range of new design options to explore. According to the Industry Analysis-consultancy, WiseGuyReports, the market of ventilated façades is projected to reach a market size of 36.49 billion in 2035, while it was valued at 21.46 billion in 2024. With this increase of about 70%, there will be much space for the company to grow in the next decade. This confirms that there is more than enough demand for new and sustainable options in this market.

The competitive landscape market of ventilated façades currently is fragmented (WiseGuyReports, 2026). The market is competitive, with a few large multinational companies battling against smaller regional or national companies. Major façade cladding and ventilated façade manufacturers, like Saint-Gobain, ArcelorMittal, Kingspan, Hunter Douglas and Sto AG compete in the market through innovating, improving the panel quality, being more sustainable, brand presence and global distribution networks. At the same time, smaller companies can improve their competition by offering more niche and customized façade solutions tailored to the local market needs.

Market advantage

To show how the *FunctionalSkin* panel compares to existing solutions, it is compared to two products that address the same problem of UHI in different ways. The first is a conventional ventilated façade. The second is a living green wall, which is currently the best known solution for (passive) cooling and UHI mitigation (Ahluwalia & Khurana, 2025).

Table 2: *Market comparison of the FunctionalSkin panel, a conventional ventilated façade, and a Living Green Wall.*

Criterion	***FunctionalSkin***	Ventilated Façade	Living Green Wall
Installed cost per m^2	€150 - € 200 (assumption)	€50 - €120 (Kurian & Karthi, 2022)	€450 - €1500 (*Cost Comparison of Different Types of Green Facades*, z.d.)
Maintenance costs per m^2 per year	€4 - €6 (assumption)	Low, optional cleaning	€50 - €150
Cooling function	Reflects solar heat, radiates absorbed heat into space, channels warm air upwards	Cavity ventilation	Shading and evapotranspiration by plants
Energy and water input	None	None	Irrigation system required
Embodied carbon	Low (biobased materials in use)	Higher (Often non-biobased materials in use)	Low

FunctionalSkin takes a middle position for the pricing, as it costs more than a conventional ventilated façade, but only a fraction of a living green wall, and unlike that one, it needs no irrigation or plant maintenance. The advantage over a conventional ventilated façade is the additional passive cooling capacity, without energy input and a lower carbon footprint. The main disadvantage is the higher price and higher maintenance costs.

Seasonal performance (assumption): In winter, the panel would provide no passive cooling and is assumed to behave like normal ventilated façade cladding. Research in a climate chamber showed that an added ventilated façade could guarantee energy savings of more than 20% in windy conditions (Gagliano & Aneli, 2020). If this would also be the case in Dutch winter conditions still has to be tested. The sales argument will therefore be based mostly on the summer performance.

5. Business Model Canvas

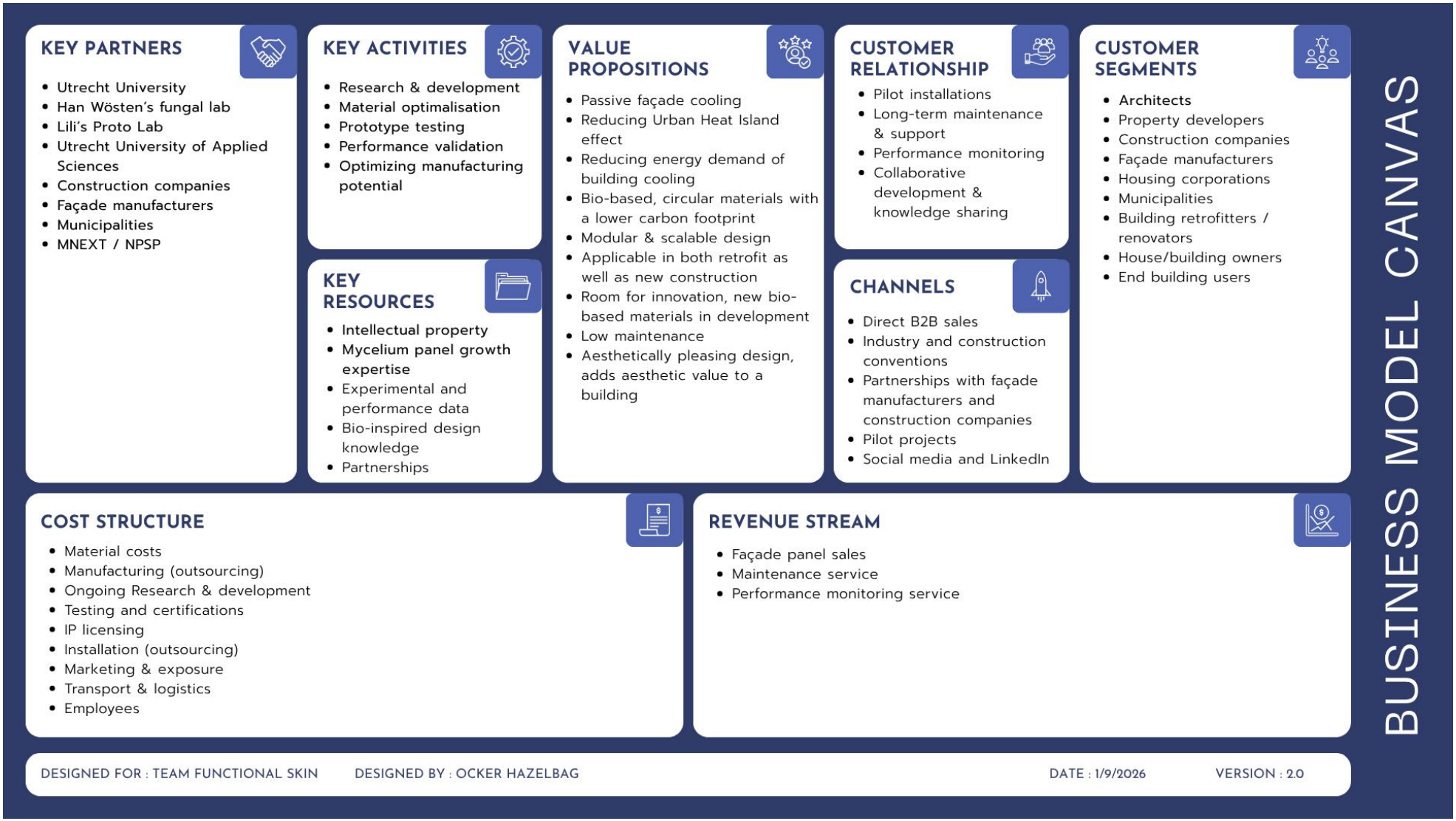


Figure 60: *Proposed Business Model Canvas for the company by 2030.*

6. Risks and Challenges

In the final Business Model Canvas, the envisioned commercial situation for the company in 2030 is considered. However, there are still some risks and challenges that need to be addressed for the company to achieve this envisioned state. The following are the key uncertainties that need to be resolved on the path towards commercialisation.

Plywood, the current material that's in use, would need an extra coating for improved moisture resistance. As the façade panel would first be introduced in the Dutch climate and be exposed to the Dutch weather, the panel will be exposed to moisture often, leading higher requirements for moisture resistance. In addition, standard plywood does not meet the required classification for the intended façade application (Appendix C). Currently, to achieve this rating, additional non-biobased coatings should be applied which may compromise the sustainability of the end panel and counter to the company's sustainability values (Appendix C).

In addition, the current scalability and manufacturalability of the product raises some more concerns. Although making a 120 x 60 cm prototype in the lab was possible, it remains unclear whether the current production process can be scaled to a five-story building. The current manual production process relies on manual CNC machining, steaming, folding and gluing and has limited output and is labour intensive, which may make it unviable at commercial scale. Outsourcing the CNCing to companies with larger-scale CNC machinery could become an option for large scale production. However, if the panel would be heat-pressed into a mould in the future, taking away all the excess steps (like in Pujadas-Gispert et al., 2020) this process could be streamlined and would make the process easier to scale.

The real-world performance of the panel requires more assessing, as the current tests have not come further than controlled environments and climate chambers. For final application of the product more tests need to be done on outdoor durability. Additionally, there's no data yet on the performance of the panel in different seasons and conditions. Variations in temperature, wind, solar radiation and humidity may affect the panel's passive cooling performance, as the current testing does not yet show how the performance is under these conditions.

Lastly, the economic value of the product has not yet been quantified and demonstrated on the market. At the assumed price of €150 - €200 per m^2 , the panel costs more than conventional cladding, and would also need extra maintenance costs to ensure and maintain good performance. Research has shown that cool coatings and ventilated façades can reduce peak electricity demand by a minimum of 5%, which over the whole year would account to significant energy cost savings (Park et al., 2024). However, if the primary customers are willing to pay for these benefits remains currently unknown and needs to be investigated.

Appendix G. Intellectual Property Analysis

Intellectual Property Analysis

Goal and scope

To obtain insights on the market and commercial opportunities for our product an Intellectual property (IP) analysis, consisting of quantitative and qualitative patent analyses, was conducted. The types of IP assets that will be included in the analysis are patents and to some extend design rights and trade secrets. Other IP assets were excluded for the following reasons: The façade panel has a technical function, therefore copyright does not apply for it in the Netherlands (Netherlands Patent office, 2019). A trademark is a form of IP dedicated to protect a brand including the brand name and logos for example (Netherlands Enterprise Agency, RVO, 2025). In the scope of the project these will not be created for our product, so it will not be included in the IP analysis either. However, once the product would be developed into a market ready product trademarks become relevant.

For the trade secret, no public search can be conducted to find out if a company possesses this form of intellectual properties, as they do not need to be registered or declared. Furthermore, unlike a patent, a trade secret does not cover independently developed products. It protects only against improper acquisition, like theft or accessing confidential information. A way a trade secret could be infringed accidentally could therefore be to hire a former employee of another company who discloses said companies trade secret to the new company. This scenario is however not yet applicable, as we are not hiring. Ensuring no trade secrets will be infringed by our product therefore requires guided principles in product development and not a scoping of which trade secrets exist so far. Trade secrets could however be a potential strategy to obtain IP for the product without applying for a patent.

As for the patents, a quantitative analysis was used to get insights on the product's market environment. The goal of the qualitative analysis was to assess the possibility to obtain IP on the product, as well as to check if the product infringes on somebody else's intellectual property. This form of IP analysis is called a Freedom-To-Operate (FTO) analysis. Since the product would be produced and first implemented in the Netherlands, the analysis was conducted based on Dutch IP law and based on patents and other IP active in the Netherlands. It is therefore important to note that while the IP analysis might suggest no infringement in the Netherlands, exporting and selling the panels into a country where a relevant patent is active can still be prohibited. The IP analysis therefore does not cover the entirety of the global market.

Intellectual property analyses are generally conducted by trained professionals, like patent attorneys. They are often reported in legal terms or business jargon with the goal of reporting being to provide information useful to the party financing the analysis. These reports are not written for reproducibility and transparency in the methodology

and are rarely publicly available meaning it is not possible to adopt a preexisting methodology for an IP analysis.

To create a methodology suitable for academic reporting an expert from the Dutch Patent Office (Octrooicentrum), Maud Kerstholt, was consulted. She recommended a combination of a quantitative analysis to get insights into the industry and a qualitative analysis of patent claims (Appendix A). With this it is possible to check if there are any immediately apparent patent claims the product could be in violation of. It also helps to get an idea of what kind of claims patents in this industry usually contain. Lastly a short reflection on which IP strategy is most suitable for us is added.

Quantitative Analysis and Filing Trends

Before analysing specific claims of existing patents and how they can affect a business it can be helpful to obtain a general understanding of the patent landscape of the product. An example of a quantitative analysis is a key player analysis, to identify companies with the most claims in a certain field, getting a better understanding of one's competition. Another example would be a geographical analysis of the distribution of patent activity, to see which regions are most active in filing new patents in the field.

To save time, the quantitative analysis was restricted to analysing patent filing trends over time in classifications relevant to the product. The selected classifications include the materials of the product, the product application and the function of the product. As an indicator for filing activity, the volume of patent families filed was assessed for relevant International Patent Codes (IPC) and search terms relating to the product.

According to Maud Kerstholt patent families are a better representation of patent activity than individual filings, which is why the results hereafter are based on the first publications of new patent families.

The table below shows the number of new patent families per period, as listed on Espacenet. In the search term column, the exact combination of IPC codes and search terms can be found. Synonyms for materials, the product application and product functions were combined with the Boolean operator "OR". At the bottom of the table combinations of materials, the function and product applications can be found. The exact search terms and IPC codes used for the search were selected in consultation with experts from the Dutch Patent Office (Octrooicentrum) and include synonyms of the keywords.

Table 1: *Number of new patent families filed per four-year period (2006-2025) (data taken from Espacenet)*



Keywords	2006-2009	2010-2013	2014-2017	2018-2021	2022-2025	Total 2006-2025
Wood	135623	234749	373650	492699	386031	1622752
Mycelium Materials	6904	12558	23783	24.506	24896	92.647
Façade cladding	10234	21315	32402	50461	51177	165589
Ventilation and Convection	4846	9011	13338	19119	23004	69318
Radiation and Reflection	6448	11157	15366	18765	24662	76398
All Passive Cooling	9773	17099	24484	32522	40993	124871
Wood+ façade cladding	2722	4998	7691	10510	7902	33823
Mycelium+façade cladding	6	5	13	17	29	70
Façade cladding+passive cooling	101	207	277	347	505	1437
Both materials+ façade+ passive cooling	36	75	83	96	146	436

Across the different search terms, patent family filings continuously increased throughout the four-year periods, with the sole exception of wood and the combination of wood and façade claddings having a decreased number of publications in the last four years, compared to the four years before.

Wood or timber remains subject of a lot more patent publications than mycelium, despite gradual increase in publications relating to mycelium over the years. Over the last 20 years there have been close to 20 times more publications for wood than for mycelium materials. This difference becomes even more pronounced when the search terms are combined with the product application in a façade. While patent families on mycelium façades published in the last 20 years led to only 70 results wooden façades gave almost 500 times as many results at 33823 patent families published in the same timeframe.

Combining the search for mycelium materials with both the product application and passive cooling function merely resulted in three results. While this is not enough to evaluate filing trends over time and was therefore excluded from table 1, the claims of these three patents could have high relevance to our freedom to operate, which is why they will be reviewed carefully in the qualitative analysis.

Qualitative Analysis and Business Integration

The following section will take a brief investigation of what Intellectual properties could be obtained, and which patent claims of other companies could potentially prevent commercialization of the product. It is important to note that a full FTO analysis or an analysis on how we could obtain patents for our product would normally be conducted by professionally trained patent attorneys and cost us between €10.000 and €50.000, according to our consultation with Maud Kerstholt (Appendix A). Our analyses are relying on free advice from the Dutch Patent Office as well as own research of patent databases and relevant publications. It therefore does not represent a complete analysis, that could be used in communication with investors and other clients. Furthermore, the focus of the analyses is on academic reporting and transparency rather than strategic market analyses.

Since the basic design was adopted from the VirtuE project by honours students of the TU Eindhoven, obtaining IP on the shape of the panel will not be possible. However, the design has been published (Pujadas-Gispert, Alsailani, Van Dijk, et al., 2020) and adopted in other projects and publications since (Summa et al., 2025). Therefore, no patent will be able to prevent us from producing this shape commercially, as the shape no longer qualifies as new. The same rule applies for registering a design right (Netherlands Enterprise Agency, RVO, 2024). A search on Espacenet revealed there are no patents registered in the names of the publishers of said design.

The search results from Espacenet containing keywords of both the material, product application and function were scanned for relevant claims. This was restricted to the first 20 results when searching for a combination of the terms wood, passive cooling, and façades, or synonyms thereof. Additionally, the three patents corresponding to an identical search with the material replaced with mycelium and synonyms thereof were also scanned.

From this brief review no immediate infringement could be identified, as all claims that were reviewed were too specific to be infringed by our design in its current form. However, it would still be possible that a part of the manufacturing process infringes a patent, even if it is applied in a different context. Four out of the twenty patents reviewed had claims about a chimney-like ventilation system, however not one patent held the claims of building a chimney ventilation system. Rather the claims had further

restrictions, like a certain structure of the airducts and attachment to the support structure, or combination with heat exchangers or solar panels.

IP Possibilities for the Product

Pursuing a patent with the product as it exists now is likely not viable. Except for the process of pressing the mycelium into the selected shape, the product only utilizes known techniques. Even the parameters used during heat-pressing for optimal heat and duration, stem from previously published literature. In an expert interview with Naud van der Padt, one of the founders of the company “Leafy”, he told us that apart from the value it has when recruiting investors, he does not think patents are beneficial for a company in the construction sector. In his experience the costs of filing and enforcing a patent tend to outweigh the benefits gained by preventing competitors to use it or licensing the technology to third parties.

Discussion

The filing trends show a consistent increase in patent families for each subsequent four-year period, except for wood patents decreasing in the last four years. While this suggests that the fields the product is located in are increasing in relevance, it is also important to note that patent publications in general have continuously increased over the last 15 years (WIPO, 2025). Nonetheless, since 2010 patent filings in most of the categories have increased by more than the global total increase of 1.9 (WIPO, 2025), with the exception again being wood and wood cladding.

While patent publications around mycelium are increasing much faster, they remain less represented in patent publications than wood, when looking at absolute numbers. The even larger gap in total patents filed between the two when it comes to the application in façades, suggests that using mycelium in this context could fill an unoccupied niche.

A limitation in a systematic patent search is that while it is possible to add a proximity search operator, which was useful to e.g. make sure the terms “passive” and “cooling” were mentioned together and did not refer to different things, it is not possible to establish the relationship between search terms. Therefore, a search combining passive cooling and mycelium does not guarantee that in the patent the mycelium material is responsible for the passive cooling. A short review of the result’s claims made it apparent that in many cases this combination of search terms rather referred to methods for mushroom or mycelium cultivation, in which at some step a passive cooling step is applied. A possible solution to this problem can be to manually review all results and filter out those where passive cooling is not performed by the mycelium. With the timing constraints placed on this analysis the decision was however made to exclude this search term from the analysis.

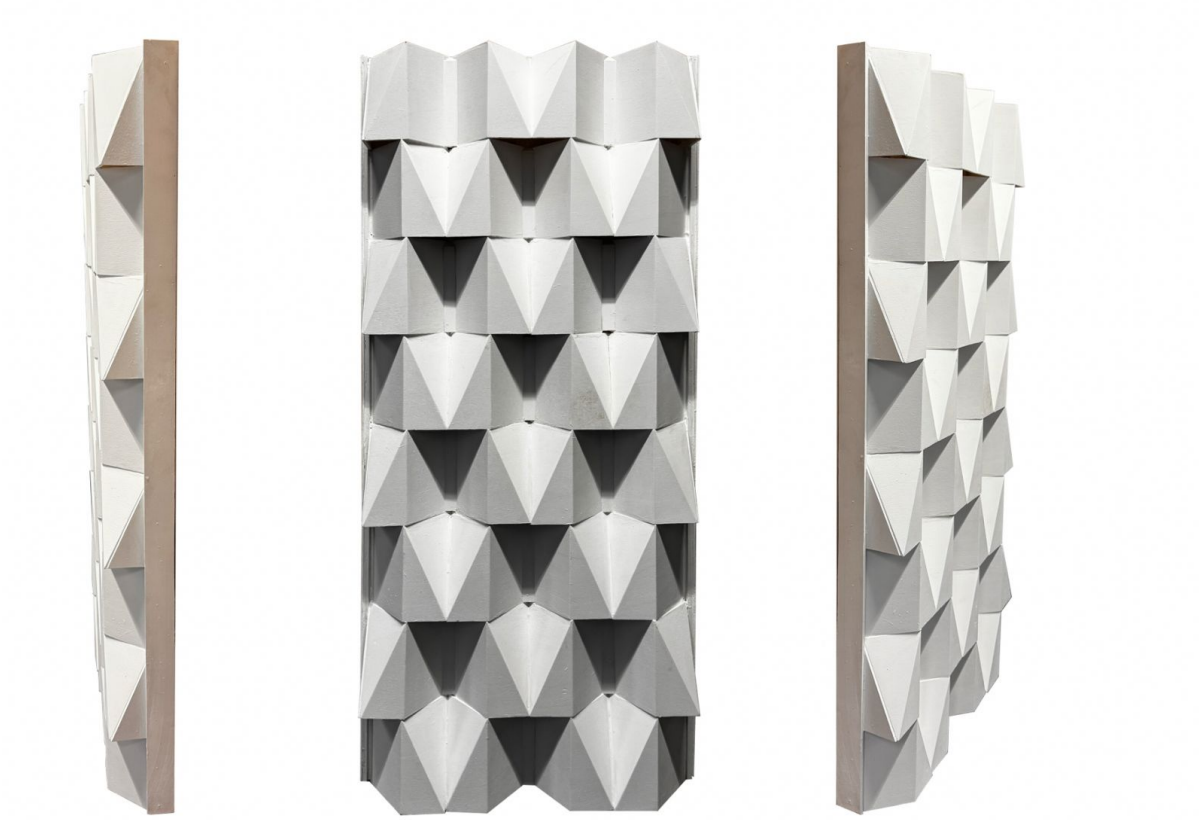
Conclusion

The product we designed was chosen because we perceived an increasing relevance of passive cooling mechanisms for the future, as the UHI effect will become an increasingly important problem. This increasing relevance of passive cooling mechanisms was also reflected in the number of patents being published over the investigated time-period. No immediately apparent patent infringements could be identified, but to establish whether we truly have freedom to operate with our product a more extensive search would be required. For our IP strategy, we will restrict our self to trying to obtain those IP rights that are obtained without investment of €10.000 to up to €50.000 (Appendix A) and time and will refrain from trying to get our own patents published.

Appendix H. Implementation plan

Implementation plan

FunctionalSkin – a passive cooling façade panel



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Ocker Hazelbag
Felipe Krebs
Jochem ten Pas

Introduction

To get from an experimental phase to a product development phase, the following question needs to be asked: What steps need to be taken to get the product functioning in the real world and keep it functioning. This is where an implementation plan comes in: it describes how the product can be used in its final context, including attachment methods, weathering advice maintenance methods, implementation hurdles, and end of life options. In essence, it is going through a full lifecycle of the product, from installation to end-of-life. Therefore, in the following sections, these subjects will be thoroughly discussed and assessed.

Attachment requirements

When the final prototype is finished and optimized, the next step is to attach the panel onto an existing façade. The goal of attaching the panel should not be to design a new attachment system, but to find which already existing attachment method is most compatible with the design of our panel and how the façade panel attachment interface should look like. To do this, first it needs to be determined what requirements the mounting system should have, as well as which parameters impact these requirements.

The panels' weight is not the driving factor for the structural load; wind is the most impactful factor. Eurocode 1 states that the wind load on a facade is impacted by the building height, the landscape (terrain roughness) and the façade location (windward/leeward). Wind exerts positive pressure on the windward façade of a building and a negative pressure, or a suction, on the edge zones and the leeward façade of a building (Abid & Abid, 2023). This leads to a need for increased mounting density at the edges and corners of buildings (Pomaranzi et al., 2022).

Even though wind is the driving factor behind the structural load of the panel, the weight is still a factor to consider. In the final prototype, 5.5 mm thick plywood with a density of 600 kg/m^3 was used. This would result in a surface density of 3.3 kg/m^2, which is considered lightweight for a façade cladding panel. However, as the panel is 3D-faceted, the actual surface area and therefore material used is higher than 1 m^2, and the weight will be somewhat higher.

Wood is a hygroscopic material; it changes dimensionally in response to moisture fluctuations (Hill et al., 2022). Additionally, wood can also increase in size when it gets hotter. Therefore, the mounting system should accommodate this movement rather than restrain it. Rigidity could lead to deformation, cracks in the coatings or paint and end up in reduced functionality. Façades are therefore conventionally attached to one fixed point with multiple sliding points to account for dimensional flexibility.

For the specific function of our panel, there are some other requirements which the attachment system should comply with. First, the final design includes a reflective

coating or radiative paint and it is important that this surface is uninterrupted. When the mounting method requires some sort of screws or front-faced attachment parts, it disrupts the continuity of the reflective/radiative function. Additionally, the mounting system should not block or interrupt the airflow on which the function of the product is based.

Lastly, it is important that the panels can also be demounted (easily) if it needs to be replaced. It was a deliberate choice to use a biodegradable material, whose benefits would diminish if permanent fixings would be used and the material therefore can't be recycled or used for other purposes.

Attachment methods and recommendations

In general, fastening systems for cladding in ventilated façade systems can be categorized in different ways: the European Technical Approval Guideline (ETAG) groups the systems into eight families based on their fastening methods (EOTA, 2012) (**Figure 1**).

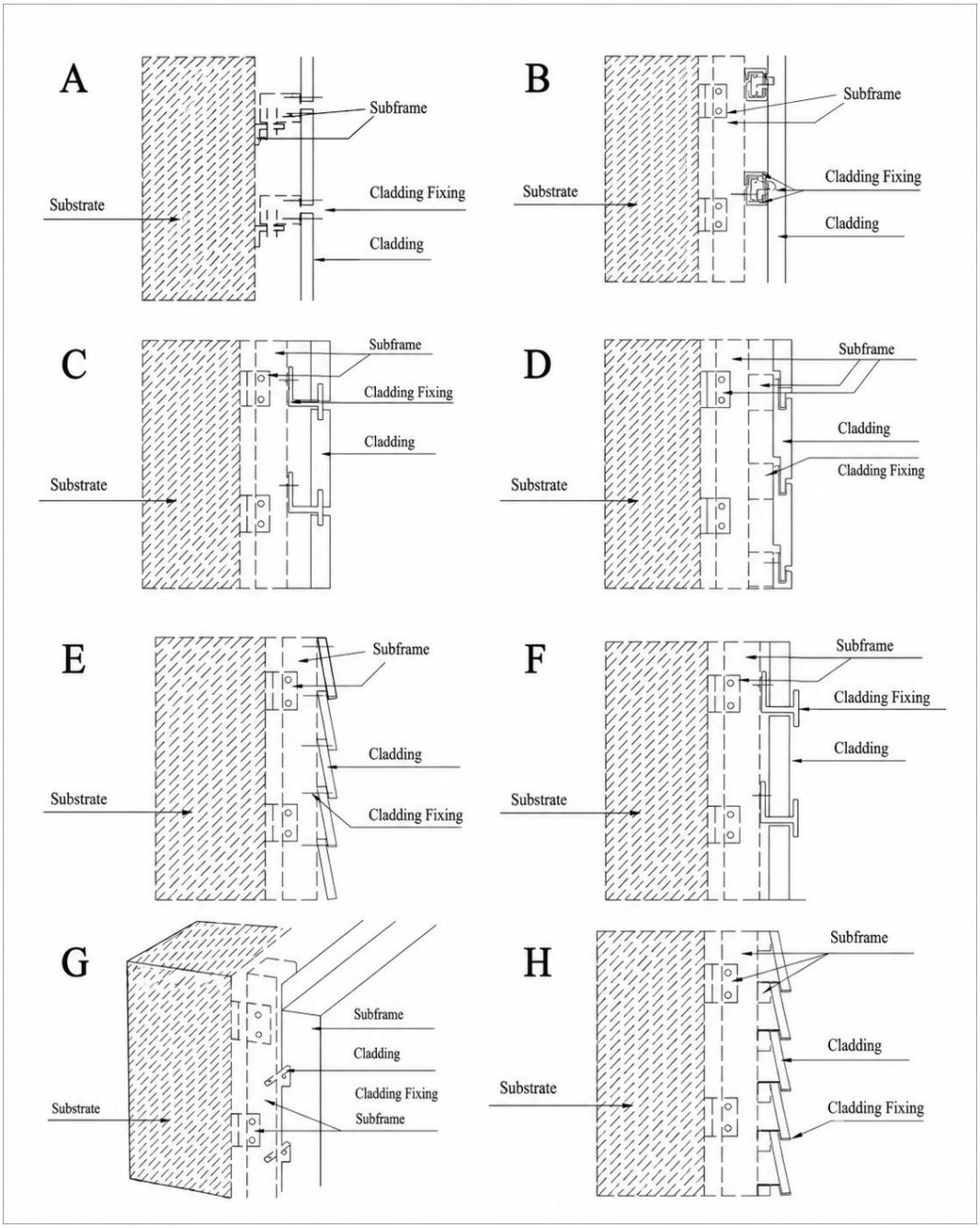


***Figure 1**: Classification of ventilated façade cladding fastening systems into eight categories , according to the fastening principle defined in ETAG 034.* Family B represents concealed point-fixing systems, which is the mounting approach recommended in this report. Adapted from EOTA (2012).

Alternatively, Bazzocchi et al. (2022) divide the fastening systems into two groups: point fixing systems and distributed fixing systems. With point fixing systems, the panels are fastened with the minimum number of anchorages, of which one is the fixed point. The other anchorages can then be detailed so movement of the panel is accommodated. Because there is no redundancy, such systems are isostatic, or "safe life".

With distributed fixing systems, façade cladding panels are constrained to the substructure via many anchorage points along a continuous line, for example by a profile running in a groove milled into the panel edge. This makes the system hyperstatic, or "fail safe": if part of the connection fails, total collapse of the element is limited. The type of fastening can be either visible or concealed. All fastening systems which are visible are point fixing systems, while concealed, or hidden fastening systems can be both point and distributed fixing systems.

For distributed fixing systems, the panel needs to be thick enough so that the edges don't become too fragile (Bazzocchi et al., 2022). For our product, which is manufactured from thin pieces of wood, this would not be the case. Additionally, ideally the front face of the panel is uninterrupted to reach the full potential of the reflective or radiative coating. For this reason, a concealed, point fixing mounting system would be the most suitable for our product.

In Bazzocchi et al. (2022), several examples of concealed, point fixing systems are given. Two in particular seem suitable for our product:

- Stainless steel clips or brackets that fit into milled slots on the back of the panel. As the individual pieces of our panel are attached to a timber frame, this provides the opportunity to mill the grooves (or slots) in the back of this timber frame and attach the panel to the substructure.
- Stainless steel inserts (threaded pins or threaded sockets) are installed into metal sleeves fixed to the back of the panel at predetermined locations. The inserts are mechanically connected to the substructure, allowing the panel to be attached to the building. Because the load is transferred on the panel's back rather than its edges, this is a suitable method for relatively thin façade panels.

An example for the second system that would work for our façade panel, is the NV3 system developed by NVELOPE®, a concealed, point fixing system (SFS Group, z.d.). The point brackets in this system allow dimensional movement and absorb the wind loading, which complies with the requirements. The brackets also allow for a range of cavity depth from 40 to 300mm, which is in correspondence to the aimed range that was composed in the Design Report (*SFS Group - NV3 installation guide*, z.d.)(Appendix).

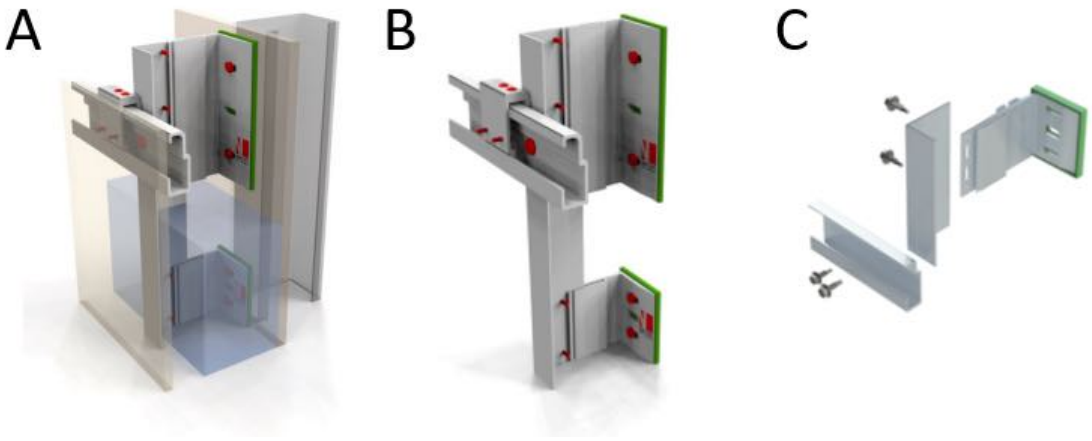


Figure 2: *Concealed fixing point mounting system, the NVELOPE ® NV3 system. A) System shown with cladding panel and backing wall. B) Bracket, hanger and rivets shown in isolation. C) All individual components of the system shown. Adapted from SFS Group - NV3 Installation Guide (n.d.)*

The NV3 system has most similarities with ETAG 034 family B (Figure 1) and looking at the list of requirements from the previous section, this system checks all boxes: it carries the structural load of the panel and provides space for movement of the panel through the flexible point brackets. Furthermore, it is easily demountable and is a concealed system, which leaves the front face uninterrupted. However, the choice of attachment system for façade panels can be flexible and is dependent on other factors, such as the specific climate (wind load, temperature) in that area, building height, panel weight and substrate type. Therefore, this system is a general recommendation which should not be considered the only option for attaching the panel to a building.

Weathering of timber

Façade cladding panels are continuously being exposed to biotic and abiotic stresses, which impact the durability and quality of the panels. Our final product is made from plywood, which is susceptible to several types of stresses. Fluctuations in moisture content result in dimensional movement in wood, which generates tensile stress and surface cracks in the wood (Hill et al., 2022). These cracks cause localised regions of elevated moisture concentrations, which are more prone to biological degradation by bacteria and fungi (Davis et al., 2021; Sandberg, 1999).

Fungi can also enter wood more easily when parts of lignin are degraded because of continuous exposure to solar radiation. The surface moulds that can colonize the wood because of this so-called photodegradation, can ruin the aesthetic appearance of the wood by discolouring it to a grey, silvery colour (Hill et al., 2022). The panel's function depends on a high-reflectance surface, which might also be degraded by these surface moulds, thereby reducing the functionality of the coating. Other wood-decaying fungi impact the wood in a different manner: they can break down lignin and (hemi-)cellulose which can reduce the structural integrity (Hill et al., 2022).

Photodegradation, moisture-induced degradation and biotic degradation are the three main forms of weathering to which wood is subject (Davis et al., 2021). All types of

weathering are connected to each other and it is therefore important to sufficiently protect the panel against these stresses, especially the vulnerable parts. Besides the durability of the panel, the functionality of the panel is impacted by weathering. When using high-albedo or radiative cooling paint, it is important that the surface is regularly maintained to sustain its function. Therefore, in the following section, treatment methods to improve the durability of the timber and maintenance requirements are discussed.

Timber treatment

As mentioned before, our prototype panel is made from plywood as it is available in the specific thickness (5-10 mm) that we need to make the panel into the specific geometry that is needed.
However, for future iterations, a solid timber like Douglas fir is suggested to be used for the production of the panel. Therefore, common wood modification methods will be assessed. In the *Material Selection Mini-Report*, a more extensive explanation is given on the mechanisms behind wood modification techniques (Appendix C). However, here, the outcomes of different modification techniques and their pros and cons will be discussed.

Acetylated wood is a chemical process which uses acetic anhydride to form stable acetyl groups from the water-attracting hydroxyl groups in wood. Commercially it is one of the most developed techniques of wood modification. Acetylated wood has several benefits, such as improved natural durability (class 1, EN 350) and dimensional stability increased by 70-80% (Sandberg et al., 2017). At the same time, the effect on the mechanical properties of acetylated wood is negligible.

The downsides of acetylated wood are mostly the cost and need for specific equipment when working with it. Acetylating wood is energy-intensive which is reflected in the price (Hill et al., 2021; Sandberg et al., 2017). Additionally, residual acetic acid in wood can lead to corrosion in screws or fasteners, resulting in the need for stainless steel screws and fasteners (Klaassen et al., 2018). Moreover, specific glues are needed as water-based adhesives are not absorbed easily (Frihart et al., 2017).

Globally, the most produced type of modified wood is thermally modified wood (D. Jones et al., 2020b). This type of treatment involves heating wood in low-oxygen conditions which alters the cell wall material composition (Sandberg et al., 2017). This modification results in increased dimensional stability and improved resistance against moulds (D. Jones et al., 2020b). Additionally, the density of wood is lowered during the thermal modification which makes it more lightweight. However, the loss in mass also decreases the strength of thermally modified wood and leads to increased brittleness (Sandberg et al., 2017). Furthermore, reduced nail withdrawal strength has been observed in thermally modified wood, an important construction property (Barcík et al., 2014).

Impregnation of wood is another common technique used to modify properties of wood. Impregnation can be described as infusing wood with an impregnating agent which spreads in the cell walls, where it polymerizes. This might happen via active modification, such as cross-linking or passive modification by bulking (Augustina et al., 2023). As a result, the dimensional stability and resistance against fungi can be improved (Sandberg et al., 2017). A downside of impregnation is that most impregnating agents are made from non-renewable, petroleum based resources, which does not align with our biobased objectives for our panel (Karthäuser et al., 2024). There is, however, a rising trend in the development of environmentally benign impregnation methods.

Furfurylated wood is the most common “green”, impregnated timber type and uses furfuryl alcohol for impregnation. It improves the physical-mechanical properties and the durability of wood, but does also lead to embrittlement and decreased impact strength and modulus of rupture (Augustina et al., 2023; Dong et al., 2020).

Other environmentally friendly impregnating agents such as natural oils and waxes are limited as they do not form any chemical bonds, resulting in a short-lived protective effect (Jebrane et al., 2017). Additionally, these compounds can negatively affect the adhesion Cryopaint, which is applied to our panel.

As the first prototype was made from plywood, the options to modify plywood are also investigated. Plywood offers improved dimensional stability compared to solid timber for thinly cut pieces. However, plywood will be vulnerable to weathering, especially at the top veneer layers and sawn edges (W. Li et al., 2016). To make plywood more suitable for exterior applications, it can be impregnated. Additionally, petroleum-based edge sealers can be used to protect the edges against moisture and decay. While this is not a biobased element of the panel, it is necessary to protect the sawn edges and therefore required to use our panel in outdoor conditions, if plywood was used in future panels.

Both acetylation and thermal modification are techniques most commonly used for solid timber, although they can be used for plywood (Aro et al., 2014; Candan et al., 2012; Lovrić et al., 2017; Slabohm et al., 2023). However, plywood made with either of these types of modified wood is currently not widely commercially available. Similarly, research on incorporating furfurylated wood veneers into plywood panels is scarce (Mubarok et al., 2025).

Because of the limited modification options of plywood, it is highly advised to use a solid timber like Douglas fir in future iterations. But to maximise the service life of our façade panel the best, recurring maintenance is needed. Therefore, in the next section, the required maintenance will be discussed.

Maintenance

To ensure the maximum service life of our timber façade panel, maintenance is required. Typically, for timber façade cladding, visual inspections are performed to assess whether the panel needs cleaning, recoating (if a coating is used) or replacement (Gustavsen et al., 2026). It is important that the frequency of inspections is optimized to the specific material and climate conditions of the panel. Too few inspections could lead to irreversible degradation which could have been prevented with more frequent maintenance, while too many inspections drive up the costs for building owners (Hill et al., 2022).

A maintenance plan for our panels is therefore dependent on the specific conditions they are installed in: climate and number of storeys are important to decide the type and frequency of maintenance that is required. Additionally, the type of coating or paint also impacts the type of maintenance. For example, if a reflective, high albedo paint was applied to our panel, it is important that accumulated dirt or other materials are regularly removed from the panel's front-face. This is especially important as it can directly disrupt the functionality of the panel. Therefore, the panel's front-face cleaning interval might be shorter than the regular façade cladding inspection interval, if a paint with added functionality is applied.

The frequency of recoating or repainting is dependent on the type of paint. Transparent stains require recoating every three to five years, while opaque paints can last for eight to twelve years (Gustavsen et al., 2026). Therefore it is important to match the frequency of recoating or repainting with the type of coating in question. Inspections could take place every two years, with cleaning and surface treatment when needed (Madureira et al., 2017). As a potential revenue stream for our company, inspections can be done by someone from the *FunctionalSkin* company.

One important remark regarding cleaning of the panel, is that wood is typically cleaned differently than other cladding materials. Where other cladding materials can be cleaned with a high-pressure cleaner, wooden cladding needs to be gently washed to mitigate the chance of damaging the surface (Gustavsen et al., 2026). The frequency of cleaning differs per country, as in Norway and Sweden annual cleaning is recommended by national guidelines, while in Germany cleaning frequency is determined by the manufacturer (Gustavsen et al., 2026). Additionally, gentle washing is required, as high-pressure cleaners impose the risk of getting moisture in cut edges. As a result of the specific geometry of our panel, it has significantly more cut edges than a regular, flat cladding panel. These edges expose the end grain and glue, which must therefore be prevented from wetting. The rear face of the panel cannot be inspected after installation, which makes it impossible to observe degradation of the backside of the panel. This highlights the necessity of easy demountability of the panel.

If weathering has resulted in degradation of the panel, such as cracks, flaking or blistering, and recoating or repainting is no longer an option, replacement of the panel

should be carried out. If only a part of the panel needs to be replaced, it can be decided to only substitute the individual 15 x 15 cm modular tiles, without renewing the full panel. Without the possibility of proper maintenance, the chance of failure of the panels increases. The replaced panel can then enter the end-of-life phase, which will be discussed.

Implementation hurdles

As a functional product on the market, this façade panel will need to adhere to industry requirements. These requirements include the following.

Fire safety

Besides weathering and the related durability of our panel, one of the main issues with it is the **fire safety** – as was widely highlighted in 2017 after the Grenfell Tower fire in London, ventilated facades can face serious risks. Besides untreated wood being a combustible material, the function of the panel also plays a role in its fire resistance. The panel transfers warmer air through little “chimneys” upwards and releases it when it has cooled down. When a potential fire starts, this mechanism can accelerate the spreading because of the convection that takes place within the panel.

As outlined in the *Material Selection Mini-Report,* the panel should meet a fire class of B or D (dependant on building type) and a smoke class of s2 or s0 (Koninkrijksrelaties, 2026). Further research is required to determine what class this panel falls under, based on the new materials proposed. However, there are some techniques available to reduce fire risk based on how they are installed.

Firstly, the placement of panels should keep this danger in mind. By compartmentalising the flow of air the spread of fire could be reduced, while maintaining an adequate flow of air for cooling purposes (Fire Protection Association, 2023). The cassette design already reduces air movement in the horizontal direction, which means the vertical movement of fire would be the focus. By allowing for a gap or barrier at intervals in the vertical direction, the movement of fire could be reduced. Another option would be to instal fire barriers. Many products can be found on the market, such as Flamro’s “Firebreather Cavity Barrier” which is designed for ventilated facades (Flamro, 2024). These barriers are designed to allow for ventilation but upon interaction with flames and heat, the barrier swells to block flames while insulating against the increased temperature. While these options are available, further research is still required.

Soundproofing

Façades must suppress at least 20db of noise according to the Besluit Bouwwerken Leefomgeving (BBL). To reduce noise there is possibility for implementing a porous surface texture to aid acoustic insulation and reduce sound transmission through the wall. Studies have shown that adding simple perforations in material can improve these aspect (Sailesh et al., 2022). Additionally, the foam-like filling of the inner portion

of the cassette could provide some acoustic insulation. However, for this project the cassette was not tested as a whole but is a promising route of acoustic potential.

Rodents

Although part of nature and part of the ecosystem, rodents, such as rats and mice, are a public health risk and can damage infrastructure. According to BBL, there should be no openings present which are larger than 1cm in residential buildings (Rijkswaterstaat, n.d.). For the implementation of the façade panel, this may require the addition of a mesh (or similar) to the lower panels to stop access by rodents. Additionally, the positioning of panels could be higher off the ground, which would make the panels more difficult to access by rodents.

Thermal performance

The degree of thermal performance required for this product depends on the context that it is applied for. For example, the Rc value (insulation) of the closed parts of a façade has to be at least 4.7 m^2 xK/W only if >25% of the façade is completely stripped and renewed (Rijkswaterstaat, n.d.). However, when installed over the top of an existing wall, this can also be lower (1.3 m^2 xK/W). Additionally, the cassette design incorporates a foam-like insulator which was not included in this project and falls outside the current scope. Because of this, thermal performance is not necessary to implement into the panel skin itself as insulation will be applied in context.

Manufacturing

The manufacturing of the panels is currently a bespoke process that is labour intensive. The key to scaling this up would be to streamline each part of the process and sourcing materials in bulk. Timber and wholesale suppliers can offer Marine grade okoume multiplex for about €27.96 p/m^2 (Houthandel Online, n.d.; Op Maat Zagen, n.d.), or Douglas fir €29.83 p/m^2 (Houthandel Jan Sok, n.d.). These timber types would be preferable for the ideal function of the panel but would have quite an influence on product price.

Companies such as Yart Factory NL can cut the material to shape, if the materials are provided by the client. Following this, the panels need to be processed and assembled in house, which requires a workshop and staff. This would be a significant cost should there be no option to outsource this final part of the production process. However, it could benefit local employment and provide opportunities to tailorize orders. Tailoring each order to the context of the final installation would provide better long-term sustainability as the product can be a best-fit version to suit that local ecosystem. Additionally, providing installation of the product is optional at this stage but would again provide opportunity for further adapting and suitability case by case.

Other hurdles to consider:

In implementing the new façade panels, the consideration for existing systems is also important. In most cases, façade are already somewhat ventilated, therefore the

accumulation of moisture between layers could become problematic if not considered when the façade is altered. To avoid this, the assessment of water and air seals by an engineer is required, as well as to assess the structural integrity/load bearing capacity of existing walls for retrofitting the new facade on top. These inspections and decisions on installation would need to be on a case-by-case basis.

Scalability

This passive cooling façade is designed to be modular for ease of installing, maintaining and deinstalling/recycling. In this sense, the product can be installed to endlessly increase the coverage of the façade as needed, and thanks to the repetitive design it can be used to cover walls of various shapes and sizes whilst maintaining the overall aesthetic and function. Logistically, scalability will be limited by the abovementioned manufacturing limits and resource availabilities. By utilising CNC as the chosen manufacturing method, enhancing the scale of production is fairly straightforward. The difficulty would lie in the final stage of assembly which would require increased labour or an alternative manufacturing/assembly method. As for the façade itself, it can also be adjusted over time to suit client needs. That is, more panels can be installed at later date or even removed or rearranged. This lends the cassette to be very flexible in its scalability.

End of life options

The initial choice for wood as material for our panel was partly based on it being a biodegradable, sustainable and renewable material. Considering that the panel is made using existing standardised materials it can be treated in the same way as other timber construction materials. An improvement lies in the possibility of using better module attachment methods (avoiding glue or using biodegradable adhesives). This would improve the deconstruction of the panel into individual parts for more efficient recycling. The coating used in this project is Cryopaint (from Cryo X Co.), which is made up of about 50-60% Barium sulfate (Cryo X, 2024). This is an insoluble compound that is prone to accumulating in waterways (ECHEMI, 2023), and as such is advised to restrict potential water or soil contamination (Cryo X, 2024). General advice on the material post use is to dispose similar to other industrial waste (this process can differ per region, but existing waste collection companies do exist). Additionally, as stated earlier, the attachment system to the wall itself would be optimised for demounting from the building in the easiest way possible. This lends again to a smoother end of life scenario should panels need to be decommissioned, as it lessens the time and resource constraints on the customer and construction industry.

Comparison with fungal counterpart

The façade panel prototyping was undertaken in parallel with a fungal counterpart. Mycelium based composite (MBC) was also investigated as an alternative material to the timber used in the façade panel. After testing MBC as a possible alternative and future material, there are four key differences when it comes to implementation: weathering, maintenance, manufacture and end of life scenario. These differences are explained below.

Weathering

The MBCs tested were uncoated as the baseline properties were the aspects of interest. However, if this material were to be used in an outdoor context, a coating would be necessary. This is due to heat-pressed MBC being quite prone to absorbing water on contact (See *Functional Skin Report* for more information)– which would not suit rainy or humid weather conditions. Such coatings as “BioSeal” from Imper Shield (the Netherlands) are already available on the market and offer both UV and water protection (Impershield, n.d.). The Cryopaint used on the timber-based prototype has not been assessed for adhesivity on mycelium at this stage but could also serve this function. However, there is a lack of research as to how long this would allow the MBC to last outdoors.

Maintenance

Maintenance would be a major factor in the use of MBC as it would likely have a smaller lifespan if the coating cannot adequately protect the material from degradation. This may require the more frequent replacement of panels however long-term research on MBC and their maintenance requirements is lacking. Since the panels are modular however, this does lend the product the ability to simply be replaced panel by panel as necessary – reducing the need for large scale renewal or maintenance. In this case, perhaps the use of MBC panels would be more suitable for temporary structures such as building sites, festivals or disaster relief housing as these structures are also often modular and are suited to planned degradation after short-term use.

Manufacture

The process of manufacture for MBCs does differ significantly to timber. In the case of timber, a producer can source timber easily from a supplier. In the Netherlands there are not many mycelium suppliers for this type of material, let alone an MBC of high-grade quality. There are some sources of waste stream MBC (that are used in artwork for example) sourced from oyster mushroom farms after harvesting the fruiting bodies (A. Hendriks, persoonlijke communicatie, 2026) (interview with Arne Hedricks, 2026). Since this project was also focussed on the understanding of the material, the MBCs were grown and processed by hand - this takes up to 19 days but can be streamlined. When ready, the material is assembled into a functioning panel quite fast, as it needs only heat pressing into shape and weatherproof coating. Additionally, MBCs can be grown on waste materials, which can in some cases be easier than for timber, as it

has larger waste streams to take from (choice of substrate can vary, giving different options for this aspect whereas timber recycling into high grade material is still lacking industrially and can be more expensive).

End of life

Finally, a benefit of using MBC's is that they can degrade easily and safely in nature. Importantly, this can only be considered a valuable aspect should the coating used also be sufficiently biodegradable. If the coating is not entirely biobased or biodegradable, then there would be some difficulty separating the components for adequate disposal. This aspect requires further investigation as it can have quite an impact to the true biodegradability of the product and what recycling of material is possible.

Conclusion

The successful implementation of this façade panel will require further investigation into the maintenance and end of life scenarios. There are some unknowns both in the disposal of the material and the methods of maintaining an MBC based panel. This is partially due to the fact that in the construction industry, mycelium has not yet become a standard. However, there is serious promise in both the timber and MBC material options, their manufacture and eventual use of the panels in Dutch cities. With more understanding, this product could be implemented at scale as a safe and effective method of passively cooling urban environments and buildings.

Appendix I. Supplementary data

The exact dimensions of the models manufactured for our prototype can be found in the table below

Table 1: *Fusion 360 Models used for the final design*

Aluminium Mold (Model for CNC)	https://a360.co/4yFomTX
Wooden panels (Model for CNC)	https://a360.co/4hJ9jBd
Assembly Jig (Model for 3D-print)	https://a360.co/4hM8A3H
Experimental results section B.2	Github/FunctionalSkin